

**SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4**

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5749 (09-2008)

Received

OCT 20 2016

Well File No.

28655

ND Oil & Gas Division

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

☐ Notice of Intent	Approximate Start Date
☒ Report of Work Done	Date Work Completed September 27, 2016
☐ Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03.	Approximate Start Date

☐ Drilling Prognosis	☐ Spill Report
☐ Redrilling or Repair	☐ Shooting
☐ Casing or Liner	☐ Acidizing
☐ Plug Well	☐ Fracture Treatment
☐ Supplemental History	☒ Change Production Method
☐ Temporarily Abandon	☐ Reclamation
☐ Other	Well is now on pump

Well Name and Number Kline Federal 5300 41-18 14BX							
Footages		Qtr-Qtr	Section	Township	Range		
368 F S L 237 F W L		LOT4	18	153 N	100 W		
Field Baker		Pool Bakken		County McKenzie			

24-HOUR PRODUCTION RATE

Before		After	
Oil	Bbls	Oil	Bbls
Water	Bbls	Water	Bbls
Gas	MCF	Gas	MCF

Name of Contractor(s)			
Address	City	State	Zip Code

DETAILS OF WORK

Effective 09/27/2016 the above referenced well was converted to rod pump.

End of Tubing: 2-7/8" L-80 tubing @ 9620.23'

Pump: 2-1/2" x 2.0" x 24' insert pump @ 9492.97'

Company Oasis Petroleum North America LLC		Telephone Number 281-404-9436	
Address 1001 Fannin, Suite 1500			
City Houston		State TX	Zip Code 77002
Signature 		Printed Name Jennifer Swenson	
Title Regulatory Specialist		Date October 19, 2016	
Email Address jswenson@oasispetroleum.com			

FOR STATE USE ONLY

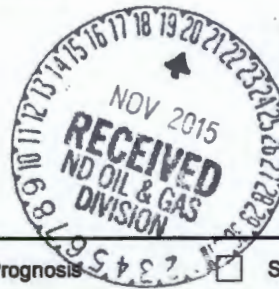
☒ Received	☐ Approved
Date 10-28-2016	
By 	
Title TAYLOR ROTH Engineering Technician	



SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5749 (09-2008)

Well File No.
28655



PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

☐ Notice of Intent	Approximate Start Date
☒ Report of Work Done	Date Work Completed November 15, 2015
☐ Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03.	Approximate Start Date

☐ Drilling Prognosis	☐ Spill Report
☐ Redrilling or Repair	☐ Shooting
☐ Casing or Liner	☐ Acidizing
☐ Plug Well	☐ Fracture Treatment
☐ Supplemental History	☒ Change Production Method
☐ Temporarily Abandon	☐ Reclamation
☐ Other	Well is now on pump

Well Name and Number Kline Federal 5300 41-18 14BX					
Footages 368 F S L 237 F W L		Qtr-Qtr LOT4	Section 18	Township 153 N	Range 100 W
Field Baker	Pool Bakken	County McKenzie			

24-HOUR PRODUCTION RATE			
Before		After	
Oil	Bbls	Oil	Bbls
Water	Bbls	Water	Bbls
Gas	MCF	Gas	MCF

Name of Contractor(s)			
Address	City	State	Zip Code

DETAILS OF WORK

Effective 11/15/2015 the above referenced well is on pump.

End of Tubing: 2-7/8" L-80 tubing @ 1019'

Pump: ESP @ 10019'

Company Oasis Petroleum North America LLC		Telephone Number 281-404-9436	
Address 1001 Fannin, Suite 1500			
City Houston		State TX	Zip Code 77002
Signature 		Printed Name Jennifer Swenson	
Title Regulatory Specialist		Date November 19, 2015	
Email Address jswenson@oasispetroleum.com			

FOR STATE USE ONLY	
☒ Received	☐ Approved
Date 11-30-2015	
By Original Signed By:	
TAYLOR ROTH	
Title Engineering Technician	

**WELL COMPLETION OR RECOMPLETION REPORT - FORM 6**

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 2468 (04-2010)



Well File No. **28655**

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

Designate Type of Completion			
☒ Oil Well	☐ EOR Well	☐ Recompletion	☐ Deepened Well
☐ Gas Well	☐ SWD Well	☐ Water Supply Well	☐ Other:
Well Name and Number Kline Federal 5300 41-18 14BX		Spacing Unit Description Sec. 18/17/19/20 T153N R100W	
Operator Oasis Petroleum North America		Telephone Number (281) 404-9591	
Address 1001 Fannin, Suite 1500		Field Baker	
City Houston		Pool Bakken	
State TX	Zip Code 77002	Permit Type ☐ Wildcat ☒ Development ☐ Extension	

LOCATION OF WELL

At Surface 368 F S L 237 F W L	Qtr-Qtr SWSW	Section 18	Township 153 N	Range 100 W	County McKenzie
Spud Date September 19, 2014	Date TD Reached February 18, 2015	Drilling Contractor and Rig Number Nabors 486		KB Elevation (Ft) 2082	Graded Elevation (Ft) 2057
Type of Electric and Other Logs Run (See Instructions) MWD/GR from KOP to TD					

CASING & TUBULARS RECORD (Report all strings set in well)

Well Bore	String Type	Size (Inch)	Top Set (MD Ft)	Depth Set (MD Ft)	Hole Size (Inch)	Weight (Lbs/Ft)	Anchor Set (MD Ft)	Packer Set (MD Ft)	Sacks Cement	Top of Cement
Surface Hole	Surface	9 5/8	0	2148	13 1/2	36			1210	0
Vertical Hole	Intermediate	7	0	6645	8 3/4	29			797	4500
Vertical Hole	Intermediate	7	6647	10310	8 3/4	32				
Vertical Hole	Intermediate	7	10310	11113	8 3/4	29				
Lateral1	Liner	4 1/2	10244	20645	6	13.5			515	10244

PERFORATION & OPEN HOLE INTERVALS

Well Bore	Well Bore TD Drillers Depth (MD Ft)	Completion Type	Open Hole/Perforated Interval (MD,Ft)		Kick-off Point (MD Ft)	Top of Casing Window (MD Ft)	Date Perf'd or Drilled	Date Isolated	Isolation Method	Sacks Cement
Lateral1	20650	Perforations	11113	20645	10228		06/09/2015			

PRODUCTION

Current Producing Open Hole or Perforated Interval(s), This Completion, Top and Bottom, (MD Ft) Lateral 1- 11113' to 20645'							Name of Zone (If Different from Pool Name)		
Date Well Completed (SEE INSTRUCTIONS) August 22, 2015			Producing Method flowing		Pumping-Size & Type of Pump		Well Status (Producing or Shut-In) Producing		
Date of Test 08/24/2015	Hours Tested 24	Choke Size 34 /64	Production for Test	Oil (Bbbls) 1270	Gas (MCF) 1209	Water (Bbbls) 4111	Oil Gravity-API (Corr.) °	Disposition of Gas Sold	
Flowing Tubing Pressure (PSI)		Flowing Casing Pressure (PSI) 1800		Calculated 24-Hour Rate	Oil (Bbbls) 1270	Gas (MCF) 1209	Water (Bbbls) 4111	Gas-Oil Ratio 952	

PLUG BACK INFORMATION

[illegible][illegible]

Top (Ft)	Bottom (Ft)	Formation	Top (Ft)	Bottom (Ft)	Formation

Test Date	Formation	Top (Ft)	Bottom (Ft)	BH Temp (°F)	CL ppm	H2S ppm	Shut-in 1 (PSIG)	Shut-in 2 (PSIG)
Drill Pipe Recovery								
Sample Chamber Recovery								
Test Date	Formation	Top (Ft)	Bottom (Ft)	BH Temp (°F)	CL ppm	H2S ppm	Shut-in 1 (PSIG)	Shut-in 2 (PSIG)
Drill Pipe Recovery								
Sample Chamber Recovery								
Test Date	Formation	Top (Ft)	Bottom (Ft)	BH Temp (°F)	CL ppm	H2S ppm	Shut-in 1 (PSIG)	Shut-in 2 (PSIG)
Drill Pipe Recovery								
Sample Chamber Recovery								
Test Date	Formation	Top (Ft)	Bottom (Ft)	BH Temp (°F)	CL ppm	H2S ppm	Shut-in 1 (PSIG)	Shut-in 2 (PSIG)
Drill Pipe Recovery								
Sample Chamber Recovery								
Test Date	Formation	Top (Ft)	Bottom (Ft)	BH Temp (°F)	CL ppm	H2S ppm	Shut-in 1 (PSIG)	Shut-in 2 (PSIG)
Drill Pipe Recovery								
Sample Chamber Recovery								
Test Date	Formation	Top (Ft)	Bottom (Ft)	BH Temp (°F)	CL ppm	H2S ppm	Shut-in 1 (PSIG)	Shut-in 2 (PSIG)
Drill Pipe Recovery								
Sample Chamber Recovery								

Well Specific Stimulations

Date Stimulated 06/09/2015	Stimulated Formation Bakken	Top (Ft) 11113	Bottom (Ft) 20645	Stimulation Stages 50	Volume 214895	Volume Units Barrels
Type Treatment Sand Frac	Acid %	Lbs Proppant 9543260	Maximum Treatment Pressure (PSI) 9411		Maximum Treatment Rate (BBLS/Min) 41.0	
Details 100 Mesh White: 235860 40/70 White: 1249960 20/40 White: 6292960 20/40 Resin Coated: 1364480						

Date Stimulated	Stimulated Formation	Top (Ft)	Bottom (Ft)	Stimulation Stages	Volume	Volume Units
Type Treatment	Acid %	Lbs Proppant	Maximum Treatment Pressure (PSI)		Maximum Treatment Rate (BBLS/Min)	
Details						

Date Stimulated	Stimulated Formation	Top (Ft)	Bottom (Ft)	Stimulation Stages	Volume	Volume Units
Type Treatment	Acid %	Lbs Proppant	Maximum Treatment Pressure (PSI)		Maximum Treatment Rate (BBLS/Min)	
Details						

Date Stimulated	Stimulated Formation	Top (Ft)	Bottom (Ft)	Stimulation Stages	Volume	Volume Units
Type Treatment	Acid %	Lbs Proppant	Maximum Treatment Pressure (PSI)		Maximum Treatment Rate (BBLS/Min)	
Details						

Date Stimulated	Stimulated Formation	Top (Ft)	Bottom (Ft)	Stimulation Stages	Volume	Volume Units
Type Treatment	Acid %	Lbs Proppant	Maximum Treatment Pressure (PSI)		Maximum Treatment Rate (BBLS/Min)	
Details						

ADDITIONAL INFORMATION AND/OR LIST OF ATTACHMENTS

--	--	--

I hereby swear or affirm that the information provided is true, complete and correct as determined from all available records.	Email Address jswenson@oasispetroleum.com		Date 09/18/2015
	Signature 	Printed Name Jennifer Swenson	Title Regulatory Specialist



AUTHORIZATION TO PURCHASE AND TRANSPORT OIL FROM LEASE -- Form 8
INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5698 (03-2000)



Well File No.
28655
NDIC CTB No.
To be assigned

228654

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND FOUR COPIES.

Well Name and Number KLINE FEDERAL 5300 41-18 14BX	Qtr-Qtr LOT4	Section 18	Township 153	Range 100	County Mckenzie
--	-----------------	---------------	-----------------	--------------	--------------------

Operator Oasis Petroleum North America LLC	Telephone Number (281) 404-9573	Field BAKER
--	---	-----------------------

Address 1001 Fannin, Suite 1500	City Houston	State TX	Zip Code 77002
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Name of First Purchaser Oasis Petroleum Marketing LLC	Telephone Number (281) 404-9627	% Purchased 100%	Date Effective August 22, 2015
---	---	----------------------------	--

Principal Place of Business 1001 Fannin, Suite 1500	City Houston	State TX	Zip Code 77002
---	------------------------	--------------------	--------------------------

Field Address	City	State	Zip Code
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Transporter Hiland Crude, LLC	Telephone Number (580) 616-2058	% Transported 75%	Date Effective August 22, 2015
---	---	-----------------------------	--

Address P.O. Box 3886	City Enid	State OK	Zip Code 73702
---------------------------------	---------------------	--------------------	--------------------------

The above named producer authorizes the above named purchaser to purchase the percentage of oil stated above which is produced from the lease designated above until further notice. The oil will be transported by the above named transporter.

Other First Purchasers Purchasing From This Lease	% Purchased	Date Effective
---	-------------	----------------

Other First Purchasers Purchasing From This Lease	% Purchased	Date Effective
---	-------------	----------------

Other Transporters Transporting From This Lease	% Transported	Date Effective
---	---------------	----------------

Power Crude Transport	25%	August 22, 2015
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Other Transporters Transporting From This Lease	% Transported	Date Effective
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		August 22, 2015
--	--	------------------------

Comments

I hereby swear or affirm that the information provided is true, complete and correct as determined from all available records.	Date August 22, 2015
--	--------------------------------

Signature <i>Dina Barron</i>	Printed Name Dina Barron	Title Mktg. Contracts Administrator
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Above Signature Witnessed By:

Signature <i>Jeremy Harris</i>	Printed Name Jeremy Harris	Title Marketing Scheduler
-----------------------------------	--------------------------------------	-------------------------------------

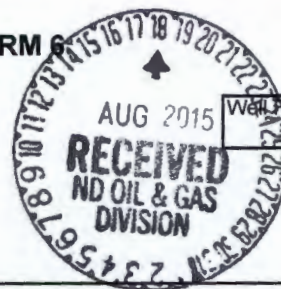
FOR STATE USE ONLY

Date Approved SEP 18 2015
By <i>Eric Roberson</i>
Title Oil & Gas Production Analyst



WELL COMPLETION OR RECOMPLETION REPORT - FORM 6

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 2468 (04-2010)



Well File No. **28655**

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.

PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

Designate Type of Completion			
☒ Oil Well	☐ EOR Well	☐ Recompletion	☐ Deepened Well
☐ Gas Well	☐ SWD Well	☐ Water Supply Well	☐ Other:
Well Name and Number Kline Federal 5300 41-18 14BX		Spacing Unit Description Sec. 18/17/19/20 T153N R100W	
Operator Oasis Petroleum North America		Telephone Number (281) 404-9591	
Address 1001 Fannin, Suite 1500		Field Baker	
City Houston		State TX	
Zip Code 77002		Permit Type ☐ Wildcat ☒ Development ☐ Extension	

LOCATION OF WELL

At Surface 368 F S L	237 F W L	Qtr-Qtr SWSW	Section 18	Township 153 N	Range 100 W	County McKenzie
Spud Date September 19, 2014	Date TD Reached February 18, 2015	Drilling Contractor and Rig Number Nabors 486		KB Elevation (Ft) 2082	Graded Elevation (Ft) 2057	
Type of Electric and Other Logs Run (See Instructions) MWD/GR from KOP to TD						

CASING & TUBULARS RECORD (Report all strings set in well)

Well Bore	String Type	Size (Inch)	Top Set (MD Ft)	Depth Set (MD Ft)	Hole Size (Inch)	Weight (Lbs/Ft)	Anchor Set (MD Ft)	Packer Set (MD Ft)	Sacks Cement	Top of Cement
Surface Hole	Surface	9 5/8	0	2148	13 1/2	36			1210	0
Vertical Hole	Intermediate	7	0	6645	8 3/4	29			797	4500
Vertical Hole	Intermediate	7	6647	10310	8 3/4	32				
Vertical Hole	Intermediate	7	10310	11113	8 3/4	29				
Lateral1	Liner	4 1/2	10244	20645	6	13.5			515	10244

PERFORATION & OPEN HOLE INTERVALS

Well Bore	Well Bore TD Drillers Depth (MD Ft)	Completion Type	Open Hole/Perforated Interval (MD, Ft)	Kick-off Point (MD Ft)	Top of Casing Window (MD Ft)	Date Perf'd or Drilled	Date Isolated	Isolation Method	Sacks Cement
Lateral1	20650	Perforations	Top Bottom	10228					

PRODUCTION

Current Producing Open Hole or Perforated Interval(s), This Completion, Top and Bottom, (MD Ft) Lateral 1-						Name of Zone (If Different from Pool Name)				
Date Well Completed (SEE INSTRUCTIONS)			Producing Method		Pumping-Size & Type of Pump			Well Status (Producing or Shut-In)		
Date of Test	Hours Tested	Choke Size /64	Production for Test		Oil (Bbbls)	Gas (MCF)	Water (Bbbls)	Oil Gravity-API (Corr.)	Disposition of Gas	
Flowing Tubing Pressure (PSI)			Flowing Casing Pressure (PSI)		Calculated 24-Hour Rate	Oil (Bbbls)	Gas (MCF)	Water (Bbbls)	Gas-Oil Ratio	

GEOLOGICAL MARKERS

[illegible]

PLUG BACK INFORMATION

[illegible]

CORES CUT

Top (Ft)	Bottom (Ft)	Formation	Top (Ft)	Bottom (Ft)	Formation

Drill Stem Test

Test Date	Formation	Top (Ft)	Bottom (Ft)	BH Temp (°F)	CL ppm	H2S ppm	Shut-in 1 (PSIG)	Shut-in 2 (PSIG)
Drill Pipe Recovery								
Sample Chamber Recovery								
Test Date	Formation	Top (Ft)	Bottom (Ft)	BH Temp (°F)	CL ppm	H2S ppm	Shut-in 1 (PSIG)	Shut-in 2 (PSIG)
Drill Pipe Recovery								
Sample Chamber Recovery								
Test Date	Formation	Top (Ft)	Bottom (Ft)	BH Temp (°F)	CL ppm	H2S ppm	Shut-in 1 (PSIG)	Shut-in 2 (PSIG)
Drill Pipe Recovery								
Sample Chamber Recovery								
Test Date	Formation	Top (Ft)	Bottom (Ft)	BH Temp (°F)	CL ppm	H2S ppm	Shut-in 1 (PSIG)	Shut-in 2 (PSIG)
Drill Pipe Recovery								
Sample Chamber Recovery								
Test Date	Formation	Top (Ft)	Bottom (Ft)	BH Temp (°F)	CL ppm	H2S ppm	Shut-in 1 (PSIG)	Shut-in 2 (PSIG)
Drill Pipe Recovery								
Sample Chamber Recovery								
Test Date	Formation	Top (Ft)	Bottom (Ft)	BH Temp (°F)	CL ppm	H2S ppm	Shut-in 1 (PSIG)	Shut-in 2 (PSIG)
Drill Pipe Recovery								
Sample Chamber Recovery								

Well Specific Stimulations

Date Stimulated	Stimulated Formation	Top (Ft)	Bottom (Ft)	Stimulation Stages	Volume	Volume Units Barrels
Type Treatment Sand Frac	Acid %	Lbs Proppant	Maximum Treatment Pressure (PSI)		Maximum Treatment Rate (BBLS/Min)	
Details						

Date Stimulated	Stimulated Formation	Top (Ft)	Bottom (Ft)	Stimulation Stages	Volume	Volume Units
Type Treatment	Acid %	Lbs Proppant	Maximum Treatment Pressure (PSI)		Maximum Treatment Rate (BBLS/Min)	
Details						

Date Stimulated	Stimulated Formation	Top (Ft)	Bottom (Ft)	Stimulation Stages	Volume	Volume Units
Type Treatment	Acid %	Lbs Proppant	Maximum Treatment Pressure (PSI)		Maximum Treatment Rate (BBLS/Min)	
Details						

Date Stimulated	Stimulated Formation	Top (Ft)	Bottom (Ft)	Stimulation Stages	Volume	Volume Units
Type Treatment	Acid %	Lbs Proppant	Maximum Treatment Pressure (PSI)		Maximum Treatment Rate (BBLS/Min)	
Details						

Date Stimulated	Stimulated Formation	Top (Ft)	Bottom (Ft)	Stimulation Stages	Volume	Volume Units
Type Treatment	Acid %	Lbs Proppant	Maximum Treatment Pressure (PSI)		Maximum Treatment Rate (BBLS/Min)	
Details						

ADDITIONAL INFORMATION AND/OR LIST OF ATTACHMENTS

This is a preliminary completion report. A supplemental report will be filed upon first production of the well.

I hereby swear or affirm that the information provided is true, complete and correct as determined from all available records.	Email Address	Date
	jswenson@oasispetroleum.com	08/17/2015
Signature	Printed Name	Title
	Jennifer Swenson	Regulatory Specialist

Industrial Commission of North Dakota
Oil and Gas Division

Well or Facility No

28655

Verbal Approval To Purchase and Transport Oil

Tight Hole **Yes**

OPERATOR

Operator	Representative	Rep Phone
OASIS PETROLEUM NORTH AMERICA LL	Todd Hanson	(701) 577-1632

WELL INFORMATION

Well Name						Inspector
KLINE FEDERAL 5300 41-18 14BX						Richard Dunn
Well Location	QQ	Sec	Twp	Rng		County
	LOT4	18	153 N	100 W		MCKENZIE
Footages	368	Feet From the	S	Line		Field
	237	Feet From the	W	Line		BAKER
						Pool
						BAKKEN
Date of First Production Through Permanent Wellhead						This Is Not The First Sales

PURCHASER / TRANSPORTER

Purchaser	Transporter
Kinder Morgan	Kinder Morgan

TANK BATTERY

Central Tank Battery Number : 228654-01
--

SALES INFORMATION This Is Not The First Sales

ESTIMATED BARRELS TO BE SOLD		ACTUAL BARRELS SOLD	DATE
15000	BBLS	BBLS	
	BBLS	BBLS	
	BBLS	BBLS	
	BBLS	BBLS	
	BBLS	BBLS	
	BBLS	BBLS	
	BBLS	BBLS	
	BBLS	BBLS	
	BBLS	BBLS	

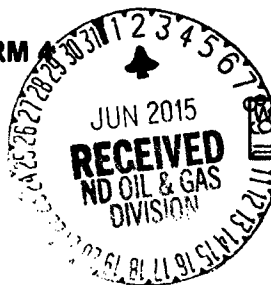
DETAILS

Must E-Mail or Call Inspector at 701-770-3554/rsdunn@nd.gov on first date of sales and report amount sold, date sold, and first date of production through the permanent wellhead. Must also forward Forms 6 & 8 to State prior to reaching 15000 Bbl estimate or no later than required time frame for submitting those forms.

Start Date **8/25/2015**
Date Approved **8/25/2015**
Approved By **Richard Dunn**

**SUNDRY NOTICES AND REPORTS ON WELLS - FORM**

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 8749 (08-2008)



Well File No.

28665

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

☒ Notice of Intent	Approximate Start Date June 2, 2015
☐ Report of Work Done	Date Work Completed
☐ Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03.	Approximate Start Date

☐ Drilling Prognosis	☐ Spill Report
☐ Redrilling or Repair	☐ Shooting
☐ Casing or Liner	☐ Acidizing
☐ Plug Well	☐ Fracture Treatment
☐ Supplemental History	☐ Change Production Method
☐ Temporarily Abandon	☐ Reclamation
☒ Other	Change well status to CONFIDENTIAL

Well Name and Number Kline Federal 5300 41-18 14BX <i>Lot 4</i>					
Footages 368 F S L	237 F W L	Qtr-Qtr SWSW	Section 18	Township 153 N	Range 100 W
Field Baker	Pool BAKKEN	County McKenzie			

24-HOUR PRODUCTION RATE			
Before		After	
Oil	Bbls	Oil	Bbls
Water	Bbls	Water	Bbls
Gas	MCF	Gas	MCF

Name of Contractor(s)			
Address	City	State	Zip Code

DETAILS OF WORK

Effective Immediately, we request CONFIDENTIAL STATUS for the above referenced well.

This well has not been completed

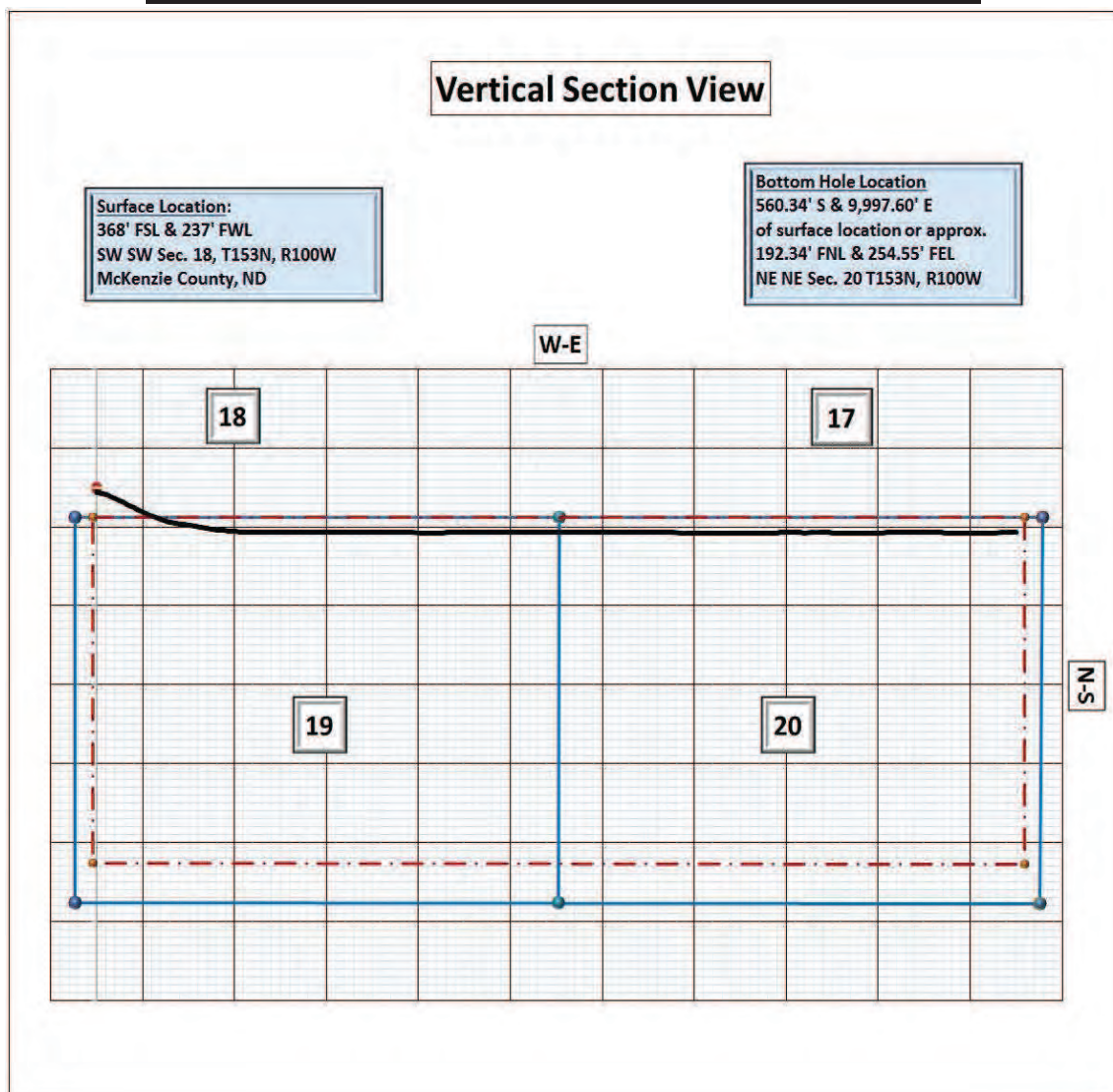
OFF CONFIDENTIAL 12/02/15.

Company Oasis Petroleum North America LLC		Telephone Number 281-404-9436	
Address 1001 Fannin, Suite 1500			
City Houston		State TX	Zip Code 77002
Signature <i>[Signature]</i>	Printed Name Jennifer Swenson		
Title Regulatory Specialist	Date June 2, 2015		
Email Address jswenson@oasispetroleum.com			

FOR STATE USE ONLY	
☐ Received	☒ Approved
Date 6/03/15	
By <i>Alice D. Wehler</i>	
Title Engineering Technician	

Oasis Petroleum North America

Kline Federal 5300 41-18 14BX



Services Performed For:

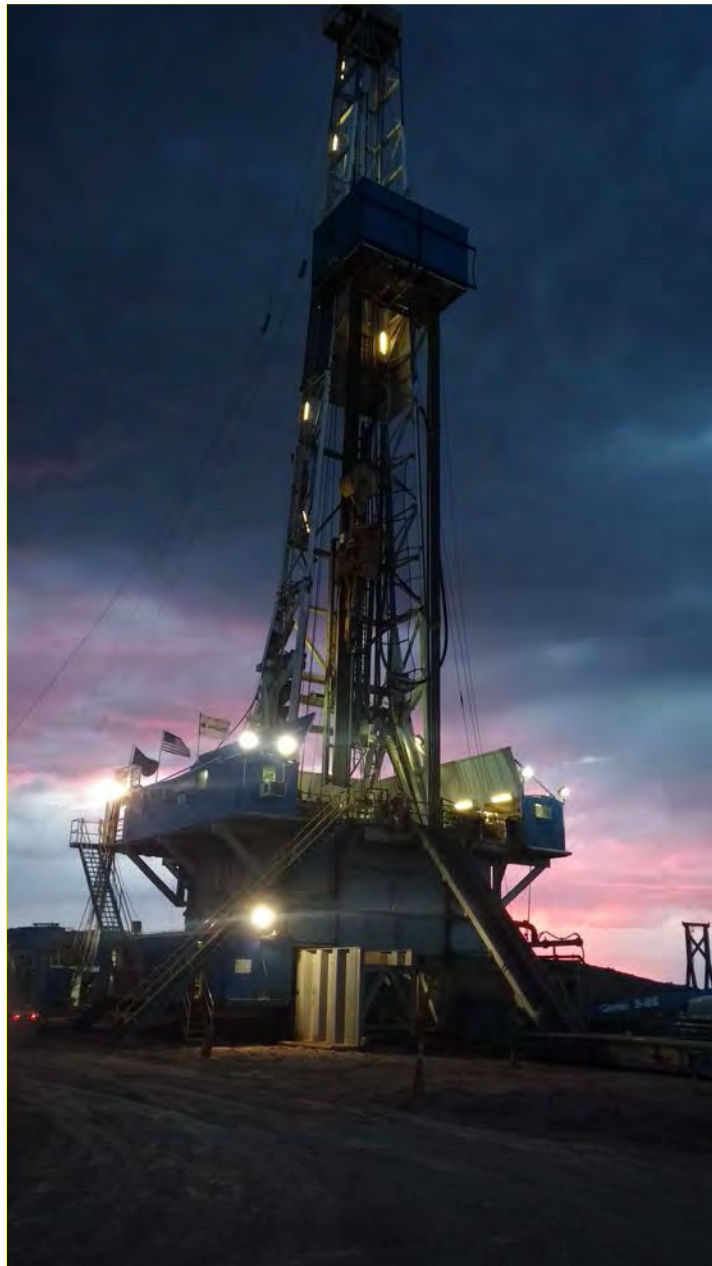
Andrew Nelson, Brendan Hargrove
Oasis Petroleum North America
1001 Fannin Suite 1500
Houston, TX 77002

Onsite Geology Performed by:

Annika Tostengard, Bob Witmer, Ryan Cox, Rob Semple
RPM Geologic
geology@rpmconsultinginc.com
(303) 595-7625

WELL EVALUATION

Oasis Petroleum
Kline Federal 5300 41-18 14BX
SW SW Sec. 18, T153N, R100W
Williams County, North Dakota



Synopsis

Oasis Petroleum *Kline Federal 5300 41-18 14BX* [SW SW Sec. 18 T153N R100W] is located ~6 miles South of the town of Williston, North Dakota. *Kline Federal 5300 41-18 14BX* is situated within the Baker Field. A single lateral leg trending 90° was proposed to be drilled from the SW SW corner of Section 18 to the NE NE corner of Section 20 targeting the porous sandstone and silty sandstone of the Middle Bakken.

Control Wells

Two completed wells were used as the control offsets to the *Kline Federal 5300 41-18 14BX*.

The primary offset considered was the *Kline Federal 5300 41-18 10B* [SW SW Sec. 18 T153N, R100W] which is a horizontal well drilled by Oasis Petroleum. *Kline Federal 5300 41-18 10B2* is located approximately 33 feet north of the *Kline Federal 5300 41-18 14BX*.

The other offset considered was the *Lynn 1* [NE SW Sec. 23 T153N, R100W] which is located approximately 4 miles East of the *Kline Federal 5300 41-18 14BX*.

During the curve, the gamma ray that was produced was constantly compared to the *Kline Federal 5300 41-18 10B* and the *Lynn 1* wells. Gamma ray was used to help determine a proper landing depth.

To assist in the choosing of the landing target, an isopach table (Table 1) was constructed to measure the distance of formation tops to the target depth determined from the *Kline Federal 5300 41-18 10B* and *Lynn 1* offset wells. Comparison of gamma ray markers showed the most consistent isopach for up-hole formations to be the Base of the Last Salt.

<i>Control Well</i>				
Operator:	Oasis Petroleum			
Well Name:	<i>Kline Federal 5300 41-18 10B</i>			
Location:	SW SW Sec. 18 T153N R100W			
	McKenzie Co.,ND			
Elevation:	KB: 2,082'			
Formation/Zone	Est. TVD	MSL Datum	Thickness	Dist to target
Kibbey Lime	8,382	-6,300'	178'	2,358'
Charles Salt	8,560	-6,478'	646'	2,180'
Base Last Salt	9,206	-7,124'	222'	1,534'
Mission Canyon	9,428	-7,346'	547'	1,312'
Lodgepole	9,975	-7,893'	730'	765'
False Bakken	10,705	-8,623'	7'	35'
Upper Bakken Shale	10,712	-8,630'	18'	28'
Middle Bakken	10,730	-8,648'	-	10'

<i>Control Well</i>				
Operator:	Oasis Petroleum			
Well Name:	<i>Lynn 1</i>			
Location:	NE SW Sec. 23 T153N R100W			
	McKenzie Co.,ND			
Elevation:	KB: 2,142'			
Formation/Zone	Est. TVD	MSL Datum	Thickness	Dist to target
Kibbey Lime	8,368'	-6,226'	150'	2,370'
Charles Salt	8,518'	-6,376'	696'	2,220'
Base Last Salt	9,214'	-7,072'	226'	1,524'
Mission Canyon	9,440'	-7,298'	548'	1,298'
Lodgepole	9,988'	-7,846'	715'	750'
False Bakken	10,703'	-8,561'	11'	35'
Upper Bakken Shale	10,714'	-8,572'	14'	24'
Middle Bakken	10,728'	-8,586'	-	10'

Table 1: Distance from Formation tops in the control wells were utilized to determine an initial drilling target.

Geologic Assessment

Methods

Geologic supervision of *Kline Federal 5300 41-18 14BX* was provided by experienced RPM well site geologists. Gas and chromatograph levels were measured using NOV WellSight Gas Detection real time gas detector and chromatograph system. The WellSight gas detection system uses non-dispersive infrared and chemical sensor gas detection to evaluate gases liberated from the formation by the drill bit and carried to the surface by the drilling fluid.

The gas detector was interfaced with a RigWatch electronic data recorder system. RigWatch provided drill rate, on-off bottom and pump strokes to the gas detector and received total gas information from the gas detector for viewing on location and remotely.

Under the direction of RPM well site geologists, rig crews were instructed to catch lagged drill cutting samples at 30' intervals in the vertical hole and curve build from 8,350' to 11,200' MD and 50' samples from 11,200' to TD (20,650').

Sampled drill cuttings were examined wet and dry under a trinocular microscope using both plain (broad spectrum) and transmitted light. Cuttings were evaluated for hydrocarbon "cut" by immersion in Acetone and inspection under a UV fluoroscope. 10% hydrochloric acid and alizarin red were used to determine the calcareous and dolomitic content of rocks and cementing.

RPM Geologic, Inc. (RPM) well site geologists also closely examined MWD gamma-ray information and penetration rates to aid in steering decisions and dip degree estimations.

Vertical Operations

Overview

The *Kline Federal 5300 41-18 14BX* was spud for surface drilling on 9 September 2014 utilizing Nabors 486 as the drilling contractor. Prior to commencement of RPM mud logging services, a 13 1/2" hole was drilled with fresh water to depth of 2,150' and isolated with 9 5/8" 54# J-55 casing cemented from 2,148' to surface. Due to a loss circulation zone present in the **Dakota** formation at the subject well location, a stage of intermediate casing was also set at 6,150' MD.

RPM well site geologists began logging the vertical interval at 8,350'.

One PDC bit was used to drill out of the first intermediate casing to vertical TD. PDC bit #3 (8 3/4" Varel R616PDG2UX) was used to drill from 6,150' to 10,288' and averaged an ROP of 62.70 ft/hr in 66.0 hours of use.

Diesel invert drilling fluid with a mud weight ranging from 9.95-11.40 ppg was used for the remainder of the vertical hole and in the curve builds sections. The vertical interval was drilled to a kick off point (KOP) of 10,288', at which point vertical operations were ceased.

Lithology

The top of the **Kibbey** [Mississippian - Madison Group] was picked at 8,382' MD/TVD (-6,300' MSL), 74' low to the *Kline Federal 5300 41-18 10B*. Samples from this interval (Figure 1) were described as:

LIMESTONE: wackestone, medium gray, friable to hard, finely crystalline, blocky, earthy to sucrosic, trace visible porosity



Figure 1: Photograph of Limestone seen in sample from the Kibbey formation.

The top of the **Charles Salt** [Mississippian - Madison Group] was logged at 8,564' MD/TVD (-6,482' MSL), 106' low to the *Kline Federal 5300 41-18 10B*. The **Base Last Salt** was drilled at 9,205' MD/TVD (-7,123' MSL), 51' low to the *Kline Federal 5300 41-18 10B*. Samples from this interval (Figure 2) were described as:

SALT: translucent, trace off white, crystalline, firm, crystalline texture, no visible porosity, no visible oil stain

LIMESTONE: mudstone, light gray to medium gray, off white to cream, microcrystalline, soft to fair, chalky to crystalline texture, trace intercrystalline porosity, slightly light brown oil stain

ANHYDRITE: off white, cream, soft to firm, amorphous texture, no visible porosity



Figure 2: Photograph of Salt seen in sample from the Charles formation.

The top of the Mission Canyon formation [Mississippian - Madison Group] was reached at 9,428' MD/TVD (-7,346' MSL), 48' low to the *Kline Federal 5300 41-18 10B*. Samples from this interval (Figure 3) were described as:

LIMESTONE: mudstone, light gray to medium gray, occasional off white to cream, firm to hard, microcrystalline, chalky to earthy texture, trace sparry calcite, possible intercrystalline porosity

DOLOMITE LIMESTONE: mudstone, dark gray to medium gray, off white in part, firm to hard, microcrystalline, occasional sparry calcite, trace intercrystalline porosity, slightly light brown spotty oil stain



Figure 3: Photograph of limestone and argillaceous limestone from the Mission Canyon formation.

The top of the **Lodgepole** formation [Mississippian - Madison Group] was logged at 9,976' MD/TVD (-7,894' MSL), 48' low to the *Kline Federal 5300 41-18 10B*. Samples from this interval (Figure 4) were described as:

ARGILLACEOUS LIMESTONE: wackestone, medium gray brown to light gray brown, firm, crystalline texture, trace sparry calcite, carbonaceous, possible intercrystalline porosity, possible slightly light brown spotty oil stain



*Figure 4: Photograph of argillaceous limestone from **Lodgepole** formation.*

Directional Operations

Gyrodata provided equipment and personnel for MWD teams, while RPM Consulting provided directional personnel. RPM geologists worked closely with RPM directional as well as Gyrodata MWD teams to formulate steering decisions to maximize the footage of borehole in the pay zone.

Curve Build

Overview

Two PDC bits were used to build the curve section. Bit #4 was an 8 ¾" Varel A09457 bit used to drill the curve from a measured depth of 10,288' to 10,422' in 16.75 hours, averaging 8.00 ft/hr. Bit #5 was an 8 ¾" Security MMD55M bit used to finish the curve from a measured depth of 10,422' to 11,125' in 23 hours, averaging 36.39 ft/hr. 7" 32 lbs casing was set to 11,117' MD upon completion of the curve.

A Chart was constructed for the curve (Chart 1) that compared the isopach of certain easily recognizable gamma markers from the **Kibbey Lime** to **Middle Bakken** on the control wells to the *Kline Federal 5300 41-18 14BX* as marker picks were made. This chart shows the **False Bakken** top being the closest in thickness to target based on the offset well, and was the main point used to make changes to the target landing depth during the curve. The curve was completed at a measured depth of 11,125' and TVD of 10,748'.

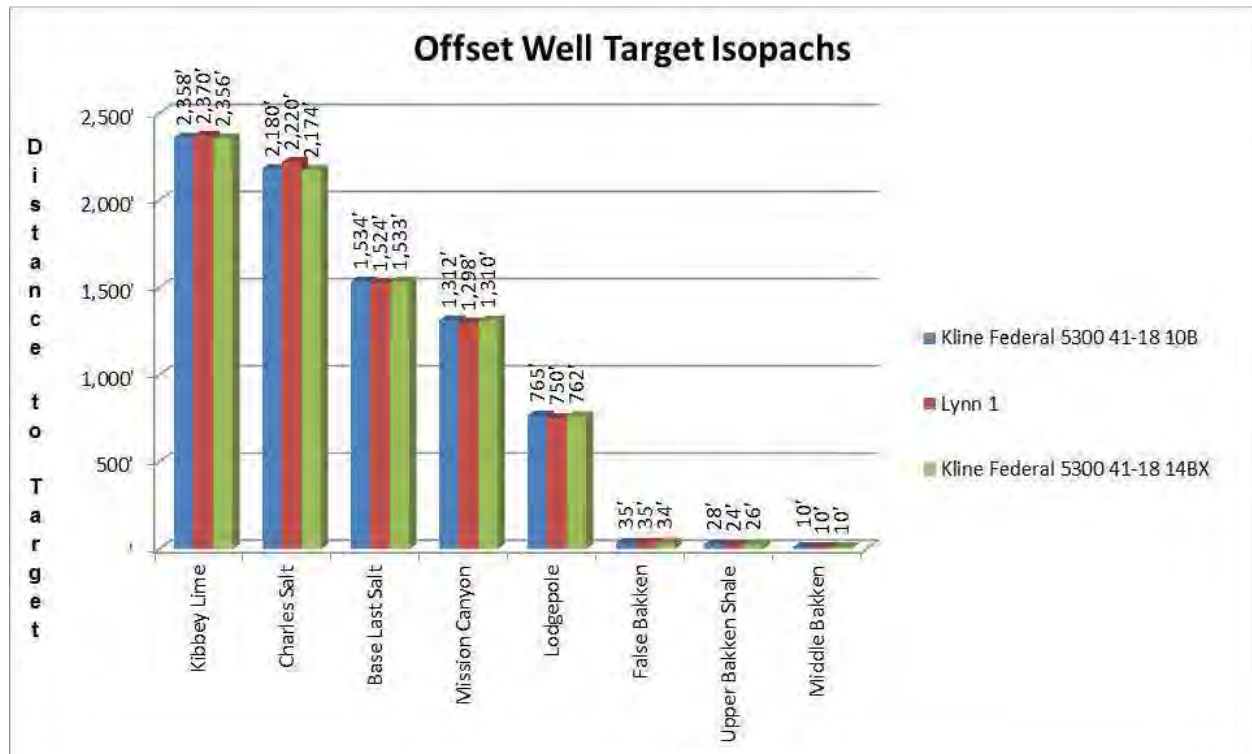


Chart 1: Comparing distances from gamma markers on the offset wells to markers in gamma in the curve.

Lithology

The top of the “**False Bakken**” [Mississippian / Devonian] was drilled at a measured depth of 10,826' (TVD 10,704') (-8,622' MSL) 61' low to the *Kline Federal 5300 41-18 10B*. Due to the low angle of penetration the “**False Bakken**” was not largely observed in sample.

The *Upper Shale* of the **Bakken** formation [Mississippian / Devonian] was drilled at 10,847' MD (10,712' TVD) (-8,630' MSL) with sample returns of black, carbonaceous, and petroliferous shale (Figure 5) characterized by gamma ray values in excess of 255 API counts. The *Upper Shale* was 58' low to the *Kline Federal 5300 41-18 10B*. Samples were described as:

SHALE: black, firm, earthy to waxy texture, pyrite, carbonaceous, platy, visible fracture porosity



Figure 5: Photograph sample from the Upper Bakken Shale.

The *Middle Member* of the **Bakken** formation [Devonian] was penetrated at 10,895' MD, (10,728' TVD) (-8,646' MSL), 60' low to the *Kline Federal 5300 41-18 10B*. The *Middle Member* contained layers of sandstone and silty sandstones with varying amounts of porosity and oil stain (Figure 6). Samples from the *Middle Member* were described as:

SILTY SANDSTONE: common salt and pepper, occasional light gray to medium gray, visible white, very fine grained, soft to firm, moderately to well sorted, rounded to subrounded, moderately to poorly cemented, visible calcareous filled fractures, possible fracture porosity, common intergranular porosity, visible light brown spotty to even oil stain



Figure 6: Photograph of sandstone and silty sandstone from the middle member of the Bakken formation.

Lateral

Overview

Drilling fluid consisting of open system salt water brine (9.9 - 10.7 ppg) was used while drilling the lateral section up to the TD of 20,650'. One 6" PDC bits was used to drill the lateral with no sidetracks required.

Bit #6 (Reed SK516) was a PDC bit used to drill the lateral from 11,125' to TD of 20,650' and averaged an ROP of 64.36 over 148 hours of use. One trip at 11,380' MD was needed for a new motor.

Drilling operations were concluded at 11:15 on Feb 17, 2015 at a bottom hole location of 560.34' S & 9,997.60' E of surface location or approx. 192.34' FNL & 254.55' FEL, NE NE Sec. 20, T153N, R100W.

Lithology

To aid in communications while drilling the lateral, a type log was created using a common gamma signature from the original offset well for the pad, the *Kline Federal 5300 41-18 13TX* (Diagram 1). The nondescript gamma "E" marker (80 api – 90 api) was the main reliable marker. The low shoal between "C" and "D" appeared to move around a few feet with zone so was only relied on to indicate the bit was getting high in zone.

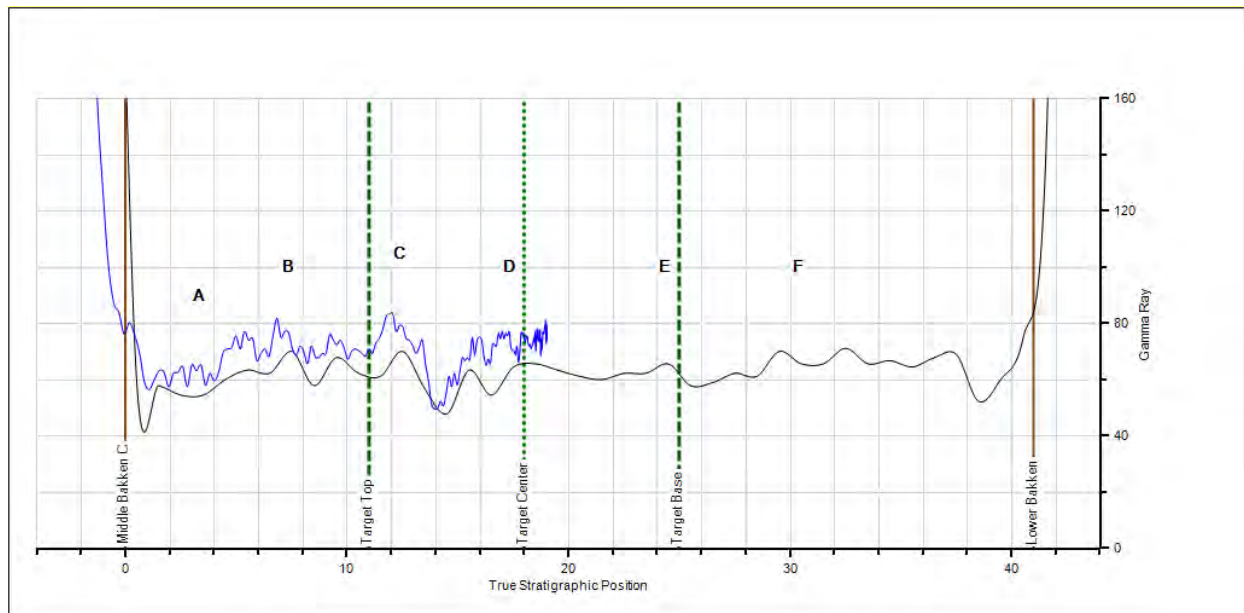
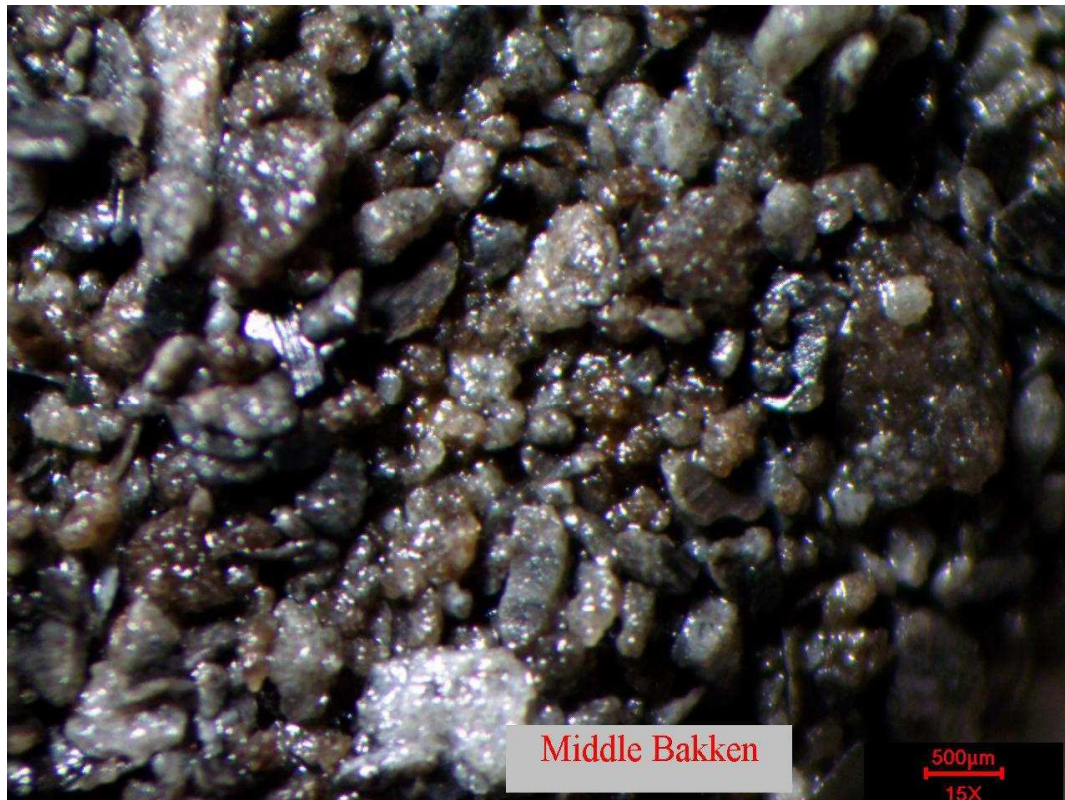


Diagram 1: Type log created from gamma log of the offset well Kline Federal 5300 41-18 13TX.

The target Zone of the **Middle Bakken** consisted of a silty sandstone with intergranular porosity, moderate gas, and moderate to abundant oil show (Figure 7). Samples toward the mid-end of well contained significant quantities of pumped lube. Samples from the target Zone were generally described as:

SLTY SANDSTONE: light gray, md gray, cream, white, friable, firm, fine grained, sub spherules, sub rounded, moderately well sorted, calcareous cement, moderately cement quality, intergranular porosity, spotty brown oil stain, moderately fast streaming milky yellow cut



*Figure 7: Photograph of sandstone from “Target Zone” on Diagram 1
6” Bit & Salt Water Brine*

Gas and hydrocarbon Shows

It was found that the gas and hydrocarbon shows were a decent indicator of placement in Zone throughout the lateral. Gas remained between 100 and 1,000 units while in the **Middle Bakken** member with increased gas as the wellbore was higher in zone.

The NOV Gas Watch showed components of methane and propane present through the entire lateral. Sample cuttings throughout the lateral showed oil stain and a fast white diffuse cut when submerged in Acetone solution and observed with a UV fluoroscope.

Summary

- 1) The *Kline Federal 5300 41-18 14BX* was initially spud for surface drilling on 19th of Sept 2014 in the Baker field of McKenzie County, North Dakota. The well was spud by Nabors 486 and the vertical hole was drilled to a total depth (KOP) of 10,288'.
- 2) A mud program consisted of diesel invert (9.95-11.4 ppg) after surface casing through the curve build sections. The drilling mud was successful in maintaining stable hole conditions and minimizing washout through the salt intervals. Brine water with a weight of 9.9-10.7 ppg was used during lateral drilling operations to maintain hydrostatic balance until a TD of 20,650'.
- 3) One 8 ¾" Varel PDC bit was used to drill out of the intermediate **Dakota** casing to a KOP of 10,288'. Two 8 ¾" PDC bits were used to drill the curve to 11,125' MD. One 6" PDC bit was used to drill the lateral to 20,650' TD.
- 4) The lateral was targeting the **Middle Bakken** member ranging from about 11' below the base of the **Upper Bakken Shale** down to approximately 18' above the **Lower Bakken Shale**.
- 5) Gas levels between 100 and 1,000 units were maintained through the lateral drilling operations while in the **Middle Bakken**, with higher gas levels apparent near the top of the target window.
- 6) The **Middle Bakken** target consisted of one observed gross lithology. The **Middle Bakken** was observed to be a silty sandstone with abundant disseminated pyrite and apparent intergranular porosity.
- 7) Trace to moderate oil stain was observed in zone associated with a fast, white, diffuse cut. Trace oil washed out of the samples into the wash water when they were cleaned.
- 8) Drilling operations were concluded at 11:15 on Feb 17th, 2015 at a bottom hole location of 560.34' S & 9,997.60' E of surface location or approx. 192.34' FNL & 254.55' FEL, NE NE Sec. 20, T153N, R100W.
- 9) Exposure to target zone was ~95% of the lateral with 100% securely within the bounds of the **Middle Bakken** member. Oasis Petroleum's *Kline Federal 5300 41-18 14BX* awaits completion operations to determine its ultimate production potential.

Respectfully submitted,
Ryan Cox
RPM Geologic LLC

Well Information

Operator: Oasis Petroleum North America

Address: 1001 Fannin Suite 1500
Houston, TX 77002

Well Name: Kline Federal 5300 41-18 14BX

Field/ Prospect: Baker

Elevation: GL: 2,056'
KB: 2,082'

Spud Date: 19-Sep-14

API #: 33-053-06029

NDIC Well File #: 28655

Surface Location: SW SW Sec. 18 T153N, R100W

Footages: 368' FSL & 237' FWL

County, State: McKenzie Co., ND

Basin: Williston

Well Type: Horizontal Bakken

Contractor: Nabors 486

Toolpushers: Dwight Knudson

Field Supervisors: Mark Lawlar/Chris Harris

Directional Drilling RPM Consulting
Mark Lawlar
Eric Lindemann

Chemical Company Halliburton

Mud Engineer Tim Vaira

H₂S MONITORING: NOV WellSight Gas Watch

MWD Gyro Data
George Kerr, David Skuzacek

Wellsite Geologist Annika Tostengard
Bob Witmer
Rob Semple
Ryan Cox

Prospect Geologist Brendan Hargrove

Sample Examination: Binocular microscope & fluoroscope

Rock Sampling: 30' from 8,380' to 11,100 MD
50' 11,100' MD to 20,650' (TD)

Gas Detector NOV WellSight Gas Watch

Sample Cuts: Acetone

Horizontal Target Middle Bakken

Key Offset Wells:

Oasis Petroleum.	Kline Federal 5300 41-18 10B	SW SW Sec. 18 T153N R100W	McKenzie Co.,ND
Oasis Petroleum	Lynn 1	NE SW Sec. 23 T153N R100W	McKenzie Co.,ND

Pumps: #1 & #2: HHF 1600 Triplex - 12" Stroke length Output: 0.0838 bbl/stk

Mud Type: Diesel invert mud 2,140'-11,125' (Curve TD)
Salt Water Brine, 11,215' - 20,650' (TD)

Casing: Surface: 13 3/8" 54# J-55 @ 2,148'
Intermediate: 9 5/8" 40# HCL80 @ 6,071', 7" 32# P-110 @ 11,117'

Hole Size: 13 1/2" to 2,151' MD, 8 3/4" to 11,125' MD 6" to 20,650' (TD)

Total Drilling Days: 24

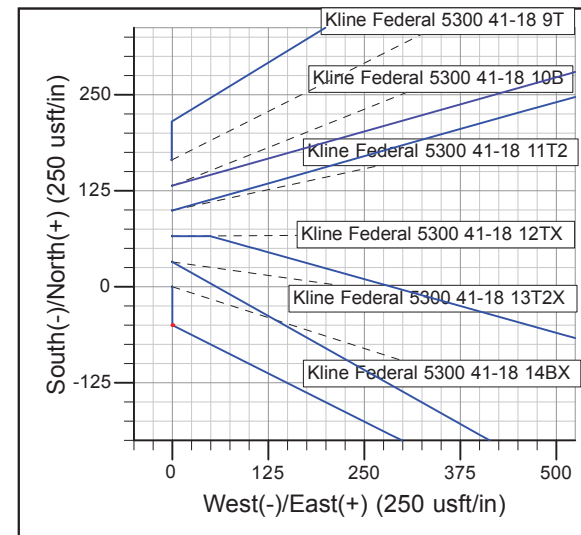
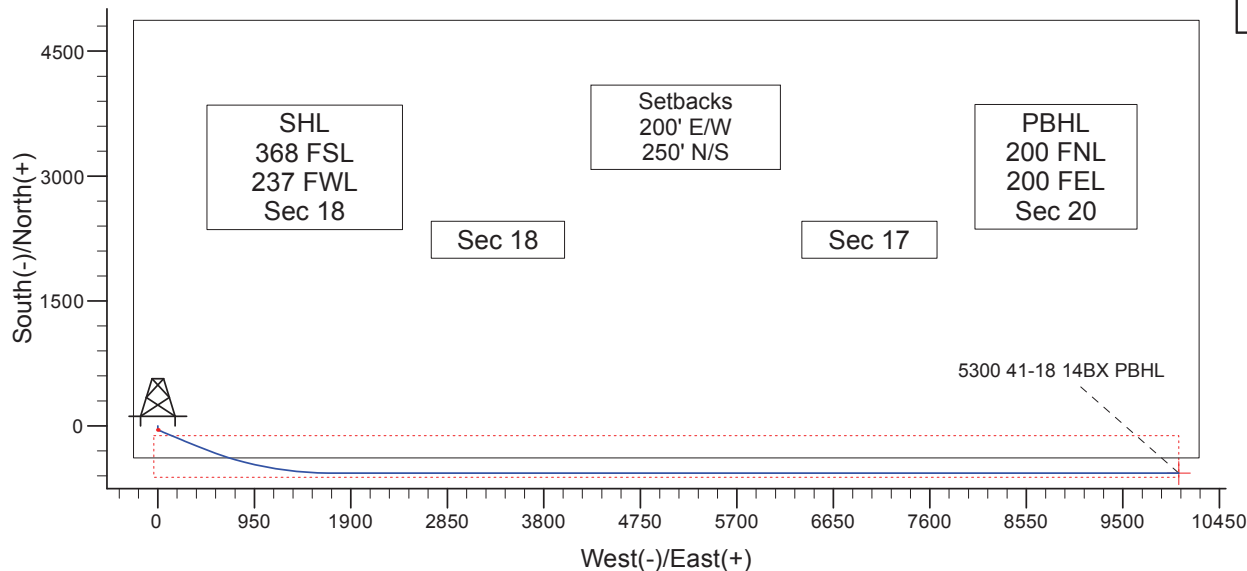
<u>Horizontal Target:</u>	<i>Middle Bakken</i>	<u>BOTTOM HOLE LOCATION:</u>
<u>Kick-Off Point / Date:</u>	<i>10,288' on Oct 1, 2014</i>	<i>560.34' S & 9,997.60' E of surface location or approx.</i>
<u>Total Depth/ Date:</u>	<i>20,650' Feb 17, 2015</i>	<i>192.34' FNL & 254.55' FEL, NE NE Sec. 20, T153N, R100W</i>
<u>Ending Vertical Section</u>	<i>9997.6</i>	
<u>Ending Azimuth</u>	<i>92.04</i>	
<u>Status of Well:</u>	<i>Complete</i>	
<u>Exposure to Formation:</u>	<i>100%</i>	

Project: Indian Hills
 Site: 153N-100W-17/18
 Well: Kline Federal 5300 41-18 14BX
 Wellbore: Kline Federal 5300 41-18 14BX
 Design: Design #2



WELL DETAILS: Kline Federal 5300 41-18 14BX

Ground Level: 2057.0
 Northing 405070.82 Easting 1209965.56 Latitude 48° 4' 6.900 N Longitude 103° 36' 11.950 W



CASING DETAILS

TVD MD Name Size
 10765.0 11037.0 7" 7

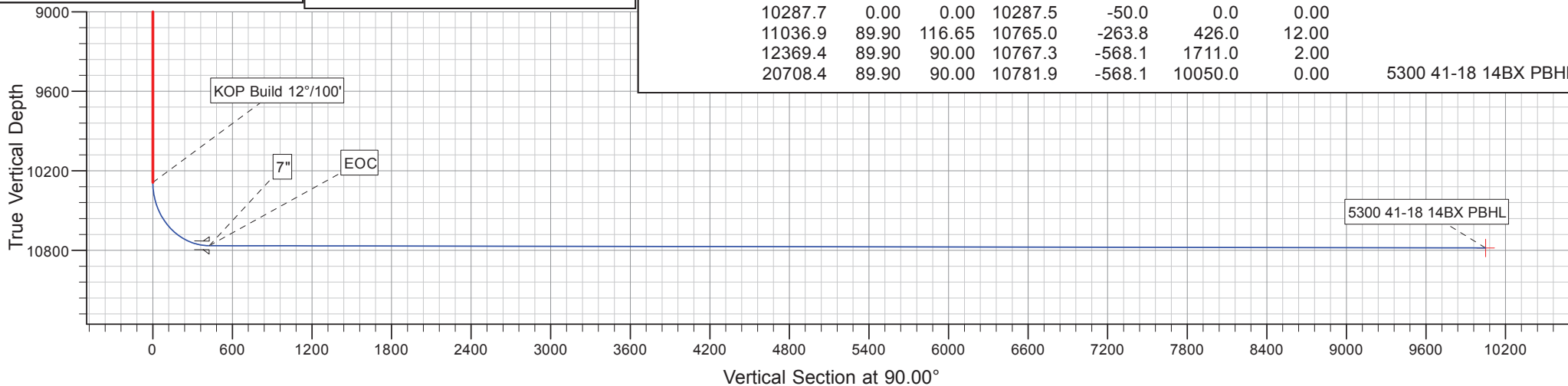


Azimuths to True North
 Magnetic North: 8.27°

Magnetic Field
 Strength: 56437.0snT
 Dip Angle: 72.99°
 Date: 5/23/2014
 Model: IGRF2010

SECTION DETAILS

MD	Inc	Azi	TVD	+N/-S	+E/-W	Dleg	Target
0.0	0.00	0.00	0.0	0.0	0.0	0.00	
2200.0	0.00	0.00	2200.0	0.0	0.0	0.00	
2216.7	0.50	180.00	2216.7	-0.1	0.0	3.00	
7926.7	0.50	180.00	7926.4	-49.9	0.0	0.00	
7943.3	0.00	0.00	7943.1	-50.0	0.0	3.00	
10287.7	0.00	0.00	10287.5	-50.0	0.0	0.00	
11036.9	89.90	116.65	10765.0	-263.8	426.0	12.00	
12369.4	89.90	90.00	10767.3	-568.1	1711.0	2.00	
20708.4	89.90	90.00	10781.9	-568.1	10050.0	0.00	5300 41-18 14BX PBHL



Daily Activity

Day	Date 2014	Depth 0600 Hrs	24 Hr Footage	Bit #	WOB (Klbs) Rotate	WOB (Klbs) Slide	RPM (RT)	Pump Pressure	SPM 1	SPM 2	GPM	24 Hr Activity	Formation
8	28-Sep	8,953'	1,236'	3	12	6	45	2970	67	67	471	Drill F/7717-8090, BOP drill, Service rig, Down time top drive could not break quill from stump, Drill F/8180-8376, Drill F/8376-8953	Charles Salt
9	29-Sep	10,132'	1,179'	3	9	23	45	2200	63	63	450	Drill F/8942-9412, Service rig, Drill F/9412-9538, Drill F/9538-10132	Mission Canyon
10	30-Sep	10,288'	156'	3	9	37	47	3420	63	63	450	Drill F/ 10132-10288, Circulate and condition, Dry job till past tue salt, TOH, Ream through salt twice, TOH, Lay down BHA, Rig up Wire line crew, Rig down wire line crew, Pick up BHA, TIH	Mission Canyon
11	1-Oct	10,422'	134'	4	0	22	0	2250	63	63	450	Change rotating head, TIH Ream and wash through salts, Rig service, Build curve F/10288/10422, Wait on orders TOH strap pipe, Lay down BHA, ick up BHA, TIH	Mission Canyon
12	2-Oct	11,126'	704'	4	25	30	25	2750	63	63	450	TIH, change rotating head, TIH, relog two stands of gamma, drill 10422-10846	Middle Bakken
13	3-Oct	11,126'	0'	-	-	-	-	-	-	-	-	Drill F/ 10892-11125, Circulate and condition trip out 9 stands, Circulate and condition Bottoms up, TOOHS slow so to high winds, Latdown BHA run casing.	Middle Bakken
14	4-Oct	11,126'	0'	-	-	-	-	-	-	-	-	Run casing, Circulate and condition, SPR@30SKS 455,Circulate and conition hole to cement, Primary cementin droppen plug at 1:30 Primary ig down cementers, Working as directed by operator, set casing slips.	Middle Bakken

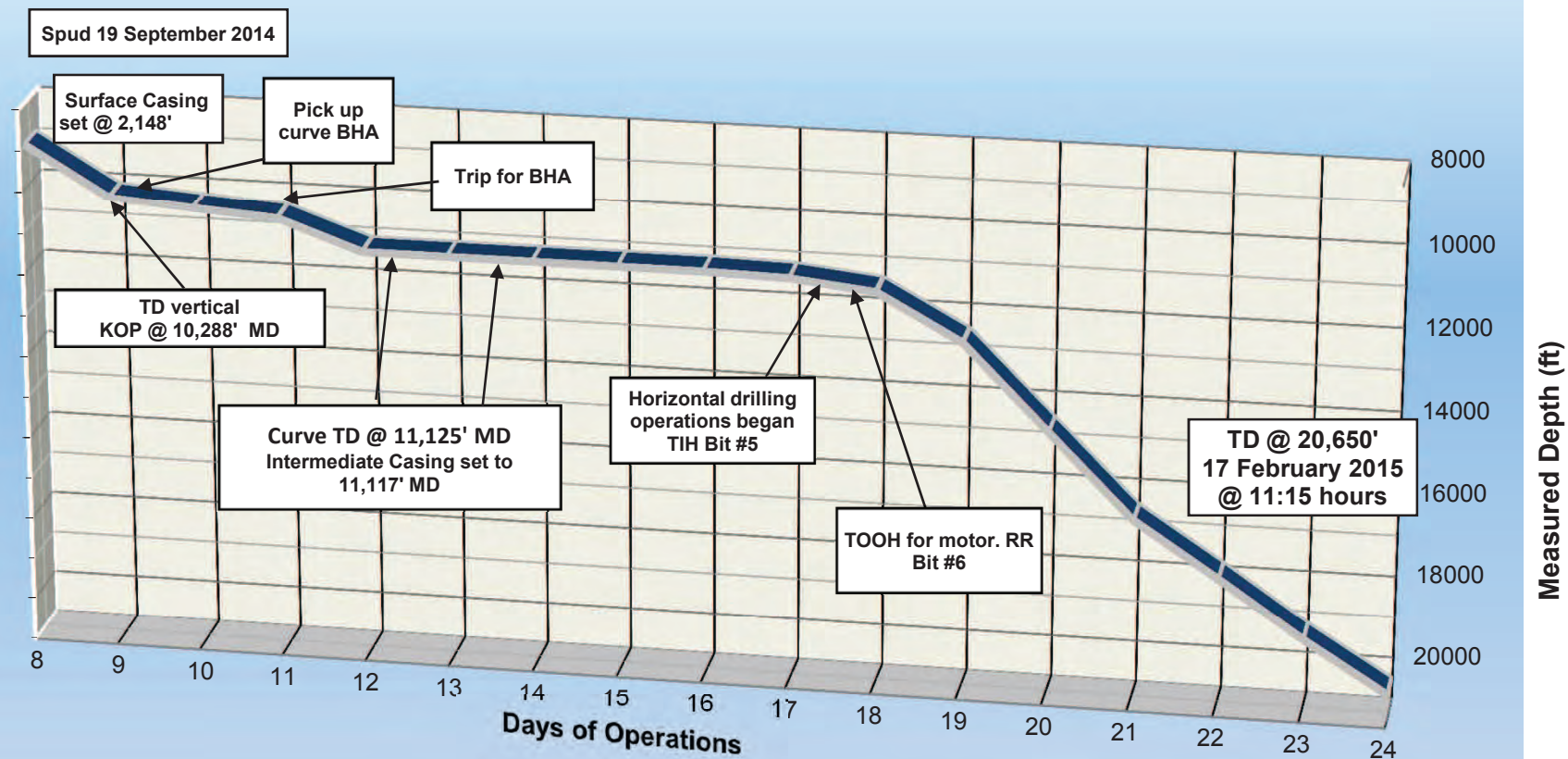
Daily Activity

Day	Date 2014	Depth 0600 Hrs	24 Hr Footage	Bit #	WOB (Klbs) Rotate	WOB (Klbs) Slide	RPM (RT)	Pump Pressure	SPM 1	SPM 2	GPM	24 Hr Activity	Formation
15	5-Oct	11,126'	0'	-	-	-	-	-	-	-	-	waiting on cement, float not holding, shut in w/1500 psi, working as directed by operator, wait for casing to drain, nipple down bop, rig down, skid rig, nipple up bop, pressure tests, pick up bha	Middle Bakken
16	9-Feb	11,126'	0'	-	-	-	-	-	-	-	-	Skid rig // Secure old well // New brake bands on draw works // Cut drilling line // Nipple up bops, choke line, flow line // Cut drilling line finished spooling up // Drum and put clamps on dead man // Test BOP // Pick up BHA // TOOH // Drill actual F/ 11,125' - 11,170'	Middle Bakken
17	10-Feb	11,170'	44'	5	13	28	45	2525	75	-		Drill actual F/ 11,170'-11380; TOOH; L/D BHA; P/U BHA; TIH	Middle Bakken
18	11-Feb	11,380'	210'	5	11	28	46	2605	72	-	272	TIH, Drill ahead, Service Rig, Down time for top drive, Drill. Service rig	Middle Bakken
19	12-Feb	12,467'	1,087'	5	10	13	46	3200	-	71	251	Rotary drilling f/ 12,460'-13,222', rig service, drill f/ 13,222'-13,537', rotary drilling f/ 13,537'-14,481'	Middle Bakken
20	13-Feb	14,481'	2,014'	5	12	17	42	3500	73	-	258	Drill f/ 14,481'-15,206', service top drive, drill f/ 15,206'-16,481'	Middle Bakken
21	14-Feb	16,481'	2,000'	5	11	14	45	2500	-	73	256	Drill f/ 16,496'-16,811', service top drive, drill f/ 16,811'-17,882'	Middle Bakken
22	15-Feb	17,782'	1,301'	5	10	25	46	3500	-	77	272	Drill f/ 17,882'-18,321', BOP drill, service top drive, drill f/ 18,321'-19,137'	Middle Bakken
23	16-Feb	19,137'	1,355'	5	10	25	46	3500	-	77	272	Drill f/ 19,137'-19,544', service top drive, drill f/ 19,544'-20,370'.	Middle Bakken
24	17-Feb	20,370'	1,233'	5	15	44	48	3500	71	-	252	Drill f/ 20,370'-20,650'	Middle Bakken



Daily Progress

Oasis Petroleum
Kline Federal 5300 41-18 14BX



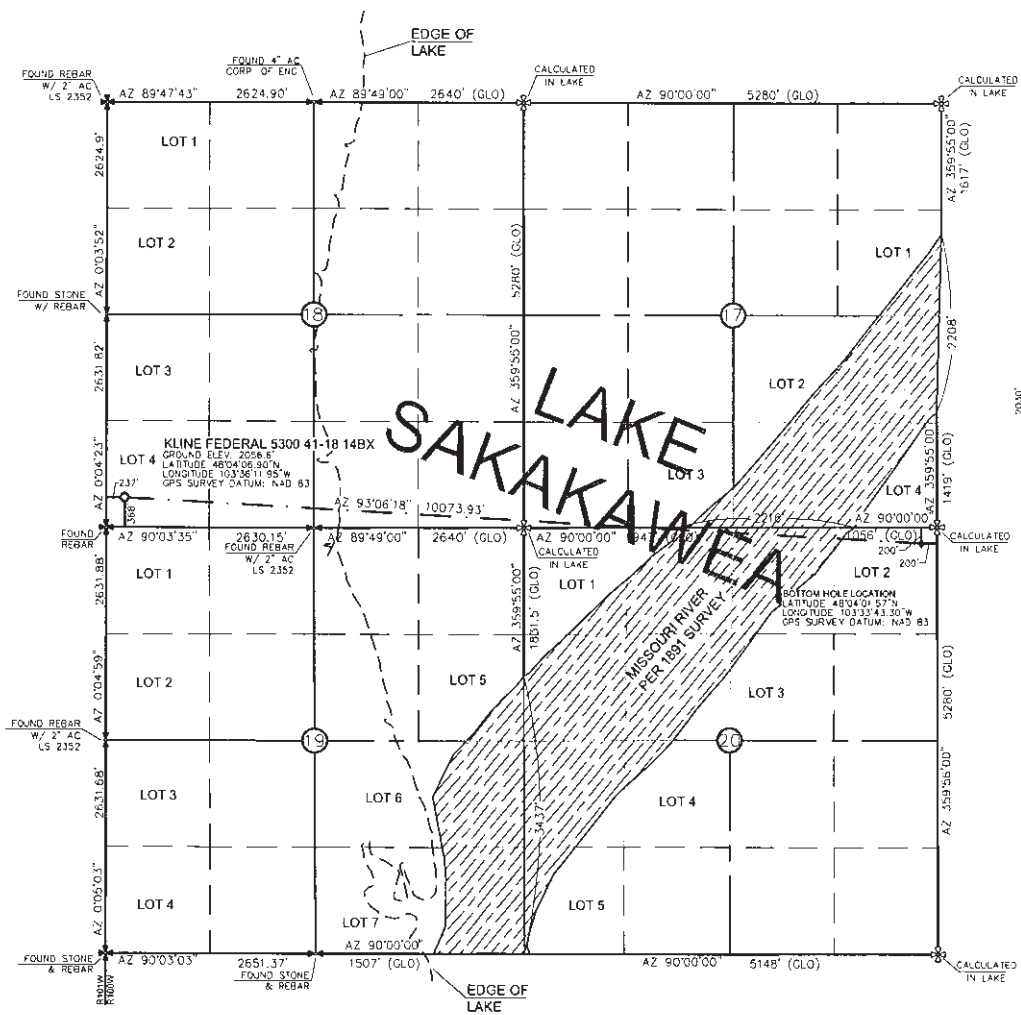
Bit Record

<i>Bit #</i>	<i>Size</i>	<i>Make</i>	<i>Model</i>	<i>Serial #</i>	<i>Jets</i>	<i>Depth In</i>	<i>Depth Out</i>	<i>Footage</i>	<i>Hours</i>	<i>Mean ROP (ft/hr)</i>	<i>Accum. Hours</i>
1	8 3/4"	Varel	R616PDG2UX	4006379	6 x 16	2,140'	9,800'	7,660'	56	136.79	56.0
2	8 3/4"	Baker	T505	7147631	5 x 16	9,800'	10,354'	554'	20	27.70	76.00
3	8 3/4"	Varel	MDSI6BPX	JJ7489	6x18	6,150'	10,288'	4,138'	66	62.70	142.00
4	8 3/4"	Varel	A09457	4007013	5x18	10,288'	10,422'	134'	16.75	8.00	158.75
5	8 3/4"	SEC	MMD55M	12545836	5x18	10,288'	11,125'	837'	23	36.39	181.75
6	6"	Reed	PDC	SK516	5x20	11,125'	20,650'	9,525'	148	64.36	329.75

Daily Mud Data

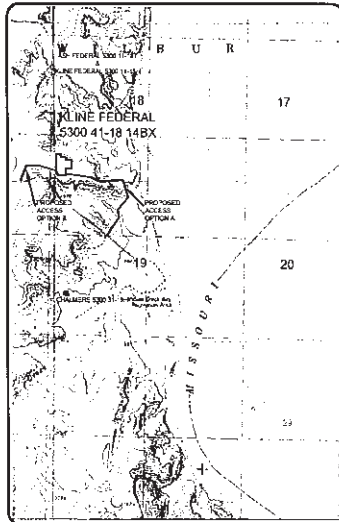
Day	Date 2014	Depth (0600 Hrs)	Mud WT (ppg)	VIS (sec)	PV (cP)	YP (lbs/ 100 ft²)	Gels (lbs/ 100 ft²)	600/ 300	HTHP (cc/30min)	NAP/ H₂O (ratio)	Cake (APT/ HT)	Solid s (%)	pH	Alk	Cl- (mg/l)	CaCl₂	ES (v)	Loss (Bbls)
3	22-Sep	2,140'	9.95	52	11	8	4/5/6	30/19	9	79.7/20.3	2	9.7	-	-	30K	-	615	-
4	23-Sep	3,939'	9.95	57	20	11	6/7/10	51/31	6	78.7/21.3	2	9.3	-	-	28K	-	640	-
5	24-Sep	5,695'	10.45	55	22	14	8/9/12	58/36	7	77.6/22.4	2	11.2	-	-	30K	-	675	-
6	25-Sep	6,150'	11.40	57	20	14	6/7/9	54/34	8	78.4/21.6	2	14.70	-	-	30K	-	675	-
7	26-Sep	6,150'	10.00	50	10	14	4/5/7	34/24	10	79.7/20.	2	9.80	-	-	28K	-	720	-
8	27-Sep	8,091'	10.05	48	16	10	4/5/7	42/26	8	79.5/20.5	2	10.30	-	-	28K	-	735	-
9	28-Sep	8,091'	10.05	48	16	10	4/5/7	42/26	8	79.5/20.5	2	10.30	-	-	28K	-	735	-
10	29-Sep	9,035'	10.05	50	50	17	6/7/9	24/16	8	77.3/22.7	2	9.90	-	-	34K	-	700	-
11	30-Sep	10,288'	10.10	53	53	23	8/9/13	62/30	6	783/21.7	2	10.20	-	-	38K	-	780	-
12	1-Oct	10,289'	10.25	66	24	14	8/10/15	27/18	6	78.9/21.1	2	10.20	-	-	37K	-	760	-
13	2-Oct	10,442'	10.05	64	22	14	7/10/15	27/18	6	78.1/21.9	2	10.25	-	-	28K	-	780	-
14	3-Oct	11,001'	10.70	49	21	18	13/15/18	32/23	6	79.1/21.9	2	10.70	-	-	34K	-	765	-
15	4-Oct	11,001'	10.70	49	21	18	13/15/19	32/24	7	79.1/21.10	3	11.70	-	-	34K	-	766	-
16	5-Oct	11,001'	10.70	49	21	18	13/15/20	32/25	8	79.1/21.11	4	12.70	-	-	34K	-	767	-
17	10-Feb	11,001'	10.70	49	21	18	13/15/21	32/26	-	79.1/21.12	-	13.70	-	-	34K	-	-	-
18	11-Feb	11,001'	10.70	49	21	18	13/15/22	32/27	-	79.1/21.13	-	14.70	-	-	34K	-	-	-
19	12-Feb	11,001'	10.70	49	21	18	13/15/23	32/28	-	79.1/21.14	-	15.70	-	-	34K	-	-	-
20	13-Feb	13,408'	10.00	28	1	2	0/1/0	4/3	-	0.0/88.5	-	9.50	9.5	-	184k	-	-	-
21	14-Feb	13,408'	10.00	28	1	2	0/1/0	4/4	-	0.0/88.6	-	10.50	10	-	184k	-	-	-
22	15-Feb	16,835'	9.90	28	1	3	1/1/1	5/4	-	0.0/88.9	-	0.30	9.5	-	180k	-	-	-
23	16-Feb	-	10.00	28	-	-	-	-	-	-	-	-	10	-	-	-	-	-
24	17-Feb	19,755'	9.90	28	1	3	1/1/1	5/4	-	0.0/89.3	-	0.60	8.5	-	170k	-	-	-

WELL LOCATION PLAT
 OASIS PETROLEUM NORTH AMERICA, LLC
 1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
 "KLINE FEDERAL 5300 41-18 148X"
 368 FEET FROM SOUTH LINE AND 237 FEET FROM WEST LINE
 SECTION 18, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



- ⊕ — MONUMENT — RECOVERED
- ⊗ — MONUMENT — NOT RECOVERED

VICINITY MAP



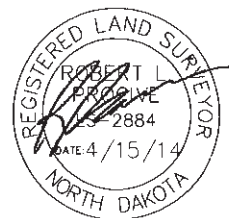
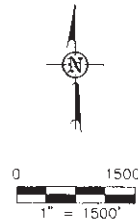
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STAKED ON 4/15/14
 VERTICAL CONTROL DATUM WAS BASED UPON
 CONTROL POINT 4 WITH AN ELEVATION OF 2090.8'

THIS SURVEY AND PLAT IS BEING PROVIDED AT THE REQUEST
 OF ERIC BAYES OF OASIS PETROLEUM. I CERTIFY THAT THIS
 PLAT CORRECTLY REPRESENTS WORK PERFORMED BY ME OR
 UNDER MY SUPERVISION AND IS TRUE AND CORRECT TO THE
 BEST OF MY KNOWLEDGE AND BELIEF.

Robert L. Procive
 ROBERT L. PROCIVE 2884LS

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Circle 100 on Reader Service Card

OASIS PETROLEUM NORTH AMERICA, LLC
 WELL LOCATION PLAT
 SECTION 18, T153N, R100W
 MCKENZIE COUNTY, NORTH DAKOTA

Drawn By: B.H.H. Project No.: 514-09-001
 Checked By: B.H.H. Date: APRIL 2014

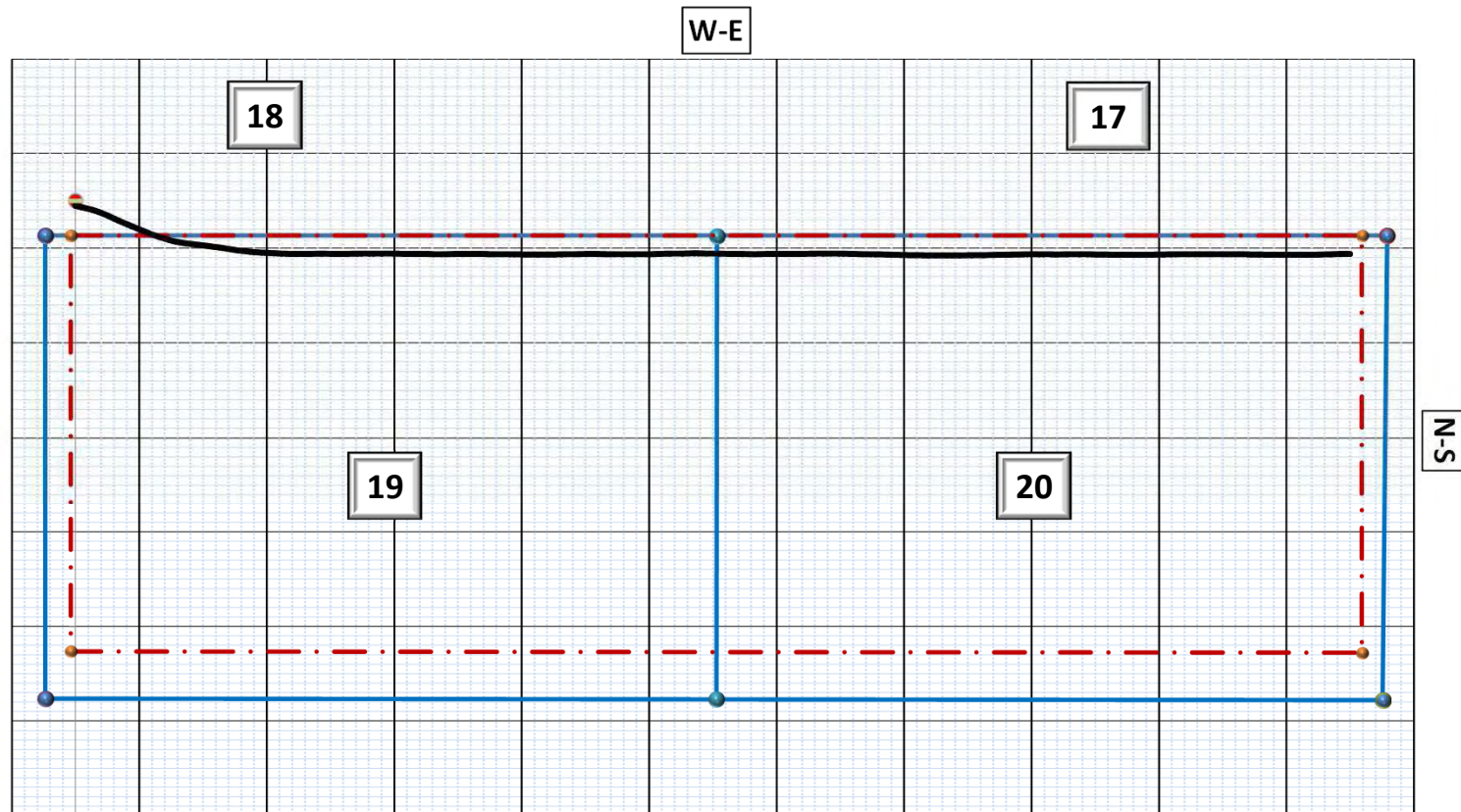
Revision	Date	By	Description



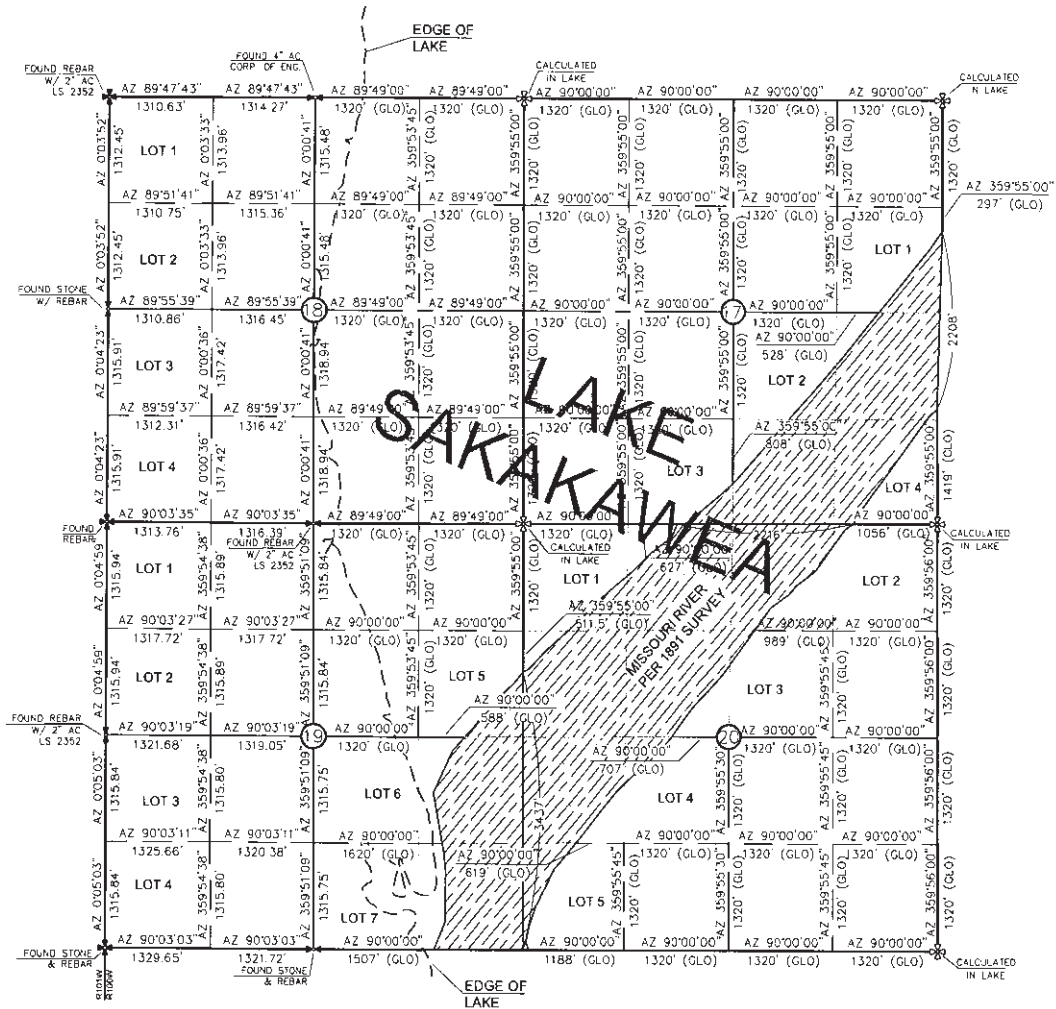
Vertical Section View

Surface Location:
368' FSL & 237' FWL
SW SW Sec. 18, T153N, R100W
McKenzie County, ND

Bottom Hole Location
560.34' S & 9,997.60' E
of surface location or approx.
192.34' FNL & 254.55' FEL
NE NE Sec. 20 T153N, R100W

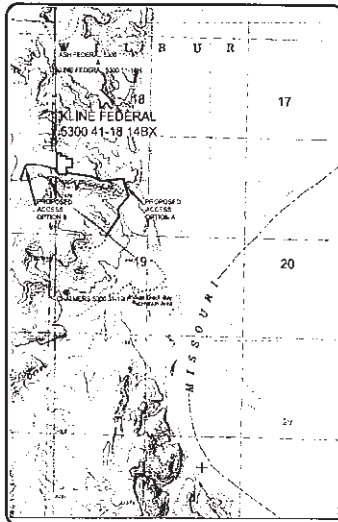


SECTION BREAKDOWN
 OASIS PETROLEUM NORTH AMERICA, LLC
 1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
 "KLINE FEDERAL 5300 41-18 148X"
 368 FEET FROM SOUTH LINE AND 237 FEET FROM WEST LINE
 SECTIONS 17, 18, 19, & 20, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



- MONUMENT - RECOVERED
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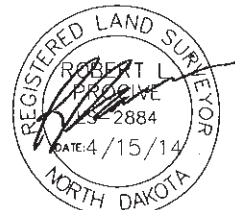
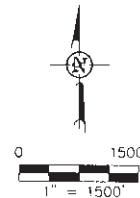
VICINITY MAP



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ALL AZIMUTHS ARE BASED ON G.P.S.
 OBSERVATIONS. THE ORIGINAL SURVEY OF THIS
 AREA FOR THE GENERAL LAND OFFICE (G.L.O.)
 WAS 1891. THE CORNERS FOUND ARE AS
 INDICATED AND ALL OTHERS ARE COMPUTED FROM
 THOSE CORNERS FOUND AND BASED ON G.L.O.
 DATA. THE MAPPING ANGLE FOR THIS AREA IS
 APPROXIMATELY 0°03'.

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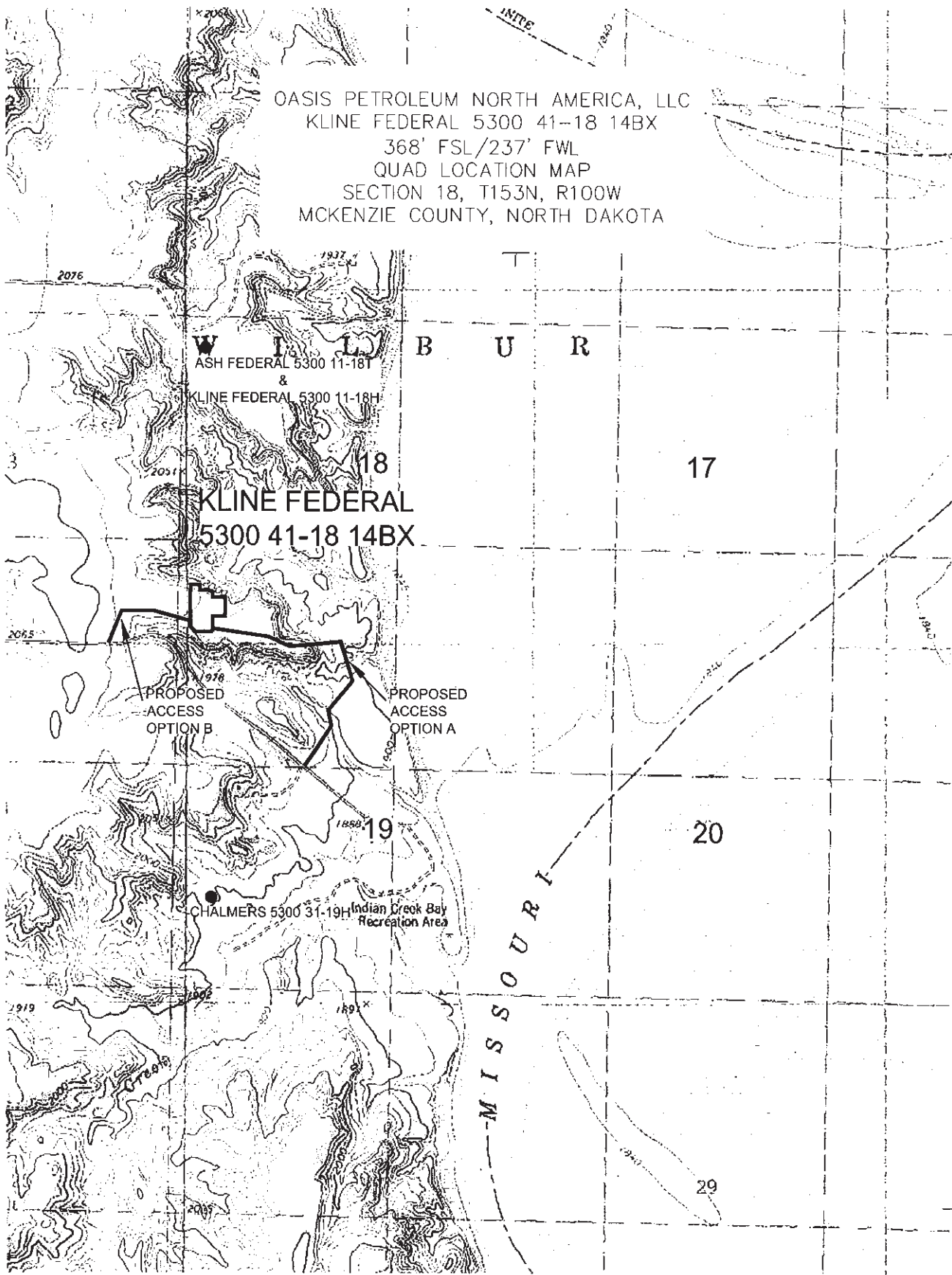


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 Under contract to McKenzie County, North Dakota and South Dakota

OASIS PETROLEUM NORTH AMERICA, LLC
 SECTION BREAKDOWN
 SECTIONS 17, 18, 19, & 20, T153N, R100W
 MCKENZIE COUNTY, NORTH DAKOTA
 Drawn By: B.H.M. Project No.: 14-09-081
 Checked By: R.A.P. Date: APRIL 2014

Revision	Date	By	Description

OASIS PETROLEUM NORTH AMERICA, LLC
 KLINE FEDERAL 5300 41-18 14BX
 368' FSL/237' FWL
 QUAD LOCATION MAP
 SECTION 18, T153N, R100W
 MCKENZIE COUNTY, NORTH DAKOTA



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OASIS PETROLEUM NORTH AMERICA, LLC
 QUAD LOCATION MAP
 SECTION 18, T153N, R100W
 MCKENZIE COUNTY, NORTH DAKOTA

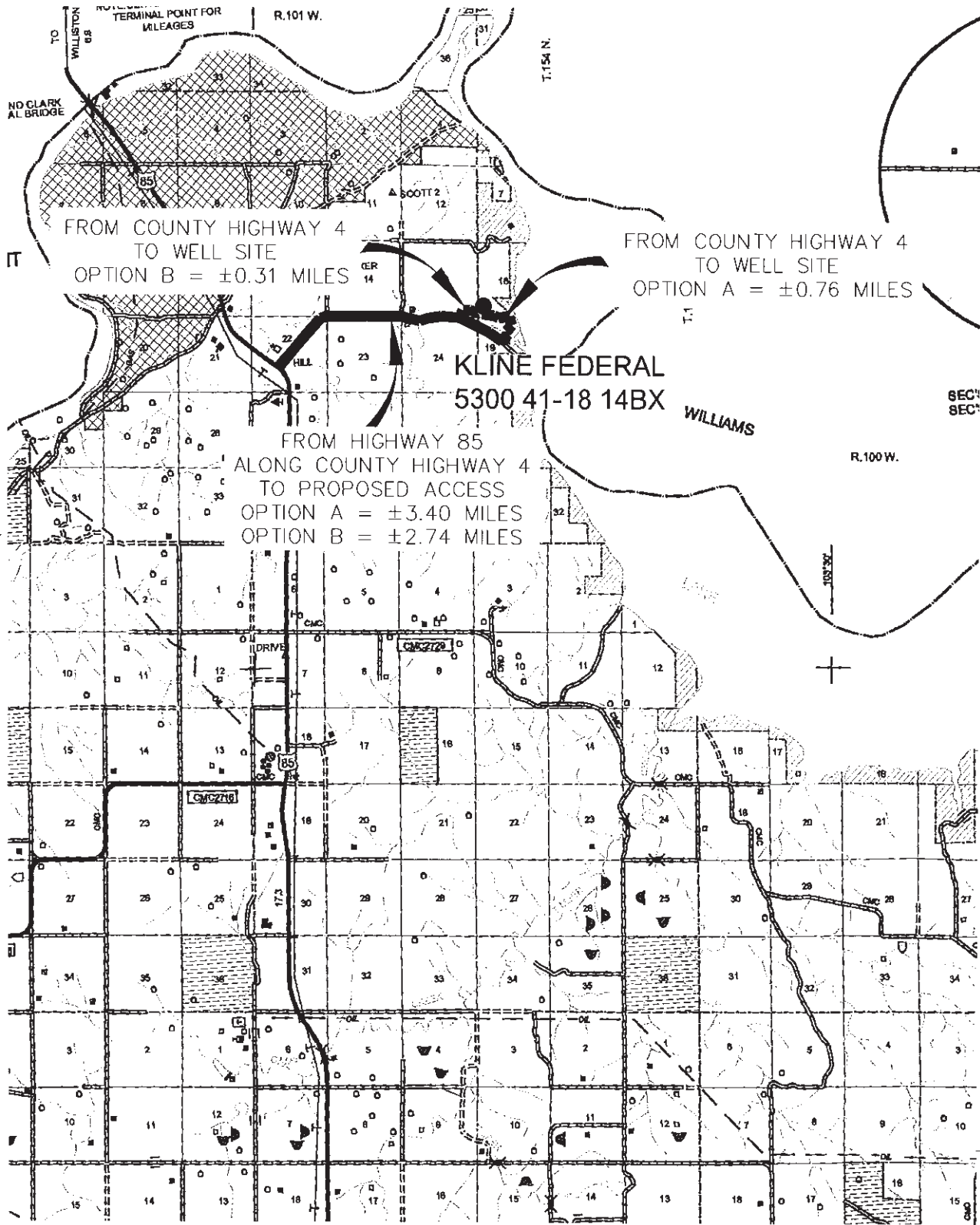
Drawn By: B.H.H. Project No.: S14-09-081
 Checked By: R.L.P. Date: APRIL 2014

Revision No.	Date	By	Description

COUNTY ROAD MAP

OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
"KLINE FEDERAL 5300 41-18 14BX"

368 FEET FROM SOUTH LINE AND 237 FEET FROM WEST LINE
SECTION 18, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



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SCALE: 1" = 2 MILE

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Professional's you need, people you trust

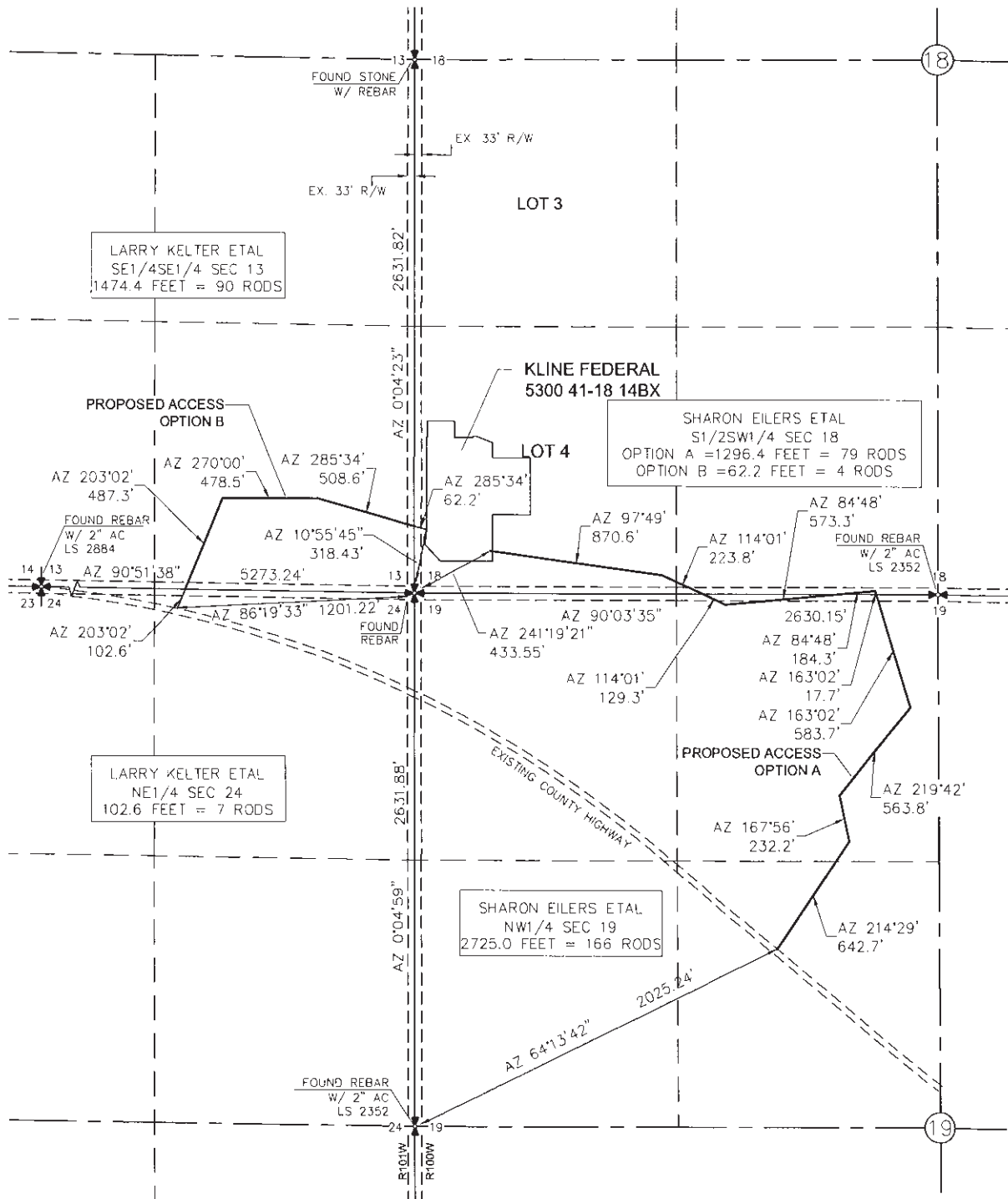
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OASIS PETROLEUM NORTH AMERICA, LLC
COUNTY ROAD MAP
SECTION 18, T153N, R100W
MCKENZIE COUNTY, NORTH DAKOTA

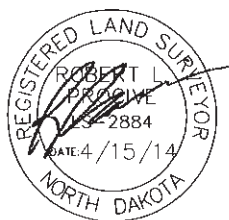
Drawn By: B.H.H. Project No.: S14-09-081
Checked By: R.L.P. Date: APRIL 2014

Revision No.	Date	By	Description

ACCESS APPROACH
 OASIS PETROLEUM NORTH AMERICA, LLC
 1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
 "KLINE FEDERAL 5300 41-18 14BX"
 368 FEET FROM SOUTH LINE AND 237 FEET FROM WEST LINE
 SECTION 18, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA

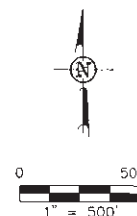


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NOTE: All utilities shown are preliminary only, a complete utilities location is recommended before construction.

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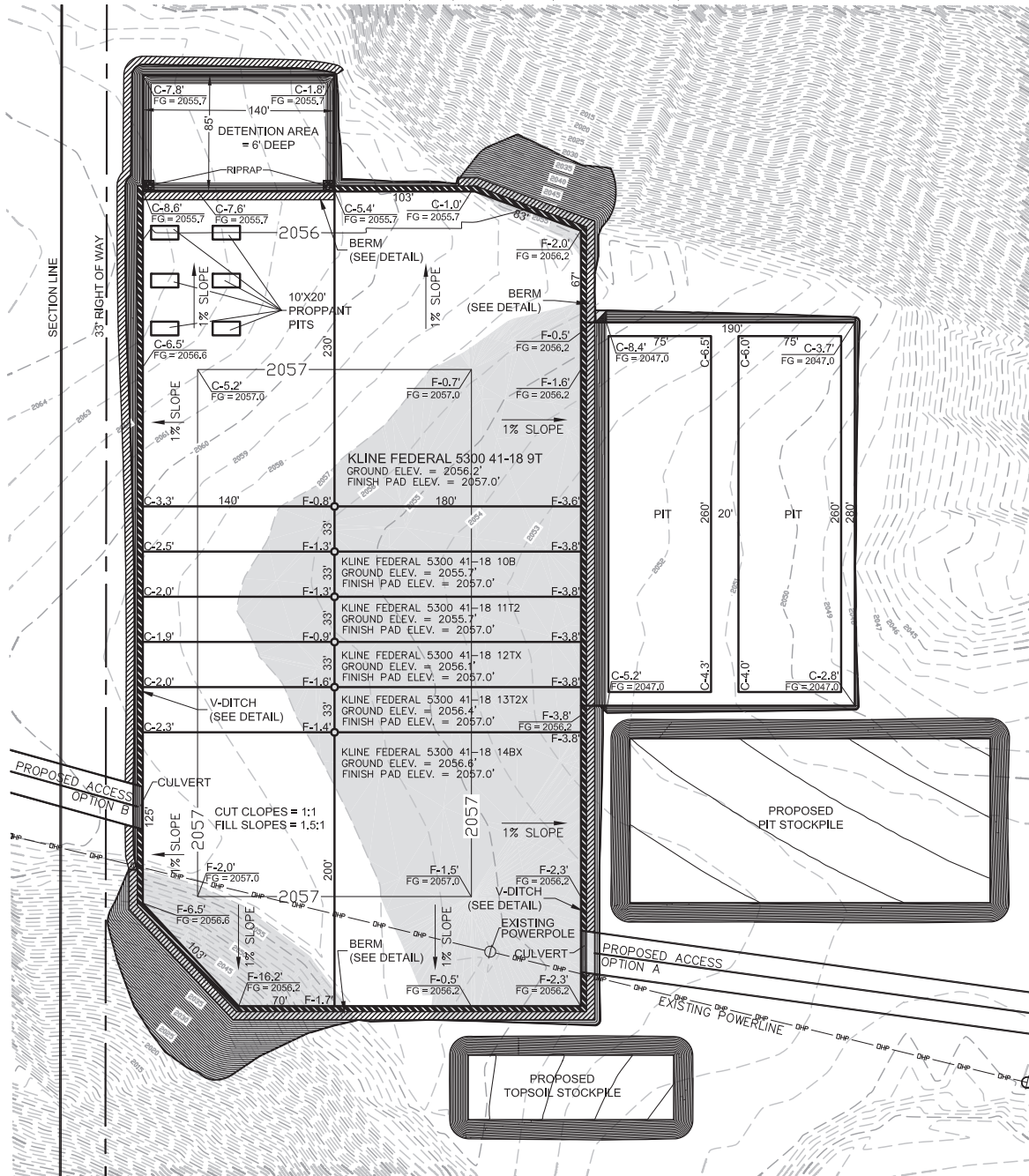


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OASIS PETROLEUM NORTH AMERICA, LLC
 ACCESS APPROACH
 SECTION 18, T153N, R100W
 MCKENZIE COUNTY, NORTH DAKOTA
 Drawn By: B.L.P.
 Checked By: B.L.P.
 Project No.: 514-09281
 Date: APRIL 2014

Revision No.	Date	By	Description

PAD LAYOUT
 OASIS PETROLEUM NORTH AMERICA, LLC
 1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
 KLINE FEDERAL 5300 41-18 9T
 533 FEET FROM SOUTH LINE AND 237 FEET FROM WEST LINE
 SECTION 18, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA

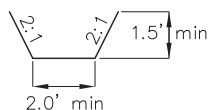


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NOTE: Pad dimensions shown are to usable area, the v-ditch and berm areas shall be built to the outside of the pad dimensions.



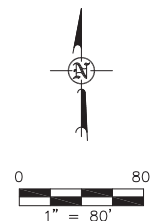
V-DITCH DETAIL



- Proposed Contours
- Original Contours
- BERM
- DITCH

NOTE: All utilities shown are preliminary only, a complete utilities location is recommended before construction.

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OASIS PETROLEUM NORTH AMERICA, LLC
 PAD LAYOUT
 SECTION 18, T153N, R100W
 MCKENZIE COUNTY, NORTH DAKOTA
 Drawn By: B.H.H. Project No.: S14-09-081.05
 Checked By: D.D.K. Date: APRIL 2014

Revision No.	Date	By	Description

U:\2014\153N-100W\153N-100W.dwg Plot Date: 4/15/14 User: B.H.H. Scale: 1"=80' Plot Size: 11" x 17" Plot Path: U:\2014\153N-100W\153N-100W.dwg

DRILLING PLAN																							
OPERATOR	Oasis Petroleum		COUNTY/STATE	McKenzie Co., ND																			
WELL NAME	Kline Federal 5300 41-18 14BX		RIG	0																			
WELL TYPE	Horizontal Middle Bakken																						
LOCATION	SWSW 18-153N-100W		Surface Location (survey plat):	368' fsl	237' fwl																		
EST. T.D.	20,708'		GROUND ELEV:	2057	Finished Pad Elev.	Sub Height: 25																	
TOTAL LATERA		9,671'		KB ELEV:	2082																		
PROGNOSIS: Based on 2,082' KB(est)			LOGS:																				
MARKER			DEPTH (Surf Loc)			DATUM (Surf Loc)																	
Pierre			NDIC MAP			1,972			110														
Greenhorn						4,646			-2564														
Mowry						5,057			-2975														
Dakota						5,471			-3389														
Rierdon						6,486			-4404														
Dunham Salt						6,814			-4732														
Dunham Salt Base						6,929			-4847														
Spearfish						7,024			-4942														
Pine Salt						7,283			-5201														
Pine Salt Base						7,318			-5236														
Opeche Salt						7,374			-5292														
Opeche Salt Base						7,403			-5321														
Broom Creek (Top of Minnelusa Gp.)						7,605			-5523														
Amsden						7,684			-5602														
Tyler						7,853			-5771														
Otter (Base of Minnelusa Gp.)						8,047			-5965														
Kibbey Lime						8,390			-6308														
Charles Salt						8,542			-6460														
UB						9,159			-7077														
Base Last Salt						9,238			-7156														
Ratcliffe						9,301			-7219														
Mission Canyon						9,454			-7372														
Lodgepole						10,022			-7940														
Lodgepole Fracture Zone						10,209			-8127														
False Bakken						10,726			-8644														
Upper Bakken						10,735			-8653														
Middle Bakken						10,751			-8669														
Middle Bakken Sand Target						10,760			-8678														
Base Middle Bakken Sand Target						10,770			-8688														
Lower Bakken						10,785			-8703														
Three Forks						10,813			-8731														
Dip Rate:			-0.1																				
Max. Anticipated BHP:			4673			Surface Formation:			Glacial till														
MUD:			Interval			Type			WT			Vis			WL			Remarks					
Surface:			0' -			2,072'			FW/Gel - Lime Sweeps			8.4-9.0			28-32			NC			Circ Mud Tanks		
Intermediate:			2,072' -			11,037'			Invert			9.5-10.4			40-50			30+HtHp			Circ Mud Tanks		
Lateral:			11,037' -			20,708'			Salt Water			9.8-10.2			28-32			NC			Circ Mud Tanks		
CASING:			Size			Wt ppf			Hole			Depth			Cement			WOC			Remarks		
Surface:			9-5/8"			36#			13-1/2"			2,072'			To Surface			12			100' into Pierre		
Intermediate:			7"			29/32#			8-3/4"			11,037'			3971			24			1500' above Dakota		
Production Liner:			4.5"			11.6#			6"			20,708'			TOL @ 10,238'						50' above KOP		
PROBABLE PLUGS, IF REQ'D:																							
OTHER:			MD			TVD			FNL/FSL			FEL/FWL			S-T-R			AZI					
Surface:			2,072			2,072			368' FSL			237' FWL			SEC 18-T153N-R100W			Survey Company:					
KOP:			10,288'			10,288'			318' FSL			237' FWL			SEC 18-T153N-R100W			Build Rate: 12 deg /100'					
EOC:			11,037'			10,765'			104' FSL			663' FWL			SEC 18-T153N-R100W								
Casing Point:			11,037'			10,765'			104' FSL			663' FWL			SEC 18-T153N-R100W								
Middle Bakken Lateral TD:			20,708'			10,782'			200' FNL			200' FEL			SEC 20-T153N-R100W								
Comments:																							
Request a Sundry for an Open Hole Log Waiver																							
Exception well: Oasis Petroleum's Kline Federal 5300 11-18H (153N 100W 18 NW NW)																							
35 packers, 35 sleeves, no frac string																							
Oasis Petroleum does not use Diesel Fuel, as defined by the US EPA in the list below, in our hydraulic fracture operations.																							
68334-30-5 (Primary Name: Fuels, diesel) 68476-34-6 (Primary Name: Fuels, diesel, No. 2)68476-30-2 (Primary Name: Fuel oil No. 2)																							
68476-31-3 (Primary Name: Fuel oil, No. 4)8008-20-6 (Primary Name: Kerosene)																							
																							
Geology: M. Steed 4/3/2014									Engineering: hbader rpm														

Formation Tops							
Operator: Well Name: Location: Elevation:	Subject Well:						
	Oasis Petroleum North America						
	Kline Federal 5300 41-18 14BX						
	368' FSL & 237' FWL SW SW Sec. 18 T153N, R100W						
	GL: 2,056'	Sub: 32'	KB: 2,082'				
Formation/ Zone	Prog. Top	Prog. MSL Datum	Est. MD Top (Gamma)	TVD Top (Gamma)	Est. MLS Datum	Thickness	Dip To Kline Federal 5300 41- 18 10B
Kibbey Lime	8,380	-6,300'	8,382'	8,382'	-6,300'	182'	-74'
Charles Salt	8,561	-6,482'	8,564'	8,564'	-6,482'	641'	-106'
Base Last Salt	9,240	-7,123'	9,205'	9,205'	-7,123'	223'	-51'
Mission Canyon	9,465	-7,346'	9,428'	9,428'	-7,346'	548'	-48'
Lodgepole	10,007	-7,849'	9,976'	9,976'	-7,894'	728'	-48'
False Bakken	10,720	-8,622'	10,826'	10,704'	-8,622'	8'	-61'
Upper Bakken Shale	10,723	-8,630'	10,847'	10,712'	-8,630'	16'	-58'
Middle Bakken	10,750	-8,646'	10,895'	10,728'	-8,646'	-	-60'

Control Well				
Operator: Well Name: Location:	Oasis Petroleum <i>Kline Federal 5300 41-18 10B</i> SW SW Sec. 18 T153N R100W McKenzie Co.,ND			
	Elevation: KB: 2,082'			
	Formation/Zone	Est. TVD	MSL Datum	Thickness
				Dist to target
Kibbey Lime	8,382	-6,300'	178'	2,358'
Charles Salt	8,560	-6,478'	646'	2,180'
Base Last Salt	9,206	-7,124'	222'	1,534'
Mission Canyon	9,428	-7,346'	547'	1,312'
Lodgepole	9,975	-7,893'	730'	765'
False Bakken	10,705	-8,623'	7'	35'
Upper Bakken Shale	10,712	-8,630'	18'	28'
Middle Bakken	10,730	-8,648'	-	10'

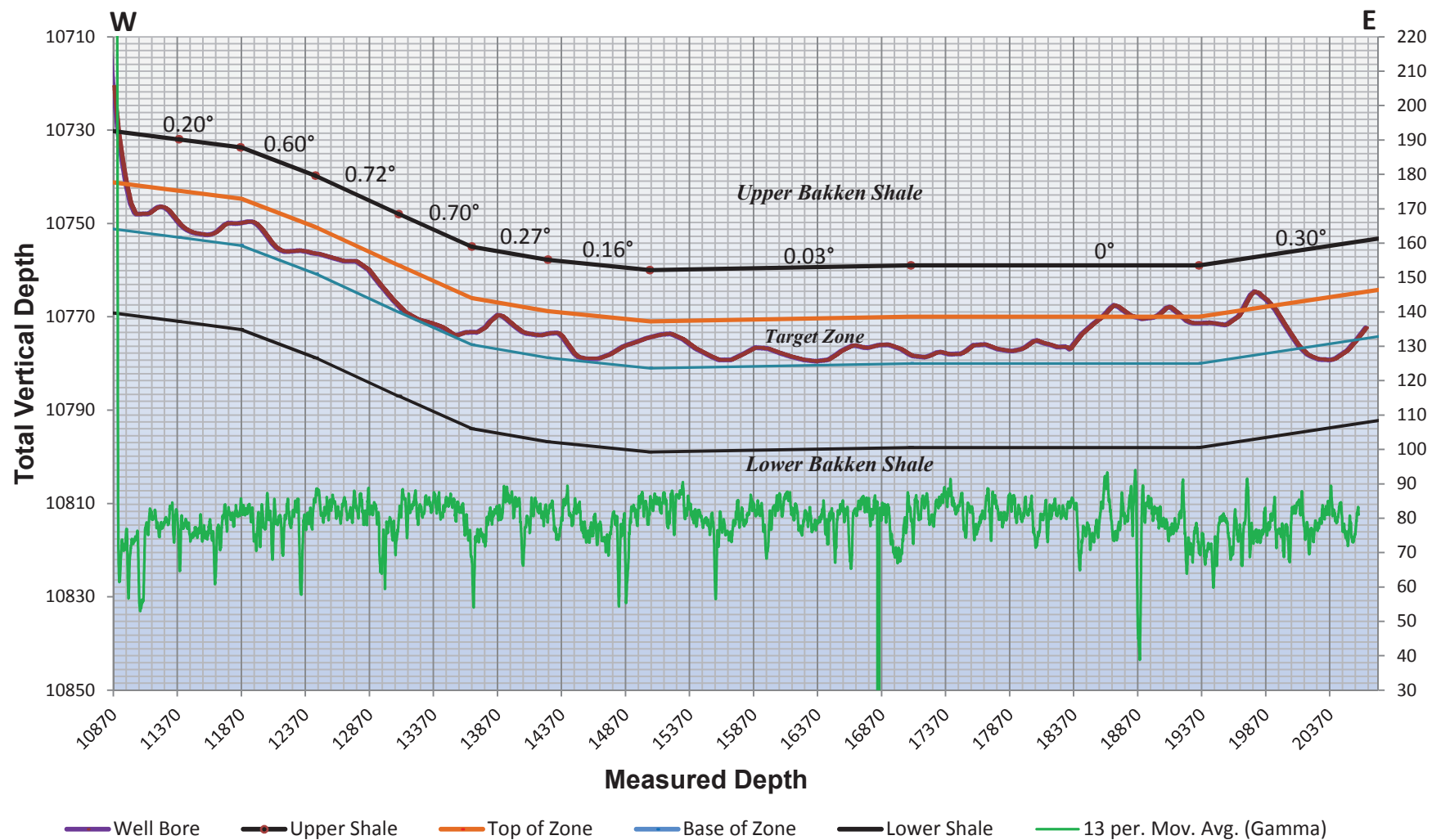
Control Well				
Operator: Well Name: Location:	Oasis Petroleum <i>Lynn 1</i> NE SW Sec. 23 T153N R100W McKenzie Co.,ND			
	Elevation: KB: 2,142'			
	Formation/Zone	Est. TVD	MSL Datum	Thickness
				Dist to target
Kibbey Lime	8,368'	-6,226'	150'	2,370'
Charles Salt	8,518'	-6,376'	696'	2,220'
Base Last Salt	9,214'	-7,072'	226'	1,524'
Mission Canyon	9,440'	-7,298'	548'	1,298'
Lodgepole	9,988'	-7,846'	715'	750'
False Bakken	10,703'	-8,561'	11'	35'
Upper Bakken Shale	10,714'	-8,572'	14'	24'
Middle Bakken	10,728'	-8,586'	-	10'

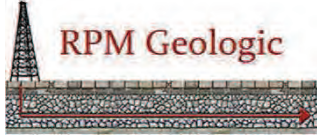


Surface: 368' FSL & 237' FWL
SW SW Sec. 18, T153N-R100W
Mckenzie County., ND

Kline Federal 5300 41-18 14BX Gamma Cross Section

Bottom Hole Location-
560.34' S & 9,997.60' E
of surface location or approx.
192.34' FNL & 254.55' FEL
NE NE Sec. 20, T153N-R100W

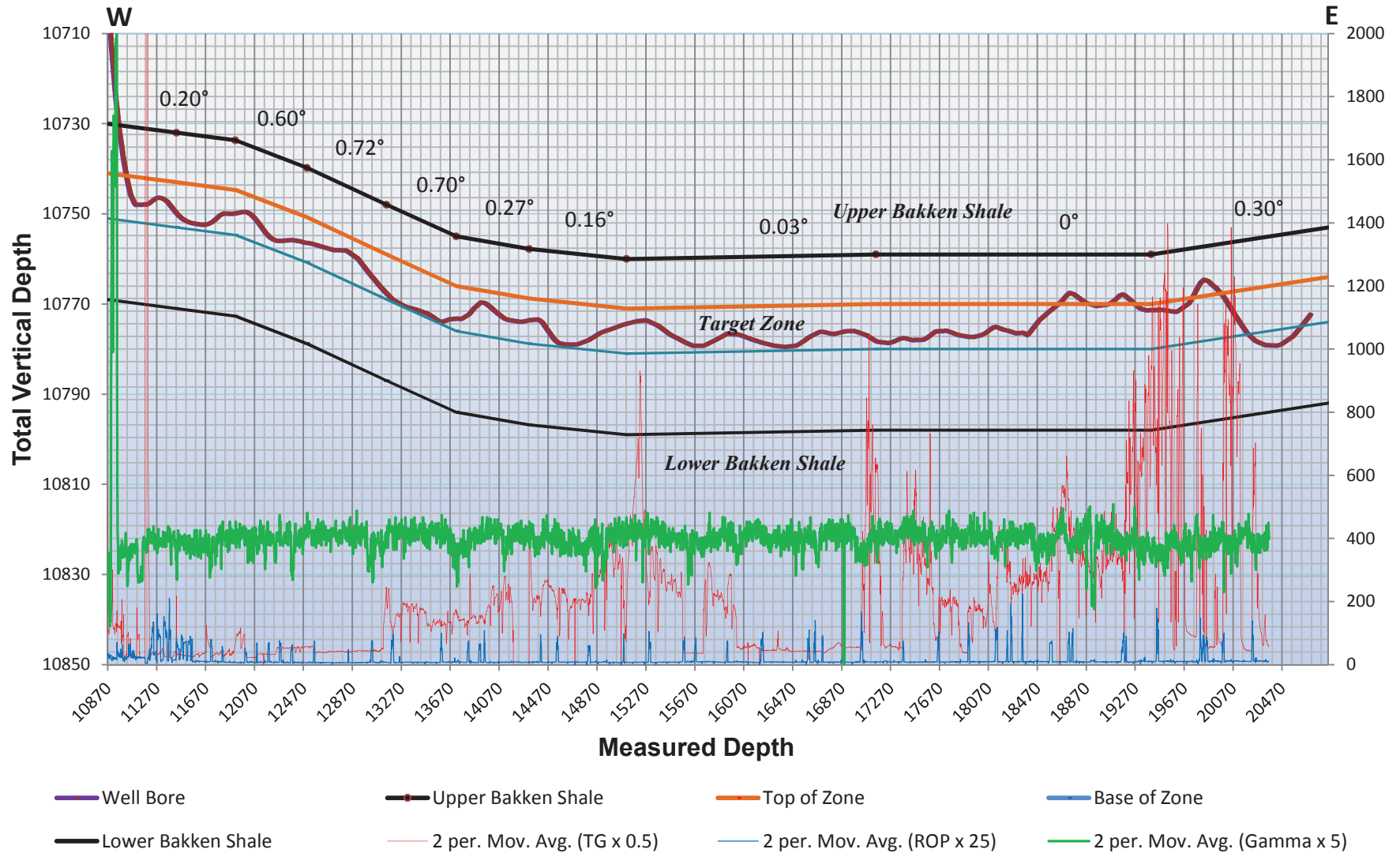




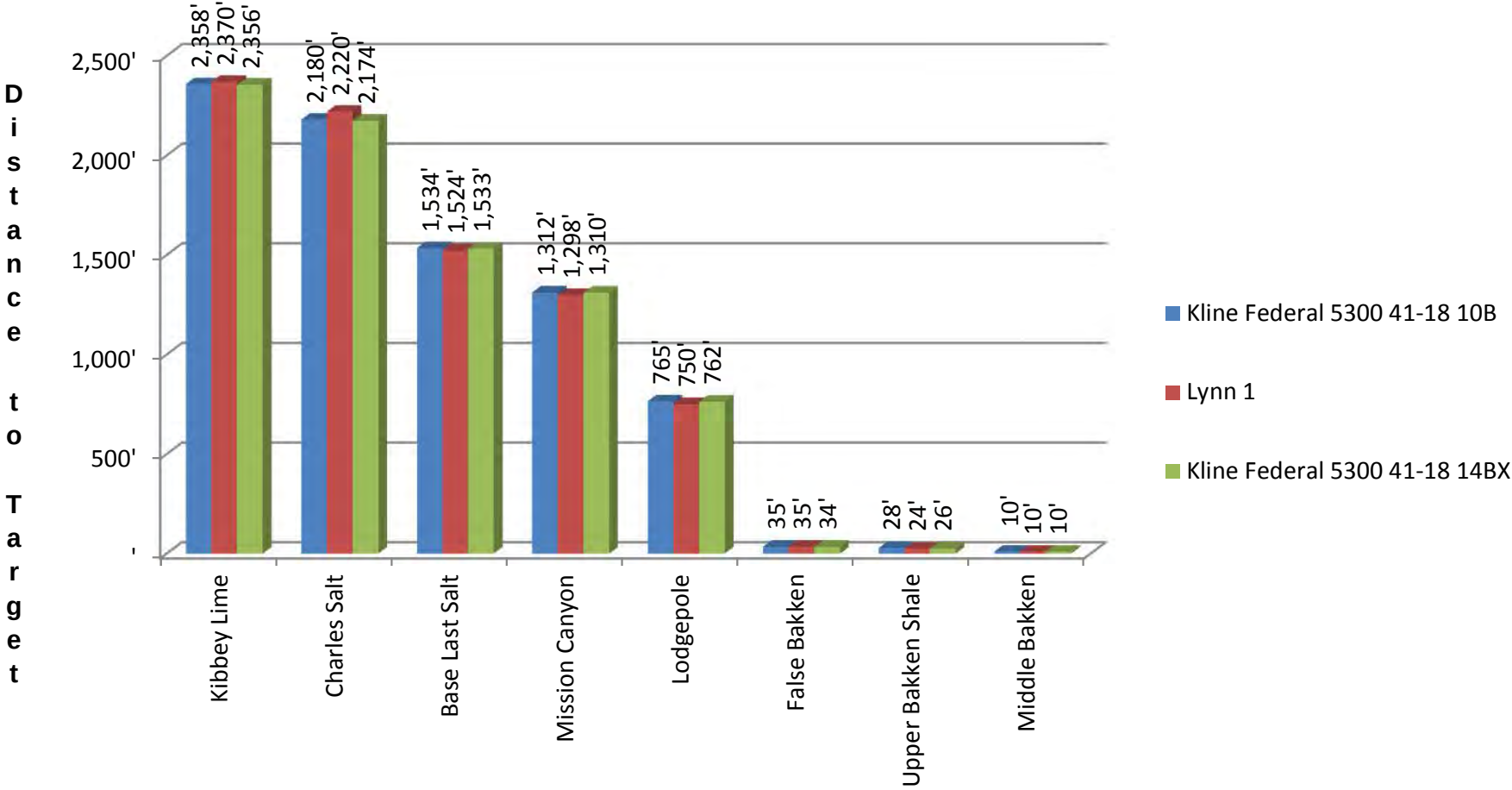
Surface: 368' FSL & 237' FWL
SW SW Sec. 18, T153N-R100W
Mckenzie County., ND

Kline Federal 5300 41-18 14BX Horizontal Cross Section

Bottom Hole Location-
560.34' S & 9,997.60' E
of surface location or approx.
192.34' FNL & 254.55' FEL
NE NE Sec. 20, T153N-R100W



Offset Well Target Isopachs





Company: Oasis Petroleum
 Company Man: Mark Lawlar
 Well: Kline Federal 5300 41-18 14BX
 Rig: Nabors 486
 Job #: RD0814M181
 Project: Indian Hills
 County: Williams, ND
 Field: Indian Hills

MWD SURVEY SHEET

Azimuth Reference: True North
 Vertical Section Azimuth: 90
 Magnetic Declination: 8.27
 Grid Correction: 0
 Total Correction Used: 8.27
 Vertical Reference: Rotary Table
 Calculation Method: Minimum Curvature
 Magnetic Model:
 Tool Error Model: Magnetic (Good)

Date: February 17, 2015
 MWD Operator: George Kerr
 API #: 33-053-06-028
 Latitude: 48.0686
 Longitude: 103.603
 Ground Elevation: 2057
 KB to Ground Level: 25
 Ideal Dip Angle: 72.99
 Ideal Total Magnetic Field: 0.564
 Geodetic System: State Plane NAD 1927
 Tool Type: MWD-Positive Pulse

Survey Type	SURVEY	MWD RUN #	BIT DEPTH	SURVEY DEPTH	INCLINATION	AZIMUTH	MAGNETIC DIP ANGLE	TOTAL FIELD (m)	GRAVITY TOTAL	D-H TEMP	TVD	Course Length	Vertical Section	North (N) South (S)	East (E) West (W)	Closure Dist (ft)	Closure Angle	DLS	Build Rate (° / 100 ft)	Walk Rate (° / 100 ft)
TIE IN				0	0	0					0.00			0.00	0.00					
MWD	1	1	194	140	0.97	48.35	75.1	0.576	1.001	63	139.99	140.00	0.9	0.79 N	0.89 E	1.2	48.4	0.69	0.69	34.54
MWD	2	1	288	234	0.70	53.45	73.2	0.568	1.001	63	233.98	94.00	1.9	1.66 N	1.94 E	2.6	49.5	0.30	-0.29	5.43
MWD	3	1	381	327	0.88	66.10	73.3	0.567	1.001	66	326.97	93.00	3.1	2.29 N	3.05 E	3.8	53.2	0.27	0.19	13.60
MWD	4	1	475	421	0.70	57.14	73	0.566	1.004	73	420.97	94.00	4.2	2.89 N	4.19 E	5.1	55.4	0.23	-0.19	-9.53
MWD	5	1	570	516	0.13	64.52	72.9	0.566	0.998	77	515.96	95.00	4.8	3.25 N	4.78 E	5.8	55.8	0.60	-0.60	7.77
MWD	6	1	661	607	0.57	64.08	73.2	0.564	1.001	82	606.96	91.00	5.3	3.49 N	5.28 E	6.3	56.5	0.48	0.48	-0.48
MWD	7	1	753	699	0.40	275.90	73	0.564	1.001	86	698.96	92.00	5.4	3.73 N	5.37 E	6.5	55.2	1.02	-0.18	230.24
MWD	8	1	845	791	0.57	257.18	73	0.566	1.001	89	790.96	92.00	4.6	3.66 N	4.60 E	5.9	51.5	0.25	0.18	-20.35
MWD	9	1	936	882	1.05	269.92	73	0.564	1.001	90	881.95	91.00	3.3	3.56 N	3.33 E	4.9	43.1	0.56	0.53	14.00
MWD	10	1	1028	974	1.19	290.93	72.9	0.565	1.001	91	973.93	92.00	1.6	3.90 N	1.59 E	4.2	22.2	0.47	0.15	22.84
MWD	11	1	1120	1066	0.53	329.77	73.1	0.566	1.001	93	1065.92	92.00	0.5	4.61 N	0.49 E	4.6	6.0	0.92	-0.72	42.22
MWD	12	1	1212	1158	0.53	310.09	72.9	0.564	1.001	95	1157.92	92.00	-0.1	5.25 N	0.05 W	5.2	359.4	0.20	0.00	-21.39
MWD	13	1	1304	1250	0.44	313.16	73.1	0.563	1.000	95	1249.91	92.00	-0.6	5.76 N	0.64 W	5.8	353.7	0.10	-0.10	3.34
MWD	14	1	1396	1342	0.48	182.65	72.7	0.579	1.001	97	1341.91	92.00	-0.9	5.62 N	0.91 W	5.7	350.8	0.91	0.04	-141.86
MWD	15	1	1487	1433	0.53	202.95	72.8	0.564	1.000	97	1432.91	91.00	-1.1	4.85 N	1.09 W	5.0	347.3	0.20	0.05	22.31
MWD	16	1	1582	1528	0.75	192.49	72.9	0.565	1.001	97	1527.90	95.00	-1.4	3.84 N	1.40 W	4.1	340.0	0.26	0.23	-11.01
MWD	17	1	1677	1623	1.14	214.81	73.2	0.563	1.001	99	1622.89	95.00	-2.1	2.46 N	2.07 W	3.2	319.8	0.56	0.41	23.49
MWD	18	1	1773	1719	1.23	196.27	74.4	0.560	1.001	95	1718.87	96.00	-2.9	0.68 N	2.91 W	3.0	283.2	0.41	0.09	-19.31
MWD	19	1	1867	1813	1.10	200.22	72.6	0.562	1.000	99	1812.85	94.00	-3.5	1.13 S	3.50 W	3.7	252.1	0.16	-0.14	4.20
MWD	20	1	1962	1908	1.27	204.18	73	0.564	1.000	100	1907.83	95.00	-4.2	2.95 S	4.25 W	5.2	235.2	0.20	0.18	4.17
MWD	21	1	2057	2003	0.18	146.43	72.9	0.563	1.001	102	2002.82	95.00	-4.6	4.03 S	4.60 W	6.1	228.7	1.25	-1.15	-60.79
MWD	22	1	2151	2097	0.26	11.79	73.3	0.561	1.001	104	2096.82	94.00	-4.5	3.95 S	4.47 W	6.0	228.6	0.43	0.09	-143.23
MWD	23	2	2240	2189	0.44	69.44	74.2	0.590	1.001	91	2188.82	92.00	-4.1	3.62 S	4.10 W	5.5	228.5	0.40	0.20	62.66
MWD	24	2	2334	2283	0.48	94.84	73.6	0.583	1.001	95	2282.81	94.00	-3.4	3.53 S	3.37 W	4.9	223.7	0.22	0.04	27.02
MWD	25	2	2428	2377	0.18	82.01	73.6	0.583	1.001	97	2376.81	94.00	-2.8	3.54 S	2.83 W	4.5	218.6	0.33	-0.32	-13.65
MWD	26	2	2522	2471	0.40	138.61	73.6	0.583	1.001	100	2470.81	94.00	-2.5	3.76 S	2.47 W	4.5	213.2	0.36	0.23	60.21
MWD	27	2	2617	2566	0.31	128.06	73.6	0.583	1.001	102	2565.81	95.00	-2.0	4.17 S	2.04 W	4.6	206.1	0.12	-0.09	-11.11
MWD	28	2	2711	2660	0.31	129.38	73.5	0.583	1.001	104	2659.81	94.00	-1.6	4.49 S	1.65 W	4.8	200.1	0.01	0.00	1.40
MWD	29	2	2807	2756	0.26	29.10	73.5	0.582	1.001	106	2755.81	96.00	-1.3	4.46 S	1.34 W	4.7	196.7	0.46	-0.05	-104.46
MWD	30	2	2902	2851	0.13	110.57	73.6	0.582	1.001	109	2850.81	95.00	-1.1	4.31 S	1.13 W	4.5	194.7	0.29	-0.14	85.76
MWD	31	2	2995	2944	0.31	347.35	73.5	0.582	1.002	111	2943.81	93.00	-1.1	4.10 S	1.09 W	4.2	194.9	0.43	0.19	254.60
MWD	32	2	3089	3038	0.31	58.54	73.5	0.582	1.002	111	3037.81	94.00	-0.9	3.72 S	0.93 W	3.8	194.0	0.38	0.00	-307.24
MWD	33	2	3184	3133	0.22	55.99	73.6	0.582	1.002	113	3132.80	95.00	-0.6	3.49 S	0.56 W	3.5	189.1	0.10	-0.09	-2.68
MWD	34	2	3277	3226	0.31	11.08	73.5	0.582	1.001	113	3225.80	93.00	-0.4	3.14 S	0.36 W	3.2	186.6	0.24	0.10	-48.29
MWD	35	2	3372	3321	0.57	51.78	73.5	0.582	1.002	113	3320.80	95.00	0.1	2.60 S	0.06 E	2.6	178.7	0.41	0.27	42.84
MWD	36	2	3467	3416	0.62	37.63	73.5	0.582	1.002	116	3415.80	95.00	0.7	1.90 S	0.74 E	2.0	158.6	0.16	0.05	-14.89
MWD	37	2	3562	3511	0.35	10.82	73.5	0.582	1.001	118	3510.79	95.00	1.1	1.20 S	1.11 E	1.6	137.3	0.36	-0.28	-28.22

MWD	38	2	3656	3605	0.92	143.45	73.5	0.582	1.001	120	3604.79	94.00	1.6	1.53 S	1.61 E	2.2	133.4	1.26	0.61	141.10
MWD	39	2	3751	3700	0.70	153.20	73.6	0.582	1.001	122	3699.78	95.00	2.3	2.66 S	2.33 E	3.5	138.8	0.27	-0.23	10.26
MWD	40	2	3845	3794	1.05	143.01	73.5	0.582	1.001	123	3793.77	94.00	3.1	3.86 S	3.11 E	5.0	141.2	0.41	0.37	-10.84
MWD	41	2	3939	3888	0.97	154.43	73.5	0.582	1.001	125	3887.75	94.00	4.0	5.27 S	3.97 E	6.6	143.0	0.23	-0.09	12.15
MWD	42	2	4033	3982	0.57	197.85	73.6	0.583	1.002	123	3981.75	94.00	4.2	6.43 S	4.17 E	7.7	147.0	0.72	-0.43	46.19
MWD	43	2	4128	4077	0.57	215.52	73.5	0.582	1.002	127	4076.74	95.00	3.7	7.26 S	3.75 E	8.2	152.7	0.18	0.00	18.60
MWD	44	2	4221	4170	0.75	221.49	73.6	0.582	1.002	129	4169.73	93.00	3.1	8.10 S	3.08 E	8.7	159.2	0.21	0.19	6.42
MWD	45	2	4315	4264	0.53	212.88	73.6	0.583	1.002	129	4263.73	94.00	2.4	8.92 S	2.43 E	9.2	164.7	0.25	-0.23	-9.16
MWD	46	2	4411	4360	0.66	219.82	73.6	0.583	1.001	131	4359.72	96.00	1.8	9.72 S	1.84 E	9.9	169.3	0.15	0.14	7.23
MWD	47	2	4506	4455	0.40	229.58	73.5	0.583	1.001	134	4454.72	95.00	1.2	10.35 S	1.24 E	10.4	173.2	0.29	-0.27	10.27
MWD	48	2	4600	4549	0.66	216.75	73.5	0.580	1.001	136	4548.72	94.00	0.7	11.00 S	0.66 E	11.0	176.6	0.30	0.28	-13.65
MWD	49	2	4695	4644	0.44	172.01	73.6	0.583	1.002	140	4643.71	95.00	0.4	11.80 S	0.39 E	11.8	178.1	0.49	-0.23	-47.09
MWD	50	2	4790	4739	0.35	211.56	73.6	0.583	1.002	144	4738.71	95.00	0.3	12.41 S	0.28 E	12.4	178.7	0.30	-0.09	41.63
MWD	51	2	4885	4834	0.13	102.93	73.6	0.583	1.000	145	4833.71	95.00	0.2	12.68 S	0.24 E	12.7	178.9	0.43	-0.23	-114.35
MWD	52	2	4980	4929	0.22	170.78	73.6	0.583	1.002	145	4928.71	95.00	0.4	12.88 S	0.37 E	12.9	178.3	0.22	0.09	71.42
MWD	53	2	5074	5023	0.26	211.91	73.6	0.583	1.002	145	5022.71	94.00	0.3	13.24 S	0.29 E	13.2	178.8	0.18	0.04	43.76
MWD	54	2	5169	5118	0.79	153.03	73.6	0.583	1.002	149	5117.70	95.00	0.5	14.01 S	0.47 E	14.0	178.1	0.73	0.56	-61.98
MWD	55	2	5264	5213	0.88	146.61	73.6	0.583	1.001	151	5212.69	95.00	1.2	15.20 S	1.17 E	15.2	175.6	0.14	0.09	-6.76
MWD	56	2	5358	5307	0.84	139.05	73.6	0.582	1.001	151	5306.68	94.00	2.0	16.33 S	2.02 E	16.5	173.0	0.13	-0.04	-8.04
MWD	57	2	5453	5402	0.97	125.78	73.6	0.583	1.001	153	5401.67	95.00	3.1	17.32 S	3.13 E	17.6	169.8	0.26	0.14	-13.97
MWD	58	2	5547	5496	0.62	103.28	73.5	0.583	1.002	153	5495.66	94.00	4.3	17.90 S	4.27 E	18.4	166.6	0.49	-0.37	-23.94
MWD	59	2	5642	5591	0.70	112.51	73.5	0.582	1.002	154	5590.66	95.00	5.3	18.24 S	5.30 E	19.0	163.8	0.14	0.08	9.72
MWD	60	2	5737	5686	0.84	150.74	73.5	0.582	1.003	155	5685.65	95.00	6.2	19.07 S	6.18 E	20.1	162.0	0.55	0.15	40.24
MWD	61	2	5831	5780	0.66	141.42	73.5	0.582	1.002	155	5779.64	94.00	6.9	20.10 S	6.86 E	21.2	161.2	0.23	-0.19	-9.91
MWD	62	2	5926	5875	0.66	170.69	73.6	0.583	1.002	131	5874.63	95.00	7.3	21.07 S	7.28 E	22.3	160.9	0.35	0.00	30.81
MWD	63	2	6021	5970	0.57	160.94	73.6	0.583	1.002	138	5969.63	95.00	7.5	22.05 S	7.53 E	23.3	161.2	0.14	-0.09	-10.26
MWD	64	2	6115	6064	0.22	128.68	73.6	0.583	1.001	145	6063.63	94.00	7.8	22.61 S	7.82 E	23.9	160.9	0.43	-0.37	-34.32
MWD	65	2	6150	6095	0.48	201.28	73.5	0.583	1.000	145	6094.63	31.00	7.8	22.77 S	7.82 E	24.1	161.0	1.50	0.84	234.19
MWD	66	3	6207	6160	0.22	168.76	72.6	0.570	1.002	141	6159.62	65.00	7.7	23.14 S	7.75 E	24.4	161.5	0.49	-0.40	-50.03
MWD	67	3	6301	6254	0.31	178.95	73.2	0.569	1.002	147	6253.62	94.00	7.8	23.57 S	7.79 E	24.8	161.7	0.11	0.10	10.84
MWD	68	3	6396	6349	0.13	240.04	73.2	0.570	1.002	153	6348.62	95.00	7.7	23.88 S	7.70 E	25.1	162.1	0.29	-0.19	64.31
MWD	69	3	6490	6443	0.18	329.51	73.2	0.570	1.002	154	6442.62	94.00	7.5	23.81 S	7.53 E	25.0	162.4	0.24	0.05	95.18
MWD	70	3	6585	6538	0.66	0.54	73.1	0.569	1.002	158	6537.62	95.00	7.5	23.13 S	7.46 E	24.3	162.1	0.54	0.51	-346.28
MWD	71	3	6680	6633	0.18	243.55	73.1	0.569	1.002	161	6632.62	95.00	7.3	22.65 S	7.33 E	23.8	162.1	0.80	-0.51	255.80
MWD	72	3	6773	6726	0.18	275.98	73.2	0.570	1.002	167	6725.62	93.00	7.1	22.70 S	7.05 E	23.8	162.7	0.11	0.00	34.87
MWD	73	3	6868	6821	0.57	259.20	73.2	0.569	1.002	170	6820.61	95.00	6.4	22.78 S	6.44 E	23.7	164.2	0.42	0.41	-17.66
MWD	74	3	6962	6915	0.40	253.75	73.1	0.570	1.002	172	6914.61	94.00	5.7	22.96 S	5.67 E	23.6	166.1	0.19	-0.18	-5.80
MWD	75	3	7057	7010	0.53	278.71	73.2	0.570	1.002	174	7009.61	95.00	4.9	22.98 S	4.92 E	23.5	167.9	0.25	0.14	26.27
MWD	76	3	7151	7104	1.14	203.21	73.2	0.570	1.002	176	7103.60	94.00	4.1	23.78 S	4.12 E	24.1	170.2	1.20	0.65	-80.32
MWD	77	3	7245	7198	1.05	209.72	73.2	0.569	1.002	178	7197.58	94.00	3.3	25.38 S	3.32 E	25.6	172.5	0.16	-0.10	6.93
MWD	78	3	7340	7293	1.14	219.82	73.2	0.570	1.002	180	7292.56	95.00	2.3	26.86 S	2.28 E	27.0	175.1	0.22	0.09	10.63
MWD	79	3	7433	7386	1.14	217.98	73.1	0.569	1.002	181	7385.55	93.00	1.1	28.30 S	1.12 E	28.3	177.7	0.04	0.00	-1.98
MWD	80	3	7528	7481	1.32	197.23	73.2	0.570	1.002	181	7480.53	95.00	0.2	30.09 S	0.22 E	30.1	179.6	0.50	0.19	-21.84
MWD	81	3	7623	7576	1.19	193.54	73.2	0.570	1.001	183	7575.50	95.00	-0.3	32.10 S	0.34 W	32.1	180.6	0.16	-0.14	-3.88
MWD	82	3	7717	7670	1.05	202.07	73.1	0.570	1.002	183	7669.48	94.00	-0.9	33.85 S	0.89 W	33.9	181.5	0.23	-0.15	9.07
MWD	83	3	7811	7764	1.19	213.06	73.1	0.569	1.002	185	7763.47	94.00	-1.7	35.46 S	1.75 W	35.5	182.8	0.27	0.15	11.69
MWD	84	3	7904	7857	0.92	202.68	73.1	0.570	1.002	185	7856.45	93.00	-2.6	36.96 S	2.56 W	37.0	184.0	0.35	-0.29	-11.16
MWD	85	3	7998	7951	1.01	216.22	73.2	0.570	1.002	186	7950.44	94.00	-3.3	38.32 S	3.34 W	38.5	185.0	0.26	0.10	14.40

MWD	86	3	8092	8045	0.70	208.84	73.2	0.570	1.002	188	8044.43	94.00	-4.1	39.50 S	4.11 W	39.7	185.9	0.35	-0.33	-7.85
MWD	87	3	8187	8140	0.66	225.01	73.2	0.570	1.002	190	8139.42	95.00	-4.8	40.39 S	4.77 W	40.7	186.7	0.21	-0.04	17.02
MWD	88	3	8281	8234	0.66	239.16	73.2	0.570	1.002	190	8233.41	94.00	-5.6	41.05 S	5.62 W	41.4	187.8	0.17	0.00	15.05
MWD	89	3	8376	8329	1.01	196.44	73.2	0.570	1.002	190	8328.40	95.00	-6.3	42.14 S	6.33 W	42.6	188.5	0.73	0.37	-44.97
MWD	90	3	8470	8423	1.23	200.14	73.2	0.571	1.002	187	8422.39	94.00	-6.9	43.88 S	6.91 W	44.4	189.0	0.25	0.23	3.94
MWD	91	3	8564	8517	1.76	180.62	73.2	0.570	1.002	189	8516.35	94.00	-7.3	46.27 S	7.27 W	46.8	188.9	0.77	0.56	-20.77
MWD	92	3	8658	8611	2.02	166.30	73.2	0.571	1.002	189	8610.30	94.00	-6.9	49.32 S	6.90 W	49.8	188.0	0.57	0.28	-15.23
MWD	93	3	8753	8706	1.45	182.29	73.2	0.571	1.002	189	8705.26	95.00	-6.5	52.15 S	6.55 W	52.6	187.2	0.78	-0.60	16.83
MWD	94	3	8848	8801	0.92	170.60	73.2	0.571	1.002	192	8800.24	95.00	-6.5	54.10 S	6.47 W	54.5	186.8	0.61	-0.56	-12.31
MWD	95	3	8942	8895	0.75	172.80	73.1	0.571	1.002	194	8894.23	94.00	-6.3	55.46 S	6.27 W	55.8	186.5	0.18	-0.18	2.34
MWD	96	3	9036	8989	0.40	73.40	73.2	0.571	1.002	194	8988.23	94.00	-5.9	55.97 S	5.88 W	56.3	186.0	0.96	-0.37	-105.74
MWD	97	3	9130	9083	0.57	75.42	73.2	0.571	1.002	197	9082.22	94.00	-5.1	55.76 S	5.11 W	56.0	185.2	0.18	0.18	2.15
MWD	98	3	9224	9177	0.35	52.39	73.2	0.571	1.002	197	9176.22	94.00	-4.4	55.47 S	4.43 W	55.6	184.6	0.30	-0.23	-24.50
MWD	99	3	9318	9271	0.40	42.55	73.2	0.571	1.002	199	9270.22	94.00	-4.0	55.05 S	3.98 W	55.2	184.1	0.09	0.05	-10.47
MWD	100	3	9413	9366	0.31	33.67	73.2	0.571	1.002	199	9365.22	95.00	-3.6	54.59 S	3.62 W	54.7	183.8	0.11	-0.09	-9.35
MWD	101	3	9508	9461	0.35	26.11	73.1	0.571	1.002	201	9460.21	95.00	-3.3	54.12 S	3.35 W	54.2	183.5	0.06	0.04	-7.96
MWD	102	3	9601	9554	0.22	19.96	73.2	0.571	1.002	201	9553.21	93.00	-3.2	53.70 S	3.16 W	53.8	183.4	0.14	-0.14	-6.61
MWD	103	3	9695	9648	0.35	57.31	73.3	0.571	1.002	203	9647.21	94.00	-2.9	53.37 S	2.86 W	53.4	183.1	0.23	0.14	39.73
MWD	104	3	9788	9741	0.35	40.88	73.2	0.571	1.002	203	9740.21	93.00	-2.4	53.00 S	2.43 W	53.1	182.6	0.11	0.00	-17.67
MWD	105	3	9882	9835	0.18	24.27	73.2	0.571	1.002	205	9834.21	94.00	-2.2	52.65 S	2.18 W	52.7	182.4	0.20	-0.18	-17.67
MWD	106	3	9976	9929	0.26	348.06	73.2	0.571	1.002	207	9928.21	94.00	-2.2	52.31 S	2.17 W	52.4	182.4	0.17	0.09	344.46
MWD	107	3	10070	10023	0.18	3.96	73.2	0.571	1.002	208	10022.21	94.00	-2.2	51.95 S	2.20 W	52.0	182.4	0.11	-0.09	-366.06
MWD	108	3	10165	10118	0.44	320.37	73.2	0.571	1.002	208	10117.21	95.00	-2.4	51.52 S	2.42 W	51.6	182.7	0.35	0.27	333.06
MWD	109	3	10260	10213	0.35	317.82	73.2	0.570	1.002	208	10212.20	95.00	-2.9	51.03 S	2.85 W	51.1	183.2	0.10	-0.09	-2.68
MWD	110	3	10291	10244	0.26	348.67	73.2	0.571	1.002	208	10243.20	31.00	-2.9	50.89 S	2.93 W	51.0	183.3	0.59	-0.29	99.52
MWD	111	4	10356	10276	0.53	100.38	73.1	0.567	1.002	195	10275.20	32.00	-2.8	50.84 S	2.80 W	50.9	183.1	2.10	0.84	-775.91
MWD	112	4	10388	10308	3.43	114.00	73	0.567	1.002	200	10307.18	32.00	-1.8	51.26 S	1.78 W	51.3	182.0	9.12	9.06	42.56
MWD	113	4	10420	10340	7.08	115.67	73.1	0.567	1.002	192	10339.04	32.00	0.9	52.50 S	0.88 E	52.5	179.0	11.42	11.41	5.22
MWD	114	5	10450	10373	10.51	112.42	73.4	0.572	1.004	185	10371.65	33.00	5.5	54.53 S	5.49 E	54.8	174.2	10.50	10.39	-9.85
MWD	115	5	10481	10404	13.63	108.55	73.4	0.572	1.004	185	10401.96	31.00	11.6	56.77 S	11.57 E	57.9	168.5	10.39	10.06	-12.48
MWD	116	5	10513	10436	17.63	108.29	73.4	0.572	1.004	190	10432.77	32.00	19.7	59.50 S	19.75 E	62.7	161.6	12.50	12.50	-0.81
MWD	117	5	10545	10468	21.85	107.76	73.4	0.571	1.004	190	10462.88	32.00	30.0	62.83 S	30.03 E	69.6	154.5	13.20	13.19	-1.66
MWD	118	5	10577	10500	25.93	106.62	73.2	0.571	1.004	192	10492.14	32.00	42.4	66.65 S	42.41 E	79.0	147.5	12.83	12.75	-3.56
MWD	119	5	10608	10531	30.46	108.11	73.3	0.573	1.003	194	10519.45	31.00	56.4	71.04 S	56.38 E	90.7	141.6	14.79	14.61	4.81
MWD	120	5	10640	10563	34.95	108.47	73.4	0.572	1.004	196	10546.37	32.00	72.8	76.46 S	72.79 E	105.6	136.4	14.04	14.03	1.13
MWD	121	5	10672	10595	39.34	108.73	73.4	0.571	1.004	196	10571.87	32.00	91.1	82.63 S	91.10 E	123.0	132.2	13.73	13.72	0.81
MWD	122	5	10703	10626	44.26	110.40	73.3	0.572	1.004	198	10594.97	31.00	110.6	89.56 S	110.55 E	142.3	129.0	16.27	15.87	5.39
MWD	123	5	10735	10658	48.88	111.81	73.3	0.574	1.003	199	10616.97	32.00	132.2	97.93 S	132.22 E	164.5	126.5	14.79	14.44	4.41
MWD	124	5	10766	10689	53.49	113.12	73.2	0.574	1.003	199	10636.39	31.00	154.5	107.17 S	154.53 E	188.1	124.7	15.23	14.87	4.23
MWD	125	5	10798	10721	57.58	115.41	72.9	0.570	1.004	201	10654.50	32.00	178.6	118.02 S	178.57 E	214.0	123.5	14.08	12.78	7.16
MWD	126	5	10829	10752	60.35	118.13	73.1	0.568	1.004	201	10670.48	31.00	202.3	129.99 S	202.28 E	240.4	122.7	11.68	8.94	8.77
MWD	127	5	10860	10783	62.37	120.42	73.3	0.573	1.003	201	10685.34	31.00	226.0	143.30 S	226.00 E	267.6	122.4	9.19	6.52	7.39
MWD	128	5	10892	10815	64.53	121.03	73.4	0.571	1.003	201	10699.65	32.00	250.6	157.92 S	250.61 E	296.2	122.2	6.96	6.75	1.91
MWD	129	5	10924	10847	67.87	121.39	73.1	0.567	1.004	203	10712.56	32.00	275.6	173.09 S	275.65 E	325.5	122.1	10.49	10.44	1.13
MWD	130	5	10954	10877	72.57	122.09	73.2	0.575	1.003	208	10722.71	30.00	299.6	187.94 S	299.65 E	353.7	122.1	15.82	15.67	2.33
MWD	131	5	10986	10909	77.63	120.07	72.5	0.572	1.003	205	10730.93	32.00	326.1	203.89 S	326.12 E	384.6	122.0	16.95	15.81	-6.31
MWD	132	5	11018	10941	80.22	120.33	73.3	0.568	1.003	203	10737.08	32.00	353.3	219.69 S	353.26 E	416.0	121.9	8.13	8.09	0.81
MWD	133	5	11049	10972	81.45	120.42	73.4	0.574	1.003	205	10742.02	31.00	379.7	235.16 S	379.67 E	446.6	121.8	3.98	3.97	0.29

MWD	134	5	11080	11003	84.66	120.24	73.3	0.57	1.003	207	10745.77	31.00	406.2	250.70 S	406.22 E	477.4	121.7	10.37	10.35	-0.58
MWD	135	5	11112	11035	88.35	119.28	73.3	0.568	1.003	207	10747.72	32.00	433.9	266.55 S	433.95 E	509.3	121.6	11.91	11.53	-3.00
MWD	136	5	11125	11048	88.97	117.96	72.6	0.568	1.003	207	10748.02	13.00	445.4	272.78 S	445.36 E	522.3	121.5	11.22	4.77	-10.15
MWD	137	6	11222	11145	91.16	119.01	73.1	0.561	0.999	221	10747.91	97.00	530.6	319.04 S	530.61 E	619.1	121.0	2.50	2.26	1.08
MWD	138	6	11253	11176	91.16	118.48	73.1	0.565	0.999	221	10747.28	31.00	557.8	333.94 S	557.78 E	650.1	120.9	1.71	0.00	-1.71
MWD	139	6	11284	11207	91.16	117.69	72.9	0.561	0.999	221	10746.66	31.00	585.1	348.53 S	585.12 E	681.1	120.8	2.55	0.00	-2.55
MWD	140	6	11315	11238	89.85	115.94	72.9	0.561	0.999	219	10746.38	31.00	612.8	362.51 S	612.79 E	712.0	120.6	7.05	-4.23	-5.65
MWD	141	6	11347	11270	89.32	114.79	72.9	0.562	1.000	231	10746.61	32.00	641.7	376.22 S	641.70 E	743.9	120.4	3.96	-1.66	-3.59
MWD	142	6	11378	11301	88.35	113.12	72.9	0.562	0.999	221	10747.24	31.00	670.0	388.81 S	670.02 E	774.7	120.1	6.23	-3.13	-5.39
MWD	143	7	11410	11332	87.87	111.81	72.9	0.562	0.999	225	10748.27	31.00	698.7	400.64 S	698.66 E	805.4	119.8	4.50	-1.55	-4.23
MWD	144	7	11440	11362	88.18	111.72	72.9	0.561	0.999	225	10749.30	30.00	726.5	411.76 S	726.50 E	835.1	119.5	1.08	1.03	-0.30
MWD	145	7	11471	11393	88.09	109.26	72.9	0.565	0.999	225	10750.31	31.00	755.5	422.61 S	755.52 E	865.7	119.2	7.94	-0.29	-7.94
MWD	146	7	11502	11424	89.23	106.44	72.9	0.565	0.999	225	10751.03	31.00	785.0	432.11 S	785.02 E	896.1	118.8	9.81	3.68	-9.10
MWD	147	7	11534	11456	88.97	103.37	73	0.562	0.999	225	10751.54	32.00	815.9	440.33 S	815.93 E	927.2	118.4	9.63	-0.81	-9.59
MWD	148	7	11564	11486	89.63	100.20	73	0.562	0.999	225	10751.90	30.00	845.3	446.46 S	845.29 E	956.0	117.8	10.79	2.20	-10.57
MWD	149	7	11595	11517	89.63	98.97	73	0.562	0.999	225	10752.10	31.00	875.9	451.62 S	875.86 E	985.4	117.3	3.97	0.00	-3.97
MWD	150	7	11627	11549	89.85	98.97	73	0.561	0.999	225	10752.25	32.00	907.5	456.61 S	907.47 E	1015.9	116.7	0.69	0.69	0.00
MWD	151	7	11658	11580	89.63	98.62	73	0.561	0.999	225	10752.39	31.00	938.1	461.35 S	938.10 E	1045.4	116.2	1.33	-0.71	-1.13
MWD	152	7	11689	11611	90.2	99.06	73	0.561	0.999	226	10752.44	31.00	968.7	466.11 S	968.74 E	1075.0	115.7	2.32	1.84	1.42
MWD	153	7	11720	11642	90.42	99.06	73	0.561	1.000	228	10752.27	31.00	999.3	471.00 S	999.35 E	1104.8	115.2	0.71	0.71	0.00
MWD	154	7	11751	11673	91.25	100.12	73	0.561	1.000	228	10751.82	31.00	1029.9	476.16 S	1029.91 E	1134.7	114.8	4.34	2.68	3.42
MWD	155	7	11783	11705	91.34	100.47	73.1	0.562	0.999	228	10751.09	32.00	1061.4	481.88 S	1061.39 E	1165.7	114.4	1.13	0.28	1.09
MWD	156	7	11814	11736	91.52	100.47	73.1	0.562	0.999	228	10750.32	31.00	1091.9	487.51 S	1091.86 E	1195.8	114.1	0.58	0.58	0.00
MWD	157	7	11845	11767	89.98	101.35	72.9	0.562	1.000	228	10749.91	31.00	1122.3	493.38 S	1122.30 E	1226.0	113.7	5.72	-4.97	2.84
MWD	158	7	11876	11798	89.8	101.43	73.1	0.562	1.000	228	10749.97	31.00	1152.7	499.50 S	1152.69 E	1256.3	113.4	0.64	-0.58	0.26
MWD	159	7	11907	11829	90.02	101.35	73.1	0.562	0.999	230	10750.02	31.00	1183.1	505.62 S	1183.08 E	1286.6	113.1	0.76	0.71	-0.26
MWD	160	7	11938	11860	90.33	101.80	73	0.561	0.999	231	10749.93	31.00	1213.4	511.84 S	1213.45 E	1317.0	112.9	1.76	1.00	1.45
MWD	161	7	11970	11892	90.2	101.08	73.1	0.561	0.999	232	10749.78	32.00	1244.8	518.19 S	1244.81 E	1348.4	112.6	2.29	-0.41	-2.25
MWD	162	7	12001	11923	90.46	100.64	73	0.561	1.000	232	10749.60	31.00	1275.3	524.03 S	1275.25 E	1378.7	112.3	1.65	0.84	-1.42
MWD	163	7	12032	11954	89.54	98.18	73	0.561	1.000	232	10749.60	31.00	1305.8	529.10 S	1305.83 E	1409.0	112.1	8.47	-2.97	-7.94
MWD	164	7	12063	11985	88.7	97.74	73.1	0.562	0.999	230	10750.08	31.00	1336.5	533.39 S	1336.53 E	1439.0	111.8	3.06	-2.71	-1.42
MWD	165	7	12094	12016	88.22	97.22	73	0.562	1.000	232	10750.91	31.00	1367.3	537.42 S	1367.26 E	1469.1	111.5	2.28	-1.55	-1.68
MWD	166	7	12125	12047	87.91	95.81	73	0.561	0.999	232	10751.96	31.00	1398.0	540.94 S	1398.04 E	1499.0	111.2	4.65	-1.00	-4.55
MWD	167	7	12156	12078	87.82	95.72	73.1	0.561	0.999	232	10753.11	31.00	1428.9	544.05 S	1428.86 E	1528.9	110.8	0.41	-0.29	-0.29
MWD	168	7	12188	12110	87.91	95.55	73.1	0.561	1.000	234	10754.30	32.00	1460.7	547.19 S	1460.68 E	1559.8	110.5	0.60	0.28	-0.53
MWD	169	7	12219	12141	88.88	94.40	72.9	0.561	0.999	234	10755.17	31.00	1491.6	549.88 S	1491.55 E	1589.7	110.2	4.85	3.13	-3.71
MWD	170	7	12250	12172	89.19	94.40	73	0.561	0.999	232	10755.69	31.00	1522.5	552.25 S	1522.46 E	1619.5	109.9	1.00	1.00	0.00
MWD	171	7	12281	12203	89.71	94.40	73	0.561	0.999	234	10755.99	31.00	1553.4	554.63 S	1553.36 E	1649.4	109.6	1.68	1.68	0.00
MWD	172	7	12312	12234	90.2	92.73	73	0.561	0.999	235	10756.02	31.00	1584.3	556.56 S	1584.30 E	1679.2	109.4	5.61	1.58	-5.39
MWD	173	7	12344	12266	90.07	92.82	73.1	0.561	0.999	234	10755.94	32.00	1616.3	558.11 S	1616.27 E	1709.9	109.1	0.49	-0.41	0.28
MWD	174	7	12375	12297	90.2	92.82	73	0.561	0.999	235	10755.87	31.00	1647.2	559.63 S	1647.23 E	1739.7	108.8	0.42	0.42	0.00
MWD	175	7	12406	12328	90.07	90.10	73	0.561	0.999	237	10755.79	31.00	1678.2	560.42 S	1678.22 E	1769.3	108.5	8.78	-0.42	-8.77
MWD	176	7	12468	12390	89.41	89.92	73	0.561	0.999	237	10756.08	62.00	1740.2	560.43 S	1740.21 E	1828.2	107.9	1.10	-1.06	-0.29
MWD	177	7	12499	12421	89.63	90.10	73	0.561	0.999	237	10756.34	31.00	1771.2	560.44 S	1771.21 E	1857.8	107.6	0.92	0.71	0.58
MWD	178	7	12560	12482	89.89	89.39	73	0.561	0.999	237	10756.59	61.00	1832.2	560.17 S	1832.21 E	1915.9	107.0	1.24	0.43	-1.16
MWD	179	7	12655	12577	89.19	90.27	73	0.561	0.999	241	10757.35	95.00	1927.2	559.89 S	1927.21 E	2006.9	106.2	1.18	-0.74	0.93
MWD	180	7	12749	12671	89.93	90.18	73	0.561	0.999	241	10758.08	94.00	2021.2	560.26 S	2021.20 E	2097.4	105.5	0.79	0.79	-0.10
MWD	181	7	12781	12703	90.02	90.27	73	0.561	0.999	243	10758.09	32.00	2053.2	560.38 S	2053.20 E	2128.3	105.3	0.40	0.28	0.28

MWD	182	7	12844	12766	89.93	89.83	73	0.561	0.999	243	10758.12	63.00	2116.2	560.44 S	2116.20 E	2189.2	104.8	0.71	-0.14	-0.70
MWD	183	7	12938	12860	87.91	88.87	73	0.561	0.999	244	10759.89	94.00	2210.2	559.37 S	2210.17 E	2279.9	104.2	2.38	-2.15	-1.02
MWD	184	7	12970	12892	87.96	89.13	73	0.561	0.999	244	10761.04	32.00	2242.1	558.81 S	2242.15 E	2310.7	104.0	0.83	0.16	0.81
MWD	185	7	13033	12955	88	88.95	73	0.561	0.999	244	10763.26	63.00	2305.1	557.76 S	2305.10 E	2371.6	103.6	0.29	0.06	-0.29
MWD	186	7	13064	12986	88.35	90.54	73	0.561	0.999	244	10764.25	31.00	2336.1	557.62 S	2336.08 E	2401.7	103.4	5.25	1.13	5.13
MWD	187	7	13128	13050	87.91	90.45	73	0.561	0.999	244	10766.34	64.00	2400.0	558.17 S	2400.05 E	2464.1	103.1	0.70	-0.69	-0.14
MWD	188	7	13222	13144	88.84	90.89	73	0.561	0.999	243	10769.00	94.00	2494.0	559.27 S	2494.00 E	2555.9	102.6	1.09	0.99	0.47
MWD	189	7	13316	13238	89.41	91.15	73	0.561	0.999	243	10770.44	94.00	2588.0	560.94 S	2587.97 E	2648.1	102.2	0.67	0.61	0.28
MWD	190	7	13348	13270	89.49	90.54	73	0.561	0.999	243	10770.75	32.00	2620.0	561.42 S	2619.97 E	2679.4	102.1	1.92	0.25	-1.91
MWD	191	7	13411	13333	89.71	90.80	73	0.565	0.999	243	10771.19	63.00	2683.0	562.15 S	2682.96 E	2741.2	101.8	0.54	0.35	0.41
MWD	192	7	13505	13427	89.19	90.54	73	0.565	0.999	250	10772.09	94.00	2777.0	563.25 S	2776.95 E	2833.5	101.5	0.62	-0.55	-0.28
MWD	193	7	13537	13459	89.05	90.36	73.1	0.561	0.999	250	10772.58	32.00	2808.9	563.50 S	2808.95 E	2864.9	101.3	0.71	-0.44	-0.56
MWD	194	7	13600	13522	88.75	90.01	73.1	0.561	0.999	248	10773.79	63.00	2871.9	563.71 S	2871.93 E	2926.7	101.1	0.73	-0.48	-0.56
MWD	195	7	13631	13553	90.55	89.57	73.1	0.561	0.999	250	10773.98	31.00	2902.9	563.59 S	2902.93 E	2957.1	101.0	5.98	5.81	-1.42
MWD	196	7	13694	13616	90.55	89.83	73	0.562	1.000	252	10773.37	63.00	2965.9	563.26 S	2965.93 E	3018.9	100.8	0.41	0.00	0.41
MWD	197	7	13757	13679	89.67	89.31	73.1	0.561	0.999	251	10773.25	63.00	3028.9	562.79 S	3028.93 E	3080.8	100.5	1.62	-1.40	-0.83
MWD	198	7	13788	13710	90.02	89.39	73	0.562	1.000	248	10773.34	31.00	3059.9	562.44 S	3059.92 E	3111.2	100.4	1.16	1.13	0.26
MWD	199	7	13851	13773	91.91	90.10	73.1	0.561	0.999	252	10772.28	63.00	3122.9	562.16 S	3122.91 E	3173.1	100.2	3.20	3.00	1.13
MWD	200	7	13883	13805	92.18	90.62	73.1	0.561	0.999	248	10771.13	32.00	3154.9	562.36 S	3154.89 E	3204.6	100.1	1.83	0.84	1.63
MWD	201	7	13946	13868	90.55	90.89	73.1	0.562	1.000	250	10769.63	63.00	3217.9	563.19 S	3217.86 E	3266.8	99.9	2.62	-2.59	0.43
MWD	202	7	13977	13899	89.05	90.98	73	0.561	0.999	250	10769.74	31.00	3248.9	563.69 S	3248.86 E	3297.4	99.8	4.85	-4.84	0.29
MWD	203	7	14009	13931	88.75	90.89	73.1	0.561	0.999	252	10770.36	32.00	3280.8	564.22 S	3280.85 E	3329.0	99.8	0.98	-0.94	-0.28
MWD	204	7	14072	13994	88.53	90.62	73.1	0.562	1.000	253	10771.85	63.00	3343.8	565.05 S	3343.83 E	3391.2	99.6	0.55	-0.35	-0.43
MWD	205	7	14135	14057	89.1	90.54	73.1	0.561	1.000	253	10773.15	63.00	3406.8	565.68 S	3406.81 E	3453.5	99.4	0.91	0.90	-0.13
MWD	206	7	14166	14088	89.63	90.80	73.2	0.562	1.000	253	10773.50	31.00	3437.8	566.05 S	3437.80 E	3484.1	99.4	1.90	1.71	0.84
MWD	207	7	14261	14183	89.85	90.71	73	0.562	0.999	255	10773.93	95.00	3532.8	567.30 S	3532.80 E	3578.1	99.1	0.25	0.23	-0.09
MWD	208	7	14292	14214	90.15	90.27	73	0.561	0.999	255	10773.93	31.00	3563.8	567.56 S	3563.79 E	3608.7	99.0	1.72	0.97	-1.42
MWD	209	7	14355	14277	90.55	90.54	73.1	0.562	0.999	255	10773.54	63.00	3626.8	568.01 S	3626.79 E	3671.0	98.9	0.77	0.63	0.43
MWD	210	7	14418	14340	89.63	89.13	73.1	0.562	0.999	255	10773.45	63.00	3689.8	567.83 S	3689.79 E	3733.2	98.7	2.67	-1.46	-2.24
MWD	211	7	14450	14372	87.69	89.22	73.1	0.562	0.999	255	10774.19	32.00	3721.8	567.37 S	3721.77 E	3764.8	98.7	6.07	-6.06	0.28
MWD	212	7	14481	14403	87.74	88.87	73.1	0.562	0.999	255	10775.43	31.00	3752.7	566.85 S	3752.75 E	3795.3	98.6	1.14	0.16	-1.13
MWD	213	7	14544	14466	87.87	88.78	73.1	0.562	1.000	255	10777.84	63.00	3815.7	565.56 S	3815.69 E	3857.4	98.4	0.25	0.21	-0.14
MWD	214	7	14576	14498	89.41	89.22	73.1	0.563	1.000	255	10778.60	32.00	3847.7	565.00 S	3847.67 E	3888.9	98.4	5.00	4.81	1.37
MWD	215	7	14639	14561	89.85	89.48	73.1	0.563	1.001	255	10779.01	63.00	3910.7	564.29 S	3910.67 E	3951.2	98.2	0.81	0.70	0.41
MWD	216	7	14733	14655	90.15	88.95	73.1	0.563	1.001	255	10779.01	94.00	4004.7	563.00 S	4004.66 E	4044.0	98.0	0.65	0.32	-0.56
MWD	217	7	14828	14750	91.03	90.71	73.1	0.563	1.001	255	10778.03	95.00	4099.6	562.72 S	4099.65 E	4138.1	97.8	2.07	0.93	1.85
MWD	218	7	14891	14813	90.9	91.06	73.1	0.563	1.001	255	10776.97	63.00	4162.6	563.69 S	4162.63 E	4200.6	97.7	0.59	-0.21	0.56
MWD	219	7	14922	14844	90.95	91.59	73.1	0.563	1.001	255	10776.47	31.00	4193.6	564.41 S	4193.62 E	4231.4	97.7	1.72	0.16	1.71
MWD	220	7	14954	14876	90.37	90.54	73.1	0.563	1.001	259	10776.10	32.00	4225.6	565.00 S	4225.61 E	4263.2	97.6	3.75	-1.81	-3.28
MWD	221	7	15017	14939	90.64	90.27	73.1	0.563	1.001	259	10775.54	63.00	4288.6	565.45 S	4288.60 E	4325.7	97.5	0.61	0.43	-0.43
MWD	222	7	15111	15033	90.51	89.57	73.1	0.563	1.001	257	10774.60	94.00	4382.6	565.32 S	4382.60 E	4418.9	97.4	0.76	-0.14	-0.74
MWD	223	7	15205	15127	90.29	89.22	73.1	0.563	1.001	257	10773.95	94.00	4476.6	564.32 S	4476.59 E	4512.0	97.2	0.44	-0.23	-0.37
MWD	224	7	15237	15159	90.33	88.69	73.1	0.563	1.001	259	10773.77	32.00	4508.6	563.74 S	4508.59 E	4543.7	97.1	1.66	0.12	-1.66
MWD	225	7	15300	15222	89.93	87.99	73	0.561	0.999	257	10773.63	63.00	4571.6	561.91 S	4571.56 E	4606.0	97.0	1.28	-0.63	-1.11
MWD	226	7	15394	15316	88.66	89.48	73	0.563	0.999	260	10774.79	94.00	4665.5	559.84 S	4665.52 E	4699.0	96.8	2.08	-1.35	1.59
MWD	227	7	15488	15410	89.01	89.22	73	0.562	0.999	262	10776.70	94.00	4759.5	558.77 S	4759.50 E	4792.2	96.7	0.46	0.37	-0.28
MWD	228	7	15583	15505	89.36	88.25	73.1	0.561	0.999	260	10778.05	95.00	4854.5	556.68 S	4854.46 E	4886.3	96.5	1.09	0.37	-1.02
MWD	229	7	15614	15536	89.01	90.27	73.2	0.562	0.999	260	10778.49	31.00	4885.5	556.28 S	4885.46 E	4917.0	96.5	6.61	-1.13	6.52

MWD	230	7	15677	15599	89.63	90.8	73.2	0.563	1.000	260	10779.24	63.00	4948.4	556.86 S	4948.45 E	4979.7	96.4	1.29	0.98	0.84
MWD	231	7	15772	15694	90.37	91.59	73	0.561	0.999	262	10779.24	95.00	5043.4	558.85 S	5043.43 E	5074.3	96.3	1.14	0.78	0.83
MWD	232	7	15866	15788	91.16	91.24	73.1	0.561	0.999	262	10777.98	94.00	5137.4	561.17 S	5137.39 E	5167.9	96.2	0.92	0.84	-0.37
MWD	233	7	15961	15883	90.51	90.89	73	0.561	0.998	262	10776.60	95.00	5232.4	562.93 S	5232.36 E	5262.6	96.1	0.78	-0.68	-0.37
MWD	234	7	16056	15978	89.27	89.92	73	0.562	0.999	262	10776.78	95.00	5327.4	563.60 S	5327.36 E	5357.1	96.0	1.66	-1.31	-1.02
MWD	235	7	16150	16072	89.58	88.6	73.2	0.562	0.999	264	10777.72	94.00	5421.3	562.39 S	5421.34 E	5450.4	95.9	1.44	0.33	-1.40
MWD	236	7	16245	16167	89.41	90.01	73	0.562	0.999	262	10778.56	95.00	5516.3	561.24 S	5516.33 E	5544.8	95.8	1.49	-0.18	1.48
MWD	237	7	16339	16261	89.8	89.74	73	0.562	0.999	264	10779.21	94.00	5610.3	561.03 S	5610.33 E	5638.3	95.7	0.50	0.41	-0.29
MWD	238	7	16433	16355	89.8	89.57	73	0.562	0.999	264	10779.54	94.00	5704.3	560.47 S	5704.32 E	5731.8	95.6	0.18	0.00	-0.18
MWD	239	7	16528	16450	90.55	88.78	73	0.562	0.999	264	10779.25	95.00	5799.3	559.10 S	5799.31 E	5826.2	95.5	1.15	0.79	-0.83
MWD	240	7	16591	16513	91.43	88.69	73	0.562	0.999	261	10778.16	63.00	5862.3	557.71 S	5862.29 E	5888.8	95.4	1.40	1.40	-0.14
MWD	241	7	16622	16544	90.95	89.48	73	0.562	0.999	261	10777.51	31.00	5893.3	557.21 S	5893.28 E	5919.6	95.4	2.98	-1.55	2.55
MWD	242	7	16654	16576	90.81	90.36	73	0.562	0.999	261	10777.02	32.00	5925.3	557.17 S	5925.27 E	5951.4	95.4	2.78	-0.44	2.75
MWD	243	7	16717	16639	90.68	91.77	73	0.562	0.999	261	10776.20	63.00	5988.3	558.34 S	5988.25 E	6014.2	95.3	2.25	-0.21	2.24
MWD	244	7	16748	16670	89.36	91.68	73	0.562	0.999	261	10776.19	31.00	6019.2	559.27 S	6019.24 E	6045.2	95.3	4.27	-4.26	-0.29
MWD	245	7	16811	16733	89.89	91.59	73	0.562	0.999	261	10776.61	63.00	6082.2	561.07 S	6082.21 E	6108.0	95.3	0.85	0.84	-0.14
MWD	246	7	16843	16765	90.02	91.24	73	0.562	0.999	261	10776.63	32.00	6114.2	561.86 S	6114.20 E	6140.0	95.3	1.17	0.41	-1.09
MWD	247	7	16906	16828	90.77	91.68	73	0.562	0.999	262	10776.20	63.00	6177.2	563.46 S	6177.18 E	6202.8	95.2	1.38	1.19	0.70
MWD	248	7	16938	16860	89.8	91.94	73	0.562	0.999	262	10776.04	32.00	6209.2	564.48 S	6209.16 E	6234.8	95.2	3.14	-3.03	0.81
MWD	249	7	16969	16891	90.2	91.41	73	0.561	0.999	264	10776.04	31.00	6240.1	565.38 S	6240.15 E	6265.7	95.2	2.14	1.29	-1.71
MWD	250	7	17001	16923	90.02	91.41	73	0.562	0.999	260	10775.98	32.00	6272.1	566.17 S	6272.14 E	6297.6	95.2	0.56	-0.56	0.00
MWD	251	7	17095	17017	88.88	92.47	73	0.562	0.999	264	10776.88	94.00	6366.1	569.35 S	6366.08 E	6391.5	95.1	1.66	-1.21	1.13
MWD	252	7	17158	17080	89.14	91.41	73.1	0.563	1.000	264	10777.97	63.00	6429.0	571.48 S	6429.03 E	6454.4	95.1	1.73	0.41	-1.68
MWD	253	7	17190	17112	89.41	91.15	73.1	0.561	0.999	264	10778.37	32.00	6461.0	572.20 S	6461.02 E	6486.3	95.1	1.17	0.84	-0.81
MWD	254	7	17284	17206	90.24	91.33	73	0.563	0.999	266	10778.66	94.00	6555.0	574.23 S	6555.00 E	6580.1	95.0	0.90	0.88	0.19
MWD	255	7	17315	17237	90.55	90.89	73.1	0.561	0.999	266	10778.44	31.00	6586.0	574.83 S	6585.99 E	6611.0	95.0	1.74	1.00	-1.42
MWD	256	7	17379	17301	90.86	91.24	73	0.562	0.999	264	10777.66	64.00	6650.0	576.02 S	6649.98 E	6674.9	95.0	0.73	0.48	0.55
MWD	257	7	17410	17332	89.54	91.06	73	0.563	0.999	264	10777.55	31.00	6681.0	576.64 S	6680.97 E	6705.8	94.9	4.30	-4.26	-0.58
MWD	258	7	17473	17395	89.63	90.1	73.1	0.562	1.000	266	10778.01	63.00	6744.0	577.28 S	6743.96 E	6768.6	94.9	1.53	0.14	-1.52
MWD	259	7	17568	17490	90.46	90.1	73	0.562	0.999	267	10777.93	95.00	6839.0	577.45 S	6838.96 E	6863.3	94.8	0.87	0.87	0.00
MWD	260	7	17599	17521	91.38	89.92	73.2	0.562	1.000	258	10777.43	31.00	6870.0	577.45 S	6869.96 E	6894.2	94.8	3.02	2.97	-0.58
MWD	261	7	17662	17584	91.03	89.57	73	0.562	0.999	264	10776.11	63.00	6932.9	577.17 S	6932.94 E	6956.9	94.8	0.79	-0.56	-0.56
MWD	262	7	17757	17679	89.19	88.78	73	0.563	0.999	267	10775.93	95.00	7027.9	575.81 S	7027.93 E	7051.5	94.7	2.11	-1.94	-0.83
MWD	263	7	17851	17773	89.63	88.25	73	0.563	0.999	267	10776.89	94.00	7121.9	573.37 S	7121.89 E	7144.9	94.6	0.73	0.47	-0.56
MWD	264	7	17945	17867	89.85	89.13	73	0.562	0.999	267	10777.32	94.00	7215.9	571.22 S	7215.86 E	7238.4	94.5	0.96	0.23	0.94
MWD	265	7	17976	17898	90.15	89.22	73	0.562	0.999	267	10777.32	31.00	7246.9	570.77 S	7246.86 E	7269.3	94.5	1.01	0.97	0.29
MWD	266	7	18039	17961	90.55	89.48	73	0.562	0.999	267	10776.94	63.00	7309.9	570.06 S	7309.86 E	7332.0	94.5	0.76	0.63	0.41
MWD	267	7	18070	17992	91.08	88.78	73	0.562	0.999	267	10776.49	31.00	7340.8	569.59 S	7340.85 E	7362.9	94.4	2.83	1.71	-2.26
MWD	268	7	18133	18055	91.21	87.64	73	0.561	0.999	267	10775.24	63.00	7403.8	567.62 S	7403.80 E	7425.5	94.4	1.82	0.21	-1.81
MWD	269	7	18164	18086	89.32	88.51	73.2	0.561	0.999	267	10775.09	31.00	7434.8	566.58 S	7434.78 E	7456.3	94.4	6.71	-6.10	2.81
MWD	270	7	18195	18117	89.27	89.13	73	0.561	0.999	262	10775.47	31.00	7465.8	565.94 S	7465.78 E	7487.2	94.3	2.01	-0.16	2.00
MWD	271	7	18227	18149	89.76	88.60	73	0.561	0.999	262	10775.74	32.00	7497.8	565.31 S	7497.77 E	7519.0	94.3	2.26	1.53	-1.66
MWD	272	7	18258	18180	89.54	89.66	73	0.561	0.999	262	10775.93	31.00	7528.8	564.84 S	7528.76 E	7549.9	94.3	3.49	-0.71	3.42
MWD	273	7	18290	18212	89.41	91.41	73	0.561	0.999	264	10776.23	32.00	7560.8	565.14 S	7560.76 E	7581.9	94.3	5.48	-0.41	5.47
MWD	274	7	18321	18243	89.54	90.98	73	0.561	0.999	264	10776.51	31.00	7591.8	565.78 S	7591.75 E	7612.8	94.3	1.45	0.42	-1.39
MWD	275	7	18384	18306	90.68	90.27	73	0.561	0.999	266	10776.39	63.00	7654.7	566.47 S	7654.75 E	7675.7	94.2	2.13	1.81	-1.13
MWD	276	7	18415	18337	90.99	90.54	73.1	0.562	0.999	266	10775.94	31.00	7685.7	566.69 S	7685.74 E	7706.6	94.2	1.33	1.00	0.87
MWD	277	7	18509	18431	91.78	88.95	73.1	0.561	0.999	267	10773.67	94.00	7779.7	566.27 S	7779.71 E	7800.3	94.2	1.89	0.84	-1.69

MWD	278	7	18604	18526	90.95	90.18	73	0.562	0.999	266	10771.40	95.00	7874.7	565.55 S	7874.68 E	7895.0	94.1	1.56	-0.87	1.29
MWD	279	7	18667	18589	91.47	90.54	73.1	0.561	0.999	266	10770.07	63.00	7937.7	565.95 S	7937.66 E	7957.8	94.1	1.00	0.83	0.57
MWD	280	7	18698	18620	91.65	90.98	73	0.562	0.999	264	10769.23	31.00	7968.6	566.36 S	7968.65 E	7988.7	94.1	1.53	0.58	1.42
MWD	281	7	18730	18652	92	90.54	73	0.562	0.999	264	10768.21	32.00	8000.6	566.78 S	8000.63 E	8020.7	94.1	1.76	1.09	-1.37
MWD	282	7	18761	18683	90.55	90.89	73.2	0.563	1.000	264	10767.52	31.00	8031.6	567.17 S	8031.62 E	8051.6	94.0	4.81	-4.68	1.13
MWD	283	7	18792	18714	88.44	91.68	73	0.562	0.999	264	10767.79	31.00	8062.6	567.86 S	8062.61 E	8082.6	94.0	7.27	-6.81	2.55
MWD	284	7	18887	18809	89.27	91.15	73	0.562	0.999	268	10769.69	95.00	8157.6	570.21 S	8157.56 E	8177.5	94.0	1.04	0.87	-0.56
MWD	285	7	18980	18902	89.8	90.10	73	0.561	0.998	268	10770.45	93.00	8250.5	571.22 S	8250.55 E	8270.3	94.0	1.26	0.57	-1.13
MWD	286	7	19074	18996	90.59	90.01	73	0.562	0.999	270	10770.13	94.00	8344.5	571.31 S	8344.55 E	8364.1	93.9	0.85	0.84	-0.10
MWD	287	7	19105	19027	91.12	89.48	73.1	0.561	0.998	270	10769.66	31.00	8375.5	571.17 S	8375.54 E	8395.0	93.9	2.42	1.71	-1.71
MWD	288	7	19137	19059	91.38	89.13	73.1	0.561	0.998	270	10768.97	32.00	8407.5	570.79 S	8407.53 E	8426.9	93.9	1.36	0.81	-1.09
MWD	289	7	19168	19090	91.69	89.04	73.1	0.561	0.998	266	10768.14	31.00	8438.5	570.29 S	8438.52 E	8457.8	93.9	1.04	1.00	-0.29
MWD	290	7	19199	19121	89.23	88.87	73	0.561	0.998	265	10767.89	31.00	8469.5	569.73 S	8469.51 E	8488.6	93.8	7.95	-7.94	-0.55
MWD	291	7	19262	19184	88.22	88.43	73	0.561	0.998	265	10769.29	63.00	8532.5	568.24 S	8532.47 E	8551.4	93.8	1.75	-1.60	-0.70
MWD	292	7	19325	19247	89.01	88.43	73	0.561	0.998	265	10770.81	63.00	8595.4	566.52 S	8595.43 E	8614.1	93.8	1.25	1.25	0.00
MWD	293	7	19356	19278	89.54	88.69	73	0.561	0.998	265	10771.20	31.00	8626.4	565.74 S	8626.42 E	8644.9	93.8	1.90	1.71	0.84
MWD	294	7	19387	19309	89.76	88.87	73	0.561	0.999	268	10771.39	31.00	8657.4	565.08 S	8657.41 E	8675.8	93.7	0.92	0.71	0.58
MWD	295	7	19450	19372	90.29	89.04	73	0.561	0.999	268	10771.36	63.00	8720.4	563.93 S	8720.40 E	8738.6	93.7	0.88	0.84	0.27
MWD	296	7	19513	19435	89.85	90.71	73	0.561	0.999	268	10771.29	63.00	8783.4	563.79 S	8783.40 E	8801.5	93.7	2.74	-0.70	2.65
MWD	297	7	19544	19466	89.71	90.18	73	0.561	0.999	268	10771.41	31.00	8814.4	564.03 S	8814.40 E	8832.4	93.7	1.77	-0.45	-1.71
MWD	298	7	19607	19529	89.8	90.27	73	0.561	0.999	268	10771.68	63.00	8877.4	564.28 S	8877.39 E	8895.3	93.6	0.20	0.14	0.14
MWD	299	7	19638	19560	89.98	89.83	73	0.561	0.999	268	10771.74	31.00	8908.4	564.31 S	8908.39 E	8926.3	93.6	1.53	0.58	-1.42
MWD	300	7	19733	19655	92.22	89.66	73	0.561	0.999	268	10769.91	95.00	9003.4	563.88 S	9003.37 E	9021.0	93.6	2.36	2.36	-0.18
MWD	301	7	19764	19686	92.88	89.92	73.2	0.563	1.000	266	10768.53	31.00	9034.3	563.77 S	9034.34 E	9051.9	93.6	2.29	2.13	0.84
MWD	302	7	19795	19717	93.27	89.74	73.1	0.562	0.999	266	10766.87	31.00	9065.3	563.68 S	9065.29 E	9082.8	93.6	1.39	1.26	-0.58
MWD	303	7	19827	19749	92.26	90.71	73.1	0.563	0.999	264	10765.33	32.00	9097.3	563.80 S	9097.26 E	9114.7	93.5	4.37	-3.16	3.03
MWD	304	7	19858	19780	90.46	90.62	73.1	0.562	0.999	266	10764.59	31.00	9128.2	564.16 S	9128.24 E	9145.7	93.5	5.81	-5.81	-0.29
MWD	305	7	19889	19811	88.48	91.94	73.1	0.563	0.999	266	10764.88	31.00	9159.2	564.86 S	9159.23 E	9176.6	93.5	7.68	-6.39	4.26
MWD	306	7	19921	19843	88.57	92.12	73	0.563	0.999	268	10765.70	32.00	9191.2	565.99 S	9191.20 E	9208.6	93.5	0.63	0.28	0.56
MWD	307	7	19952	19874	88.97	91.41	73	0.562	0.999	264	10766.37	31.00	9222.2	566.94 S	9222.18 E	9239.6	93.5	2.63	1.29	-2.29
MWD	308	7	20015	19937	87.56	90.98	73.1	0.563	0.999	267	10768.27	63.00	9285.1	568.26 S	9285.14 E	9302.5	93.5	2.34	-2.24	-0.68
MWD	309	7	20109	20031	87.43	91.41	73.1	0.563	1.000	270	10772.38	94.00	9379.0	570.22 S	9379.03 E	9396.3	93.5	0.48	-0.14	0.46
MWD	310	7	20172	20094	87.78	90.36	73.2	0.563	1.000	271	10775.02	63.00	9442.0	571.19 S	9441.96 E	9459.2	93.5	1.76	0.56	-1.67
MWD	311	7	20204	20126	87.74	90.54	73	0.562	0.999	266	10776.27	32.00	9473.9	571.44 S	9473.94 E	9491.2	93.5	0.58	-0.13	0.56
MWD	312	7	20267	20189	89.14	90.45	73	0.562	0.999	266	10777.98	63.00	9536.9	571.98 S	9536.91 E	9554.0	93.4	2.23	2.22	-0.14
MWD	313	7	20298	20220	89.14	90.1	73	0.562	0.999	266	10778.45	31.00	9567.9	572.13 S	9567.90 E	9585.0	93.4	1.13	0.00	-1.13
MWD	314	7	20361	20283	89.63	89.66	73	0.561	0.999	266	10779.12	63.00	9630.9	572.00 S	9630.90 E	9647.9	93.4	1.05	0.78	-0.70
MWD	315	7	20393	20315	89.76	89.04	73	0.561	0.999	266	10779.29	32.00	9662.9	571.64 S	9662.90 E	9679.8	93.4	1.98	0.41	-1.94
MWD	316	7	20456	20378	90.33	88.34	73	0.561	0.999	266	10779.24	63.00	9725.9	570.20 S	9725.88 E	9742.6	93.4	1.43	0.90	-1.11
MWD	317	7	20487	20409	90.51	88.16	73	0.561	0.999	266	10779.02	31.00	9756.9	569.25 S	9756.87 E	9773.5	93.3	0.82	0.58	-0.58
MWD	318	7	20582	20504	91.6	87.64	73	0.561	0.999	266	10777.27	95.00	9851.8	565.77 S	9851.78 E	9868.0	93.3	1.27	1.15	-0.55
MWD	319	7	20650	20572	92.04	87.9	73	0.561	0.999	266	10775.11	68.00	9919.7	563.13 S	9919.70 E	9935.7	93.2	0.75	0.65	0.38



Operator:	Oasis Petroleum
Well :	Kline Federal 5300 41-18 14BX
MWD Providers	GyroData
Directional Supervision:	RPM

Section:	18	QQ:	SW SW	County:	McKenzie	State:	ND
Township:	153	N/S:	N	Footages:	368	FN/SL:	S
Range:	100	E/W:	W		237	FE/WL:	W

Vertical Section Plane:	90
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#	MD	Inc.	Azm.	T.V.D.	Ver. Sect.	Coordinates		DLS
						+N / -S	+E / -W	
Tie	10244.00	0.26	348.67	10243.20	-2.93	-50.89	-2.93	0.59
1	10276.00	0.53	100.38	10275.20	-2.80	-50.85	-2.80	2.10
2	10308.00	3.43	114.00	10307.18	-1.78	-51.26	-1.78	9.12
3	10340.00	7.08	115.67	10339.04	0.87	-52.51	0.87	11.42
4	10373.00	10.51	112.42	10371.64	5.49	-54.54	5.49	10.50
5	10404.00	13.63	108.55	10401.96	11.57	-56.78	11.57	10.39
6	10436.00	17.63	108.29	10432.77	19.75	-59.50	19.75	12.50
7	10468.00	21.85	107.76	10462.88	30.02	-62.84	30.02	13.20
8	10500.00	25.93	106.62	10492.13	42.41	-66.66	42.41	12.83
9	10531.00	30.46	108.11	10519.45	56.38	-71.04	56.38	14.79
10	10563.00	34.95	108.47	10546.37	72.79	-76.47	72.79	14.04
11	10595.00	39.34	108.73	10571.87	91.10	-82.63	91.10	13.73
12	10626.00	44.26	110.40	10594.97	110.55	-89.56	110.55	16.27
13	10658.00	48.88	111.81	10616.96	132.22	-97.94	132.22	14.79
14	10689.00	53.49	113.12	10636.39	154.53	-107.17	154.53	15.23
15	10721.00	57.58	115.41	10654.50	178.57	-118.02	178.57	14.08
16	10752.00	60.35	118.13	10670.48	202.28	-129.99	202.28	11.68
17	10783.00	62.37	120.42	10685.34	226.00	-143.30	226.00	9.19
18	10815.00	64.53	121.03	10699.64	250.61	-157.92	250.61	6.96
19	10847.00	67.87	121.39	10712.55	275.65	-173.09	275.65	10.49
20	10877.00	72.57	122.09	10722.70	299.65	-187.94	299.65	15.82
21	10909.00	77.63	120.07	10730.93	326.12	-203.89	326.12	16.95
22	10941.00	80.22	120.33	10737.08	353.26	-219.69	353.26	8.13
23	10972.00	81.45	120.42	10742.01	379.67	-235.17	379.67	3.98
24	11003.00	84.66	120.24	10745.76	406.22	-250.70	406.22	10.37
25	11035.00	88.35	119.28	10747.71	433.95	-266.55	433.95	11.91
26	11048.00	88.97	117.96	10748.02	445.35	-272.78	445.35	11.22
27	11145.00	91.16	119.01	10747.91	530.60	-319.04	530.60	2.50
28	11176.00	91.16	118.48	10747.28	557.78	-333.94	557.78	1.71
29	11207.00	91.16	117.69	10746.65	585.12	-348.53	585.12	2.55
30	11238.00	89.85	115.94	10746.38	612.79	-362.52	612.79	7.05
31	11270.00	89.32	114.79	10746.61	641.70	-376.22	641.70	3.96

32	11301.00	88.35	113.12	10747.24	670.02	-388.81	670.02	6.23
33	11332.00	87.87	111.81	10748.26	698.65	-400.65	698.65	4.50
34	11362.00	88.18	111.72	10749.30	726.50	-411.76	726.50	1.08
35	11393.00	88.09	109.26	10750.31	755.52	-422.61	755.52	7.94
36	11424.00	89.23	106.44	10751.03	785.02	-432.11	785.02	9.81
37	11456.00	88.97	103.37	10751.53	815.93	-440.34	815.93	9.63
38	11486.00	89.63	100.20	10751.90	845.29	-446.46	845.29	10.79
39	11517.00	89.63	98.97	10752.10	875.86	-451.62	875.86	3.97
40	11549.00	89.85	98.97	10752.25	907.47	-456.61	907.47	0.69
41	11580.00	89.63	98.62	10752.39	938.10	-461.35	938.10	1.33
42	11611.00	90.20	99.06	10752.43	968.73	-466.12	968.73	2.32
43	11642.00	90.42	99.06	10752.26	999.35	-471.00	999.35	0.71
44	11673.00	91.25	100.12	10751.81	1029.91	-476.16	1029.91	4.34
45	11705.00	91.34	100.47	10751.09	1061.39	-481.88	1061.39	1.13
46	11736.00	91.52	100.47	10750.32	1091.86	-487.51	1091.86	0.58
47	11767.00	89.98	101.35	10749.91	1122.30	-493.38	1122.30	5.72
48	11798.00	89.80	101.43	10749.97	1152.69	-499.50	1152.69	0.64
49	11829.00	90.02	101.35	10750.02	1183.08	-505.62	1183.08	0.76
50	11860.00	90.33	101.80	10749.92	1213.45	-511.84	1213.45	1.76
51	11892.00	90.20	101.08	10749.78	1244.81	-518.19	1244.81	2.29
52	11923.00	90.46	100.64	10749.60	1275.25	-524.03	1275.25	1.65
53	11954.00	89.54	98.18	10749.60	1305.83	-529.10	1305.83	8.47
54	11985.00	88.70	97.74	10750.07	1336.53	-533.39	1336.53	3.06
55	12016.00	88.22	97.22	10750.91	1367.26	-537.42	1367.26	2.28
56	12047.00	87.91	95.81	10751.95	1398.04	-540.94	1398.04	4.65
57	12078.00	87.82	95.72	10753.11	1428.86	-544.05	1428.86	0.41
58	12110.00	87.91	95.55	10754.30	1460.68	-547.19	1460.68	0.60
59	12141.00	88.88	94.40	10755.17	1491.55	-549.88	1491.55	4.85
60	12172.00	89.19	94.40	10755.69	1522.46	-552.26	1522.46	1.00
61	12203.00	89.71	94.40	10755.99	1553.36	-554.63	1553.36	1.68
62	12234.00	90.20	92.73	10756.01	1584.30	-556.56	1584.30	5.61
63	12266.00	90.07	92.82	10755.94	1616.26	-558.11	1616.26	0.49
64	12297.00	90.20	92.82	10755.86	1647.23	-559.64	1647.23	0.42
65	12328.00	90.07	90.10	10755.79	1678.21	-560.43	1678.21	8.78
66	12390.00	89.41	89.92	10756.07	1740.21	-560.44	1740.21	1.10
67	12421.00	89.63	90.10	10756.33	1771.21	-560.44	1771.21	0.92
68	12482.00	89.89	89.39	10756.59	1832.21	-560.17	1832.21	1.24
69	12577.00	89.19	90.27	10757.35	1927.21	-559.89	1927.21	1.18
70	12671.00	89.93	90.18	10758.07	2021.20	-560.26	2021.20	0.79
71	12703.00	90.02	90.27	10758.09	2053.20	-560.38	2053.20	0.40
72	12766.00	89.93	89.83	10758.11	2116.20	-560.44	2116.20	0.71
73	12860.00	87.91	88.87	10759.89	2210.17	-559.37	2210.17	2.38
74	12892.00	87.96	89.13	10761.04	2242.15	-558.81	2242.15	0.83
75	12955.00	88.00	88.95	10763.26	2305.10	-557.76	2305.10	0.29

76	12986.00	88.35	90.54	10764.25	2336.08	-557.62	2336.08	5.25
77	13050.00	87.91	90.45	10766.33	2400.04	-558.17	2400.04	0.70
78	13144.00	88.84	90.89	10769.00	2494.00	-559.27	2494.00	1.09
79	13238.00	89.41	91.15	10770.44	2587.97	-560.95	2587.97	0.67
80	13270.00	89.49	90.54	10770.74	2619.97	-561.42	2619.97	1.92
81	13333.00	89.71	90.80	10771.18	2682.96	-562.16	2682.96	0.54
82	13427.00	89.19	90.54	10772.08	2776.95	-563.25	2776.95	0.62
83	13459.00	89.05	90.36	10772.58	2808.95	-563.51	2808.95	0.71
84	13522.00	88.75	90.01	10773.79	2871.93	-563.71	2871.93	0.73
85	13553.00	90.55	89.57	10773.98	2902.93	-563.60	2902.93	5.98
86	13616.00	90.55	89.83	10773.37	2965.93	-563.27	2965.93	0.41
87	13679.00	89.67	89.31	10773.25	3028.93	-562.79	3028.93	1.62
88	13710.00	90.02	89.39	10773.33	3059.92	-562.44	3059.92	1.16
89	13773.00	91.91	90.10	10772.27	3122.91	-562.16	3122.91	3.20
90	13805.00	92.18	90.62	10771.13	3154.89	-562.36	3154.89	1.83
91	13868.00	90.55	90.89	10769.63	3217.86	-563.19	3217.86	2.62
92	13899.00	89.05	90.98	10769.74	3248.86	-563.70	3248.86	4.85
93	13931.00	88.75	90.89	10770.35	3280.85	-564.22	3280.85	0.98
94	13994.00	88.53	90.62	10771.85	3343.82	-565.05	3343.82	0.55
95	14057.00	89.10	90.54	10773.15	3406.81	-565.69	3406.81	0.91
96	14088.00	89.63	90.80	10773.49	3437.80	-566.05	3437.80	1.90
97	14183.00	89.85	90.71	10773.93	3532.79	-567.30	3532.79	0.25
98	14214.00	90.15	90.27	10773.93	3563.79	-567.57	3563.79	1.72
99	14277.00	90.55	90.54	10773.54	3626.79	-568.01	3626.79	0.77
100	14340.00	89.63	89.13	10773.44	3689.79	-567.83	3689.79	2.67
101	14372.00	87.69	89.22	10774.19	3721.77	-567.37	3721.77	6.07
102	14403.00	87.74	88.87	10775.43	3752.74	-566.85	3752.74	1.14
103	14466.00	87.87	88.78	10777.84	3815.69	-565.56	3815.69	0.25
104	14498.00	89.41	89.22	10778.60	3847.67	-565.00	3847.67	5.00
105	14561.00	89.85	89.48	10779.00	3910.66	-564.29	3910.66	0.81
106	14655.00	90.15	88.95	10779.00	4004.66	-563.00	4004.66	0.65
107	14750.00	91.03	90.71	10778.03	4099.65	-562.72	4099.65	2.07
108	14813.00	90.90	91.06	10776.97	4162.63	-563.69	4162.63	0.59
109	14844.00	90.95	91.59	10776.46	4193.62	-564.41	4193.62	1.72
110	14876.00	90.37	90.54	10776.10	4225.61	-565.00	4225.61	3.75
111	14939.00	90.64	90.27	10775.54	4288.60	-565.45	4288.60	0.61
112	15033.00	90.51	89.57	10774.60	4382.60	-565.32	4382.60	0.76
113	15127.00	90.29	89.22	10773.94	4476.59	-564.33	4476.59	0.44
114	15159.00	90.33	88.69	10773.77	4508.59	-563.74	4508.59	1.66
115	15222.00	89.93	87.99	10773.63	4571.56	-561.92	4571.56	1.28
116	15316.00	88.66	89.48	10774.78	4665.52	-559.84	4665.52	2.08
117	15410.00	89.01	89.22	10776.69	4759.50	-558.78	4759.50	0.46
118	15505.00	89.36	88.25	10778.04	4854.46	-556.68	4854.46	1.09
119	15536.00	89.01	90.27	10778.49	4885.46	-556.28	4885.46	6.61

120	15599.00	89.63	90.80	10779.23	4948.45	-556.87	4948.45	1.29
121	15694.00	90.37	91.59	10779.23	5043.43	-558.85	5043.43	1.14
122	15788.00	91.16	91.24	10777.98	5137.39	-561.17	5137.39	0.92
123	15883.00	90.51	90.89	10776.59	5232.36	-562.93	5232.36	0.78
124	15978.00	89.27	89.92	10776.78	5327.36	-563.61	5327.36	1.66
125	16072.00	89.58	88.60	10777.72	5421.34	-562.39	5421.34	1.44
126	16167.00	89.41	90.01	10778.56	5516.33	-561.24	5516.33	1.49
127	16261.00	89.80	89.74	10779.20	5610.33	-561.03	5610.33	0.50
128	16355.00	89.80	89.57	10779.53	5704.32	-560.47	5704.32	0.18
129	16450.00	90.55	88.78	10779.24	5799.31	-559.10	5799.31	1.15
130	16513.00	91.43	88.69	10778.15	5862.29	-557.71	5862.29	1.40
131	16544.00	90.95	89.48	10777.51	5893.27	-557.21	5893.27	2.98
132	16576.00	90.81	90.36	10777.02	5925.27	-557.17	5925.27	2.78
133	16639.00	90.68	91.77	10776.20	5988.25	-558.34	5988.25	2.25
134	16670.00	89.36	91.68	10776.19	6019.24	-559.27	6019.24	4.27
135	16733.00	89.89	91.59	10776.60	6082.21	-561.07	6082.21	0.85
136	16765.00	90.02	91.24	10776.63	6114.20	-561.86	6114.20	1.17
137	16828.00	90.77	91.68	10776.19	6177.18	-563.47	6177.18	1.38
138	16860.00	89.80	91.94	10776.03	6209.16	-564.48	6209.16	3.14
139	16891.00	90.20	91.41	10776.03	6240.15	-565.38	6240.15	2.14
140	16923.00	90.02	91.41	10775.97	6272.14	-566.17	6272.14	0.56
141	17017.00	88.88	92.47	10776.87	6366.08	-569.35	6366.08	1.66
142	17080.00	89.14	91.41	10777.96	6429.03	-571.49	6429.03	1.73
143	17112.00	89.41	91.15	10778.37	6461.02	-572.20	6461.02	1.17
144	17206.00	90.24	91.33	10778.65	6555.00	-574.23	6555.00	0.90
145	17237.00	90.55	90.89	10778.44	6585.99	-574.84	6585.99	1.74
146	17301.00	90.86	91.24	10777.65	6649.97	-576.02	6649.97	0.73
147	17332.00	89.54	91.06	10777.55	6680.97	-576.65	6680.97	4.30
148	17395.00	89.63	90.10	10778.00	6743.96	-577.28	6743.96	1.53
149	17490.00	90.46	90.10	10777.93	6838.96	-577.45	6838.96	0.87
150	17521.00	91.38	89.92	10777.43	6869.96	-577.46	6869.96	3.02
151	17584.00	91.03	89.57	10776.10	6932.94	-577.18	6932.94	0.79
152	17679.00	89.19	88.78	10775.92	7027.93	-575.81	7027.93	2.11
153	17773.00	89.63	88.25	10776.89	7121.89	-573.37	7121.89	0.73
154	17867.00	89.85	89.13	10777.32	7215.86	-571.22	7215.86	0.96
155	17898.00	90.15	89.22	10777.32	7246.86	-570.78	7246.86	1.01
156	17961.00	90.55	89.48	10776.93	7309.85	-570.06	7309.85	0.76
157	17992.00	91.08	88.78	10776.49	7340.85	-569.59	7340.85	2.83
158	18055.00	91.21	87.64	10775.23	7403.80	-567.62	7403.80	1.82
159	18086.00	89.32	88.51	10775.09	7434.78	-566.58	7434.78	6.71
160	18117.00	89.27	89.13	10775.47	7465.77	-565.94	7465.77	2.01
161	18149.00	89.76	88.60	10775.74	7497.77	-565.31	7497.77	2.26
162	18180.00	89.54	89.66	10775.93	7528.76	-564.84	7528.76	3.49
163	18212.00	89.41	91.41	10776.22	7560.76	-565.14	7560.76	5.48

164	18243.00	89.54	90.98	10776.51	7591.75	-565.78	7591.75	1.45
165	18306.00	90.68	90.27	10776.39	7654.74	-566.47	7654.74	2.13
166	18337.00	90.99	90.54	10775.93	7685.74	-566.69	7685.74	1.33
167	18431.00	91.78	88.95	10773.66	7779.71	-566.27	7779.71	1.89
168	18526.00	90.95	90.18	10771.40	7874.68	-565.55	7874.68	1.56
169	18589.00	91.47	90.54	10770.07	7937.66	-565.95	7937.66	1.00
170	18620.00	91.65	90.98	10769.22	7968.65	-566.36	7968.65	1.53
171	18652.00	92.00	90.54	10768.21	8000.63	-566.78	8000.63	1.76
172	18683.00	90.55	90.89	10767.52	8031.62	-567.17	8031.62	4.81
173	18714.00	88.44	91.68	10767.79	8062.61	-567.86	8062.61	7.27
174	18809.00	89.27	91.15	10769.69	8157.56	-570.21	8157.56	1.04
175	18902.00	89.80	90.10	10770.44	8250.55	-571.22	8250.55	1.26
176	18996.00	90.59	90.01	10770.12	8344.54	-571.31	8344.54	0.85
177	19027.00	91.12	89.48	10769.66	8375.54	-571.18	8375.54	2.42
178	19059.00	91.38	89.13	10768.96	8407.53	-570.79	8407.53	1.36
179	19090.00	91.69	89.04	10768.13	8438.52	-570.29	8438.52	1.04
180	19121.00	89.23	88.87	10767.88	8469.51	-569.73	8469.51	7.95
181	19184.00	88.22	88.43	10769.28	8532.47	-568.24	8532.47	1.75
182	19247.00	89.01	88.43	10770.81	8595.43	-566.52	8595.43	1.25
183	19278.00	89.54	88.69	10771.20	8626.42	-565.74	8626.42	1.90
184	19309.00	89.76	88.87	10771.39	8657.41	-565.08	8657.41	0.92
185	19372.00	90.29	89.04	10771.36	8720.40	-563.93	8720.40	0.88
186	19435.00	89.85	90.71	10771.28	8783.40	-563.79	8783.40	2.74
187	19466.00	89.71	90.18	10771.40	8814.40	-564.03	8814.40	1.77
188	19529.00	89.80	90.27	10771.67	8877.39	-564.28	8877.39	0.20
189	19560.00	89.98	89.83	10771.73	8908.39	-564.31	8908.39	1.53
190	19655.00	92.22	89.66	10769.91	9003.37	-563.89	9003.37	2.36
191	19686.00	92.88	89.92	10768.53	9034.34	-563.77	9034.34	2.29
192	19717.00	93.27	89.74	10766.87	9065.29	-563.68	9065.29	1.39
193	19749.00	92.26	90.71	10765.32	9097.26	-563.81	9097.26	4.37
194	19780.00	90.46	90.62	10764.59	9128.24	-564.17	9128.24	5.81
195	19811.00	88.48	91.94	10764.87	9159.23	-564.86	9159.23	7.68
196	19843.00	88.57	92.12	10765.70	9191.20	-565.99	9191.20	0.63
197	19874.00	88.97	91.41	10766.36	9222.18	-566.95	9222.18	2.63
198	19937.00	87.56	90.98	10768.27	9285.13	-568.26	9285.13	2.34
199	20031.00	87.43	91.41	10772.38	9379.02	-570.22	9379.02	0.48
200	20094.00	87.78	90.36	10775.01	9441.96	-571.19	9441.96	1.76
201	20126.00	87.74	90.54	10776.26	9473.94	-571.44	9473.94	0.58
202	20189.00	89.14	90.45	10777.98	9536.91	-571.99	9536.91	2.23
203	20220.00	89.14	90.10	10778.44	9567.90	-572.13	9567.90	1.13
204	20283.00	89.63	89.66	10779.12	9630.90	-572.00	9630.90	1.05
205	20315.00	89.76	89.04	10779.29	9662.90	-571.64	9662.90	1.98
206	20378.00	90.33	88.34	10779.24	9725.88	-570.20	9725.88	1.43
207	20409.00	90.51	88.16	10779.01	9756.87	-569.25	9756.87	0.82

208	20504.00	91.60	87.64	10777.26	9851.78	-565.77	9851.78	1.27
209	20572.00	92.04	87.90	10775.10	9919.70	-563.13	9919.70	0.75
Project	20650.00	92.00	88.00	10772.35	9997.60	-560.34	9997.60	0.14

Kline Federal 5300 41-18 11T2 LITHOLOGY

Rig crews caught lagged samples in 30' intervals in the vertical hole, curve build and landing from 8,350' to 11,288' MD; 50' samples in the lateral from 11,288' to TD at 20,650'.

*Electric geophysical log, sample and/or MWD gamma ray markers and tops are included in the sample descriptions below for reference. Samples were examined wet and dry under a binocular microscope in approximately 30' to 50' intervals. Sample descriptions begin just above the **Kibbey** formation. The drilling fluid was diesel invert during vertical and curve build sections and salt water brine in the lateral to TD.*

8350-8382 SILTSTONE: reddish brown, occasional, gray to light gray, brown to light brown gray, soft, sub blocky, earthy texture, no visible porosity, no visible oil stain

Kibbey

8,382' MD / 8,382' TVD (-6,300')

8382-8410 SILTSTONE: reddish brown, occasional, gray to light gray, brown to light brown gray, soft, sub blocky, earthy texture, no visible porosity, no visible oil stain; LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

8410-8440 SILTY SANDSTONE: medium gray, fine grained, firm to friable, rounded, well sorted, moderately cemented, calcite cement, possible intergranular porosity, no visible oil stain; SILTSTONE: reddish brown, occasional, gray to light gray, brown to light brown gray, soft, sub blocky, earthy texture, no visible porosity, no visible oil stain

8440-8470 SILTY SANDSTONE: medium gray, fine grained, firm to friable, rounded, well sorted, moderately cemented, calcite cement, possible intergranular porosity, no visible oil stain; SILTSTONE: reddish brown, occasional, gray to light gray, brown to light brown gray, soft, sub blocky, earthy texture, no visible porosity, no visible oil stain

8470-8500 SILTY SANDSTONE: medium gray, fine grained, firm to friable, rounded, well sorted, moderately cemented, calcite cement, possible intergranular porosity, no visible oil stain; SILTSTONE: reddish brown, occasional, gray to light gray, brown to light brown gray, soft, sub blocky, earthy texture, no visible porosity, no visible oil stain

8500-8530 SILTY SANDSTONE: medium gray, fine grained, firm to friable, rounded, well sorted, moderately cemented, calcite cement, possible intergranular porosity, no visible oil stain; SILTSTONE: reddish brown, occasional, gray to light gray, brown to light brown gray, soft, sub blocky, earthy texture, no visible porosity, no visible oil stain

8530-8564 SILTY SANDSTONE: medium gray, fine grained, firm to friable, rounded, well sorted, moderately cemented, calcite cement, possible intergranular porosity, no visible oil stain; SILTSTONE: reddish brown, occasional, gray to light gray, brown to light brown gray, soft, sub blocky, earthy texture, no visible porosity, no visible oil stain

Charles Salt

8,564' MD / 8,564' TVD (-6,482')

8564-8590 SILTY SANDSTONE: medium gray, fine grained, firm to friable, rounded, well sorted, moderately cemented, calcite cement, possible intergranular porosity, no visible oil stain; SILTSTONE: reddish brown, occasional, gray to light gray, brown to light brown gray, soft, sub blocky, earthy texture, SALT: translucent, trace off white, crystalline, firm, crystalline texture, no visible porosity, no visible oil stain

8590-9620 SALT: translucent, trace off white, crystalline, firm, crystalline texture, no visible porosity, no visible oil stain

9620-9650 SALT: translucent, trace off white, crystalline, firm, crystalline texture, no visible porosity, no visible oil stain, LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

9650-8680 SALT: translucent, trace off white, crystalline, firm, crystalline texture, no visible porosity, no visible oil stain

8680-8710 SALT: translucent, trace off white, crystalline, firm, crystalline texture, no visible porosity, no visible oil stain

8710-8740 SALT: translucent, trace off white, crystalline, firm, crystalline texture, no visible porosity, no visible oil stain, LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

8740-8770 SALT: translucent, trace off white, crystalline, firm, crystalline texture, no visible porosity, no visible oil stain

8770-8800 SALT: translucent, trace off white, crystalline, firm, crystalline texture, no visible porosity, no visible oil stain, ANHYDRITE: light to medium gray, amorphous, very soft to soft, earthy texture, no visible porosity or oil stain

8800-8830 SALT: translucent, trace off white, crystalline, firm, crystalline texture, no visible porosity, no visible oil stain, ANHYDRITE: light to medium gray, amorphous, very soft to soft, earthy texture, no visible porosity or oil stain

8830-8860 SALT: translucent, trace off white, crystalline, firm, crystalline texture, no visible porosity, no visible oil stain, LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

8860-8890 SALT: translucent, trace off white, crystalline, firm, crystalline texture, no visible porosity, no visible oil stain

8890-8920 SALT: translucent, trace off white, crystalline, firm, crystalline texture, no visible porosity, no visible oil stain

8920-8950 SALT: translucent, trace off white, crystalline, firm, crystalline texture, no visible porosity, no visible oil stain

8950-8980 SALT: translucent, trace off white, crystalline, firm, crystalline texture, no visible porosity, no visible oil stain

8980-9010 SALT: translucent, trace off white, crystalline, firm, crystalline texture, no visible porosity, no visible oil stain, LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

9010-9040 LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline; ANHYDRITE: light to medium gray, amorphous, very soft to soft, earthy texture, no visible porosity or oil stain

9040-9070 LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline; ANHYDRITE: light to medium gray, amorphous, very soft to soft, earthy texture, no visible porosity or oil stain

9070-9100 LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

9100-9130 LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

9130-9160 LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

9160-9190 SALT: translucent, trace off white, crystalline, firm, crystalline texture, no visible porosity, no visible oil stain, LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

9190-9205 SALT: translucent, trace off white, crystalline, firm, crystalline texture, no visible porosity, no visible oil stain

Base Charles Salt ***9,205' MD / 9,205' TVD (-7,123')***

9205-9250 LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

9250-9280 LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

9280-9310 LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

9310-9340 LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

9340-9370 LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

9370-9400 LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

9400-9428 LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

Mission Canyon ***9,428' MD / 9,428' TVD (-7,346')***

9428-9460 LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

9460-9490 LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

9490-9520 LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

9520-9550 LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

9550-9580 LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

9580-9610 LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

9610-9640 LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

9640-9670 LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

9670-9700 LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

9700-9730 LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

9730-9760 LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

9760-9790 LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

9790-9820 LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

9820-9850 LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

9850-9880 LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

9880-9910 LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

9910-9940 LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

9940-9976 LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

Lodgepole ***9,976' MD / 9,976' TVD (-7,894')***

9976-10000 LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

10000-10030 ARGILLACEOUS LIMESTONE: wackestone, medium gray brown to light gray brown, firm, crystalline texture, trace sparry calcareous, carbonaceous, possible intercrystalline porosity, possible slightly light brown spotty oil stain

10030-10060 ARGILLACEOUS LIMESTONE: wackestone, medium gray brown to light gray brown, firm, crystalline texture, trace sparry calcareous, carbonaceous, possible intercrystalline porosity, possible slightly light brown spotty oil stain

10060-10090 ARGILLACEOUS LIMESTONE: wackestone, medium gray brown to light gray brown, firm, crystalline texture, trace sparry calcareous, carbonaceous, possible intercrystalline porosity, possible slightly light brown spotty oil stain

10480-10510 ARGILLACEOUS LIMESTONE: wackestone, medium gray brown to light gray brown, firm, crystalline texture, trace sparry calcareous, carbonaceous, possible intercrystalline porosity, possible slightly light brown spotty oil stain

10510-10540 ARGILLACEOUS LIMESTONE: wackestone, medium gray brown to light gray brown, firm, crystalline texture, trace sparry calcareous, carbonaceous, possible intercrystalline porosity, possible slightly light brown spotty oil stain

10540-10570 ARGILLACEOUS LIMESTONE: wackestone, medium gray brown to light gray brown, firm, crystalline texture, trace sparry calcareous, carbonaceous, possible intercrystalline porosity, possible slightly light brown spotty oil stain

10570-10600 ARGILLACEOUS LIMESTONE: wackestone, medium gray brown to light gray brown, firm, crystalline texture, trace sparry calcareous, carbonaceous, possible intercrystalline porosity, possible slightly light brown spotty oil stain

10600-10630 ARGILLACEOUS LIMESTONE: wackestone, medium gray brown to light gray brown, firm, crystalline texture, trace sparry calcareous, carbonaceous, possible intercrystalline porosity, possible slightly light brown spotty oil stain

10630-10660 ARGILLACEOUS LIMESTONE: wackestone, medium gray brown to light gray brown, firm, crystalline texture, trace sparry calcareous, carbonaceous, possible intercrystalline porosity, possible slightly light brown spotty oil stain

10660-10690 ARGILLACEOUS LIMESTONE: wackestone, medium gray brown to light gray brown, firm, crystalline texture, trace sparry calcareous, carbonaceous, possible intercrystalline porosity, possible slightly light brown spotty oil stain

10690-10720 ARGILLACEOUS LIMESTONE: wackestone, medium gray brown to light gray brown, firm, crystalline texture, trace sparry calcareous, carbonaceous, possible intercrystalline porosity, possible slightly light brown spotty oil stain

10720-10750 ARGILLACEOUS LIMESTONE: wackestone, medium gray brown to light gray brown, firm, crystalline texture, trace sparry calcareous, carbonaceous, possible intercrystalline porosity, possible slightly light brown spotty oil stain

10750-10780 ARGILLACEOUS LIMESTONE: wackestone, medium gray brown to light gray brown, firm, crystalline texture, trace sparry calcareous, carbonaceous, possible intercrystalline porosity, possible slightly light brown spotty oil stain

10780-10810 ARGILLACEOUS LIMESTONE: wackestone, medium gray brown to light gray brown, firm, crystalline texture, trace sparry calcareous, carbonaceous, possible intercrystalline porosity, possible slightly light brown spotty oil stain

10810-10847 ARGILLACEOUS LIMESTONE: wackestone, medium gray brown to light gray brown, firm, crystalline texture, trace sparry calcareous, carbonaceous, possible intercrystalline porosity, possible slightly light brown spotty oil stain

Bakken Formation, Upper Shale

10,847' MD / 10,712' TVD (-8,630')

10847-10895 SHALE: black, dark gray black, firm, sub blocky, earthy, calcareous cement, petro, carbonaceous, abundant fine disseminated pyrite, fine laminated

Bakken Formation, Middle Member**10,895' MD / 10,728' TVD (-8,646')**

10895-10930 SILTY SANDSTONE: common salt and pepper, occasional light gray to medium gray, visible white, very fine grained, soft to firm, moderately to well sorted, rounded to subrounded, moderately to poorly cemented, visible calcareous filled fractures, possible fracture porosity, common intergranular porosity, visible light brown spotty to even oil stain

10930-10960 SILTY SANDSTONE: common salt and pepper, occasional light gray to medium gray, visible white, very fine grained, soft to firm, moderately to well sorted, rounded to subrounded, moderately to poorly cemented, visible calcareous filled fractures, possible fracture porosity, common intergranular porosity, visible light brown spotty to even oil stain

10960-10990 SILTY SANDSTONE: common salt and pepper, occasional light gray to medium gray, visible white, very fine grained, soft to firm, moderately to well sorted, rounded to subrounded, moderately to poorly cemented, visible calcareous filled fractures, possible fracture porosity, common intergranular porosity, visible light brown spotty to even oil stain

10990-11020 SILTY SANDSTONE: common salt and pepper, occasional light gray to medium gray, visible white, very fine grained, soft to firm, moderately to well sorted, rounded to subrounded, moderately to poorly cemented, visible calcareous filled fractures, possible fracture porosity, common intergranular porosity, visible light brown spotty to even oil stain

11020-11050 SILTY SANDSTONE: common salt and pepper, occasional light gray to medium gray, visible white, very fine grained, soft to firm, moderately to well sorted, rounded to subrounded, moderately to poorly cemented, visible calcareous filled fractures, possible fracture porosity, common intergranular porosity, visible light brown spotty to even oil stain

11050-11080 SILTY SANDSTONE: common salt and pepper, occasional light gray to medium gray, visible white, very fine grained, soft to firm, moderately to well sorted, rounded to subrounded, moderately to poorly cemented, visible calcareous filled fractures, possible fracture porosity, common intergranular porosity, visible light brown spotty to even oil stain

11080-11110 SILTY SANDSTONE: common salt and pepper, occasional light gray to medium gray, visible white, very fine grained, soft to firm, moderately to well sorted, rounded to subrounded, moderately to poorly cemented, visible calcareous filled fractures, possible fracture porosity, common intergranular porosity, visible light brown spotty to even oil stain

11110-11150 SILTY SANDSTONE: common salt and pepper, occasional light gray to medium gray, visible white, very fine grained, soft to firm, moderately to well sorted, rounded to subrounded, moderately to poorly cemented, visible calcareous filled fractures, possible fracture porosity, common intergranular porosity, visible light brown spotty to even oil stain

11150-11200 SILTY SANDSTONE: common salt and pepper, occasional light gray to medium gray, visible white, very fine grained, soft to firm, moderately to well sorted, rounded to subrounded, moderately to poorly cemented, visible calcareous filled fractures, possible fracture porosity, common intergranular porosity, visible light brown spotty to even oil stain

11200-11250 SILTY SANDSTONE: common salt and pepper, occasional light gray to medium gray, visible white, very fine grained, soft to firm, moderately to well sorted, rounded to subrounded, moderately to poorly cemented, visible calcareous filled fractures, possible fracture porosity, common intergranular porosity, visible light brown spotty to even oil stain

11250-11300 SILTY SANDSTONE: light gray to medium gray, visible white, very fine grained, soft to firm, moderately to well sorted, rounded to subrounded, moderately to poorly cemented, visible calcareous filled fractures, possible fracture porosity, common intergranular porosity, visible light brown spotty to even oil stain

11950-12000 SILTY SANDSTONE: light gray, md gray, cream, white, friable, firm, fine grained, sub spherules, sub rounded, moderately well sorted, calcareous cemented, moderately cemented quality, intergranular porosity, spotty brown oil stain, moderately fast streaming milky yellow cut

12000-12050 SILTY SANDSTONE: light gray, md gray, cream, white, friable, firm, fine grained, sub spherules, sub rounded, moderately well sorted, calcareous cemented, moderately cemented quality, intergranular porosity, spotty brown oil stain, moderately fast streaming milky yellow cut

12050-12100 SILTY SANDSTONE: light gray, md gray, cream, white, friable, firm, fine grained, sub spherules, sub rounded, moderately well sorted, calcareous cemented, moderately cemented quality, intergranular porosity, spotty brown oil stain, moderately fast streaming milky yellow cut

12100-12150 SILTY SANDSTONE: light gray, md gray, cream, white, friable, firm, fine grained, sub spherules, sub rounded, moderately well sorted, calcareous cemented, moderately cemented quality, intergranular porosity, spotty brown oil stain, moderately fast streaming milky yellow cut

12150-12200 SILTY SANDSTONE: light gray, md gray, cream, white, friable, firm, fine grained, sub spherules, sub rounded, moderately well sorted, calcareous cemented, moderately cemented quality, intergranular porosity, spotty brown oil stain, moderately fast streaming milky yellow cut

12200-12250 SILTY SANDSTONE: light gray, md gray, cream, white, friable, firm, fine grained, sub spherules, sub rounded, moderately well sorted, calcareous cemented, moderately cemented quality, intergranular porosity, spotty brown oil stain, moderately fast streaming milky yellow cut

12250-12300 SILTY SANDSTONE: light gray, md gray, cream, white, friable, firm, fine grained, sub spherules, sub rounded, moderately well sorted, calcareous cemented, moderately cemented quality, intergranular porosity, spotty brown oil stain, moderately fast streaming milky yellow cut

12300-12350 SILTY SANDSTONE: light gray, md gray, cream, white, friable, firm, fine grained, sub spherules, sub rounded, moderately well sorted, calcareous cemented, moderately cemented quality, intergranular porosity, spotty brown oil stain, moderately fast streaming milky yellow cut

12350-12400 SILTY SANDSTONE: md to dark gray, light gray, off white, friable, firm, brittle in part, fine grained, sub spherules, sub rounded, well sorted, calcareous cemented, moderately cemented quality, intergranular porosity, rare spotty brown oil stain, moderately streaming blue/white cut

12400-12450 SILTY SANDSTONE: md to dark gray, light gray, off white, friable, firm, brittle in part, fine grained, sub spherules, sub rounded, well sorted, calcareous cemented, moderately cemented quality, intergranular porosity, rare spotty brown oil stain, moderately streaming blue/white cut

12450-12500 SILTY SANDSTONE: md to dark gray, light gray, off white, friable, firm, brittle in part, fine grained, sub spherules, sub rounded, well sorted, calcareous cemented, moderately cemented quality, intergranular porosity, rare spotty brown oil stain, moderately streaming blue/white cut

12500-12550 SILTY SANDSTONE: md to dark gray, light gray, off white, rounded brown in part, friable, soft to firm, brittle, fine grained, sub spherules, sub rounded, well sorted, calcareous cemented, moderately cemented quality, intergranular porosity, rare spotty brown oil stain, moderately streaming blue/white cut

12550-12600 SILTY SANDSTONE: md to dark gray, light gray, off white, rounded brown in part, friable, soft to firm, brittle, fine grained, sub spherules, sub rounded, well sorted, calcareous cemented, moderately cemented quality, intergranular porosity, rare spotty brown oil stain, moderately streaming blue/white cut

12600-12650 SILTY SANDSTONE: md to light gray, off white, friable, soft to firm, brittle, fine grained, sub spherules, sub rounded, well sorted, calcareous cemented, moderately cemented quality, intergranular porosity, rare spotty brown oil stain, moderately streaming blue/white cut

12650-12700 SILTY SANDSTONE: md to light gray, off white, friable, soft to firm, brittle, fine grained, sub spherules, sub rounded, well sorted, calcareous cemented, moderately cemented quality, intergranular porosity, rare spotty brown oil stain, moderately streaming blue/white cut

12700-12750 SILTY SANDSTONE: md to light gray, off white, friable, soft to firm, brittle, fine grained, sub spherules, sub rounded, well sorted, calcareous cemented, moderately cemented quality, intergranular porosity, rare spotty brown oil stain, moderately streaming blue/white cut

12750-12800 SILTY SANDSTONE: light to md gray, dark gray in part, vitreous grain, friable, soft to firm, fine to very fine grained, sub spherules, sub rounded, moderately sorted, calcareous cemented, moderately cemented quality, trace disseminated pyrite, intergranular porosity, spotty brown oil stain, moderately slow streaming milky blue cut, moderately bright yellow res ring

12800-12850 SILTY SANDSTONE: light to md gray, dark gray in part, vitreous grain, friable, soft to firm, fine to very fine grained, sub spherules, sub rounded, moderately sorted, calcareous cemented, moderately cemented quality, trace disseminated pyrite, intergranular porosity, spotty brown oil stain, moderately slow streaming milky blue cut, moderately bright yellow res ring

12850-12900 SILTY SANDSTONE: light to md gray, dark gray in part, vitreous grain, friable, soft to firm, fine to very fine grained, sub spherules, sub rounded, moderately sorted, calcareous cemented, moderately cemented quality, trace disseminated pyrite, intergranular porosity, spotty brown oil stain, moderately slow streaming milky blue cut, moderately bright yellow res ring

12900-12950 SILTY SANDSTONE: light to md gray, dark gray in part, vitreous grain, friable, soft to firm, fine to very fine grained, sub spherules, sub rounded, moderately sorted, calcareous cemented, moderately cemented quality, trace disseminated pyrite, intergranular porosity, spotty brown oil stain, moderately slow streaming milky blue cut, moderately bright yellow res ring

12950-13000 SILTY SANDSTONE: md gray, light gray, vitreous grains, friable, firm, fine to very fine grained, sub spherules, sub rounded, moderately sorted, calcareous cemented, moderately good cemented quality, trace disseminated pyrite, intergranular porosity, rare to common spotty brown oil stain, slow streaming milky blue cut, moderately bright yellow/grain res ring

13000-13050 SILTY SANDSTONE: md gray, light gray, vitreous grains, friable, firm, fine to very fine grained, sub spherules, sub rounded, moderately sorted, calcareous cemented, moderately good cemented quality, trace disseminated pyrite, intergranular porosity, rare to common spotty brown oil stain, slow streaming milky blue cut, moderately bright yellow/grain res ring

13050-13100 SILTY SANDSTONE: md gray, light gray, vitreous grains, friable, firm, fine to very fine grained, sub spherules, sub rounded, moderately sorted, calcareous cemented, moderately good cemented quality, trace disseminated pyrite, intergranular porosity, rare to common spotty brown oil stain, slow streaming milky blue cut, moderately bright yellow/grain res ring

13100-13150 SILTY SANDSTONE: md gray, light gray, vitreous grains, friable, firm, fine to very fine grained, sub spherules, sub rounded, moderately sorted, calcareous cemented, moderately good cemented quality, trace disseminated pyrite, intergranular porosity, rare to common spotty brown oil stain, slow streaming milky blue cut, moderately bright yellow/grain res ring

13150-13200 SILTY SANDSTONE: light to md gray, dark gray in part, vitreous grain, friable, soft to firm, fine to very fine grained, sub spherules, sub rounded, moderately sorted, calcareous cemented, moderately cemented quality, trace disseminated pyrite, intergranular porosity, spotty brown oil stain, moderately slow streaming milky blue cut, moderately bright yellow res ring

13200-13250 SILTY SANDSTONE: light to md gray, dark gray in part, vitreous grain, friable, soft to firm, fine to very fine grained, sub spherules, sub rounded, moderately sorted, calcareous cemented, moderately cemented quality, trace disseminated pyrite, intergranular porosity, spotty brown oil stain, moderately slow streaming milky blue cut, moderately bright yellow res ring

13250-13300 SILTY SANDSTONE: light to md gray, dark gray in part, vitreous grain, friable, soft to firm, fine to very fine grained, sub spherules, sub rounded, moderately sorted, calcareous cemented, moderately cemented quality, trace disseminated pyrite, intergranular porosity, spotty brown oil stain, moderately slow streaming milky blue cut, moderately bright yellow res ring

13300-13350 SILTY SANDSTONE: light to md gray, dark gray in part, vitreous grain, friable, soft to firm, fine to very fine grained, sub spherules, sub rounded, moderately sorted, calcareous cemented, moderately cemented quality, trace disseminated pyrite, intergranular porosity, spotty brown oil stain, moderately slow streaming milky blue cut, moderately bright yellow res ring

13350-13400 SILTY SANDSTONE: light to md gray, dark gray in part, vitreous grain, friable, soft to firm, fine to very fine grained, sub spherules, sub rounded, moderately sorted, calcareous cemented, moderately cemented quality, trace disseminated pyrite, intergranular porosity, spotty brown oil stain, moderately slow streaming milky blue cut, moderately bright yellow res ring

13400-13450 SILTY SANDSTONE: light to md gray, dark gray in part, vitreous grain, friable, soft to firm, fine to very fine grained, sub spherules, sub rounded, moderately sorted, calcareous cemented, moderately cemented quality, trace disseminated pyrite, intergranular porosity, spotty brown oil stain, moderately slow streaming milky blue cut, moderately bright yellow res ring

13450-13500 SILTY SANDSTONE: white to off white, translucent, light gray, friable, very fine to fine grained, subangular to sun rounded, moderately to well sorted, moderately calcite cemented, common disseminated pyrite, occasional tan to light brown spotty oil stain, occasional yellow fluorescence, slightly white streaming cut

13500-13550 SILTY SANDSTONE: white to off white, translucent, light gray, friable, very fine to fine grained, subangular to sun rounded, moderately to well sorted, moderately calcite cemented, common disseminated pyrite, occasional tan to light brown spotty oil stain, occasional yellow fluorescence, slightly white streaming cut

13550-13600 SILTY SANDSTONE: white to off white, translucent, light gray, friable, very fine to fine grained, subangular to sun rounded, moderately to well sorted, moderately calcite cemented, common disseminated pyrite, occasional tan to light brown spotty oil stain, occasional yellow fluorescence, slightly white streaming cut

13600-13650 SILTY SANDSTONE: white to off white, translucent, light gray, friable, very fine to fine grained, subangular to sun rounded, moderately to well sorted, moderately calcite cemented, common disseminated pyrite, occasional tan to light brown spotty oil stain, common bright yellow fluorescence, fast gray white streaming cut

13650-13700 SILTY SANDSTONE: white to off white, translucent, light gray, friable, very fine to fine grained, subangular to sun rounded, moderately to well sorted, moderately calcite cemented, common disseminated pyrite, occasional tan to light brown spotty oil stain, common bright yellow fluorescence, fast gray white streaming cut

13700-13750 SILTY SANDSTONE: white to off white, translucent, light gray, occasional md gray, friable, very fine to fine grained, subangular to sun rounded, moderately to well sorted, moderately calcite cemented, abundant disseminated pyrite, occasional tan to light brown spotty oil stain, common bright yellow fluorescence, fast light gray streaming cut

13750-13800 SILTY SANDSTONE: white to off white, translucent, light gray, decreasing md gray, friable, very fine to fine grained, subangular to sun rounded, moderately to well sorted, moderately calcite cemented, occasional disseminated pyrite, occasional tan to light brown spotty oil stain, common bright yellow fluorescence, fast grain white streaming cut

13800-13850 SILTY SANDSTONE: white to off white, translucent, light gray, occasional md gray, friable, very fine to fine grained, subangular to sun rounded, moderately to well sorted, moderately calcite cemented, abundant disseminated pyrite, occasional tan to light brown spotty oil stain, common bright yellow fluorescence, fast light gray streaming cut

13850-13900 SILTY SANDSTONE: white to off white, translucent, light gray, occasional md gray, friable, very fine to fine grained, subangular to sun rounded, moderately to well sorted, moderately calcite cemented, abundant disseminated pyrite, occasional tan to light brown spotty oil stain, common bright yellow fluorescence, fast light gray streaming cut

13900-13950 SILTY SANDSTONE: white to off white, translucent, light gray, decreasing md gray, friable, very fine to fine grained, subangular to sun rounded, moderately to well sorted, moderately calcite cemented, occasional disseminated pyrite, occasional tan to light brown spotty oil stain, common bright yellow fluorescence, fast grain white streaming cut

13950-14000 SILTY SANDSTONE: white to off white, translucent, light gray, occasional md gray, friable, very fine to fine grained, subangular to sun rounded, moderately to well sorted, moderately calcite cemented, abundant disseminated pyrite, occasional tan to light brown spotty oil stain, common bright yellow fluorescence, fast light gray streaming cut

14000-14050 SILTY SANDSTONE: white to off white, translucent, light gray, occasional md gray, friable, very fine to fine grained, subangular to sun rounded, moderately to well sorted, moderately calcite cemented, abundant disseminated pyrite, occasional tan to light brown spotty oil stain, common bright yellow fluorescence, fast light gray streaming cut

14050-14100 SILTY SANDSTONE: white to off white, translucent, light gray, occasional md gray, friable, very fine to fine grained, subangular to sun rounded, moderately to well sorted, moderately calcite cemented, abundant disseminated pyrite, occasional tan to light brown spotty oil stain, common bright yellow fluorescence, fast light gray streaming cut

14100-14150 SILTY SANDSTONE: white to off white, translucent, light gray, occasional md gray, friable, very fine to fine grained, subangular to sun rounded, moderately to well sorted, moderately calcite cemented, abundant disseminated pyrite, occasional tan to light brown spotty oil stain, common bright yellow fluorescence, fast light gray streaming cut

14150-14200 SILTY SANDSTONE: white to off white, translucent, light gray, occasional md gray, friable, very fine to fine grained, subangular to sun rounded, moderately to well sorted, moderately calcite cemented, abundant disseminated pyrite, occasional tan to light brown spotty oil stain, common bright yellow fluorescence, fast light gray streaming cut

14200-14250 SILTY SANDSTONE: white to off white, translucent, light gray, occasional md gray, friable, very fine to fine grained, subangular to sun rounded, moderately to well sorted, moderately calcite cemented, abundant disseminated pyrite, occasional tan to light brown spotty oil stain, common bright yellow fluorescence, fast light gray streaming cut

14250-14300 SILTY SANDSTONE: white to off white, translucent, light gray, occasional md gray, friable, very fine to fine grained, subangular to sun rounded, moderately to well sorted, moderately calcite cemented, abundant disseminated pyrite, occasional tan to light brown spotty oil stain, common bright yellow fluorescence, fast light gray streaming cut

14300-14350 SILTY SANDSTONE: white to off white, translucent, light gray, occasional md gray, friable, very fine to fine grained, subangular to sun rounded, moderately to well sorted, moderately calcite cemented, abundant disseminated pyrite, occasional tan to light brown spotty oil stain, common bright yellow fluorescence, fast light gray streaming cut

14350-14400 SILTY SANDSTONE: md gray, dark gray, off white, cream, transparent, friable, firm, brittle, lams, fine to very fine grained, sub spherules, sub rounded to sub angular, moderately sorted, calcareous cemented, moderately cemented quality, rare disseminated pyrite, intergranular porosity, occasional light brown oil stain, fast streaming milky blue cut

14400-14450 SILTY SANDSTONE: md gray, dark gray, off white, cream, transparent, friable, firm, brittle, lams, fine to very fine grained, sub spherules, sub rounded to sub angular, moderately sorted, calcareous cemented, moderately cemented quality, rare disseminated pyrite, intergranular porosity, occasional light brown oil stain, fast streaming milky blue cut

14450-14500 SILTY SANDSTONE: md gray, dark gray, off white, cream, transparent, friable, firm, brittle, lams, fine to very fine grained, sub spherules, sub rounded to sub angular, moderately sorted, calcareous cemented, moderately cemented quality, rare disseminated pyrite, intergranular porosity, common patchy light brown oil stain, fast streaming milky blue cut, bright yellow res ring

14500-14550 SILTY SANDSTONE: md gray, dark gray, off white, cream, transparent, friable, firm, brittle, lams, fine to very fine grained, sub spherules, sub rounded to sub angular, moderately sorted, calcareous cemented, moderately cemented quality, rare disseminated pyrite, intergranular porosity, common patchy light brown oil stain, fast streaming milky blue cut, bright yellow res ring

14550-14600 SILTY SANDSTONE: md gray, dark gray, off white, cream, transparent, friable, soft to firm, lams, very fine grained, sub spherules, sub rounded to sub angular, moderately well sorted, calcareous cemented, moderately cemented quality, occasional disseminated pyrite, intergranular porosity, common patchy light brown oil stain, fast streaming milky blue cut, bright yellow res ring

14600-14650 SILTY SANDSTONE: md gray, dark gray, off white, cream, transparent, friable, soft to firm, lams, very fine grained, sub spherules, sub rounded to sub angular, moderately well sorted, calcareous cemented, moderately cemented quality, occasional disseminated pyrite, intergranular porosity, common patchy light brown oil stain, fast streaming milky blue cut, bright yellow res ring

14650-14700 SILTY SANDSTONE: md gray, dark gray, off white, transparent, friable, firm, lams, very fine grained, sub spherules, sub rounded to sub angular, moderately well sorted, calcareous cemented, moderately cemented quality, trace disseminated pyrite, intergranular porosity, occasional patchy light brown oil stain, fast streaming milky blue cut, bright yellow res ring

14700-14750 SILTY SANDSTONE: md gray, dark gray, off white, transparent, friable, firm, lams, very fine grained, sub spherules, sub rounded to sub angular, moderately well sorted, calcareous cemented, moderately cemented quality, trace disseminated pyrite, intergranular porosity, occasional patchy light brown oil stain, fast streaming milky blue cut, bright yellow res ring

14750-14800 SILTY SANDSTONE: md gray, dark gray, off white, transparent, friable, firm, lams, very fine grained, sub spherules, sub rounded to sub angular, moderately well sorted, calcareous cemented, moderately cemented quality, trace disseminated pyrite, intergranular porosity, occasional patchy light brown oil stain, fast streaming milky blue cut, bright yellow/grain res ring

14800-14850 SILTY SANDSTONE: light gray, dark gray, white, cream, transparent, friable, firm, brittle, lams, very fine grained, sub spherules, sub rounded to sub angular, moderately well sorted, calcareous cemented, moderately cemented quality, trace disseminated pyrite, intergranular porosity, rare spotty light brown oil stain, moderately streaming milky blue cut, bright yellow/grain res ring

14850-14900 SILTY SANDSTONE: light gray, dark gray, white, cream, transparent, friable, firm, brittle, lams, very fine grained, sub spherules, sub rounded to sub angular, moderately well sorted, calcareous cemented, moderately cemented quality, trace disseminated pyrite, intergranular porosity, rare spotty light brown oil stain, moderately streaming milky blue cut, bright yellow/grain res ring

14950-15000 SILTY SANDSTONE: light gray, dark gray, white, cream, transparent, friable, firm, brittle, lams, very fine grained, sub spherules, sub rounded to sub angular, moderately well sorted, calcareous cemented, moderately cemented quality, trace disseminated pyrite, intergranular porosity, rare spotty light brown oil stain, moderately streaming milky blue cut, bright yellow/grain res ring

15000-15050 SILTY SANDSTONE: light gray, dark gray, white, cream, transparent, friable, firm, brittle, lams, very fine grained, sub spherules, sub rounded to sub angular, moderately well sorted, calcareous cemented, moderately cemented quality, trace disseminated pyrite, intergranular porosity, rare spotty light brown oil stain, moderately streaming milky blue cut, bright yellow/grain res ring

15050-15100 SILTY SANDSTONE: light gray, dark gray, white, cream, transparent, friable, firm, brittle, lams, very fine grained, sub spherules, sub rounded to sub angular, moderately well sorted, calcareous cemented, moderately cemented quality, trace disseminated pyrite, intergranular porosity, rare spotty light brown oil stain, moderately streaming milky blue cut, bright yellow/grain res ring

15100-15150 SILTY SANDSTONE: light gray, dark gray, white, cream, transparent, friable, firm, brittle, lams, very fine grained, sub spherules, sub rounded to sub angular, moderately well sorted, calcareous cemented, moderately cemented quality, trace disseminated pyrite, intergranular porosity, rare spotty light brown oil stain, moderately streaming milky blue cut, bright yellow/grain res ring

15150-15200 SILTY SANDSTONE: light gray, dark gray, white, cream, transparent, friable, firm, brittle, lams, very fine grained, sub spherules, sub rounded to sub angular, moderately well sorted, calcareous cemented, moderately cemented quality, trace disseminated pyrite, intergranular porosity, rare spotty light brown oil stain, moderately streaming milky blue cut, bright yellow/grain res ring

15200-15250 SILTY SANDSTONE: light gray, dark gray, white, cream, transparent, friable, firm, brittle, lams, very fine grained, sub spherules, sub rounded to sub angular, moderately well sorted, calcareous cemented, moderately cemented quality, trace disseminated pyrite, intergranular porosity, rare spotty light brown oil stain, moderately streaming milky blue cut, bright yellow/grain res ring

15250-15300 SILTY SANDSTONE: light gray, dark gray, white, cream, transparent, friable, firm, brittle, lams, very fine grained, sub spherules, sub rounded to sub angular, moderately well sorted, calcareous cemented, moderately cemented quality, trace disseminated pyrite, intergranular porosity, rare spotty light brown oil stain, moderately streaming milky blue cut, bright yellow/grain res ring

15300-15350 SILTY SANDSTONE: light to md gray, translucent, white to off white, friable, very fine to fine grained, angular to subangular, moderately sorted, moderately to well calcite cemented, trace disseminated pyrite, occasional tan to light brown spotty oil stain, occasional yellow fluorescence, moderately white streaming cut

15350-15400 SILTY SANDSTONE: light to md gray, translucent, white to off white, friable to firm, very fine to fine grained, angular to subangular, moderately sorted, well calcite cemented, trace disseminated pyrite, occasional tan to light brown spotty oil stain, occasional yellow fluorescence, slightly dull white diffuse and moderately white streaming cut

15400-15450 SILTY SANDSTONE: translucent, white, off white, light gray, loose, fine grained, sun rounded, moderately sorted, poorly calcite cemented, common disseminated pyrite, trace dark tan spotty oil stain, occasional yellow fluorescence, moderately bright white diffuse and moderately bright white streaming cut

15450-15500 SILTY SANDSTONE: translucent, white, off white, light gray, loose, fine grained, sun rounded, moderately sorted, poorly calcite cemented, common disseminated pyrite, trace dark tan spotty oil stain, occasional yellow fluorescence, moderately bright white diffuse and moderately bright white streaming cut

15500-15550 SILTY SANDSTONE: light to md gray, translucent, white to off white, friable to firm, very fine to fine grained, angular to subangular, moderately sorted, well calcite cemented, trace disseminated pyrite, occasional tan to light brown spotty oil stain, occasional yellow fluorescence, slightly dull white diffuse and moderately white streaming cut

15550-15600 SILTY SANDSTONE: light to md gray, translucent, white to off white, friable to firm, very fine to fine grained, angular to subangular, moderately sorted, well calcite cemented, trace disseminated pyrite, occasional tan to light brown spotty oil stain, occasional yellow fluorescence, slightly dull white diffuse and moderately white streaming cut

15600-15650 SILTY SANDSTONE: white to off white, translucent, light gray, friable, very fine to fine grained, subangular to sun rounded, moderately sorted, moderately calcite cemented, rare disseminated pyrite, common light brown spotty oil stain, moderately slightly pale white diffuse cut, occasional bright yellow fluorescence

15650-15700 SILTY SANDSTONE: white to off white, translucent, light gray, friable, very fine to fine grained, subangular to sun rounded, moderately sorted, moderately calcite cemented, rare disseminated pyrite, common light brown spotty oil stain, moderately slightly pale white diffuse cut, occasional bright yellow fluorescence

15700-15750 SILTY SANDSTONE: white to off white, translucent, light to md gray, friable, very fine to fine grained, subangular to sun rounded, moderately sorted, moderately calcite cemented, rare disseminated pyrite, common light brown spotty oil stain, slightly pale white diffuse cut, rare bright yellow fluorescence

15750-15800 SILTY SANDSTONE: white to off white, translucent, light to md gray, friable, very fine to fine grained, subangular to sun rounded, moderately sorted, moderately calcite cemented, rare disseminated pyrite, common light brown spotty oil stain, very fast bright white diffuse and very fast bright white streaming cut, occasional yellow fluorescence

15800-15850 SILTY SANDSTONE: translucent, white, off white, light gray, loose, fine grained, sun rounded, moderately sorted, poorly calcite cemented, trace disseminated pyrite, common tan spotty oil stain, very fast bright white diffuse and very fast bright white streaming cut, occasional yellow fluorescence

15850-15900 SILTY SANDSTONE: translucent, white, off white, light gray, loose, fine grained, sun rounded, moderately sorted, poorly calcite cemented, trace disseminated pyrite, common tan spotty oil stain, slightly pale white diffuse cut, rare bright yellow fluorescence

15900-15950 SILTY SANDSTONE: white to off white, translucent, light to md gray, friable, very fine to fine grained, subangular to sun rounded, moderately sorted, moderately calcite cemented, rare disseminated pyrite, common light brown spotty oil stain, slightly pale white diffuse cut, rare bright yellow fluorescence

15950-16000 SILTY SANDSTONE: white to off white, translucent, light to md gray, friable, very fine to fine grained, subangular to sun rounded, moderately sorted, moderately calcite cemented, rare disseminated pyrite, common light brown spotty oil stain, very fast very bright white diffuse and very fast and long very bright white streaming cut, abundant bright yellow fluorescence

16000-16050 SILTY SANDSTONE: white to off white, translucent, light to md gray, friable, very fine to fine grained, subangular to sun rounded, moderately sorted, moderately calcite cemented, rare disseminated pyrite, common light brown spotty oil stain, very fast very bright white diffuse and very fast and long very bright white streaming cut, abundant bright yellow fluorescence

16050-16100 SILTY SANDSTONE: white to off white, translucent, light to md gray, friable, very fine to fine grained, subangular to sun rounded, moderately sorted, moderately calcite cemented, rare disseminated pyrite, common light brown spotty oil stain, very fast very bright white diffuse and very fast and long very bright white streaming cut, abundant bright yellow fluorescence

16100-16150 SILTY SANDSTONE: white to off white, translucent, light to md gray, friable, very fine to fine grained, subangular to sun rounded, moderately sorted, moderately calcite cemented, rare disseminated pyrite, common light brown spotty oil stain, very fast very bright white diffuse and very fast and long very bright white streaming cut, abundant bright yellow fluorescence

16150-16200 SILTY SANDSTONE: white to off white, translucent, light to md gray, friable, very fine to fine grained, subangular to sun rounded, moderately sorted, moderately calcite cemented, rare disseminated pyrite, common light brown spotty oil stain, slightly pale white diffuse cut, rare bright yellow fluorescence

16200-16250 SILTY SANDSTONE: white to off white, translucent, light to md gray, friable, very fine to fine grained, subangular to sun rounded, moderately sorted, moderately calcite cemented, rare disseminated pyrite, common light brown spotty oil stain, slightly pale white diffuse cut, rare bright yellow fluorescence

16250-16300 SILTY SANDSTONE: white to off white, translucent, light to md gray, friable, very fine to fine grained, subangular to sun rounded, moderately sorted, moderately calcite cemented, rare disseminated pyrite, common light brown spotty oil stain, slightly pale white diffuse cut, rare bright yellow fluorescence

16300-16350 SILTY SANDSTONE: white to off white, translucent, light to md gray, friable, very fine to fine grained, subangular to sun rounded, moderately sorted, moderately calcite cemented, rare disseminated pyrite, common light brown spotty oil stain, slightly pale white diffuse cut, rare bright yellow fluorescence

16350-16400 SILTY SANDSTONE: light to md gray, white to off white, translucent, friable, firm, brittle in part, very fine to fine grained, sub spherules, subangular to sun rounded, moderately sorted, calcareous cemented, rare disseminated pyrite, rare light brown spotty oil stain, fast white/blue diffuse cut, bright yellow res ring

16400-16450 SILTY SANDSTONE: light to md gray, white to off white, translucent, friable, firm, brittle in part, very fine to fine grained, sub spherules, subangular to sun rounded, moderately sorted, calcareous cemented, rare disseminated pyrite, rare light brown spotty oil stain, fast white/blue diffuse cut, bright yellow res ring

16450-16500 SILTY SANDSTONE: light to md gray, white to off white, translucent, friable, firm, brittle in part, very fine to fine grained, sub spherules, subangular to sun rounded, moderately sorted, calcareous cemented, rare disseminated pyrite, rare light brown spotty oil stain, fast white/blue diffuse cut, bright yellow res ring

16500-16550 SILTY SANDSTONE: light to md gray, white to off white, translucent, friable, firm, brittle in part, fine grained, sub spherules, sun rounded, moderately well sorted, calcareous cemented, rare disseminated pyrite, rare light brown spotty oil stain, fast white/blue diffuse cut, bright yellow res ring

16550-16600 SILTY SANDSTONE: light to md gray, white to off white, translucent, friable, firm, brittle in part, fine grained, sub spherules, sun rounded, moderately well sorted, calcareous cemented, rare disseminated pyrite, rare light brown spotty oil stain, fast white/blue diffuse cut, bright yellow res ring

16600-16650 SILTY SANDSTONE: light to md gray, white to off white, translucent, friable, firm, brittle in part, fine grained, sub spherules, sun rounded, moderately well sorted, calcareous cemented, rare disseminated pyrite, rare light brown spotty oil stain, fast white/blue diffuse cut, bright yellow res ring

16650-16700 SILTY SANDSTONE: light to md gray, white to off white, translucent, friable, firm, brittle in part, fine grained, sub spherules, sun rounded, moderately well sorted, calcareous cemented, rare disseminated pyrite, rare light brown spotty oil stain, fast white/blue diffuse cut, bright yellow res ring

16700-16750 SILTY SANDSTONE: light to md gray, white to off white, translucent, friable, firm, brittle in part, very fine to fine grained, sub spherules, subangular to sun rounded, moderately sorted, calcareous cemented, rare disseminated pyrite, rare light brown spotty oil stain, fast white/blue diffuse cut, bright yellow res ring

16750-16800 SILTY SANDSTONE: light to md gray, white to off white, translucent, friable, firm, brittle in part, very fine to fine grained, sub spherules, subangular to sun rounded, moderately sorted, calcareous cemented, rare disseminated pyrite, rare light brown spotty oil stain, fast white/blue diffuse cut, bright yellow res ring

16800-16850 SILTY SANDSTONE: light to md gray, white to off white, translucent, friable, firm, brittle in part, very fine to fine grained, sub spherules, subangular to sun rounded, moderately sorted, calcareous cemented, rare disseminated pyrite, rare light brown spotty oil stain, fast white/blue diffuse cut, bright yellow res ring

16850-16900 SILTY SANDSTONE: light to md gray, white to off white, translucent, friable, firm, brittle in part, fine grained, sub spherules, sun rounded, moderately well sorted, calcareous cemented, rare disseminated pyrite, rare light brown spotty oil stain, fast white/blue diffuse cut, bright yellow res ring

16900-16950 SILTY SANDSTONE: light to md gray, white to off white, translucent, friable, fine to very fine grained, subangular to sun rounded, moderately to well sorted, moderately calcite cemented, rare disseminated pyrite, rare light brown spotty oil stain, fast blue white diffuse cut, occasional bright yellow fluorescence, sample quality poor to lube

16950-17000 SILTY SANDSTONE: light to md gray, white to off white, translucent, friable, fine to very fine grained, subangular to sun rounded, moderately to well sorted, moderately calcite cemented, rare disseminated pyrite, rare light brown spotty oil stain, fast blue white diffuse cut, occasional bright yellow fluorescence, sample quality poor to lube

17000-17050 SILTY SANDSTONE: light to md gray, white to off white, translucent, friable, fine to very fine grained, subangular to sun rounded, moderately to well sorted, moderately calcite cemented, rare disseminated pyrite, rare light brown spotty oil stain, fast blue white diffuse cut, occasional bright yellow fluorescence, sample quality poor to lube

17050-17100 SILTY SANDSTONE: light to md gray, white to off white, translucent, friable, fine to very fine grained, subangular to sun rounded, moderately to well sorted, moderately calcite cemented, rare disseminated pyrite, rare light brown spotty oil stain, fast blue white diffuse cut, occasional bright yellow fluorescence, sample quality poor to lube

17100-17150 SILTY SANDSTONE: light to md gray, white to off white, translucent, friable, fine to very fine grained, subangular to sun rounded, moderately to well sorted, moderately calcite cemented, rare disseminated pyrite, rare light brown spotty oil stain, fast blue white diffuse cut, occasional bright yellow fluorescence, sample quality poor to lube

17150-17200 SILTY SANDSTONE: light to md gray, white to off white, translucent, friable, fine to very fine grained, subangular to sun rounded, moderately to well sorted, moderately calcite cemented, rare disseminated pyrite, rare light brown spotty oil stain, fast blue white diffuse cut, occasional bright yellow fluorescence, sample quality poor to lube

17200-17250 SILTY SANDSTONE: light to md gray, white to off white, translucent, friable, fine to very fine grained, subangular to sun rounded, moderately to well sorted, moderately calcite cemented, rare disseminated pyrite, rare light brown spotty oil stain, fast blue white diffuse cut, occasional bright yellow fluorescence

17250-17300 SILTY SANDSTONE: light to md gray, white to off white, translucent, friable, fine to very fine grained, subangular to sun rounded, moderately to well sorted, moderately calcite cemented, rare disseminated pyrite, rare light brown spotty oil stain, fast blue white diffuse cut, occasional bright yellow fluorescence

17300-17350 SILTY SANDSTONE: light to md gray, white to off white, translucent, friable, fine to very fine grained, subangular to sun rounded, moderately to well sorted, moderately calcite cemented, rare disseminated pyrite, rare light brown spotty oil stain, fast blue white diffuse cut, occasional bright yellow fluorescence

17350-17400 SILTY SANDSTONE: light to md gray, white to off white, translucent, friable, fine to very fine grained, subangular to sun rounded, moderately to well sorted, moderately calcite cemented, rare disseminated pyrite, rare light brown spotty oil stain, fast blue white diffuse cut, occasional bright yellow fluorescence

17400-17450 SILTY SANDSTONE: light to md gray, white to off white, translucent, friable, fine to very fine grained, subangular to sun rounded, moderately to well sorted, moderately calcite cemented, rare disseminated pyrite, rare light brown spotty oil stain, fast blue white diffuse cut, occasional bright yellow fluorescence

17450-17500 SILTY SANDSTONE: light to md gray, white to off white, translucent, friable, fine to very fine grained, subangular to sun rounded, moderately to well sorted, moderately calcite cemented, rare disseminated pyrite, rare light brown spotty oil stain, fast blue white diffuse cut, occasional bright yellow fluorescence

17500-17550 SILTY SANDSTONE: light to md gray, white to off white, translucent, friable, fine to very fine grained, subangular to sun rounded, moderately to well sorted, moderately calcite cemented, rare disseminated pyrite, rare light brown spotty oil stain, fast blue white diffuse cut, occasional bright yellow fluorescence

17550-17600 SILTY SANDSTONE: light to md gray, white to off white, translucent, friable, fine to very fine grained, subangular to sun rounded, moderately to well sorted, moderately calcite cemented, rare disseminated pyrite, rare light brown spotty oil stain, fast blue white diffuse cut, occasional bright yellow fluorescence

17600-17650 SILTY SANDSTONE: light to md gray, white to off white, translucent, friable, fine to very fine grained, subangular to sun rounded, moderately to well sorted, moderately calcite cemented, rare disseminated pyrite, rare light brown spotty oil stain, fast blue white diffuse cut, occasional bright yellow fluorescence

17650-17700 SILTY SANDSTONE: light to md gray, white to off white, translucent, friable, fine to very fine grained, subangular to sun rounded, moderately to well sorted, moderately calcite cemented, rare disseminated pyrite, rare light brown spotty oil stain, fast blue white diffuse cut, occasional bright yellow fluorescence

17700-17750 SILTY SANDSTONE: light to md gray, white to off white, translucent, friable, fine to very fine grained, subangular to sun rounded, moderately to well sorted, moderately calcite cemented, rare disseminated pyrite, rare light brown spotty oil stain, fast blue white diffuse cut, occasional bright yellow fluorescence

17750-17800 SILTY SANDSTONE: light to md gray, white to off white, translucent, friable, fine to very fine grained, subangular to sun rounded, moderately to well sorted, moderately calcite cemented, rare disseminated pyrite, rare light brown spotty oil stain, fast blue white diffuse cut, occasional bright yellow fluorescence

17800-17850 SILTY SANDSTONE: light to md gray, cream, off white, transparent, friable, soft, very fine grained, sun rounded, moderately to well sorted, calcite cemented, rare disseminated pyrite, rare light brown spotty oil stain, abundant lube in sample

17850-17900 SILTY SANDSTONE: light to md gray, cream, off white, transparent, friable, soft, very fine grained, sun rounded, moderately to well sorted, calcite cemented, rare disseminated pyrite, rare light brown spotty oil stain, abundant lube in sample

17900-17950 SILTY SANDSTONE: light to md gray, cream, off white, transparent, friable, soft, very fine grained, sun rounded, moderately to well sorted, calcite cemented, rare disseminated pyrite, rare light brown spotty oil stain, abundant lube in sample

17950-18000 SILTY SANDSTONE: dark to md gray, cream, translucent, transparent, friable, soft to firm, brittle, very fine to fine grained, sub spherical, sun rounded, moderately sorted, calcite cemented, trace disseminated pyrite, rare light brown spotty oil stain, abundant lube in sample

18000-18050 SILTY SANDSTONE: dark to md gray, cream, translucent, transparent, friable, soft to firm, brittle, very fine to fine grained, sub spherical, sun rounded, moderately sorted, calcite cemented, trace disseminated pyrite, rare light brown spotty oil stain, abundant lube in sample

18050-18100 SILTY SANDSTONE: dark to md gray, cream, translucent, transparent, friable, soft to firm, brittle, very fine to fine grained, sub spherical, sun rounded, moderately sorted, calcite cemented, trace disseminated pyrite, rare light brown spotty oil stain, abundant lube in sample

18100-18150 SILTY SANDSTONE: dark to md gray, cream, translucent, transparent, friable, soft to firm, brittle, very fine to fine grained, sub spherical, sun rounded, moderately sorted, calcite cemented, trace disseminated pyrite, rare light brown spotty oil stain, abundant lube in sample

18150-18200 SILTY SANDSTONE: dark to md gray, cream, translucent, transparent, friable, soft to firm, brittle, very fine to fine grained, sub spherical, sun rounded, moderately sorted, calcite cemented, trace disseminated pyrite, rare light brown spotty oil stain, abundant lube in sample

18200-18250 SILTY SANDSTONE: dark to md gray, cream, translucent, transparent, friable, soft to firm, brittle, very fine to fine grained, sub spherical, sun rounded, moderately sorted, calcite cemented, trace disseminated pyrite, rare light brown spotty oil stain, abundant lube in sample

18250-18300 SILTY SANDSTONE: dark to md gray, cream, translucent, transparent, friable, soft to firm, brittle, very fine to fine grained, sub spherical, sun rounded, moderately sorted, calcite cemented, trace disseminated pyrite, rare light brown spotty oil stain, abundant lube in sample

18300-18350 SILTY SANDSTONE: light to md gray, cream, off white, transparent, friable, soft, very fine grained, sun rounded, moderately to well sorted, calcite cemented, rare disseminated pyrite, rare light brown spotty oil stain, abundant lube in sample

18350-18400 SILTY SANDSTONE: light to md gray, cream, off white, transparent, friable, soft, very fine grained, sun rounded, moderately to well sorted, calcite cemented, rare disseminated pyrite, rare light brown spotty oil stain, abundant lube in sample

18400-18450 SILTY SANDSTONE: light to md gray, white to off white, translucent, friable, fine to very fine grained, subangular to sun rounded, moderately to well sorted, moderately calcite cemented, rare disseminated pyrite, rare light brown spotty oil stain, fast blue white diffuse cut, occasional bright yellow fluorescence

18450-18500 SILTY SANDSTONE: light to md gray, white to off white, translucent, friable, fine to very fine grained, subangular to sun rounded, moderately to well sorted, moderately calcite cemented, rare disseminated pyrite, rare light brown spotty oil stain, fast blue white diffuse cut, occasional bright yellow fluorescence

18500-18550 SILTY SANDSTONE: light to md gray, cream, off white, transparent, friable, soft, very fine grained, sun rounded, moderately to well sorted, calcite cemented, rare disseminated pyrite, rare light brown spotty oil stain, abundant lube in sample

18550-18600 SILTY SANDSTONE: light to md gray, cream, off white, transparent, friable, soft, very fine grained, sun rounded, moderately to well sorted, calcite cemented, rare disseminated pyrite, rare light brown spotty oil stain, abundant lube in sample

18600-18650 SILTY SANDSTONE: light to md gray, white to off white, translucent, friable, very fine grained, subangular to sun rounded, moderately to well sorted, moderately calcite cemented, rare disseminated pyrite, occasional light brown spotty oil stain, abundant lube in sample

18650-18700 SILTY SANDSTONE: light gray, translucent, off white to white, friable, fine to very fine grained, subangular to sun rounded, moderately to well sorted, moderately calcite cemented, rare disseminated pyrite, occasional light brown spotty oil stain, abundant lube in sample

18700-18750 SILTY SANDSTONE: light gray, translucent, off white to white, friable, fine to very fine grained, subangular to sun rounded, moderately to well sorted, moderately calcite cemented, rare disseminated pyrite, occasional light brown spotty oil stain, abundant lube in sample

18750-18800 SILTY SANDSTONE: light to md gray, white to off white, translucent, friable, very fine grained, subangular to sun rounded, moderately to well sorted, moderately calcite cemented, rare disseminated pyrite, occasional light brown spotty oil stain, abundant lube in sample

18800-18850 SILTY SANDSTONE: light to md gray, white to off white, translucent, friable, very fine grained, subangular to sun rounded, moderately to well sorted, moderately calcite cemented, rare disseminated pyrite, occasional light brown spotty oil stain, abundant lube in sample

18850-18900 SILTY SANDSTONE: translucent, off white to white, occasional md gray, loose to friable, fine to very fine grained, sun rounded, moderately sorted, poorly calcite cemented, occasional disseminated pyrite, common light brown even oil stain, fast bright white diffuse and moderately bright white streaming cut, common plant yellow fluorescence

18900-18950 SILTY SANDSTONE: translucent, off white to white, occasional md gray, loose to friable, fine to very fine grained, sun rounded, moderately sorted, poorly calcite cemented, occasional disseminated pyrite, common light brown even oil stain, fast bright white diffuse and moderately bright white streaming cut, common plant yellow fluorescence

18950-19000 SILTY SANDSTONE: transparent, off white, light gray, friable, firm, fine grained, sub spherules, sun rounded, moderately well sorted, calcite cemented, moderately cemented quality, occasional disseminated pyrite, common light brown even oil stain, fast bright white diffuse and moderately bright white streaming cut

19000-19050 SILTY SANDSTONE: transparent, off white, light gray, friable, firm, fine grained, sub spherules, sun rounded, moderately well sorted, calcite cemented, moderately cemented quality, occasional disseminated pyrite, common light brown even oil stain, fast bright white diffuse and moderately bright white streaming cut

19050-19100 SILTY SANDSTONE: transparent, off white, light gray, friable, firm, fine grained, sub spherules, sun rounded, moderately well sorted, calcite cemented, moderately cemented quality, occasional disseminated pyrite, common light brown even oil stain, fast bright white diffuse and moderately bright white streaming cut

19100-19150 SILTY SANDSTONE: transparent, off white, light gray, friable, firm, fine grained, sub spherules, sun rounded, moderately well sorted, calcite cemented, moderately cemented quality, occasional disseminated pyrite, common light brown even oil stain, fast bright white diffuse and moderately bright white streaming cut

19150-19200 SILTY SANDSTONE: transparent to translucent, off white, light to md gray, friable, soft to firm, lams, fine grained, sub spherules, sun rounded, moderately well sorted, calcite cemented, moderately cemented quality, occasional disseminated pyrite, common light brown even oil stain, fast bright white diffuse and moderately bright white streaming cut

19200-19250 SILTY SANDSTONE: transparent to translucent, off white, light to md gray, friable, soft to firm, lams, fine grained, sub spherules, sun rounded, moderately well sorted, calcite cemented, moderately cemented quality, occasional disseminated pyrite, common light brown even oil stain, fast bright white diffuse and moderately bright white streaming cut

19250-19300 SILTY SANDSTONE: transparent to translucent, off white, light to md gray, friable, soft to firm, lams, fine grained, sub spherules, sun rounded, moderately well sorted, calcite cemented, moderately cemented quality, occasional disseminated pyrite, common light brown even oil stain, fast bright white diffuse and moderately bright white streaming cut

19300-19350 SILTY SANDSTONE: transparent to translucent, off white, light to md gray, friable, soft to firm, lams, fine grained, sub spherules, sun rounded, moderately well sorted, calcite cemented, moderately cemented quality, occasional disseminated pyrite, common light brown even oil stain, fast bright white diffuse and moderately bright white streaming cut

19350-19400 SILTY SANDSTONE: transparent to translucent, off white, light to md gray, friable, soft to firm, lams, fine grained, sub spherules, sun rounded, moderately well sorted, calcite cemented, moderately cemented quality, occasional disseminated pyrite, common light brown even oil stain, fast bright white diffuse and moderately bright white streaming cut

19400-19450 SILTY SANDSTONE: transparent to translucent, off white, light to md gray, friable, soft to firm, lams, fine grained, sub spherules, sun rounded, moderately well sorted, calcite cemented, moderately cemented quality, occasional disseminated pyrite, common light brown even oil stain, fast bright white diffuse and moderately bright white streaming cut

19450-19500 SILTY SANDSTONE: translucent, off white, light to md gray, friable, soft to firm, smooth lams, fine grained, sub spherules, sun rounded, moderately well sorted, calcite cemented, moderately poor cemented quality, trace disseminated pyrite, rare light brown spotty oil stain, fast bright white diffuse and moderately bright white streaming cut

19500-19550 SILTY SANDSTONE: translucent, off white, light to md gray, friable, soft to firm, smooth lams, fine grained, sub spherules, sun rounded, moderately well sorted, calcite cemented, moderately poor cemented quality, trace disseminated pyrite, rare light brown spotty oil stain, fast bright white diffuse and moderately bright white streaming cut

19550-19600 SILTY SANDSTONE: translucent, off white, light to md gray, friable, soft to firm, smooth lams, fine grained, sub spherules, sun rounded, moderately well sorted, calcite cemented, moderately poor cemented quality, trace disseminated pyrite, rare light brown spotty oil stain, fast bright white diffuse and moderately bright white streaming cut

19600-19650 SILTY SANDSTONE: translucent, off white, light to md gray, friable, soft to firm, smooth lams, fine grained, sub spherules, sun rounded, moderately well sorted, calcite cemented, moderately poor cemented quality, trace disseminated pyrite, rare light brown spotty oil stain, fast bright white diffuse and moderately bright white streaming cut

19650-19700 SILTY SANDSTONE: translucent, off white, light to md gray, friable, soft to firm, smooth lams, fine grained, sub spherules, sun rounded, moderately well sorted, calcite cemented, moderately poor cemented quality, trace disseminated pyrite, rare light brown spotty oil stain, fast bright white diffuse and moderately bright white streaming cut

19700-19750 SILTY SANDSTONE: translucent, off white, light gray, friable, firm, very fine to fine grained, sun rounded, moderately to well sorted, moderately calcite cemented, rare disseminated pyrite, poor sample quality to abundant lube

19750-19800 SILTY SANDSTONE: translucent, off white, light gray, friable, firm, very fine to fine grained, sun rounded, moderately to well sorted, moderately calcite cemented, rare disseminated pyrite, poor sample quality to abundant lube

19800-19850 SILTY SANDSTONE: translucent, off white, light gray, friable, firm, very fine to fine grained, sun rounded, moderately to well sorted, moderately calcite cemented, rare disseminated pyrite, poor sample quality to abundant lube

19850-19900 SILTY SANDSTONE: off white, light gray, translucent, friable, firm, very fine to fine grained, sun rounded, moderately to well sorted, moderately calcite cemented, rare disseminated pyrite, poor sample quality to abundant lube

19900-19950 SILTY SANDSTONE: off white, light gray, translucent, friable, firm, very fine to fine grained, sun rounded, moderately to well sorted, moderately calcite cemented, rare disseminated pyrite, poor sample quality to abundant lube

19950-20000 SILTY SANDSTONE: off white, light gray, translucent, friable, firm, very fine to fine grained, sun rounded, moderately to well sorted, moderately calcite cemented, rare disseminated pyrite, poor sample quality to abundant lube

20000-20050 SILTY SANDSTONE: translucent, off white to white, occasional md gray, loose to friable, fine to very fine grained, sun rounded, moderately sorted, poorly calcite cemented, occasional disseminated pyrite, common light brown even oil stain, fast bright white diffuse and moderately bright white streaming cut, common plant yellow fluorescence

20050-20100 SILTY SANDSTONE: translucent, off white to white, occasional md gray, loose to friable, fine to very fine grained, sun rounded, moderately sorted, poorly calcite cemented, occasional disseminated pyrite, common light brown even oil stain, fast bright white diffuse and moderately bright white streaming cut, common plant yellow fluorescence

20100-20150 SILTY SANDSTONE: off white, light gray, translucent, friable, firm, very fine to fine grained, sun rounded, moderately to well sorted, moderately calcite cemented, rare disseminated pyrite, poor sample quality to abundant lube

20150-20200 SILTY SANDSTONE: off white, light gray, translucent, friable, firm, very fine to fine grained, sun rounded, moderately to well sorted, moderately calcite cemented, rare disseminated pyrite, poor sample quality to abundant lube

20200-20250 SILTY SANDSTONE: off white, light gray, translucent, friable, firm, very fine to fine grained, sun rounded, moderately to well sorted, moderately calcite cemented, rare disseminated pyrite, poor sample quality to abundant lube

20250-20300 SILTY SANDSTONE: off white, light gray, translucent, friable, firm, very fine to fine grained, sun rounded, moderately to well sorted, moderately calcite cemented, rare disseminated pyrite, poor sample quality to abundant lube

20300-20350 SILTY SANDSTONE: off white, light gray, translucent, friable, firm, very fine to fine grained, sun rounded, moderately to well sorted, moderately calcite cemented, rare disseminated pyrite, poor sample quality to abundant lube

20350-20400 SILTY SANDSTONE: off white, light gray, translucent, friable, firm, very fine to fine grained, sun rounded, moderately to well sorted, moderately calcite cemented, rare disseminated pyrite, poor sample quality to abundant lube

20400-20450 SILTY SANDSTONE: off white, light gray, translucent, friable, firm, very fine to fine grained, sun rounded, moderately to well sorted, moderately calcite cemented, rare disseminated pyrite, poor sample quality to abundant lube

20450-20500 SILTY SANDSTONE: off white, light gray, translucent, friable, firm, very fine to fine grained, sun rounded, moderately to well sorted, moderately calcite cemented, rare disseminated pyrite, poor sample quality to abundant lube

20500-20550 SILTY SANDSTONE: off white, light gray, translucent, friable, firm, very fine to fine grained, sun rounded, moderately to well sorted, moderately calcite cemented, rare disseminated pyrite, poor sample quality to abundant lube

20550-20600 SILTY SANDSTONE: off white, light gray, translucent, friable, firm, very fine to fine grained, sun rounded, moderately to well sorted, moderately calcite cemented, rare disseminated pyrite, poor sample quality to abundant lube

20600-20650 SILTY SANDSTONE: off white, light gray, translucent, friable, firm, very fine to fine grained, sun rounded, moderately to well sorted, moderately calcite cemented, rare disseminated pyrite, poor sample quality to abundant lube

TD of 20,650' Reached 11:15 hrs on February 17, 2015





Gyrodata Incorporated
301 Thelma Dr. #413
Casper, Wyoming 82601

307-472-0182
Fax: 307-472-0204

DIRECTIONAL SURVEY CERTIFICATION

RE: _____ Oasis Petroleum _____ (Operator)
_____ Kline Federal 5300 41-18 14BX _____ (Well Name & No.)
_____ 8.27 _____ (Magnetic Declination)

Surveyor: George Kerr

I, Kristopher Bertsch, having personal knowledge of all the facts, hereby certify that the attached directional survey run from measured depth of 140 feet to a measured depth of 20572 feet and is projected to 20650 feet, is true and correct as determined from all available records.


Signature

MWD Coordinator
Title

Gyrodata, Inc.
Company

*All measurements recorded with a manned MWD Survey Tool.

A Gyrodata Directional Survey

FINAL DEFINITIVE COPY

for

OASIS PETROLEUM

Lease: Kline Federal Well: 5300 41-18 14BX, 8 3/4" Case, and 6" Hole

Location: Nabors #486 , McKenzie County, North Dakota

Job Number: RD0814M181

Run Date: 03 Mar 2015

Surveyor: George Kerr

Calculation Method: MINIMUM CURVATURE

Survey Latitude: 48.068610 deg. N Longitude: 103.602344 deg. W

Magnetic Declination: 8.2700 deg

Total Correction: 8.2700 deg

Proposed Well Direction: 90.000 deg

Depth Reference: Rotary Table

Air Gap (RKB to Ground / RKB to MSL): 25.00

Ground Elevation (Ground to MSL / MSL to Mudline): 2057.00

Vertical Section Calculated from Well Head Location

Closure Calculated from Well Head Location

Horizontal Coordinates Calculated from Well Head Location

A Gyrodata Directional Survey

Oasis Petroleum

Lease: Kline Federal Well: 5300 41-18 14BX, 8 3/4" Case, and 6" Hole

Location: Nabors #486 , McKenzie County, North Dakota

Job Number: RD0814M181

MEASURED DEPTH feet	INCL deg.	AZIMUTH deg.	DOGLEG SEVERITY deg./ 100 ft.	VERTICAL DEPTH feet	CLOSURE DIST. feet	AZIMUTH deg.	HORIZONTAL COORDINATES feet
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0.00	0.00	0.00	0.00	0.00	0.0	0.0	0.00 N 0.00 E
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NO TIE IN DATA PROVIDED. TIE INTO SURFACE.

ALL AZIMUTHS ARE REFERENCED TO TRUE NORTH.

ALL MEASURED DEPTHS AND COORDINATES REFERENCED TO NABORS 486 R.K.B. AT 25 FT.

140-20572 FT. MD ARE MAGNETIC MEASUREMENT WHILE DRILLING SURVEYS.

STRAIGHT LINE PROJECTION TO BIT DEPTH AT 20650 FT. MD.

140.00	0.97	48.35	0.69	139.99	1.2	48.4	0.79 N 0.89 E
234.00	0.70	53.45	0.30	233.98	2.6	49.5	1.66 N 1.94 E
327.00	0.88	66.10	0.27	326.97	3.8	53.2	2.29 N 3.05 E
421.00	0.70	57.14	0.23	420.97	5.1	55.4	2.89 N 4.19 E
516.00	0.13	64.52	0.60	515.96	5.8	55.8	3.25 N 4.78 E
607.00	0.57	64.08	0.48	606.96	6.3	56.5	3.49 N 5.28 E
699.00	0.40	275.90	1.02	698.96	6.5	55.2	3.73 N 5.37 E
791.00	0.57	257.18	0.25	790.96	5.9	51.5	3.66 N 4.60 E
882.00	1.05	269.92	0.56	881.95	4.9	43.1	3.56 N 3.33 E
974.00	1.19	290.93	0.47	973.93	4.2	22.2	3.90 N 1.59 E
1066.00	0.53	329.77	0.92	1065.92	4.6	6.0	4.61 N 0.49 E
1158.00	0.53	310.09	0.20	1157.92	5.2	359.4	5.25 N 0.05 W
1250.00	0.44	313.16	0.10	1249.91	5.8	353.7	5.76 N 0.64 W
1342.00	0.48	182.65	0.91	1341.91	5.7	350.8	5.62 N 0.91 W

A Gyrodata Directional Survey

Oasis Petroleum

Lease: Kline Federal Well: 5300 41-18 14BX, 8 3/4" Case, and 6" Hole

Location: Nabors #486 , McKenzie County, North Dakota

Job Number: RD0814M181

MEASURED DEPTH feet	INCL deg.	AZIMUTH deg.	DOGLEG SEVERITY deg./ 100 ft.	VERTICAL DEPTH feet	CLOSURE DIST. AZIMUTH feet deg.	HORIZONTAL COORDINATES feet
1433.00	0.53	202.95	0.20	1432.91	5.0 347.3	4.85 N 1.09 W
1528.00	0.75	192.49	0.26	1527.90	4.1 340.0	3.84 N 1.40 W
1623.00	1.14	214.81	0.56	1622.89	3.2 319.8	2.46 N 2.07 W
1719.00	1.23	196.27	0.41	1718.87	3.0 283.2	0.68 N 2.91 W
1813.00	1.10	200.22	0.16	1812.85	3.7 252.1	1.13 S 3.50 W
1908.00	1.27	204.18	0.20	1907.83	5.2 235.2	2.95 S 4.25 W
2003.00	0.18	146.43	1.25	2002.82	6.1 228.7	4.03 S 4.60 W
2097.00	0.26	11.79	0.43	2096.82	6.0 228.6	3.95 S 4.47 W
2189.00	0.44	69.44	0.40	2188.82	5.5 228.5	3.62 S 4.10 W
2283.00	0.48	94.84	0.22	2282.82	4.9 223.7	3.53 S 3.37 W
2377.00	0.18	82.01	0.33	2376.81	4.5 218.6	3.54 S 2.83 W
2471.00	0.40	138.61	0.36	2470.81	4.5 213.2	3.76 S 2.47 W
2566.00	0.31	128.06	0.12	2565.81	4.6 206.1	4.17 S 2.04 W
2660.00	0.31	129.38	0.01	2659.81	4.8 200.1	4.49 S 1.65 W
2756.00	0.26	29.10	0.46	2755.81	4.7 196.7	4.46 S 1.34 W
2851.00	0.13	110.57	0.29	2850.81	4.5 194.7	4.31 S 1.13 W
2944.00	0.31	347.35	0.43	2943.81	4.2 194.9	4.10 S 1.09 W
3038.00	0.31	58.54	0.38	3037.81	3.8 194.0	3.72 S 0.93 W
3133.00	0.22	55.99	0.10	3132.81	3.5 189.1	3.49 S 0.56 W
3226.00	0.31	11.08	0.24	3225.80	3.2 186.6	3.14 S 0.36 W
3321.00	0.57	51.78	0.41	3320.80	2.6 178.7	2.60 S 0.06 E
3416.00	0.62	37.63	0.16	3415.80	2.0 158.6	1.90 S 0.74 E

A Gyrodata Directional Survey

Oasis Petroleum

Lease: Kline Federal Well: 5300 41-18 14BX, 8 3/4" Case, and 6" Hole

Location: Nabors #486 , McKenzie County, North Dakota

Job Number: RD0814M181

MEASURED DEPTH feet	INCL deg.	AZIMUTH deg.	DOGLEG SEVERITY deg./ 100 ft.	VERTICAL DEPTH feet	CLOSURE DIST. AZIMUTH feet deg.	HORIZONTAL COORDINATES feet
3511.00	0.35	10.82	0.36	3510.79	1.6 137.3	1.20 S 1.11 E
3605.00	0.92	143.45	1.26	3604.79	2.2 133.4	1.53 S 1.61 E
3700.00	0.70	153.20	0.27	3699.78	3.5 138.8	2.66 S 2.33 E
3794.00	1.05	143.01	0.41	3793.77	5.0 141.2	3.86 S 3.11 E
3888.00	0.97	154.43	0.23	3887.76	6.6 143.0	5.27 S 3.97 E
3982.00	0.57	197.85	0.72	3981.75	7.7 147.0	6.43 S 4.17 E
4077.00	0.57	215.52	0.18	4076.74	8.2 152.7	7.26 S 3.75 E
4170.00	0.75	221.49	0.21	4169.74	8.7 159.2	8.10 S 3.08 E
4264.00	0.53	212.88	0.25	4263.73	9.2 164.7	8.92 S 2.43 E
4360.00	0.66	219.82	0.15	4359.73	9.9 169.3	9.72 S 1.84 E
4455.00	0.40	229.58	0.29	4454.72	10.4 173.2	10.35 S 1.24 E
4549.00	0.66	216.75	0.30	4548.72	11.0 176.6	11.00 S 0.66 E
4644.00	0.44	172.01	0.49	4643.71	11.8 178.1	11.80 S 0.39 E
4739.00	0.35	211.56	0.30	4738.71	12.4 178.7	12.41 S 0.28 E
4834.00	0.13	102.93	0.43	4833.71	12.7 178.9	12.68 S 0.24 E
4929.00	0.22	170.78	0.22	4928.71	12.9 178.3	12.88 S 0.37 E
5023.00	0.26	211.91	0.18	5022.71	13.2 178.8	13.24 S 0.29 E
5118.00	0.79	153.03	0.73	5117.71	14.0 178.1	14.01 S 0.47 E
5213.00	0.88	146.61	0.14	5212.70	15.2 175.6	15.20 S 1.17 E
5307.00	0.84	139.05	0.13	5306.68	16.5 173.0	16.33 S 2.02 E
5402.00	0.97	125.78	0.26	5401.67	17.6 169.8	17.32 S 3.13 E
5496.00	0.62	103.28	0.49	5495.66	18.4 166.6	17.90 S 4.27 E

A Gyrodata Directional Survey

Oasis Petroleum

Lease: Kline Federal Well: 5300 41-18 14BX, 8 3/4" Case, and 6" Hole

Location: Nabors #486 , McKenzie County, North Dakota

Job Number: RD0814M181

MEASURED DEPTH feet	INCL deg.	AZIMUTH deg.	DOGLEG SEVERITY deg./ 100 ft.	VERTICAL DEPTH feet	CLOSURE DIST. feet	AZIMUTH deg.	HORIZONTAL COORDINATES feet
5591.00	0.70	112.51	0.14	5590.66	19.0	163.8	18.24 S 5.30 E
5686.00	0.84	150.74	0.55	5685.65	20.1	162.0	19.07 S 6.18 E
5780.00	0.66	141.42	0.23	5779.64	21.2	161.2	20.10 S 6.86 E
5875.00	0.66	170.69	0.35	5874.64	22.3	160.9	21.07 S 7.28 E
5970.00	0.57	160.94	0.14	5969.63	23.3	161.2	22.05 S 7.53 E
6064.00	0.22	128.68	0.43	6063.63	23.9	160.9	22.61 S 7.82 E
6095.00	0.48	201.28	1.50	6094.63	24.1	161.0	22.77 S 7.82 E
6160.00	0.22	168.76	0.49	6159.63	24.4	161.5	23.14 S 7.75 E
6254.00	0.31	178.95	0.11	6253.63	24.8	161.7	23.57 S 7.79 E
6349.00	0.13	240.04	0.29	6348.62	25.1	162.1	23.88 S 7.70 E
6443.00	0.18	329.51	0.24	6442.62	25.0	162.4	23.81 S 7.53 E
6538.00	0.66	0.54	0.54	6537.62	24.3	162.1	23.13 S 7.46 E
6633.00	0.18	243.55	0.80	6632.62	23.8	162.1	22.65 S 7.33 E
6726.00	0.18	275.98	0.11	6725.62	23.8	162.7	22.70 S 7.05 E
6821.00	0.57	259.20	0.42	6820.62	23.7	164.2	22.78 S 6.44 E
6915.00	0.40	253.75	0.19	6914.61	23.6	166.1	22.96 S 5.67 E
7010.00	0.53	278.71	0.25	7009.61	23.5	167.9	22.98 S 4.92 E
7104.00	1.14	203.21	1.20	7103.60	24.1	170.2	23.78 S 4.12 E
7198.00	1.05	209.72	0.16	7197.58	25.6	172.5	25.38 S 3.32 E
7293.00	1.14	219.82	0.22	7292.57	27.0	175.1	26.86 S 2.28 E
7386.00	1.14	217.98	0.04	7385.55	28.3	177.7	28.30 S 1.12 E

A Gyrodata Directional Survey

Oasis Petroleum

Lease: Kline Federal Well: 5300 41-18 14BX, 8 3/4" Case, and 6" Hole

Location: Nabors #486 , McKenzie County, North Dakota

Job Number: RD0814M181

MEASURED DEPTH feet	INCL deg.	AZIMUTH deg.	DOGLEG SEVERITY deg./ 100 ft.	VERTICAL DEPTH feet	CLOSURE DIST. AZIMUTH feet deg.	HORIZONTAL COORDINATES feet
7481.00	1.32	197.23	0.50	7480.53	30.1 179.6	30.09 S 0.22 E
7576.00	1.19	193.54	0.16	7575.51	32.1 180.6	32.10 S 0.34 W
7670.00	1.05	202.07	0.23	7669.49	33.9 181.5	33.85 S 0.89 W
7764.00	1.19	213.06	0.27	7763.47	35.5 182.8	35.46 S 1.75 W
7857.00	0.92	202.68	0.35	7856.45	37.0 184.0	36.96 S 2.56 W
7951.00	1.01	216.22	0.26	7950.44	38.5 185.0	38.32 S 3.34 W
8045.00	0.70	208.84	0.35	8044.43	39.7 185.9	39.50 S 4.11 W
8140.00	0.66	225.01	0.21	8139.42	40.7 186.7	40.39 S 4.77 W
8234.00	0.66	239.16	0.17	8233.42	41.4 187.8	41.05 S 5.62 W
8329.00	1.01	196.44	0.73	8328.41	42.6 188.5	42.14 S 6.33 W
8423.00	1.23	200.14	0.25	8422.39	44.4 189.0	43.88 S 6.91 W
8517.00	1.76	180.62	0.77	8516.36	46.8 188.9	46.27 S 7.27 W
8611.00	2.02	166.30	0.57	8610.31	49.8 188.0	49.32 S 6.90 W
8706.00	1.45	182.29	0.78	8705.26	52.6 187.2	52.15 S 6.55 W
8801.00	0.92	170.60	0.61	8800.24	54.5 186.8	54.10 S 6.47 W
8895.00	0.75	172.80	0.18	8894.23	55.8 186.5	55.46 S 6.27 W
8989.00	0.40	73.40	0.96	8988.23	56.3 186.0	55.97 S 5.88 W
9083.00	0.57	75.42	0.18	9082.23	56.0 185.2	55.76 S 5.11 W
9177.00	0.35	52.39	0.30	9176.22	55.6 184.6	55.47 S 4.43 W
9271.00	0.40	42.55	0.09	9270.22	55.2 184.1	55.05 S 3.98 W
9366.00	0.31	33.67	0.11	9365.22	54.7 183.8	54.59 S 3.62 W
9461.00	0.35	26.11	0.06	9460.22	54.2 183.5	54.12 S 3.35 W

A Gyrodata Directional Survey

Oasis Petroleum

Lease: Kline Federal Well: 5300 41-18 14BX, 8 3/4" Case, and 6" Hole

Location: Nabors #486 , McKenzie County, North Dakota

Job Number: RD0814M181

MEASURED DEPTH feet	INCL deg.	AZIMUTH deg.	DOGLEG SEVERITY deg./ 100 ft.	VERTICAL DEPTH feet	CLOSURE DIST. AZIMUTH feet deg.	HORIZONTAL COORDINATES feet
9554.00	0.22	19.96	0.14	9553.22	53.8 183.4	53.70 S 3.16 W
9648.00	0.35	57.31	0.23	9647.22	53.4 183.1	53.37 S 2.86 W
9741.00	0.35	40.88	0.11	9740.21	53.1 182.6	53.00 S 2.43 W
9835.00	0.18	24.27	0.20	9834.21	52.7 182.4	52.65 S 2.18 W
9929.00	0.26	348.06	0.17	9928.21	52.4 182.4	52.31 S 2.17 W
10023.00	0.18	3.96	0.11	10022.21	52.0 182.4	51.95 S 2.20 W
10118.00	0.44	320.37	0.35	10117.21	51.6 182.7	51.52 S 2.42 W
10213.00	0.35	317.82	0.10	10212.21	51.1 183.2	51.03 S 2.85 W
10244.00	0.26	348.67	0.59	10243.21	51.0 183.3	50.89 S 2.93 W
10276.00	0.53	100.38	2.10	10275.21	50.9 183.1	50.84 S 2.80 W
10308.00	3.43	114.00	9.12	10307.18	51.3 182.0	51.26 S 1.78 W
10340.00	7.08	115.67	11.42	10339.04	52.5 179.0	52.50 S 0.88 E
10373.00	10.51	112.42	10.50	10371.65	54.8 174.2	54.53 S 5.49 E
10404.00	13.63	108.55	10.39	10401.96	57.9 168.5	56.77 S 11.57 E
10436.00	17.63	108.29	12.50	10432.77	62.7 161.6	59.50 S 19.75 E
10468.00	21.85	107.76	13.20	10462.89	69.6 154.5	62.83 S 30.03 E
10500.00	25.93	106.62	12.83	10492.14	79.0 147.5	66.65 S 42.41 E
10531.00	30.46	108.11	14.79	10519.45	90.7 141.6	71.04 S 56.38 E
10563.00	34.95	108.47	14.04	10546.37	105.6 136.4	76.46 S 72.79 E
10595.00	39.34	108.73	13.73	10571.88	123.0 132.2	82.63 S 91.10 E
10626.00	44.26	110.40	16.27	10594.98	142.3 129.0	89.56 S 110.55 E
10658.00	48.88	111.81	14.79	10616.97	164.5 126.5	97.93 S 132.22 E

A Gyrodata Directional Survey

Oasis Petroleum

Lease: Kline Federal Well: 5300 41-18 14BX, 8 3/4" Case, and 6" Hole

Location: Nabors #486 , McKenzie County, North Dakota

Job Number: RD0814M181

MEASURED DEPTH feet	INCL deg.	AZIMUTH deg.	DOGLEG SEVERITY deg./ 100 ft.	VERTICAL DEPTH feet	CLOSURE DIST. AZIMUTH feet deg.	HORIZONTAL COORDINATES feet
10689.00	53.49	113.12	15.23	10636.40	188.1 124.7	107.17 S 154.53 E
10721.00	57.58	115.41	14.08	10654.51	214.0 123.5	118.02 S 178.57 E
10752.00	60.35	118.13	11.68	10670.49	240.4 122.7	129.99 S 202.28 E
10783.00	62.37	120.42	9.19	10685.35	267.6 122.4	143.30 S 226.00 E
10815.00	64.53	121.03	6.96	10699.65	296.2 122.2	157.92 S 250.61 E
10847.00	67.87	121.39	10.49	10712.56	325.5 122.1	173.09 S 275.65 E
10877.00	72.57	122.09	15.82	10722.71	353.7 122.1	187.94 S 299.65 E
10909.00	77.63	120.07	16.95	10730.94	384.6 122.0	203.89 S 326.12 E
10941.00	80.22	120.33	8.13	10737.08	416.0 121.9	219.69 S 353.26 E
10972.00	81.45	120.42	3.98	10742.02	446.6 121.8	235.16 S 379.67 E
11003.00	84.66	120.24	10.37	10745.77	477.4 121.7	250.70 S 406.22 E
11035.00	88.35	119.28	11.91	10747.72	509.3 121.6	266.55 S 433.95 E
11048.00	88.97	117.96	11.22	10748.02	522.3 121.5	272.78 S 445.36 E
11145.00	91.16	119.01	2.50	10747.91	619.1 121.0	319.04 S 530.61 E
11176.00	91.16	118.48	1.71	10747.29	650.1 120.9	333.94 S 557.78 E
11207.00	91.16	117.69	2.55	10746.66	681.1 120.8	348.53 S 585.12 E
11238.00	89.85	115.94	7.05	10746.39	712.0 120.6	362.51 S 612.79 E
11270.00	89.32	114.79	3.96	10746.62	743.9 120.4	376.22 S 641.70 E
11301.00	88.35	113.12	6.23	10747.25	774.7 120.1	388.81 S 670.02 E
11332.00	87.87	111.81	4.50	10748.27	805.4 119.8	400.64 S 698.66 E
11362.00	88.18	111.72	1.08	10749.30	835.1 119.5	411.76 S 726.50 E

A Gyrodata Directional Survey

Oasis Petroleum

Lease: Kline Federal Well: 5300 41-18 14BX, 8 3/4" Case, and 6" Hole

Location: Nabors #486 , McKenzie County, North Dakota

Job Number: RD0814M181

MEASURED DEPTH feet	INCL deg.	AZIMUTH deg.	DOGLEG SEVERITY deg./ 100 ft.	VERTICAL DEPTH feet	CLOSURE DIST. AZIMUTH feet deg.	HORIZONTAL COORDINATES feet
11393.00	88.09	109.26	7.94	10750.31	865.7 119.2	422.61 S 755.52 E
11424.00	89.23	106.44	9.81	10751.04	896.1 118.8	432.11 S 785.02 E
11456.00	88.97	103.37	9.63	10751.54	927.2 118.4	440.33 S 815.93 E
11486.00	89.63	100.20	10.79	10751.91	956.0 117.8	446.46 S 845.29 E
11517.00	89.63	98.97	3.97	10752.11	985.4 117.3	451.62 S 875.86 E
11549.00	89.85	98.97	0.69	10752.25	1015.9 116.7	456.61 S 907.47 E
11580.00	89.63	98.62	1.33	10752.39	1045.4 116.2	461.35 S 938.10 E
11611.00	90.20	99.06	2.32	10752.44	1075.0 115.7	466.11 S 968.74 E
11642.00	90.42	99.06	0.71	10752.27	1104.8 115.2	471.00 S 999.35 E
11673.00	91.25	100.12	4.34	10751.82	1134.7 114.8	476.16 S 1029.91 E
11705.00	91.34	100.47	1.13	10751.10	1165.7 114.4	481.88 S 1061.39 E
11736.00	91.52	100.47	0.58	10750.32	1195.8 114.1	487.51 S 1091.86 E
11767.00	89.98	101.35	5.72	10749.92	1226.0 113.7	493.38 S 1122.30 E
11798.00	89.80	101.43	0.64	10749.98	1256.3 113.4	499.50 S 1152.69 E
11829.00	90.02	101.35	0.76	10750.03	1286.6 113.1	505.62 S 1183.08 E
11860.00	90.33	101.80	1.76	10749.93	1317.0 112.9	511.84 S 1213.45 E
11892.00	90.20	101.08	2.29	10749.78	1348.4 112.6	518.19 S 1244.81 E
11923.00	90.46	100.64	1.65	10749.60	1378.7 112.3	524.03 S 1275.25 E
11954.00	89.54	98.18	8.47	10749.60	1409.0 112.1	529.10 S 1305.83 E
11985.00	88.70	97.74	3.06	10750.08	1439.0 111.8	533.39 S 1336.53 E
12016.00	88.22	97.22	2.28	10750.91	1469.1 111.5	537.42 S 1367.26 E
12047.00	87.91	95.81	4.65	10751.96	1499.0 111.2	540.94 S 1398.04 E

A Gyrodata Directional Survey

Oasis Petroleum

Lease: Kline Federal Well: 5300 41-18 14BX, 8 3/4" Case, and 6" Hole

Location: Nabors #486 , McKenzie County, North Dakota

Job Number: RD0814M181

MEASURED DEPTH feet	INCL deg.	AZIMUTH deg.	DOGLEG SEVERITY deg./ 100 ft.	VERTICAL DEPTH feet	CLOSURE DIST. AZIMUTH feet deg.	HORIZONTAL COORDINATES feet
12078.00	87.82	95.72	0.41	10753.12	1528.9 110.8	544.05 S 1428.86 E
12110.00	87.91	95.55	0.60	10754.31	1559.8 110.5	547.19 S 1460.68 E
12141.00	88.88	94.40	4.85	10755.18	1589.7 110.2	549.88 S 1491.55 E
12172.00	89.19	94.40	1.00	10755.70	1619.5 109.9	552.25 S 1522.46 E
12203.00	89.71	94.40	1.68	10756.00	1649.4 109.6	554.63 S 1553.36 E
12234.00	90.20	92.73	5.61	10756.02	1679.2 109.4	556.56 S 1584.30 E
12266.00	90.07	92.82	0.49	10755.94	1709.9 109.1	558.11 S 1616.27 E
12297.00	90.20	92.82	0.42	10755.87	1739.7 108.8	559.63 S 1647.23 E
12328.00	90.07	90.10	8.78	10755.80	1769.3 108.5	560.42 S 1678.22 E
12390.00	89.41	89.92	1.10	10756.08	1828.2 107.9	560.43 S 1740.21 E
12421.00	89.63	90.10	0.92	10756.34	1857.8 107.6	560.44 S 1771.21 E
12482.00	89.89	89.39	1.24	10756.60	1915.9 107.0	560.17 S 1832.21 E
12577.00	89.19	90.27	1.18	10757.36	2006.9 106.2	559.89 S 1927.21 E
12671.00	89.93	90.18	0.79	10758.08	2097.4 105.5	560.26 S 2021.20 E
12703.00	90.02	90.27	0.40	10758.09	2128.3 105.3	560.38 S 2053.20 E
12766.00	89.93	89.83	0.71	10758.12	2189.2 104.8	560.44 S 2116.20 E
12860.00	87.91	88.87	2.38	10759.89	2279.9 104.2	559.37 S 2210.17 E
12892.00	87.96	89.13	0.83	10761.05	2310.7 104.0	558.81 S 2242.15 E
12955.00	88.00	88.95	0.29	10763.27	2371.6 103.6	557.76 S 2305.10 E
12986.00	88.35	90.54	5.25	10764.25	2401.7 103.4	557.62 S 2336.08 E
13050.00	87.91	90.45	0.70	10766.34	2464.1 103.1	558.17 S 2400.05 E
13144.00	88.84	90.89	1.09	10769.01	2555.9 102.6	559.27 S 2494.00 E

A Gyrodata Directional Survey

Oasis Petroleum

Lease: Kline Federal Well: 5300 41-18 14BX, 8 3/4" Case, and 6" Hole

Location: Nabors #486 , McKenzie County, North Dakota

Job Number: RD0814M181

MEASURED DEPTH feet	INCL deg.	AZIMUTH deg.	DOGLEG SEVERITY deg./ 100 ft.	VERTICAL DEPTH feet	CLOSURE DIST. AZIMUTH feet deg.	HORIZONTAL COORDINATES feet
13238.00	89.41	91.15	0.67	10770.44	2648.1 102.2	560.94 S 2587.97 E
13270.00	89.49	90.54	1.92	10770.75	2679.4 102.1	561.42 S 2619.97 E
13333.00	89.71	90.80	0.54	10771.19	2741.2 101.8	562.15 S 2682.96 E
13427.00	89.19	90.54	0.62	10772.09	2833.5 101.5	563.25 S 2776.95 E
13459.00	89.05	90.36	0.71	10772.58	2864.9 101.3	563.50 S 2808.95 E
13522.00	88.75	90.01	0.73	10773.79	2926.7 101.1	563.71 S 2871.94 E
13553.00	90.55	89.57	5.98	10773.98	2957.1 101.0	563.59 S 2902.93 E
13616.00	90.55	89.83	0.41	10773.38	3018.9 100.8	563.26 S 2965.93 E
13679.00	89.67	89.31	1.62	10773.26	3080.8 100.5	562.79 S 3028.93 E
13710.00	90.02	89.39	1.16	10773.34	3111.2 100.4	562.44 S 3059.92 E
13773.00	91.91	90.10	3.20	10772.28	3173.1 100.2	562.16 S 3122.91 E
13805.00	92.18	90.62	1.83	10771.14	3204.6 100.1	562.36 S 3154.89 E
13868.00	90.55	90.89	2.62	10769.64	3266.8 99.9	563.19 S 3217.86 E
13899.00	89.05	90.98	4.85	10769.75	3297.4 99.8	563.70 S 3248.86 E
13931.00	88.75	90.89	0.98	10770.36	3329.0 99.8	564.22 S 3280.85 E
13994.00	88.53	90.62	0.55	10771.86	3391.2 99.6	565.05 S 3343.83 E
14057.00	89.10	90.54	0.91	10773.16	3453.5 99.4	565.68 S 3406.81 E
14088.00	89.63	90.80	1.90	10773.50	3484.1 99.4	566.05 S 3437.80 E
14183.00	89.85	90.71	0.25	10773.93	3578.1 99.1	567.30 S 3532.80 E
14214.00	90.15	90.27	1.72	10773.93	3608.7 99.0	567.56 S 3563.79 E
14277.00	90.55	90.54	0.77	10773.55	3671.0 98.9	568.01 S 3626.79 E

A Gyrodata Directional Survey

Oasis Petroleum

Lease: Kline Federal Well: 5300 41-18 14BX, 8 3/4" Case, and 6" Hole

Location: Nabors #486 , McKenzie County, North Dakota

Job Number: RD0814M181

MEASURED DEPTH feet	INCL deg.	AZIMUTH deg.	DOGLEG SEVERITY deg./ 100 ft.	VERTICAL DEPTH feet	CLOSURE DIST. AZIMUTH feet deg.	HORIZONTAL COORDINATES feet
14340.00	89.63	89.13	2.67	10773.45	3733.2 98.7	567.83 S 3689.79 E
14372.00	87.69	89.22	6.07	10774.20	3764.8 98.7	567.37 S 3721.78 E
14403.00	87.74	88.87	1.14	10775.43	3795.3 98.6	566.85 S 3752.75 E
14466.00	87.87	88.78	0.25	10777.85	3857.4 98.4	565.56 S 3815.69 E
14498.00	89.41	89.22	5.00	10778.61	3888.9 98.4	565.00 S 3847.67 E
14561.00	89.85	89.48	0.81	10779.01	3951.2 98.2	564.29 S 3910.67 E
14655.00	90.15	88.95	0.65	10779.01	4044.0 98.0	563.00 S 4004.66 E
14750.00	91.03	90.71	2.07	10778.03	4138.1 97.8	562.72 S 4099.65 E
14813.00	90.90	91.06	0.59	10776.97	4200.6 97.7	563.69 S 4162.63 E
14844.00	90.95	91.59	1.72	10776.47	4231.4 97.7	564.41 S 4193.62 E
14876.00	90.37	90.54	3.75	10776.10	4263.2 97.6	565.00 S 4225.61 E
14939.00	90.64	90.27	0.61	10775.55	4325.7 97.5	565.45 S 4288.61 E
15033.00	90.51	89.57	0.76	10774.61	4418.9 97.4	565.32 S 4382.60 E
15127.00	90.29	89.22	0.44	10773.95	4512.0 97.2	564.32 S 4476.59 E
15159.00	90.33	88.69	1.66	10773.78	4543.7 97.1	563.74 S 4508.59 E
15222.00	89.93	87.99	1.28	10773.63	4606.0 97.0	561.91 S 4571.56 E
15316.00	88.66	89.48	2.08	10774.79	4699.0 96.8	559.84 S 4665.52 E
15410.00	89.01	89.22	0.46	10776.70	4792.2 96.7	558.77 S 4759.50 E
15505.00	89.36	88.25	1.09	10778.05	4886.3 96.5	556.68 S 4854.46 E
15536.00	89.01	90.27	6.61	10778.49	4917.0 96.5	556.28 S 4885.46 E
15599.00	89.63	90.80	1.29	10779.24	4979.7 96.4	556.86 S 4948.45 E
15694.00	90.37	91.59	1.14	10779.24	5074.3 96.3	558.85 S 5043.43 E

A Gyrodata Directional Survey

Oasis Petroleum

Lease: Kline Federal Well: 5300 41-18 14BX, 8 3/4" Case, and 6" Hole

Location: Nabors #486 , McKenzie County, North Dakota

Job Number: RD0814M181

MEASURED DEPTH feet	INCL deg.	AZIMUTH deg.	DOGLEG SEVERITY deg./ 100 ft.	VERTICAL DEPTH feet	CLOSURE DIST. AZIMUTH feet deg.	HORIZONTAL COORDINATES feet
15788.00	91.16	91.24	0.92	10777.99	5167.9 96.2	561.17 S 5137.39 E
15883.00	90.51	90.89	0.78	10776.60	5262.6 96.1	562.93 S 5232.36 E
15978.00	89.27	89.92	1.66	10776.78	5357.1 96.0	563.60 S 5327.36 E
16072.00	89.58	88.60	1.44	10777.73	5450.4 95.9	562.39 S 5421.34 E
16167.00	89.41	90.01	1.49	10778.56	5544.8 95.8	561.24 S 5516.33 E
16261.00	89.80	89.74	0.50	10779.21	5638.3 95.7	561.03 S 5610.33 E
16355.00	89.80	89.57	0.18	10779.54	5731.8 95.6	560.47 S 5704.32 E
16450.00	90.55	88.78	1.15	10779.25	5826.2 95.5	559.10 S 5799.31 E
16513.00	91.43	88.69	1.40	10778.16	5888.8 95.4	557.71 S 5862.29 E
16544.00	90.95	89.48	2.98	10777.52	5919.6 95.4	557.21 S 5893.28 E
16576.00	90.81	90.36	2.78	10777.03	5951.4 95.4	557.17 S 5925.27 E
16639.00	90.68	91.77	2.25	10776.21	6014.2 95.3	558.34 S 5988.25 E
16670.00	89.36	91.68	4.27	10776.20	6045.2 95.3	559.27 S 6019.24 E
16733.00	89.89	91.59	0.85	10776.61	6108.0 95.3	561.07 S 6082.21 E
16765.00	90.02	91.24	1.17	10776.63	6140.0 95.3	561.86 S 6114.20 E
16828.00	90.77	91.68	1.38	10776.20	6202.8 95.2	563.46 S 6177.18 E
16860.00	89.80	91.94	3.14	10776.04	6234.8 95.2	564.48 S 6209.16 E
16891.00	90.20	91.41	2.14	10776.04	6265.7 95.2	565.38 S 6240.15 E
16923.00	90.02	91.41	0.56	10775.98	6297.6 95.2	566.17 S 6272.14 E
17017.00	88.88	92.47	1.66	10776.88	6391.5 95.1	569.35 S 6366.08 E
17080.00	89.14	91.41	1.73	10777.97	6454.4 95.1	571.48 S 6429.03 E
17112.00	89.41	91.15	1.17	10778.38	6486.3 95.1	572.20 S 6461.02 E

A Gyrodata Directional Survey

Oasis Petroleum

Lease: Kline Federal Well: 5300 41-18 14BX, 8 3/4" Case, and 6" Hole

Location: Nabors #486 , McKenzie County, North Dakota

Job Number: RD0814M181

MEASURED DEPTH feet	INCL deg.	AZIMUTH deg.	DOGLEG SEVERITY deg./ 100 ft.	VERTICAL DEPTH feet	CLOSURE DIST. AZIMUTH feet deg.	HORIZONTAL COORDINATES feet
17206.00	90.24	91.33	0.90	10778.66	6580.1 95.0	574.23 S 6555.00 E
17237.00	90.55	90.89	1.74	10778.45	6611.0 95.0	574.83 S 6585.99 E
17301.00	90.86	91.24	0.73	10777.66	6674.9 95.0	576.02 S 6649.98 E
17332.00	89.54	91.06	4.30	10777.55	6705.8 94.9	576.64 S 6680.97 E
17395.00	89.63	90.10	1.53	10778.01	6768.6 94.9	577.28 S 6743.96 E
17490.00	90.46	90.10	0.87	10777.93	6863.3 94.8	577.45 S 6838.96 E
17521.00	91.38	89.92	3.02	10777.44	6894.2 94.8	577.45 S 6869.96 E
17584.00	91.03	89.57	0.79	10776.11	6956.9 94.8	577.17 S 6932.94 E
17679.00	89.19	88.78	2.11	10775.93	7051.5 94.7	575.81 S 7027.93 E
17773.00	89.63	88.25	0.73	10776.90	7144.9 94.6	573.37 S 7121.89 E
17867.00	89.85	89.13	0.96	10777.32	7238.4 94.5	571.22 S 7215.86 E
17898.00	90.15	89.22	1.01	10777.32	7269.3 94.5	570.77 S 7246.86 E
17961.00	90.55	89.48	0.76	10776.94	7332.0 94.5	570.06 S 7309.86 E
17992.00	91.08	88.78	2.83	10776.50	7362.9 94.4	569.59 S 7340.85 E
18055.00	91.21	87.64	1.82	10775.24	7425.5 94.4	567.62 S 7403.80 E
18086.00	89.32	88.51	6.71	10775.10	7456.3 94.4	566.58 S 7434.78 E
18117.00	89.27	89.13	2.01	10775.48	7487.2 94.3	565.94 S 7465.78 E
18149.00	89.76	88.60	2.26	10775.75	7519.0 94.3	565.31 S 7497.77 E
18180.00	89.54	89.66	3.49	10775.94	7549.9 94.3	564.84 S 7528.76 E
18212.00	89.41	91.41	5.48	10776.23	7581.9 94.3	565.14 S 7560.76 E
18243.00	89.54	90.98	1.45	10776.51	7612.8 94.3	565.78 S 7591.75 E

A Gyrodata Directional Survey

Oasis Petroleum

Lease: Kline Federal Well: 5300 41-18 14BX, 8 3/4" Case, and 6" Hole

Location: Nabors #486 , McKenzie County, North Dakota

Job Number: RD0814M181

MEASURED DEPTH feet	INCL deg.	AZIMUTH deg.	DOGLEG SEVERITY deg./ 100 ft.	VERTICAL DEPTH feet	CLOSURE DIST. AZIMUTH feet deg.	HORIZONTAL COORDINATES feet
18306.00	90.68	90.27	2.13	10776.39	7675.7 94.2	566.47 S 7654.75 E
18337.00	90.99	90.54	1.33	10775.94	7706.6 94.2	566.69 S 7685.74 E
18431.00	91.78	88.95	1.89	10773.67	7800.3 94.2	566.27 S 7779.71 E
18526.00	90.95	90.18	1.56	10771.41	7895.0 94.1	565.55 S 7874.68 E
18589.00	91.47	90.54	1.00	10770.08	7957.8 94.1	565.95 S 7937.66 E
18620.00	91.65	90.98	1.53	10769.23	7988.7 94.1	566.36 S 7968.65 E
18652.00	92.00	90.54	1.76	10768.21	8020.7 94.1	566.78 S 8000.63 E
18683.00	90.55	90.89	4.81	10767.52	8051.6 94.0	567.17 S 8031.62 E
18714.00	88.44	91.68	7.27	10767.80	8082.6 94.0	567.86 S 8062.61 E
18809.00	89.27	91.15	1.04	10769.70	8177.5 94.0	570.21 S 8157.56 E
18902.00	89.80	90.10	1.26	10770.45	8270.3 94.0	571.22 S 8250.55 E
18996.00	90.59	90.01	0.85	10770.13	8364.1 93.9	571.31 S 8344.55 E
19027.00	91.12	89.48	2.42	10769.67	8395.0 93.9	571.17 S 8375.54 E
19059.00	91.38	89.13	1.36	10768.97	8426.9 93.9	570.79 S 8407.53 E
19090.00	91.69	89.04	1.04	10768.14	8457.8 93.9	570.29 S 8438.52 E
19121.00	89.23	88.87	7.95	10767.89	8488.6 93.8	569.73 S 8469.51 E
19184.00	88.22	88.43	1.75	10769.29	8551.4 93.8	568.24 S 8532.47 E
19247.00	89.01	88.43	1.25	10770.81	8614.1 93.8	566.52 S 8595.43 E
19278.00	89.54	88.69	1.90	10771.21	8645.0 93.8	565.74 S 8626.42 E
19309.00	89.76	88.87	0.92	10771.40	8675.8 93.7	565.08 S 8657.41 E
19372.00	90.29	89.04	0.88	10771.37	8738.6 93.7	563.93 S 8720.40 E
19435.00	89.85	90.71	2.74	10771.29	8801.5 93.7	563.79 S 8783.40 E

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Oasis Petroleum

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Location: Nabors #486 , McKenzie County, North Dakota

Job Number: RD0814M181

MEASURED DEPTH feet	INCL deg.	AZIMUTH deg.	DOGLEG SEVERITY deg./ 100 ft.	VERTICAL DEPTH feet	CLOSURE DIST. AZIMUTH feet deg.	HORIZONTAL COORDINATES feet
19466.00	89.71	90.18	1.77	10771.41	8832.4 93.7	564.03 S 8814.40 E
19529.00	89.80	90.27	0.20	10771.68	8895.3 93.6	564.28 S 8877.40 E
19560.00	89.98	89.83	1.53	10771.74	8926.3 93.6	564.31 S 8908.40 E
19655.00	92.22	89.66	2.36	10769.92	9021.0 93.6	563.88 S 9003.37 E
19686.00	92.88	89.92	2.29	10768.54	9051.9 93.6	563.77 S 9034.34 E
19717.00	93.27	89.74	1.39	10766.87	9082.8 93.6	563.68 S 9065.29 E
19749.00	92.26	90.71	4.37	10765.33	9114.7 93.5	563.80 S 9097.26 E
19780.00	90.46	90.62	5.81	10764.59	9145.7 93.5	564.16 S 9128.24 E
19811.00	88.48	91.94	7.68	10764.88	9176.6 93.5	564.86 S 9159.23 E
19843.00	88.57	92.12	0.63	10765.71	9208.6 93.5	565.99 S 9191.20 E
19874.00	88.97	91.41	2.63	10766.37	9239.6 93.5	566.94 S 9222.18 E
19937.00	87.56	90.98	2.34	10768.28	9302.5 93.5	568.26 S 9285.14 E
20031.00	87.43	91.41	0.48	10772.39	9396.3 93.5	570.22 S 9379.03 E
20094.00	87.78	90.36	1.76	10775.02	9459.2 93.5	571.19 S 9441.96 E
20126.00	87.74	90.54	0.58	10776.27	9491.2 93.5	571.44 S 9473.94 E
20189.00	89.14	90.45	2.23	10777.98	9554.0 93.4	571.98 S 9536.91 E
20220.00	89.14	90.10	1.13	10778.45	9585.0 93.4	572.13 S 9567.91 E
20283.00	89.63	89.66	1.05	10779.13	9647.9 93.4	572.00 S 9630.90 E
20315.00	89.76	89.04	1.98	10779.30	9679.8 93.4	571.64 S 9662.90 E
20378.00	90.33	88.34	1.43	10779.25	9742.6 93.4	570.20 S 9725.88 E
20409.00	90.51	88.16	0.82	10779.02	9773.5 93.3	569.25 S 9756.87 E
20504.00	91.60	87.64	1.27	10777.27	9868.0 93.3	565.77 S 9851.78 E

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Oasis Petroleum

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Location: Nabors #486 , McKenzie County, North Dakota

Job Number: RD0814M181

MEASURED DEPTH feet	INCL deg.	AZIMUTH deg.	DOGLEG SEVERITY deg./ 100 ft.	VERTICAL DEPTH feet	CLOSURE DIST. AZIMUTH feet deg.	HORIZONTAL COORDINATES feet
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20572.00	92.04	87.90	0.75	10775.11	9935.7 93.2	563.13 S 9919.70 E
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STRAIGHT LINE PROJECTION TO BIT DEPTH AT 20650 FT. MD.

20650.00	92.04	87.90	0.00	10772.33	10013.3 93.2	560.27 S 9997.60 E
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Final Station Closure: Distance: 10013.28 ft Az: 93.21 deg.

**SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4**

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5749 (09-2006)



Well File No.

28655

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

☒ Notice of Intent	Approximate Start Date December 15, 2014
☐ Report of Work Done	Date Work Completed
☐ Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03.	Approximate Start Date

☐ Drilling Prognosis	☐ Spill Report
☐ Redrilling or Repair	☐ Shooting
☐ Casing or Liner	☐ Acidizing
☐ Plug Well	☐ Fracture Treatment
☐ Supplemental History	☐ Change Production Method
☐ Temporarily Abandon	☐ Reclamation
☐ Other	<u>Waiver from tubing/packer requirement</u>

Well Name and Number Kline Federal 5300 41-18 14BX					
Footages 368 F S L 237 F W L		Qtr-Qtr LOT 4	Section 18	Township 153 N	Range 100 W
Field BAKER	Pool Bakken	County MCKENZIE			

24-HOUR PRODUCTION RATE			
Before		After	
Oil	Bbls	Oil	Bbls
Water	Bbls	Water	Bbls
Gas	MCF	Gas	MCF

Name of Contractor(s)			
Address	City	State	Zip Code

DETAILS OF WORK

Oasis Petroleum North America LLC requests a variance to NDAC 43-02-03-21 for the tubing/packer requirement: Casing, tubing, and cementing requirements during the completion period immediately following the upcoming fracture stimulation.

The following assurances apply:

1. the well is equipped with new 29# and 32# casing at surface with an API burst rating of 11,220 psi;
2. The Frac design will use a safety factor of 0.85 API burst rating to determine the maximum pressure;
3. Damage to the casing during the frac would be detected immediately by monitoring equipment;
4. The casing is exposed to significantly lower rates and pressures during flowback than during the frac job;
5. The frac fluid and formation fluids have very low corrosion and erosion rates;
6. Production equipment will be installed as soon as possible after the well ceases flowing;
7. A 300# gauge will be installed on the surface casing during the flowback period

Company Oasis Petroleum North America LLC		Telephone Number 281-404-9436	
Address 1001 Fannin, Suite 1500			
City Houston	State TX	Zip Code 77002	
Signature 	Printed Name Jennifer Swenson		
Title Regulatory Assistant	Date December 15, 2014		
Email Address jswenson@oasispetroleum.com			

FOR STATE USE ONLY	
☐ Received	☒ Approved
Date December 22, 2014	
By 	
Title PETROLEUM ENGINEER	

**SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4**

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5749 (09-2006)



Well File No.
28655

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

☒ Notice of Intent	Approximate Start Date July 28, 2014
☐ Report of Work Done	Date Work Completed
☐ Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03.	
Approximate Start Date	

☐ Drilling Prognosis	☐ Spill Report
☐ Redrilling or Repair	☐ Shooting
☒ Casing or Liner	☐ Acidizing
☐ Plug Well	☐ Fracture Treatment
☐ Supplemental History	☐ Change Production Method
☐ Temporarily Abandon	☐ Reclamation
☐ Other	Change casing

Well Name and Number Kline Federal 5300 41-18 14BX					
Footages 368 F S L 237 F W L		Qtr-Qtr SWSW	Section 18	Township 153 N	Range 100 W
Field Baker		Pool Bakken		County McKenzie	

24-HOUR PRODUCTION RATE			
Before		After	
Oil	Bbls	Oil	Bbls
Water	Bbls	Water	Bbls
Gas	MCF	Gas	MCF

Name of Contractor(s)			
Address	City	State	Zip Code

DETAILS OF WORK

Oasis Petroleum respectfully requests permission to make the following changes to the above referenced well:

Surface Casing: 13-3/8, 54.5#, 17-1/2" Hole, 2,072' MD
Dakota Contingency: 9-5/8, 40#, 12-1/4" Hole, 6,101' MD

Attached are revised plats, drill plan, well summary, directional plan and plot

Company Oasis Petroleum North America LLC		Telephone Number 281-404-9563	
Address 1001 Fannin, Suite 1500			
City Houston		State TX	Zip Code 77002
Signature 		Printed Name Heather McCowan	
Title Regulatory Assistant		Date July 28, 2014	
Email Address hmccowan@oasispetroleum.com			

FOR STATE USE ONLY	
☐ Received	☒ Approved
Date 8-5-14	
By 	
Title Petroleum Resource Specialist	



Oasis Petroleum

Indian Hills

153N-100W-17/18

Kline Federal 5300 41-18 14BX

Kline Federal 5300 41-18 14BX

Plan: Design #2

Standard Planning Report

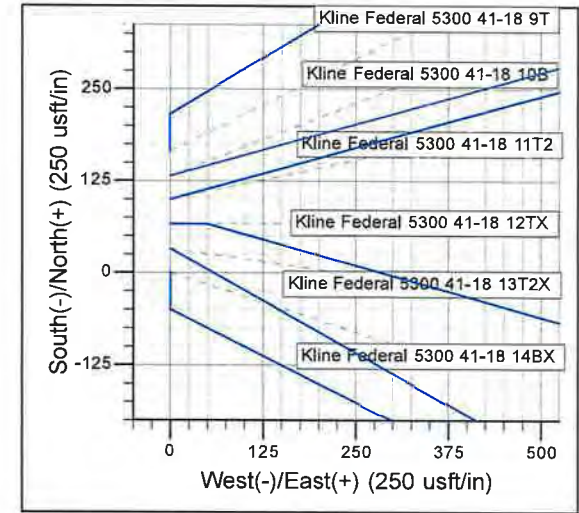
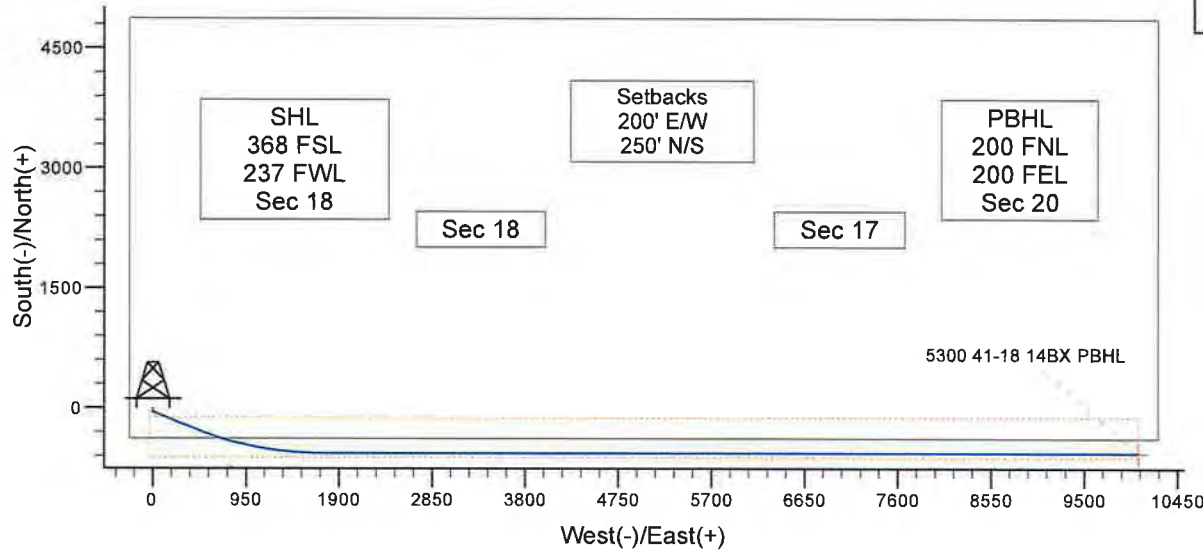
28 May, 2014

Project: Indian Hills
 Site: 153N-100W-17/18
 Well: Kline Federal 5300 41-18 14BX
 Wellbore: Kline Federal 5300 41-18 14BX
 Design: Design #2



WELL DETAILS: Kline Federal 5300 41-18 14BX

Ground Level: 2057.0
 Northing 405070.82 Easting 1209965.56 Latitude 48° 4' 6.900 N Longitude 103° 36' 11.950 W



CASING DETAILS

TVD MD Name Size
 10765.0 11037.0 7" 7"

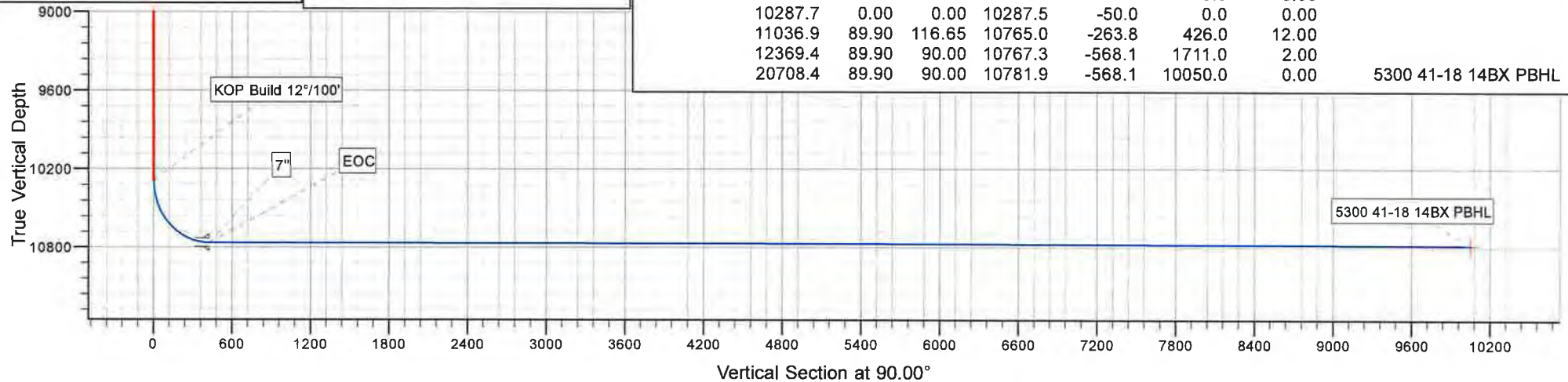


Azimuths to True North
 Magnetic North: 8.27°

Magnetic Field
 Strength: 56437.0snT
 Dip Angle: 72.99°
 Date: 5/23/2014
 Model: IGRF2010

SECTION DETAILS

MD	Inc	Azi	TVD	+N/-S	+E/-W	Dleg	Target
0.0	0.00	0.00	0.0	0.0	0.0	0.00	
2200.0	0.00	0.00	2200.0	0.0	0.0	0.00	
2216.7	0.50	180.00	2216.7	-0.1	0.0	3.00	
7926.7	0.50	180.00	7926.4	-49.9	0.0	0.00	
7943.3	0.00	0.00	7943.1	-50.0	0.0	3.00	
10287.7	0.00	0.00	10287.5	-50.0	0.0	0.00	
11036.9	89.90	116.65	10765.0	-263.8	426.0	12.00	
12369.4	89.90	90.00	10767.3	-568.1	1711.0	2.00	
20708.4	89.90	90.00	10781.9	-568.1	10050.0	0.00	5300 41-18 14BX PBHL





Ryan Directional Services Planning Report



Database: EDM 5000.1 Single User Db
Company: Oasis Petroleum
Project: Indian Hills
Site: 153N-100W-17/18
Well: Kline Federal 5300 41-18 14BX
Wellbore: Kline Federal 5300 41-18 14BX
Design: Design #2

Local Co-ordinate Reference: Well Kline Federal 5300 41-18 14BX
TVD Reference: WELL @ 2082.0usft (Original Well Elev)
MD Reference: WELL @ 2082.0usft (Original Well Elev)
North Reference: True
Survey Calculation Method: Minimum Curvature

Project	Indian Hills		
Map System:	US State Plane 1983	System Datum:	Mean Sea Level
Geo Datum:	North American Datum 1983		
Map Zone:	North Dakota Northern Zone		

Site	153N-100W-17/18		
Site Position:		Northing:	408,962.44 usft
From:	Lat/Long	Easting:	1,210,229.18 usft
Position Uncertainty:	0.0 usft	Slot Radius:	13-3/16 "
		Latitude:	48° 4' 45.380 N
		Longitude:	103° 36' 10.380 W
		Grid Convergence:	-2.31 °

Well	Kline Federal 5300 41-18 14BX		
Well Position	+N/-S	-3,899.1 usft	Northing: 405,070.82 usft
	+E/-W	-106.6 usft	Easting: 1,209,965.56 usft
Position Uncertainty	2.0 usft	Wellhead Elevation:	
		Latitude:	48° 4' 6.900 N
		Longitude:	103° 36' 11.950 W
		Ground Level:	2,057.0 usft

Wellbore	Kline Federal 5300 41-18 14BX				
Magnetics	Model Name	Sample Date	Declination (°)	Dip Angle (°)	Field Strength (nT)
	IGRF2010	5/23/2014	8.27	72.99	56,437

Design	Design #2			
Audit Notes:				
Version:	Phase:	PROTOTYPE	Tie On Depth:	0.0
Vertical Section:	Depth From (TVD) (usft)	+N/-S (usft)	+E/-W (usft)	Direction (°)
	0.0	0.0	0.0	90.00

Plan Sections										
Measured Depth (usft)	Inclination (°)	Azimuth (°)	Vertical Depth (usft)	+N/-S (usft)	+E/-W (usft)	Dogleg Rate (°/100usft)	Build Rate (°/100usft)	Turn Rate (°/100usft)	TFO (°)	Target
0.0	0.00	0.00	0.0	0.0	0.0	0.00	0.00	0.00	0.00	
2,200.0	0.00	0.00	2,200.0	0.0	0.0	0.00	0.00	0.00	0.00	
2,216.7	0.50	180.00	2,216.7	-0.1	0.0	3.00	3.00	0.00	180.00	
7,926.7	0.50	180.00	7,926.4	-49.9	0.0	0.00	0.00	0.00	0.00	
7,943.3	0.00	0.00	7,943.1	-50.0	0.0	3.00	-3.00	0.00	180.00	
10,287.7	0.00	0.00	10,287.5	-50.0	0.0	0.00	0.00	0.00	0.00	
11,036.9	89.90	116.65	10,765.0	-263.8	426.0	12.00	12.00	0.00	116.65	
12,369.4	89.90	90.00	10,767.3	-568.1	1,711.0	2.00	0.00	-2.00	-90.02	
20,708.4	89.90	90.00	10,781.9	-568.1	10,050.0	0.00	0.00	0.00	0.00	5300 41-18 14BX PBI



Ryan Directional Services

Planning Report



Database:	EDM 5000.1 Single User Db	Local Co-ordinate Reference:	Well Kline Federal 5300 41-18 14BX
Company:	Oasis Petroleum	TVD Reference:	WELL @ 2082.0usft (Original Well Elev)
Project:	Indian Hills	MD Reference:	WELL @ 2082.0usft (Original Well Elev)
Site:	153N-100W-17/18	North Reference:	True
Well:	Kline Federal 5300 41-18 14BX	Survey Calculation Method:	Minimum Curvature
Wellbore:	Kline Federal 5300 41-18 14BX		
Design:	Design #2		

Planned Survey									
Measured Depth (usft)	Inclination (°)	Azimuth (°)	Vertical Depth (usft)	+N/-S (usft)	+E/-W (usft)	Vertical Section (usft)	Dogleg Rate (°/100usft)	Build Rate (°/100usft)	Turn Rate (°/100usft)
2,200.0	0.00	0.00	2,200.0	0.0	0.0	0.0	0.00	0.00	0.00
2,216.7	0.50	180.00	2,216.7	-0.1	0.0	0.0	3.00	3.00	0.00
2,300.0	0.50	180.00	2,300.0	-0.8	0.0	0.0	0.00	0.00	0.00
2,400.0	0.50	180.00	2,400.0	-1.7	0.0	0.0	0.00	0.00	0.00
2,500.0	0.50	180.00	2,500.0	-2.5	0.0	0.0	0.00	0.00	0.00
2,600.0	0.50	180.00	2,600.0	-3.4	0.0	0.0	0.00	0.00	0.00
2,700.0	0.50	180.00	2,700.0	-4.3	0.0	0.0	0.00	0.00	0.00
2,800.0	0.50	180.00	2,800.0	-5.2	0.0	0.0	0.00	0.00	0.00
2,900.0	0.50	180.00	2,900.0	-6.0	0.0	0.0	0.00	0.00	0.00
3,000.0	0.50	180.00	3,000.0	-6.9	0.0	0.0	0.00	0.00	0.00
3,100.0	0.50	180.00	3,100.0	-7.8	0.0	0.0	0.00	0.00	0.00
3,200.0	0.50	180.00	3,200.0	-8.7	0.0	0.0	0.00	0.00	0.00
3,300.0	0.50	180.00	3,300.0	-9.5	0.0	0.0	0.00	0.00	0.00
3,400.0	0.50	180.00	3,400.0	-10.4	0.0	0.0	0.00	0.00	0.00
3,500.0	0.50	180.00	3,500.0	-11.3	0.0	0.0	0.00	0.00	0.00
3,600.0	0.50	180.00	3,599.9	-12.1	0.0	0.0	0.00	0.00	0.00
3,700.0	0.50	180.00	3,699.9	-13.0	0.0	0.0	0.00	0.00	0.00
3,800.0	0.50	180.00	3,799.9	-13.9	0.0	0.0	0.00	0.00	0.00
3,900.0	0.50	180.00	3,899.9	-14.8	0.0	0.0	0.00	0.00	0.00
4,000.0	0.50	180.00	3,999.9	-15.6	0.0	0.0	0.00	0.00	0.00
4,100.0	0.50	180.00	4,099.9	-16.5	0.0	0.0	0.00	0.00	0.00
4,200.0	0.50	180.00	4,199.9	-17.4	0.0	0.0	0.00	0.00	0.00
4,300.0	0.50	180.00	4,299.9	-18.3	0.0	0.0	0.00	0.00	0.00
4,400.0	0.50	180.00	4,399.9	-19.1	0.0	0.0	0.00	0.00	0.00
4,500.0	0.50	180.00	4,499.9	-20.0	0.0	0.0	0.00	0.00	0.00
4,600.0	0.50	180.00	4,599.9	-20.9	0.0	0.0	0.00	0.00	0.00
4,646.1	0.50	180.00	4,646.0	-21.3	0.0	0.0	0.00	0.00	0.00
Greenhorn									
4,700.0	0.50	180.00	4,699.9	-21.7	0.0	0.0	0.00	0.00	0.00
4,800.0	0.50	180.00	4,799.9	-22.6	0.0	0.0	0.00	0.00	0.00
4,900.0	0.50	180.00	4,899.9	-23.5	0.0	0.0	0.00	0.00	0.00
5,000.0	0.50	180.00	4,999.9	-24.4	0.0	0.0	0.00	0.00	0.00
5,057.1	0.50	180.00	5,057.0	-24.9	0.0	0.0	0.00	0.00	0.00
Mowry									
5,100.0	0.50	180.00	5,099.9	-25.2	0.0	0.0	0.00	0.00	0.00
5,200.0	0.50	180.00	5,199.9	-26.1	0.0	0.0	0.00	0.00	0.00
5,300.0	0.50	180.00	5,299.9	-27.0	0.0	0.0	0.00	0.00	0.00
5,400.0	0.50	180.00	5,399.9	-27.9	0.0	0.0	0.00	0.00	0.00
5,471.1	0.50	180.00	5,471.0	-28.5	0.0	0.0	0.00	0.00	0.00
Dakota									
5,500.0	0.50	180.00	5,499.9	-28.7	0.0	0.0	0.00	0.00	0.00
5,600.0	0.50	180.00	5,599.9	-29.6	0.0	0.0	0.00	0.00	0.00
5,700.0	0.50	180.00	5,699.9	-30.5	0.0	0.0	0.00	0.00	0.00
5,800.0	0.50	180.00	5,799.9	-31.3	0.0	0.0	0.00	0.00	0.00
5,900.0	0.50	180.00	5,899.9	-32.2	0.0	0.0	0.00	0.00	0.00
6,000.0	0.50	180.00	5,999.9	-33.1	0.0	0.0	0.00	0.00	0.00
6,100.0	0.50	180.00	6,099.9	-34.0	0.0	0.0	0.00	0.00	0.00
6,200.0	0.50	180.00	6,199.8	-34.8	0.0	0.0	0.00	0.00	0.00
6,300.0	0.50	180.00	6,299.8	-35.7	0.0	0.0	0.00	0.00	0.00
6,400.0	0.50	180.00	6,399.8	-36.6	0.0	0.0	0.00	0.00	0.00
6,486.2	0.50	180.00	6,486.0	-37.3	0.0	0.0	0.00	0.00	0.00
Rierdon									
6,500.0	0.50	180.00	6,499.8	-37.5	0.0	0.0	0.00	0.00	0.00



Ryan Directional Services

Planning Report



Database:	EDM 5000.1 Single User Db	Local Co-ordinate Reference:	Well Kline Federal 5300 41-18 14BX
Company:	Oasis Petroleum	TVD Reference:	WELL @ 2082.0usft (Original Well Elev)
Project:	Indian Hills	MD Reference:	WELL @ 2082.0usft (Original Well Elev)
Site:	153N-100W-17/18	North Reference:	True
Well:	Kline Federal 5300 41-18 14BX	Survey Calculation Method:	Minimum Curvature
Wellbore:	Kline Federal 5300 41-18 14BX		
Design:	Design #2		

Planned Survey										
Measured Depth (usft)	Inclination (°)	Azimuth (°)	Vertical Depth (usft)	+N/-S (usft)	+E/-W (usft)	Vertical Section (usft)	Dogleg Rate (°/100usft)	Build Rate (°/100usft)	Turn Rate (°/100usft)	
6,600.0	0.50	180.00	6,599.8	-38.3	0.0	0.0	0.00	0.00	0.00	
6,700.0	0.50	180.00	6,699.8	-39.2	0.0	0.0	0.00	0.00	0.00	
6,800.0	0.50	180.00	6,799.8	-40.1	0.0	0.0	0.00	0.00	0.00	
6,814.2	0.50	180.00	6,814.0	-40.2	0.0	0.0	0.00	0.00	0.00	
Dunham Salt										
6,900.0	0.50	180.00	6,899.8	-40.9	0.0	0.0	0.00	0.00	0.00	
6,929.2	0.50	180.00	6,929.0	-41.2	0.0	0.0	0.00	0.00	0.00	
Dunham Salt Base										
7,000.0	0.50	180.00	6,999.8	-41.8	0.0	0.0	0.00	0.00	0.00	
7,024.2	0.50	180.00	7,024.0	-42.0	0.0	0.0	0.00	0.00	0.00	
Spearfish										
7,100.0	0.50	180.00	7,099.8	-42.7	0.0	0.0	0.00	0.00	0.00	
7,200.0	0.50	180.00	7,199.8	-43.6	0.0	0.0	0.00	0.00	0.00	
7,283.2	0.50	180.00	7,283.0	-44.3	0.0	0.0	0.00	0.00	0.00	
Pine Salt										
7,300.0	0.50	180.00	7,299.8	-44.4	0.0	0.0	0.00	0.00	0.00	
7,318.2	0.50	180.00	7,318.0	-44.6	0.0	0.0	0.00	0.00	0.00	
Pine Salt Base										
7,374.2	0.50	180.00	7,374.0	-45.1	0.0	0.0	0.00	0.00	0.00	
Opeche Salt										
7,400.0	0.50	180.00	7,399.8	-45.3	0.0	0.0	0.00	0.00	0.00	
7,403.2	0.50	180.00	7,403.0	-45.3	0.0	0.0	0.00	0.00	0.00	
Opeche Salt Base										
7,500.0	0.50	180.00	7,499.8	-46.2	0.0	0.0	0.00	0.00	0.00	
7,600.0	0.50	180.00	7,599.8	-47.1	0.0	0.0	0.00	0.00	0.00	
7,605.2	0.50	180.00	7,605.0	-47.1	0.0	0.0	0.00	0.00	0.00	
Broom Creek (Top of Minnelusa Gp.)										
7,684.2	0.50	180.00	7,684.0	-47.8	0.0	0.0	0.00	0.00	0.00	
Amsden										
7,700.0	0.50	180.00	7,699.8	-47.9	0.0	0.0	0.00	0.00	0.00	
7,800.0	0.50	180.00	7,799.8	-48.8	0.0	0.0	0.00	0.00	0.00	
7,853.2	0.50	180.00	7,853.0	-49.3	0.0	0.0	0.00	0.00	0.00	
Tyler										
7,900.0	0.50	180.00	7,899.8	-49.7	0.0	0.0	0.00	0.00	0.00	
7,926.7	0.50	180.00	7,926.4	-49.9	0.0	0.0	0.00	0.00	0.00	
7,943.3	0.00	0.00	7,943.1	-50.0	0.0	0.0	3.00	-3.00	0.00	
8,000.0	0.00	0.00	7,999.8	-50.0	0.0	0.0	0.00	0.00	0.00	
8,047.2	0.00	0.00	8,047.0	-50.0	0.0	0.0	0.00	0.00	0.00	
Otter (Base of Minnelusa Gp.)										
8,100.0	0.00	0.00	8,099.8	-50.0	0.0	0.0	0.00	0.00	0.00	
8,200.0	0.00	0.00	8,199.8	-50.0	0.0	0.0	0.00	0.00	0.00	
8,300.0	0.00	0.00	8,299.8	-50.0	0.0	0.0	0.00	0.00	0.00	
8,390.2	0.00	0.00	8,390.0	-50.0	0.0	0.0	0.00	0.00	0.00	
Kibbey Lime										
8,400.0	0.00	0.00	8,399.8	-50.0	0.0	0.0	0.00	0.00	0.00	
8,500.0	0.00	0.00	8,499.8	-50.0	0.0	0.0	0.00	0.00	0.00	
8,542.2	0.00	0.00	8,542.0	-50.0	0.0	0.0	0.00	0.00	0.00	
Charles Salt										
8,600.0	0.00	0.00	8,599.8	-50.0	0.0	0.0	0.00	0.00	0.00	
8,700.0	0.00	0.00	8,699.8	-50.0	0.0	0.0	0.00	0.00	0.00	
8,800.0	0.00	0.00	8,799.8	-50.0	0.0	0.0	0.00	0.00	0.00	
8,900.0	0.00	0.00	8,899.8	-50.0	0.0	0.0	0.00	0.00	0.00	



Ryan Directional Services

Planning Report



Database:	EDM 5000.1 Single User Db	Local Co-ordinate Reference:	Well Kline Federal 5300 41-18 14BX
Company:	Oasis Petroleum	TVD Reference:	WELL @ 2082.0usft (Original Well Elev)
Project:	Indian Hills	MD Reference:	WELL @ 2082.0usft (Original Well Elev)
Site:	153N-100W-17/18	North Reference:	True
Well:	Kline Federal 5300 41-18 14BX	Survey Calculation Method:	Minimum Curvature
Wellbore:	Kline Federal 5300 41-18 14BX		
Design:	Design #2		

Planned Survey

Measured Depth (usft)	Inclination (°)	Azimuth (°)	Vertical Depth (usft)	+N/-S (usft)	+E/-W (usft)	Vertical Section (usft)	Dogleg Rate (°/100usft)	Build Rate (°/100usft)	Turn Rate (°/100usft)
9,000.0	0.00	0.00	8,999.8	-50.0	0.0	0.0	0.00	0.00	0.00
9,100.0	0.00	0.00	9,099.8	-50.0	0.0	0.0	0.00	0.00	0.00
9,159.2	0.00	0.00	9,159.0	-50.0	0.0	0.0	0.00	0.00	0.00
UB									
9,200.0	0.00	0.00	9,199.8	-50.0	0.0	0.0	0.00	0.00	0.00
9,238.2	0.00	0.00	9,238.0	-50.0	0.0	0.0	0.00	0.00	0.00
Base Last Salt									
9,300.0	0.00	0.00	9,299.8	-50.0	0.0	0.0	0.00	0.00	0.00
9,301.2	0.00	0.00	9,301.0	-50.0	0.0	0.0	0.00	0.00	0.00
Ratcliffe									
9,400.0	0.00	0.00	9,399.8	-50.0	0.0	0.0	0.00	0.00	0.00
9,454.2	0.00	0.00	9,454.0	-50.0	0.0	0.0	0.00	0.00	0.00
Mission Canyon									
9,500.0	0.00	0.00	9,499.8	-50.0	0.0	0.0	0.00	0.00	0.00
9,600.0	0.00	0.00	9,599.8	-50.0	0.0	0.0	0.00	0.00	0.00
9,700.0	0.00	0.00	9,699.8	-50.0	0.0	0.0	0.00	0.00	0.00
9,800.0	0.00	0.00	9,799.8	-50.0	0.0	0.0	0.00	0.00	0.00
9,900.0	0.00	0.00	9,899.8	-50.0	0.0	0.0	0.00	0.00	0.00
10,000.0	0.00	0.00	9,999.8	-50.0	0.0	0.0	0.00	0.00	0.00
10,022.2	0.00	0.00	10,022.0	-50.0	0.0	0.0	0.00	0.00	0.00
Lodgepole									
10,100.0	0.00	0.00	10,099.8	-50.0	0.0	0.0	0.00	0.00	0.00
10,200.0	0.00	0.00	10,199.8	-50.0	0.0	0.0	0.00	0.00	0.00
10,209.2	0.00	0.00	10,209.0	-50.0	0.0	0.0	0.00	0.00	0.00
Lodgepole Fracture Zone									
10,287.7	0.00	0.00	10,287.5	-50.0	0.0	0.0	0.00	0.00	0.00
10,287.8	0.00	116.65	10,287.6	-50.0	0.0	0.0	0.00	0.00	0.00
KOP Build 12°/100'									
10,300.0	1.47	116.65	10,299.8	-50.0	0.1	0.1	12.07	12.07	0.00
10,325.0	4.47	116.65	10,324.7	-50.6	1.3	1.3	12.00	12.00	0.00
10,350.0	7.47	116.65	10,349.6	-51.8	3.6	3.6	12.00	12.00	0.00
10,375.0	10.47	116.65	10,374.3	-53.5	7.1	7.1	12.00	12.00	0.00
10,400.0	13.47	116.65	10,398.8	-55.9	11.7	11.7	12.00	12.00	0.00
10,425.0	16.47	116.65	10,422.9	-58.8	17.5	17.5	12.00	12.00	0.00
10,450.0	19.47	116.65	10,446.7	-62.2	24.4	24.4	12.00	12.00	0.00
10,475.0	22.47	116.65	10,470.0	-66.2	32.4	32.4	12.00	12.00	0.00
10,500.0	25.47	116.65	10,492.9	-70.8	41.5	41.5	12.00	12.00	0.00
10,525.0	28.47	116.65	10,515.1	-75.9	51.6	51.6	12.00	12.00	0.00
10,550.0	31.47	116.65	10,536.8	-81.5	62.8	62.8	12.00	12.00	0.00
10,575.0	34.47	116.65	10,557.8	-87.6	74.9	74.9	12.00	12.00	0.00
10,600.0	37.47	116.65	10,578.0	-94.2	88.1	88.1	12.00	12.00	0.00
10,625.0	40.47	116.65	10,597.4	-101.2	102.1	102.1	12.00	12.00	0.00
10,650.0	43.47	116.65	10,616.0	-108.7	117.1	117.1	12.00	12.00	0.00
10,675.0	46.47	116.65	10,633.7	-116.6	132.8	132.8	12.00	12.00	0.00
10,700.0	49.47	116.65	10,650.4	-125.0	149.4	149.4	12.00	12.00	0.00
10,725.0	52.47	116.65	10,666.2	-133.7	166.8	166.8	12.00	12.00	0.00
10,750.0	55.47	116.65	10,680.9	-142.7	184.9	184.9	12.00	12.00	0.00
10,775.0	58.47	116.65	10,694.5	-152.1	203.6	203.6	12.00	12.00	0.00
10,800.0	61.47	116.65	10,707.0	-161.9	222.9	222.9	12.00	12.00	0.00
10,825.0	64.47	116.65	10,718.4	-171.8	242.8	242.8	12.00	12.00	0.00
10,843.5	66.69	116.65	10,726.0	-179.4	257.9	257.9	12.00	12.00	0.00
False Bakken									
10,850.0	67.47	116.65	10,728.5	-182.1	263.2	263.2	12.00	12.00	0.00



Ryan Directional Services Planning Report



Database: EDM 5000.1 Single User Db
Company: Oasis Petroleum
Project: Indian Hills
Site: 153N-100W-17/18
Well: Kline Federal 5300 41-18 14BX
Wellbore: Kline Federal 5300 41-18 14BX
Design: Design #2

Local Co-ordinate Reference:
TVD Reference:
MD Reference:
North Reference:
Survey Calculation Method:

Well Kline Federal 5300 41-18 14BX
 WELL @ 2082.0usft (Original Well Elev)
 WELL @ 2082.0usft (Original Well Elev)
 True
 Minimum Curvature

Planned Survey

Measured Depth (usft)	Inclination (°)	Azimuth (°)	Vertical Depth (usft)	+N/-S (usft)	+E/-W (usft)	Vertical Section (usft)	Dogleg Rate (°/100usft)	Build Rate (°/100usft)	Turn Rate (°/100usft)
10,867.6	69.59	116.65	10,735.0	-189.4	277.9	277.9	12.00	12.00	0.00
Upper Bakken									
10,875.0	70.47	116.65	10,737.5	-192.5	284.1	284.1	12.00	12.00	0.00
10,900.0	73.47	116.65	10,745.3	-203.2	305.3	305.3	12.00	12.00	0.00
10,921.9	76.10	116.65	10,751.0	-212.7	324.2	324.2	12.00	12.00	0.00
Middle Bakken									
10,925.0	76.47	116.65	10,751.7	-214.0	326.9	326.9	12.00	12.00	0.00
10,950.0	79.47	116.65	10,756.9	-225.0	348.8	348.8	12.00	12.00	0.00
10,968.7	81.72	116.65	10,760.0	-233.3	365.3	365.3	12.00	12.00	0.00
Middle Bakken Sand Target									
10,975.0	82.47	116.65	10,760.9	-236.1	370.8	370.8	12.00	12.00	0.00
11,000.0	85.47	116.65	10,763.5	-247.2	393.1	393.1	12.00	12.00	0.00
11,025.0	88.47	116.65	10,764.8	-258.4	415.4	415.4	12.00	12.00	0.00
11,036.9	89.90	116.65	10,765.0	-263.8	426.0	426.0	12.00	12.00	0.00
EOC									
11,037.0	89.90	116.65	10,765.0	-263.8	426.1	426.1	0.00	0.00	0.00
7"									
11,100.0	89.90	115.39	10,765.1	-291.4	482.7	482.7	2.00	0.00	-2.00
11,200.0	89.90	113.39	10,765.3	-332.7	573.8	573.8	2.00	0.00	-2.00
11,300.0	89.90	111.39	10,765.4	-370.8	666.2	666.2	2.00	0.00	-2.00
11,400.0	89.90	109.39	10,765.6	-405.7	760.0	760.0	2.00	0.00	-2.00
11,500.0	89.90	107.39	10,765.8	-437.2	854.8	854.8	2.00	0.00	-2.00
11,600.0	89.90	105.39	10,766.0	-465.4	950.8	950.8	2.00	0.00	-2.00
11,700.0	89.90	103.39	10,766.2	-490.3	1,047.6	1,047.6	2.00	0.00	-2.00
11,800.0	89.90	101.39	10,766.3	-511.7	1,145.3	1,145.3	2.00	0.00	-2.00
11,900.0	89.90	99.39	10,766.5	-529.7	1,243.7	1,243.7	2.00	0.00	-2.00
12,000.0	89.90	97.39	10,766.7	-544.3	1,342.6	1,342.6	2.00	0.00	-2.00
12,100.0	89.90	95.39	10,766.9	-555.4	1,442.0	1,442.0	2.00	0.00	-2.00
12,200.0	89.90	93.39	10,767.1	-563.1	1,541.7	1,541.7	2.00	0.00	-2.00
12,300.0	89.90	91.39	10,767.2	-567.3	1,641.6	1,641.6	2.00	0.00	-2.00
12,369.4	89.90	90.00	10,767.3	-568.1	1,711.0	1,711.0	2.00	0.00	-2.00
12,400.0	89.90	90.00	10,767.4	-568.1	1,741.6	1,741.6	0.00	0.00	0.00
12,500.0	89.90	90.00	10,767.6	-568.1	1,841.6	1,841.6	0.00	0.00	0.00
12,600.0	89.90	90.00	10,767.8	-568.1	1,941.6	1,941.6	0.00	0.00	0.00
12,700.0	89.90	90.00	10,767.9	-568.1	2,041.6	2,041.6	0.00	0.00	0.00
12,800.0	89.90	90.00	10,768.1	-568.1	2,141.6	2,141.6	0.00	0.00	0.00
12,900.0	89.90	90.00	10,768.3	-568.1	2,241.6	2,241.6	0.00	0.00	0.00
13,000.0	89.90	90.00	10,768.4	-568.1	2,341.6	2,341.6	0.00	0.00	0.00
13,100.0	89.90	90.00	10,768.6	-568.1	2,441.6	2,441.6	0.00	0.00	0.00
13,200.0	89.90	90.00	10,768.8	-568.1	2,541.6	2,541.6	0.00	0.00	0.00
13,300.0	89.90	90.00	10,769.0	-568.1	2,641.6	2,641.6	0.00	0.00	0.00
13,400.0	89.90	90.00	10,769.1	-568.1	2,741.6	2,741.6	0.00	0.00	0.00
13,500.0	89.90	90.00	10,769.3	-568.1	2,841.6	2,841.6	0.00	0.00	0.00
13,600.0	89.90	90.00	10,769.5	-568.1	2,941.6	2,941.6	0.00	0.00	0.00
13,700.0	89.90	90.00	10,769.7	-568.1	3,041.6	3,041.6	0.00	0.00	0.00
13,800.0	89.90	90.00	10,769.8	-568.1	3,141.6	3,141.6	0.00	0.00	0.00
13,888.8	89.90	90.00	10,770.0	-568.1	3,230.4	3,230.4	0.00	0.00	0.00
Base Middle Bakken Sand Target									
13,900.0	89.90	90.00	10,770.0	-568.1	3,241.6	3,241.6	0.00	0.00	0.00
14,000.0	89.90	90.00	10,770.2	-568.1	3,341.6	3,341.6	0.00	0.00	0.00
14,100.0	89.90	90.00	10,770.4	-568.1	3,441.6	3,441.6	0.00	0.00	0.00
14,200.0	89.90	90.00	10,770.5	-568.1	3,541.6	3,541.6	0.00	0.00	0.00
14,300.0	89.90	90.00	10,770.7	-568.1	3,641.6	3,641.6	0.00	0.00	0.00



Ryan Directional Services

Planning Report



Database: EDM 5000.1 Single User Db
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Project: Indian Hills
Site: 153N-100W-17/18
Well: Kline Federal 5300 41-18 14BX
Wellbore: Kline Federal 5300 41-18 14BX
Design: Design #2

Local Co-ordinate Reference:

TVD Reference:

MD Reference:

North Reference:

Survey Calculation Method:

Well Kline Federal 5300 41-18 14BX
WELL @ 2082.0usft (Original Well Elev)
WELL @ 2082.0usft (Original Well Elev)
True
Minimum Curvature

Planned Survey

Measured Depth (usft)	Inclination (°)	Azimuth (°)	Vertical Depth (usft)	+N/-S (usft)	+E/-W (usft)	Vertical Section (usft)	Dogleg Rate (°/100usft)	Build Rate (°/100usft)	Turn Rate (°/100usft)
14,400.0	89.90	90.00	10,770.9	-568.1	3,741.6	3,741.6	0.00	0.00	0.00
14,500.0	89.90	90.00	10,771.1	-568.1	3,841.6	3,841.6	0.00	0.00	0.00
14,600.0	89.90	90.00	10,771.2	-568.1	3,941.6	3,941.6	0.00	0.00	0.00
14,700.0	89.90	90.00	10,771.4	-568.1	4,041.6	4,041.6	0.00	0.00	0.00
14,800.0	89.90	90.00	10,771.6	-568.1	4,141.6	4,141.6	0.00	0.00	0.00
14,900.0	89.90	90.00	10,771.8	-568.1	4,241.6	4,241.6	0.00	0.00	0.00
15,000.0	89.90	90.00	10,771.9	-568.1	4,341.6	4,341.6	0.00	0.00	0.00
15,100.0	89.90	90.00	10,772.1	-568.1	4,441.6	4,441.6	0.00	0.00	0.00
15,200.0	89.90	90.00	10,772.3	-568.1	4,541.6	4,541.6	0.00	0.00	0.00
15,300.0	89.90	90.00	10,772.5	-568.1	4,641.6	4,641.6	0.00	0.00	0.00
15,400.0	89.90	90.00	10,772.6	-568.1	4,741.6	4,741.6	0.00	0.00	0.00
15,500.0	89.90	90.00	10,772.8	-568.1	4,841.6	4,841.6	0.00	0.00	0.00
15,600.0	89.90	90.00	10,773.0	-568.1	4,941.6	4,941.6	0.00	0.00	0.00
15,700.0	89.90	90.00	10,773.2	-568.1	5,041.6	5,041.6	0.00	0.00	0.00
15,800.0	89.90	90.00	10,773.3	-568.1	5,141.6	5,141.6	0.00	0.00	0.00
15,900.0	89.90	90.00	10,773.5	-568.1	5,241.6	5,241.6	0.00	0.00	0.00
16,000.0	89.90	90.00	10,773.7	-568.1	5,341.6	5,341.6	0.00	0.00	0.00
16,100.0	89.90	90.00	10,773.9	-568.1	5,441.6	5,441.6	0.00	0.00	0.00
16,200.0	89.90	90.00	10,774.0	-568.1	5,541.6	5,541.6	0.00	0.00	0.00
16,300.0	89.90	90.00	10,774.2	-568.1	5,641.6	5,641.6	0.00	0.00	0.00
16,400.0	89.90	90.00	10,774.4	-568.1	5,741.6	5,741.6	0.00	0.00	0.00
16,500.0	89.90	90.00	10,774.6	-568.1	5,841.6	5,841.6	0.00	0.00	0.00
16,600.0	89.90	90.00	10,774.7	-568.1	5,941.6	5,941.6	0.00	0.00	0.00
16,700.0	89.90	90.00	10,774.9	-568.1	6,041.6	6,041.6	0.00	0.00	0.00
16,800.0	89.90	90.00	10,775.1	-568.1	6,141.6	6,141.6	0.00	0.00	0.00
16,900.0	89.90	90.00	10,775.3	-568.1	6,241.6	6,241.6	0.00	0.00	0.00
17,000.0	89.90	90.00	10,775.4	-568.1	6,341.6	6,341.6	0.00	0.00	0.00
17,100.0	89.90	90.00	10,775.6	-568.1	6,441.6	6,441.6	0.00	0.00	0.00
17,200.0	89.90	90.00	10,775.8	-568.1	6,541.6	6,541.6	0.00	0.00	0.00
17,300.0	89.90	90.00	10,776.0	-568.1	6,641.6	6,641.6	0.00	0.00	0.00
17,400.0	89.90	90.00	10,776.1	-568.1	6,741.6	6,741.6	0.00	0.00	0.00
17,500.0	89.90	90.00	10,776.3	-568.1	6,841.6	6,841.6	0.00	0.00	0.00
17,600.0	89.90	90.00	10,776.5	-568.1	6,941.6	6,941.6	0.00	0.00	0.00
17,700.0	89.90	90.00	10,776.7	-568.1	7,041.6	7,041.6	0.00	0.00	0.00
17,800.0	89.90	90.00	10,776.8	-568.1	7,141.6	7,141.6	0.00	0.00	0.00
17,900.0	89.90	90.00	10,777.0	-568.1	7,241.6	7,241.6	0.00	0.00	0.00
18,000.0	89.90	90.00	10,777.2	-568.1	7,341.6	7,341.6	0.00	0.00	0.00
18,100.0	89.90	90.00	10,777.3	-568.1	7,441.6	7,441.6	0.00	0.00	0.00
18,200.0	89.90	90.00	10,777.5	-568.1	7,541.6	7,541.6	0.00	0.00	0.00
18,300.0	89.90	90.00	10,777.7	-568.1	7,641.6	7,641.6	0.00	0.00	0.00
18,400.0	89.90	90.00	10,777.9	-568.1	7,741.6	7,741.6	0.00	0.00	0.00
18,500.0	89.90	90.00	10,778.0	-568.1	7,841.6	7,841.6	0.00	0.00	0.00
18,600.0	89.90	90.00	10,778.2	-568.1	7,941.6	7,941.6	0.00	0.00	0.00
18,700.0	89.90	90.00	10,778.4	-568.1	8,041.6	8,041.6	0.00	0.00	0.00
18,800.0	89.90	90.00	10,778.6	-568.1	8,141.6	8,141.6	0.00	0.00	0.00
18,900.0	89.90	90.00	10,778.7	-568.1	8,241.6	8,241.6	0.00	0.00	0.00
19,000.0	89.90	90.00	10,778.9	-568.1	8,341.6	8,341.6	0.00	0.00	0.00
19,100.0	89.90	90.00	10,779.1	-568.1	8,441.6	8,441.6	0.00	0.00	0.00
19,200.0	89.90	90.00	10,779.3	-568.1	8,541.6	8,541.6	0.00	0.00	0.00
19,300.0	89.90	90.00	10,779.4	-568.1	8,641.6	8,641.6	0.00	0.00	0.00
19,400.0	89.90	90.00	10,779.6	-568.1	8,741.6	8,741.6	0.00	0.00	0.00
19,500.0	89.90	90.00	10,779.8	-568.1	8,841.6	8,841.6	0.00	0.00	0.00
19,600.0	89.90	90.00	10,780.0	-568.1	8,941.6	8,941.6	0.00	0.00	0.00
19,700.0	89.90	90.00	10,780.1	-568.1	9,041.6	9,041.6	0.00	0.00	0.00



Ryan Directional Services Planning Report



Database:	EDM 5000.1 Single User Db	Local Co-ordinate Reference:	Well Kline Federal 5300 41-18 14BX
Company:	Oasis Petroleum	TVD Reference:	WELL @ 2082.0usft (Original Well Elev)
Project:	Indian Hills	MD Reference:	WELL @ 2082.0usft (Original Well Elev)
Site:	153N-100W-17/18	North Reference:	True
Well:	Kline Federal 5300 41-18 14BX	Survey Calculation Method:	Minimum Curvature
Wellbore:	Kline Federal 5300 41-18 14BX		
Design:	Design #2		

Planned Survey									
Measured Depth (usft)	Inclination (°)	Azimuth (°)	Vertical Depth (usft)	+N/-S (usft)	+E/-W (usft)	Vertical Section (usft)	Dogleg Rate (°/100usft)	Build Rate (°/100usft)	Turn Rate (°/100usft)
19,800.0	89.90	90.00	10,780.3	-568.1	9,141.6	9,141.6	0.00	0.00	0.00
19,900.0	89.90	90.00	10,780.5	-568.1	9,241.6	9,241.6	0.00	0.00	0.00
20,000.0	89.90	90.00	10,780.7	-568.1	9,341.6	9,341.6	0.00	0.00	0.00
20,100.0	89.90	90.00	10,780.8	-568.1	9,441.6	9,441.6	0.00	0.00	0.00
20,200.0	89.90	90.00	10,781.0	-568.1	9,541.6	9,541.6	0.00	0.00	0.00
20,300.0	89.90	90.00	10,781.2	-568.1	9,641.6	9,641.6	0.00	0.00	0.00
20,400.0	89.90	90.00	10,781.4	-568.1	9,741.6	9,741.6	0.00	0.00	0.00
20,500.0	89.90	90.00	10,781.5	-568.1	9,841.6	9,841.6	0.00	0.00	0.00
20,600.0	89.90	90.00	10,781.7	-568.1	9,941.6	9,941.6	0.00	0.00	0.00
20,708.4	89.90	90.00	10,781.9	-568.1	10,050.0	10,050.0	0.00	0.00	0.00
5300 41-18 14BX PBHL									

Design Targets									
Target Name	Dip Angle (°)	Dip Dir. (°)	TVD (usft)	+N/-S (usft)	+E/-W (usft)	Northing (usft)	Easting (usft)	Latitude	Longitude
- hit/miss target									
- Shape									
5300 41-18 14BX PBHL	0.00	0.00	10,781.9	-568.0	10,050.0	404,098.33	1,219,984.51	48° 4' 1.268 N	103° 33' 43.975 W
- plan misses target center by 0.1usft at 20708.4usft MD (10781.9 TVD, -568.1 N, 10050.0 E)									
- Point									

Casing Points					
Measured Depth (usft)	Vertical Depth (usft)	Name	Casing Diameter (")	Hole Diameter (")	
11,037.0	10,765.0	7"	7	8-3/4	



Ryan Directional Services

Planning Report



Database:	EDM 5000.1 Single User Db	Local Co-ordinate Reference:	Well Kline Federal 5300 41-18 14BX
Company:	Oasis Petroleum	TVD Reference:	WELL @ 2082.0usft (Original Well Elev)
Project:	Indian Hills	MD Reference:	WELL @ 2082.0usft (Original Well Elev)
Site:	153N-100W-17/18	North Reference:	True
Well:	Kline Federal 5300 41-18 14BX	Survey Calculation Method:	Minimum Curvature
Wellbore:	Kline Federal 5300 41-18 14BX		
Design:	Design #2		

Formations						
Measured Depth (usft)	Vertical Depth (usft)	Name	Lithology	Dip (°)	Dip Direction (°)	
1,972.0	1,972.0	Pierre				
4,646.1	4,646.0	Greenhorn				
5,057.1	5,057.0	Mowry				
5,471.1	5,471.0	Dakota				
6,486.2	6,486.0	Rierdon				
6,814.2	6,814.0	Dunham Salt				
6,929.2	6,929.0	Dunham Salt Base				
7,024.2	7,024.0	Spearfish				
7,283.2	7,283.0	Pine Salt				
7,318.2	7,318.0	Pine Salt Base				
7,374.2	7,374.0	Opeche Salt				
7,403.2	7,403.0	Opeche Salt Base				
7,605.2	7,605.0	Broom Creek (Top of Minnelusa Gp.)				
7,684.2	7,684.0	Amsden				
7,853.2	7,853.0	Tyler				
8,047.2	8,047.0	Otter (Base of Minnelusa Gp.)				
8,390.2	8,390.0	Kibbey Lime				
8,542.2	8,542.0	Charles Salt				
9,159.2	9,159.0	UB				
9,238.2	9,238.0	Base Last Salt				
9,301.2	9,301.0	Ratcliffe				
9,454.2	9,454.0	Mission Canyon				
10,022.2	10,022.0	Lodgepole				
10,209.2	10,209.0	Lodgepole Fracture Zone				
10,843.5	10,726.0	False Bakken				
10,867.6	10,735.0	Upper Bakken				
10,921.9	10,751.0	Middle Bakken				
10,968.7	10,760.0	Middle Bakken Sand Target				
13,888.8	10,770.0	Base Middle Bakken Sand Target				

Plan Annotations					
Measured Depth (usft)	Vertical Depth (usft)	Local Coordinates			
		+N/-S (usft)	+E/-W (usft)	Comment	
10,287.8	10,287.6	-50.0	0.0	KOP Build 12°/100'	
11,036.9	10,765.0	-263.8	426.0	EOC	

DRILLING PLAN									
OPERATOR		Oasis Petroleum			COUNTY/STATE		McKenzie Co., ND		
WELL NAME		Kline Federal 5300 41-18 148X			RIG		0		
WELL TYPE		Horizontal Middle Bakken							
LOCATION		SWSW 18-153N-100W							
EST, T.D.		20,708'			Surface Location (survey plat):		358° (pl)		237' NW
TOTAL LATERAL		9,671'			GROUND ELEV:		2057' Finished Pad Elev		Sub Height: 25'
PROGNOSIS:		Based on 2,082' KB(est)			KB ELEV:		2082		
LOGS:		Type OH Logs: Triple Combo KOP to Kibby (or min run of 1800' whichever is greater); GR/Res to BSC; GR to surf; CND through the Dakota CBL/GR: Above top of cement/GR to base of casing MWD GR: KOP to lateral TD							
DEVIATION:		Interval Surf: 3 deg max, 1 deg / 100'; svy every 500' Prod: 5 deg max, 1 deg / 100'; svy every 100'							
DST'S:		None planned							
CORES:		None planned							
MUDLOGGING:		Two-Man: 8,342' ~200' above the Charles (Kibbey) to Casing point; Casing point to TD 30' samples at direction of wellsite geologist; 10' through target @ curve land							
BOP:		11" 5000 psi blind, pipe & annular							
Dip Rate:		0.1							
Max. Anticipated BHP:		4673			Surface Formation:		Glacial till		
MUD:		Interval		Type		WT		Vis	
Surface:		0' - 2,072'		FW/Gel - Lime Sweeps		8.4-9.0		28-32	
Intermediate:		2,072' - 11,037'		Invert		9.5-10.4		40-50	
Lateral:		11,037' - 20,708'		Salt Water		9.8-10.2		28-32	
CASING:		Size		Wt ppf		Hole		Depth	
Surface:		13-3/8"		54.5#		17-1/2"		2,072'	
Dakota Contingency:		9-5/8"		40#		12-1/4"		6,101'	
Intermediate:		7"		29/32#		8-3/4"		11,037'	
Production Liner:		4.5"		13.5#		6"		20,708'	
PROBABLE PLUGS, IF REQ'D:		TCL @ 10,238'		50' above KOP					
OTHER:		MD		TVD		FNL/FSL		FEL/FWL	
Surface:		2,072		2,072		368' FSL		237' FWL	
KOP:		10,288'		10,288'		318' FSL		237' FWL	
EOC:		11,037'		10,765'		104' FSL		663' FWL	
Casing Point:		11,037'		10,765'		104' FSL		663' FWL	
Middle Bakken Lateral TD:		20,708'		10,782'		200' FNL		200' FEL	
S-T-R		SEC 18-T153N-R100W		SEC 18-T153N-R100W		SEC 18-T153N-R100W		SEC 18-T153N-R100W	
AZI		116.65		116.65		116.65		90.00	
Survey Company:		Build Rate: 12 deg / 100'							
Comments: Request a Sundry for an Open Hole Log Waiver Exception well: Oasis Petroleum's Kline Federal 5300 11-18H (153N 100W 18 NW NW) 35 packers, 35 sleeves, no frac string Oasis Petroleum does not use Diesel Fuel, as defined by the US EPA in the list below, in our hydraulic fracture operations. 68334-30-5 (Primary Name: Fuels, diesel) 68476-34-6 (Primary Name: Fuels, diesel, No. 2) 68476-30-2 (Primary Name: Fuel oil No. 2) 68476-31-3 (Primary Name: Fuel oil, No. 4) 8008-20-6 (Primary Name: Kerosene)									
									
Geology: M. Steed 4/3/2014					Engineering: hbader rpm 7/20/14				



Oasis Petroleum

Indian Hills

153N-100W-17/18

Kline Federal 5300 41-18 14BX

Kline Federal 5300 41-18 14BX

Plan: Design #2

Standard Planning Report

28 May, 2014



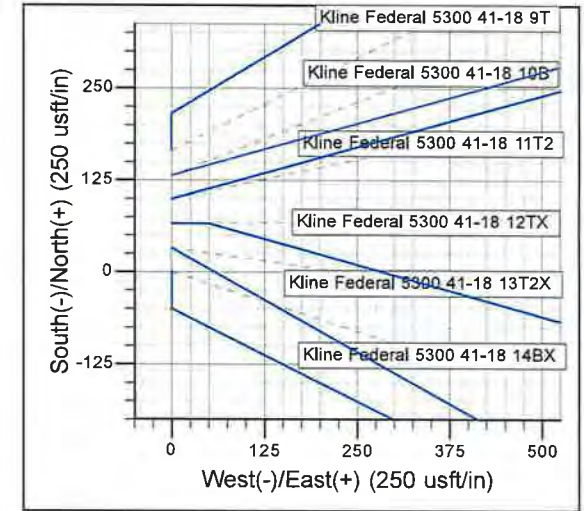
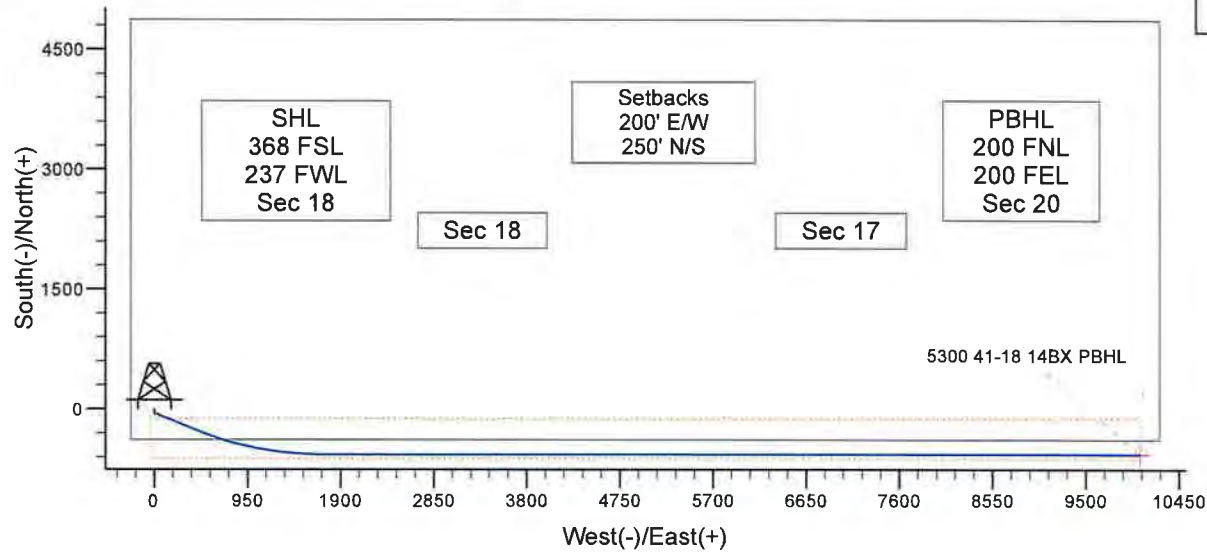
**RYAN DIRECTIONAL
SERVICES**
A NABORS COMPANY

Project: Indian Hills
 Site: 153N-100W-17/18
 Well: Kline Federal 5300 41-18 14BX
 Wellbore: Kline Federal 5300 41-18 14BX
 Design: Design #2



WELL DETAILS: Kline Federal 5300 41-18 14BX

Ground Level: 2057.0
 Northing: 405070.82 Easting: 1209965.56 Latitude: 48° 4' 6.900 N Longitude: 103° 36' 11.950 W



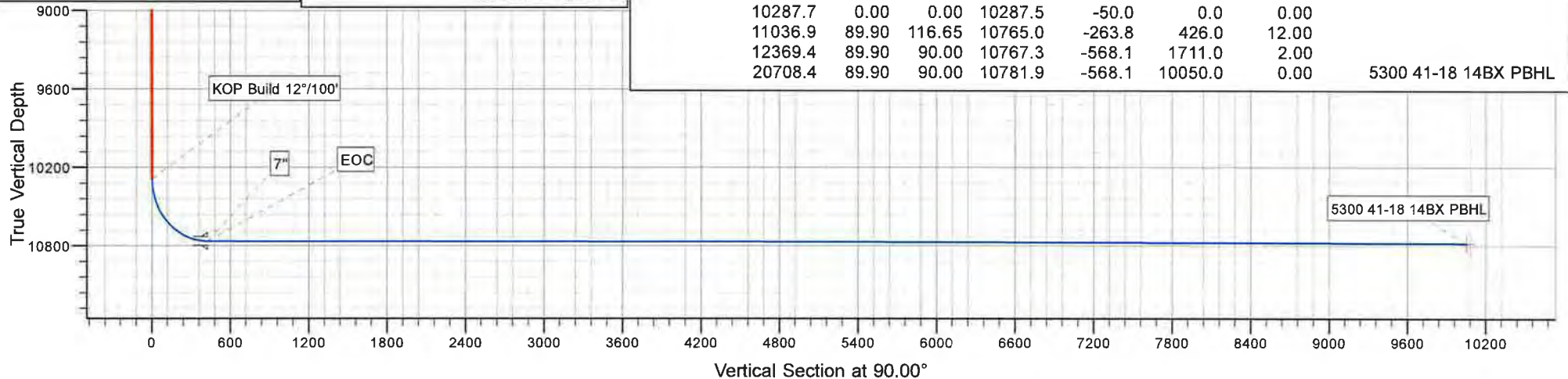
CASING DETAILS			
TVD	MD	Name	Size
10765.0	11037.0	7"	7"



Azimuths to True North
 Magnetic North: 8.27°
 Magnetic Field
 Strength: 56437.0snT
 Dip Angle: 72.99°
 Date: 5/23/2014
 Model: IGRF2010

SECTION DETAILS

MD	Inc	Azi	TVD	+N/-S	+E/-W	Dleg	Target
0.0	0.00	0.00	0.0	0.0	0.0	0.00	
2200.0	0.00	0.00	2200.0	0.0	0.0	0.00	
2216.7	0.50	180.00	2216.7	-0.1	0.0	3.00	
7926.7	0.50	180.00	7926.4	-49.9	0.0	0.00	
7943.3	0.00	0.00	7943.1	-50.0	0.0	3.00	
10287.7	0.00	0.00	10287.5	-50.0	0.0	0.00	
11036.9	89.90	116.65	10765.0	-263.8	426.0	12.00	
12369.4	89.90	90.00	10767.3	-568.1	1711.0	2.00	
20708.4	89.90	90.00	10781.9	-568.1	10050.0	0.00	5300 41-18 14BX PBHL





Ryan Directional Services
Planning Report



Database: EDM 5000.1 Single User Db
Company: Oasis Petroleum
Project: Indian Hills
Site: 153N-100W-17/18
Well: Kline Federal 5300 41-18 14BX
Wellbore: Kline Federal 5300 41-18 14BX
Design: Design #2

Local Co-ordinate Reference: Well Kline Federal 5300 41-18 14BX
TVD Reference: WELL @ 2082.0usft (Original Well Elev)
MD Reference: WELL @ 2082.0usft (Original Well Elev)
North Reference: True
Survey Calculation Method: Minimum Curvature

Project	Indian Hills		
Map System:	US State Plane 1983	System Datum:	Mean Sea Level
Geo Datum:	North American Datum 1983		
Map Zone:	North Dakota Northern Zone		

Site	153N-100W-17/18		
Site Position:		Northing:	408,962.44 usft
From:	Lat/Long	Easting:	1,210,229.18 usft
Position Uncertainty:	0.0 usft	Slot Radius:	13-3/16 "
		Latitude:	48° 4' 45.380 N
		Longitude:	103° 36' 10.380 W
		Grid Convergence:	-2.31 °

Well	Kline Federal 5300 41-18 14BX		
Well Position	+N/-S	-3,899.1 usft	Northing: 405,070.82 usft
	+E/-W	-106.6 usft	Easting: 1,209,965.56 usft
Position Uncertainty	2.0 usft	Wellhead Elevation:	Latitude: 48° 4' 6.900 N
			Longitude: 103° 36' 11.950 W
			Ground Level: 2,057.0 usft

Wellbore	Kline Federal 5300 41-18 14BX				
Magnetics	Model Name	Sample Date	Declination (°)	Dip Angle (°)	Field Strength (nT)
	IGRF2010	5/23/2014	8.27	72.99	56,437

Design	Design #2			
Audit Notes:				
Version:	Phase:	PROTOTYPE	Tie On Depth:	0.0
Vertical Section:	Depth From (TVD) (usft)	+N/-S (usft)	+E/-W (usft)	Direction (°)
	0.0	0.0	0.0	90.00

Plan Sections										
Measured Depth (usft)	Inclination (°)	Azimuth (°)	Vertical Depth (usft)	+N/-S (usft)	+E/-W (usft)	Dogleg Rate (°/100usft)	Build Rate (°/100usft)	Turn Rate (°/100usft)	TFO (°)	Target
0.0	0.00	0.00	0.0	0.0	0.0	0.00	0.00	0.00	0.00	
2,200.0	0.00	0.00	2,200.0	0.0	0.0	0.00	0.00	0.00	0.00	
2,216.7	0.50	180.00	2,216.7	-0.1	0.0	3.00	3.00	0.00	180.00	
7,926.7	0.50	180.00	7,926.4	-49.9	0.0	0.00	0.00	0.00	0.00	
7,943.3	0.00	0.00	7,943.1	-50.0	0.0	3.00	-3.00	0.00	180.00	
10,287.7	0.00	0.00	10,287.5	-50.0	0.0	0.00	0.00	0.00	0.00	
11,036.9	89.90	116.65	10,765.0	-263.8	426.0	12.00	12.00	0.00	116.65	
12,369.4	89.90	90.00	10,767.3	-568.1	1,711.0	2.00	0.00	-2.00	-90.02	
20,708.4	89.90	90.00	10,781.9	-568.1	10,050.0	0.00	0.00	0.00	0.00	5300 41-18 14BX PBI



Ryan Directional Services Planning Report



Database: EDM 5000.1 Single User Db
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Project: Indian Hills
Site: 153N-100W-17/18
Well: Kline Federal 5300 41-18 14BX
Wellbore: Kline Federal 5300 41-18 14BX
Design: Design #2

Local Co-ordinate Reference: Well Kline Federal 5300 41-18 14BX
TVD Reference: WELL @ 2082.0usft (Original Well Elev)
MD Reference: WELL @ 2082.0usft (Original Well Elev)
North Reference: True
Survey Calculation Method: Minimum Curvature

Planned Survey

Measured Depth (usft)	Inclination (°)	Azimuth (°)	Vertical Depth (usft)	+N/-S (usft)	+E/-W (usft)	Vertical Section (usft)	Dogleg Rate (°/100usft)	Build Rate (°/100usft)	Turn Rate (°/100usft)
2,200.0	0.00	0.00	2,200.0	0.0	0.0	0.0	0.00	0.00	0.00
2,216.7	0.50	180.00	2,216.7	-0.1	0.0	0.0	3.00	3.00	0.00
2,300.0	0.50	180.00	2,300.0	-0.8	0.0	0.0	0.00	0.00	0.00
2,400.0	0.50	180.00	2,400.0	-1.7	0.0	0.0	0.00	0.00	0.00
2,500.0	0.50	180.00	2,500.0	-2.5	0.0	0.0	0.00	0.00	0.00
2,600.0	0.50	180.00	2,600.0	-3.4	0.0	0.0	0.00	0.00	0.00
2,700.0	0.50	180.00	2,700.0	-4.3	0.0	0.0	0.00	0.00	0.00
2,800.0	0.50	180.00	2,800.0	-5.2	0.0	0.0	0.00	0.00	0.00
2,900.0	0.50	180.00	2,900.0	-6.0	0.0	0.0	0.00	0.00	0.00
3,000.0	0.50	180.00	3,000.0	-6.9	0.0	0.0	0.00	0.00	0.00
3,100.0	0.50	180.00	3,100.0	-7.8	0.0	0.0	0.00	0.00	0.00
3,200.0	0.50	180.00	3,200.0	-8.7	0.0	0.0	0.00	0.00	0.00
3,300.0	0.50	180.00	3,300.0	-9.5	0.0	0.0	0.00	0.00	0.00
3,400.0	0.50	180.00	3,400.0	-10.4	0.0	0.0	0.00	0.00	0.00
3,500.0	0.50	180.00	3,500.0	-11.3	0.0	0.0	0.00	0.00	0.00
3,600.0	0.50	180.00	3,599.9	-12.1	0.0	0.0	0.00	0.00	0.00
3,700.0	0.50	180.00	3,699.9	-13.0	0.0	0.0	0.00	0.00	0.00
3,800.0	0.50	180.00	3,799.9	-13.9	0.0	0.0	0.00	0.00	0.00
3,900.0	0.50	180.00	3,899.9	-14.8	0.0	0.0	0.00	0.00	0.00
4,000.0	0.50	180.00	3,999.9	-15.6	0.0	0.0	0.00	0.00	0.00
4,100.0	0.50	180.00	4,099.9	-16.5	0.0	0.0	0.00	0.00	0.00
4,200.0	0.50	180.00	4,199.9	-17.4	0.0	0.0	0.00	0.00	0.00
4,300.0	0.50	180.00	4,299.9	-18.3	0.0	0.0	0.00	0.00	0.00
4,400.0	0.50	180.00	4,399.9	-19.1	0.0	0.0	0.00	0.00	0.00
4,500.0	0.50	180.00	4,499.9	-20.0	0.0	0.0	0.00	0.00	0.00
4,600.0	0.50	180.00	4,599.9	-20.9	0.0	0.0	0.00	0.00	0.00
4,646.1	0.50	180.00	4,646.0	-21.3	0.0	0.0	0.00	0.00	0.00
Greenhorn									
4,700.0	0.50	180.00	4,699.9	-21.7	0.0	0.0	0.00	0.00	0.00
4,800.0	0.50	180.00	4,799.9	-22.6	0.0	0.0	0.00	0.00	0.00
4,900.0	0.50	180.00	4,899.9	-23.5	0.0	0.0	0.00	0.00	0.00
5,000.0	0.50	180.00	4,999.9	-24.4	0.0	0.0	0.00	0.00	0.00
5,057.1	0.50	180.00	5,057.0	-24.9	0.0	0.0	0.00	0.00	0.00
Mowry									
5,100.0	0.50	180.00	5,099.9	-25.2	0.0	0.0	0.00	0.00	0.00
5,200.0	0.50	180.00	5,199.9	-26.1	0.0	0.0	0.00	0.00	0.00
5,300.0	0.50	180.00	5,299.9	-27.0	0.0	0.0	0.00	0.00	0.00
5,400.0	0.50	180.00	5,399.9	-27.9	0.0	0.0	0.00	0.00	0.00
5,471.1	0.50	180.00	5,471.0	-28.5	0.0	0.0	0.00	0.00	0.00
Dakota									
5,500.0	0.50	180.00	5,499.9	-28.7	0.0	0.0	0.00	0.00	0.00
5,600.0	0.50	180.00	5,599.9	-29.6	0.0	0.0	0.00	0.00	0.00
5,700.0	0.50	180.00	5,699.9	-30.5	0.0	0.0	0.00	0.00	0.00
5,800.0	0.50	180.00	5,799.9	-31.3	0.0	0.0	0.00	0.00	0.00
5,900.0	0.50	180.00	5,899.9	-32.2	0.0	0.0	0.00	0.00	0.00
6,000.0	0.50	180.00	5,999.9	-33.1	0.0	0.0	0.00	0.00	0.00
6,100.0	0.50	180.00	6,099.9	-34.0	0.0	0.0	0.00	0.00	0.00
6,200.0	0.50	180.00	6,199.8	-34.8	0.0	0.0	0.00	0.00	0.00
6,300.0	0.50	180.00	6,299.8	-35.7	0.0	0.0	0.00	0.00	0.00
6,400.0	0.50	180.00	6,399.8	-36.6	0.0	0.0	0.00	0.00	0.00
6,486.2	0.50	180.00	6,486.0	-37.3	0.0	0.0	0.00	0.00	0.00
Rierdon									
6,500.0	0.50	180.00	6,499.8	-37.5	0.0	0.0	0.00	0.00	0.00



Ryan Directional Services

Planning Report



Database:	EDM 5000.1 Single User Db	Local Co-ordinate Reference:	Well Kline Federal 5300 41-18 14BX
Company:	Oasis Petroleum	TVD Reference:	WELL @ 2082.0usft (Original Well Elev)
Project:	Indian Hills	MD Reference:	WELL @ 2082.0usft (Original Well Elev)
Site:	153N-100W-17/18	North Reference:	True
Well:	Kline Federal 5300 41-18 14BX	Survey Calculation Method:	Minimum Curvature
Wellbore:	Kline Federal 5300 41-18 14BX		
Design:	Design #2		

Planned Survey										
Measured Depth (usft)	Inclination (°)	Azimuth (°)	Vertical Depth (usft)	+N/-S (usft)	+E/-W (usft)	Vertical Section (usft)	Dogleg Rate (°/100usft)	Build Rate (°/100usft)	Turn Rate (°/100usft)	
6,600.0	0.50	180.00	6,599.8	-38.3	0.0	0.0	0.00	0.00	0.00	
6,700.0	0.50	180.00	6,699.8	-39.2	0.0	0.0	0.00	0.00	0.00	
6,800.0	0.50	180.00	6,799.8	-40.1	0.0	0.0	0.00	0.00	0.00	
6,814.2	0.50	180.00	6,814.0	-40.2	0.0	0.0	0.00	0.00	0.00	
Dunham Salt										
6,900.0	0.50	180.00	6,899.8	-40.9	0.0	0.0	0.00	0.00	0.00	
6,929.2	0.50	180.00	6,929.0	-41.2	0.0	0.0	0.00	0.00	0.00	
Dunham Salt Base										
7,000.0	0.50	180.00	6,999.8	-41.8	0.0	0.0	0.00	0.00	0.00	
7,024.2	0.50	180.00	7,024.0	-42.0	0.0	0.0	0.00	0.00	0.00	
Spearfish										
7,100.0	0.50	180.00	7,099.8	-42.7	0.0	0.0	0.00	0.00	0.00	
7,200.0	0.50	180.00	7,199.8	-43.6	0.0	0.0	0.00	0.00	0.00	
7,283.2	0.50	180.00	7,283.0	-44.3	0.0	0.0	0.00	0.00	0.00	
Pine Salt										
7,300.0	0.50	180.00	7,299.8	-44.4	0.0	0.0	0.00	0.00	0.00	
7,318.2	0.50	180.00	7,318.0	-44.6	0.0	0.0	0.00	0.00	0.00	
Pine Salt Base										
7,374.2	0.50	180.00	7,374.0	-45.1	0.0	0.0	0.00	0.00	0.00	
Opeche Salt										
7,400.0	0.50	180.00	7,399.8	-45.3	0.0	0.0	0.00	0.00	0.00	
7,403.2	0.50	180.00	7,403.0	-45.3	0.0	0.0	0.00	0.00	0.00	
Opeche Salt Base										
7,500.0	0.50	180.00	7,499.8	-46.2	0.0	0.0	0.00	0.00	0.00	
7,600.0	0.50	180.00	7,599.8	-47.1	0.0	0.0	0.00	0.00	0.00	
7,605.2	0.50	180.00	7,605.0	-47.1	0.0	0.0	0.00	0.00	0.00	
Broom Creek (Top of Minnelusa Gp.)										
7,684.2	0.50	180.00	7,684.0	-47.8	0.0	0.0	0.00	0.00	0.00	
Amsden										
7,700.0	0.50	180.00	7,699.8	-47.9	0.0	0.0	0.00	0.00	0.00	
7,800.0	0.50	180.00	7,799.8	-48.8	0.0	0.0	0.00	0.00	0.00	
7,853.2	0.50	180.00	7,853.0	-49.3	0.0	0.0	0.00	0.00	0.00	
Tyler										
7,900.0	0.50	180.00	7,899.8	-49.7	0.0	0.0	0.00	0.00	0.00	
7,926.7	0.50	180.00	7,926.4	-49.9	0.0	0.0	0.00	0.00	0.00	
7,943.3	0.00	0.00	7,943.1	-50.0	0.0	0.0	3.00	-3.00	0.00	
8,000.0	0.00	0.00	7,999.8	-50.0	0.0	0.0	0.00	0.00	0.00	
8,047.2	0.00	0.00	8,047.0	-50.0	0.0	0.0	0.00	0.00	0.00	
Otter (Base of Minnelusa Gp.)										
8,100.0	0.00	0.00	8,099.8	-50.0	0.0	0.0	0.00	0.00	0.00	
8,200.0	0.00	0.00	8,199.8	-50.0	0.0	0.0	0.00	0.00	0.00	
8,300.0	0.00	0.00	8,299.8	-50.0	0.0	0.0	0.00	0.00	0.00	
8,390.2	0.00	0.00	8,390.0	-50.0	0.0	0.0	0.00	0.00	0.00	
Kibbey Lime										
8,400.0	0.00	0.00	8,399.8	-50.0	0.0	0.0	0.00	0.00	0.00	
8,500.0	0.00	0.00	8,499.8	-50.0	0.0	0.0	0.00	0.00	0.00	
8,542.2	0.00	0.00	8,542.0	-50.0	0.0	0.0	0.00	0.00	0.00	
Charles Salt										
8,600.0	0.00	0.00	8,599.8	-50.0	0.0	0.0	0.00	0.00	0.00	
8,700.0	0.00	0.00	8,699.8	-50.0	0.0	0.0	0.00	0.00	0.00	
8,800.0	0.00	0.00	8,799.8	-50.0	0.0	0.0	0.00	0.00	0.00	
8,900.0	0.00	0.00	8,899.8	-50.0	0.0	0.0	0.00	0.00	0.00	



Ryan Directional Services

Planning Report



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Company: Oasis Petroleum
Project: Indian Hills
Site: 153N-100W-17/18
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TVD Reference: WELL @ 2082.0usft (Original Well Elev)
MD Reference: WELL @ 2082.0usft (Original Well Elev)
North Reference: True
Survey Calculation Method: Minimum Curvature

Planned Survey

Measured Depth (usft)	Inclination (°)	Azimuth (°)	Vertical Depth (usft)	+N/-S (usft)	+E/-W (usft)	Vertical Section (usft)	Dogleg Rate (°/100usft)	Build Rate (°/100usft)	Turn Rate (°/100usft)
9,000.0	0.00	0.00	8,999.8	-50.0	0.0	0.0	0.00	0.00	0.00
9,100.0	0.00	0.00	9,099.8	-50.0	0.0	0.0	0.00	0.00	0.00
9,159.2	0.00	0.00	9,159.0	-50.0	0.0	0.0	0.00	0.00	0.00
UB									
9,200.0	0.00	0.00	9,199.8	-50.0	0.0	0.0	0.00	0.00	0.00
9,238.2	0.00	0.00	9,238.0	-50.0	0.0	0.0	0.00	0.00	0.00
Base Last Salt									
9,300.0	0.00	0.00	9,299.8	-50.0	0.0	0.0	0.00	0.00	0.00
9,301.2	0.00	0.00	9,301.0	-50.0	0.0	0.0	0.00	0.00	0.00
Ratcliffe									
9,400.0	0.00	0.00	9,399.8	-50.0	0.0	0.0	0.00	0.00	0.00
9,454.2	0.00	0.00	9,454.0	-50.0	0.0	0.0	0.00	0.00	0.00
Mission Canyon									
9,500.0	0.00	0.00	9,499.8	-50.0	0.0	0.0	0.00	0.00	0.00
9,600.0	0.00	0.00	9,599.8	-50.0	0.0	0.0	0.00	0.00	0.00
9,700.0	0.00	0.00	9,699.8	-50.0	0.0	0.0	0.00	0.00	0.00
9,800.0	0.00	0.00	9,799.8	-50.0	0.0	0.0	0.00	0.00	0.00
9,900.0	0.00	0.00	9,899.8	-50.0	0.0	0.0	0.00	0.00	0.00
10,000.0	0.00	0.00	9,999.8	-50.0	0.0	0.0	0.00	0.00	0.00
10,022.2	0.00	0.00	10,022.0	-50.0	0.0	0.0	0.00	0.00	0.00
Lodgepole									
10,100.0	0.00	0.00	10,099.8	-50.0	0.0	0.0	0.00	0.00	0.00
10,200.0	0.00	0.00	10,199.8	-50.0	0.0	0.0	0.00	0.00	0.00
10,209.2	0.00	0.00	10,209.0	-50.0	0.0	0.0	0.00	0.00	0.00
Lodgepole Fracture Zone									
10,287.7	0.00	0.00	10,287.5	-50.0	0.0	0.0	0.00	0.00	0.00
10,287.8	0.00	116.65	10,287.6	-50.0	0.0	0.0	0.00	0.00	0.00
KOP Build 12°/100'									
10,300.0	1.47	116.65	10,299.8	-50.0	0.1	0.1	12.07	12.07	0.00
10,325.0	4.47	116.65	10,324.7	-50.6	1.3	1.3	12.00	12.00	0.00
10,350.0	7.47	116.65	10,349.6	-51.8	3.6	3.6	12.00	12.00	0.00
10,375.0	10.47	116.65	10,374.3	-53.5	7.1	7.1	12.00	12.00	0.00
10,400.0	13.47	116.65	10,398.8	-55.9	11.7	11.7	12.00	12.00	0.00
10,425.0	16.47	116.65	10,422.9	-58.8	17.5	17.5	12.00	12.00	0.00
10,450.0	19.47	116.65	10,446.7	-62.2	24.4	24.4	12.00	12.00	0.00
10,475.0	22.47	116.65	10,470.0	-66.2	32.4	32.4	12.00	12.00	0.00
10,500.0	25.47	116.65	10,492.9	-70.8	41.5	41.5	12.00	12.00	0.00
10,525.0	28.47	116.65	10,515.1	-75.9	51.6	51.6	12.00	12.00	0.00
10,550.0	31.47	116.65	10,536.8	-81.5	62.8	62.8	12.00	12.00	0.00
10,575.0	34.47	116.65	10,557.8	-87.6	74.9	74.9	12.00	12.00	0.00
10,600.0	37.47	116.65	10,578.0	-94.2	88.1	88.1	12.00	12.00	0.00
10,625.0	40.47	116.65	10,597.4	-101.2	102.1	102.1	12.00	12.00	0.00
10,650.0	43.47	116.65	10,616.0	-108.7	117.1	117.1	12.00	12.00	0.00
10,675.0	46.47	116.65	10,633.7	-116.6	132.8	132.8	12.00	12.00	0.00
10,700.0	49.47	116.65	10,650.4	-125.0	149.4	149.4	12.00	12.00	0.00
10,725.0	52.47	116.65	10,666.2	-133.7	166.8	166.8	12.00	12.00	0.00
10,750.0	55.47	116.65	10,680.9	-142.7	184.9	184.9	12.00	12.00	0.00
10,775.0	58.47	116.65	10,694.5	-152.1	203.6	203.6	12.00	12.00	0.00
10,800.0	61.47	116.65	10,707.0	-161.9	222.9	222.9	12.00	12.00	0.00
10,825.0	64.47	116.65	10,718.4	-171.8	242.8	242.8	12.00	12.00	0.00
10,843.5	66.69	116.65	10,726.0	-179.4	257.9	257.9	12.00	12.00	0.00
False Bakken									
10,850.0	67.47	116.65	10,728.5	-182.1	263.2	263.2	12.00	12.00	0.00



Ryan Directional Services

Planning Report



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Wellbore: Kline Federal 5300 41-18 14BX
Design: Design #2

Local Co-ordinate Reference:
TVD Reference:
MD Reference:
North Reference:
Survey Calculation Method:

Well Kline Federal 5300 41-18 14BX
 WELL @ 2082.0usft (Original Well Elev)
 WELL @ 2082.0usft (Original Well Elev)
 True
 Minimum Curvature

Planned Survey

Measured Depth (usft)	Inclination (°)	Azimuth (°)	Vertical Depth (usft)	+N/-S (usft)	+E/-W (usft)	Vertical Section (usft)	Dogleg Rate (°/100usft)	Build Rate (°/100usft)	Turn Rate (°/100usft)
10,867.6	69.59	116.65	10,735.0	-189.4	277.9	277.9	12.00	12.00	0.00
Upper Bakken									
10,875.0	70.47	116.65	10,737.5	-192.5	284.1	284.1	12.00	12.00	0.00
10,900.0	73.47	116.65	10,745.3	-203.2	305.3	305.3	12.00	12.00	0.00
10,921.9	76.10	116.65	10,751.0	-212.7	324.2	324.2	12.00	12.00	0.00
Middle Bakken									
10,925.0	76.47	116.65	10,751.7	-214.0	326.9	326.9	12.00	12.00	0.00
10,950.0	79.47	116.65	10,756.9	-225.0	348.8	348.8	12.00	12.00	0.00
10,968.7	81.72	116.65	10,760.0	-233.3	365.3	365.3	12.00	12.00	0.00
Middle Bakken Sand Target									
10,975.0	82.47	116.65	10,760.9	-236.1	370.8	370.8	12.00	12.00	0.00
11,000.0	85.47	116.65	10,763.5	-247.2	393.1	393.1	12.00	12.00	0.00
11,025.0	88.47	116.65	10,764.8	-258.4	415.4	415.4	12.00	12.00	0.00
11,036.9	89.90	116.65	10,765.0	-263.8	426.0	426.0	12.00	12.00	0.00
EOC									
11,037.0	89.90	116.65	10,765.0	-263.8	426.1	426.1	0.00	0.00	0.00
7"									
11,100.0	89.90	115.39	10,765.1	-291.4	482.7	482.7	2.00	0.00	-2.00
11,200.0	89.90	113.39	10,765.3	-332.7	573.8	573.8	2.00	0.00	-2.00
11,300.0	89.90	111.39	10,765.4	-370.8	666.2	666.2	2.00	0.00	-2.00
11,400.0	89.90	109.39	10,765.6	-405.7	760.0	760.0	2.00	0.00	-2.00
11,500.0	89.90	107.39	10,765.8	-437.2	854.8	854.8	2.00	0.00	-2.00
11,600.0	89.90	105.39	10,766.0	-465.4	950.8	950.8	2.00	0.00	-2.00
11,700.0	89.90	103.39	10,766.2	-490.3	1,047.6	1,047.6	2.00	0.00	-2.00
11,800.0	89.90	101.39	10,766.3	-511.7	1,145.3	1,145.3	2.00	0.00	-2.00
11,900.0	89.90	99.39	10,766.5	-529.7	1,243.7	1,243.7	2.00	0.00	-2.00
12,000.0	89.90	97.39	10,766.7	-544.3	1,342.6	1,342.6	2.00	0.00	-2.00
12,100.0	89.90	95.39	10,766.9	-555.4	1,442.0	1,442.0	2.00	0.00	-2.00
12,200.0	89.90	93.39	10,767.1	-563.1	1,541.7	1,541.7	2.00	0.00	-2.00
12,300.0	89.90	91.39	10,767.2	-567.3	1,641.6	1,641.6	2.00	0.00	-2.00
12,369.4	89.90	90.00	10,767.3	-568.1	1,711.0	1,711.0	2.00	0.00	-2.00
12,400.0	89.90	90.00	10,767.4	-568.1	1,741.6	1,741.6	0.00	0.00	0.00
12,500.0	89.90	90.00	10,767.6	-568.1	1,841.6	1,841.6	0.00	0.00	0.00
12,600.0	89.90	90.00	10,767.8	-568.1	1,941.6	1,941.6	0.00	0.00	0.00
12,700.0	89.90	90.00	10,767.9	-568.1	2,041.6	2,041.6	0.00	0.00	0.00
12,800.0	89.90	90.00	10,768.1	-568.1	2,141.6	2,141.6	0.00	0.00	0.00
12,900.0	89.90	90.00	10,768.3	-568.1	2,241.6	2,241.6	0.00	0.00	0.00
13,000.0	89.90	90.00	10,768.4	-568.1	2,341.6	2,341.6	0.00	0.00	0.00
13,100.0	89.90	90.00	10,768.6	-568.1	2,441.6	2,441.6	0.00	0.00	0.00
13,200.0	89.90	90.00	10,768.8	-568.1	2,541.6	2,541.6	0.00	0.00	0.00
13,300.0	89.90	90.00	10,769.0	-568.1	2,641.6	2,641.6	0.00	0.00	0.00
13,400.0	89.90	90.00	10,769.1	-568.1	2,741.6	2,741.6	0.00	0.00	0.00
13,500.0	89.90	90.00	10,769.3	-568.1	2,841.6	2,841.6	0.00	0.00	0.00
13,600.0	89.90	90.00	10,769.5	-568.1	2,941.6	2,941.6	0.00	0.00	0.00
13,700.0	89.90	90.00	10,769.7	-568.1	3,041.6	3,041.6	0.00	0.00	0.00
13,800.0	89.90	90.00	10,769.8	-568.1	3,141.6	3,141.6	0.00	0.00	0.00
13,888.8	89.90	90.00	10,770.0	-568.1	3,230.4	3,230.4	0.00	0.00	0.00
Base Middle Bakken Sand Target									
13,900.0	89.90	90.00	10,770.0	-568.1	3,241.6	3,241.6	0.00	0.00	0.00
14,000.0	89.90	90.00	10,770.2	-568.1	3,341.6	3,341.6	0.00	0.00	0.00
14,100.0	89.90	90.00	10,770.4	-568.1	3,441.6	3,441.6	0.00	0.00	0.00
14,200.0	89.90	90.00	10,770.5	-568.1	3,541.6	3,541.6	0.00	0.00	0.00
14,300.0	89.90	90.00	10,770.7	-568.1	3,641.6	3,641.6	0.00	0.00	0.00



Ryan Directional Services

Planning Report



Database: EDM 5000.1 Single User Db
Company: Oasis Petroleum
Project: Indian Hills
Site: 153N-100W-17/18
Well: Kline Federal 5300 41-18 14BX
Wellbore: Kline Federal 5300 41-18 14BX
Design: Design #2

Local Co-ordinate Reference: Well Kline Federal 5300 41-18 14BX
TVD Reference: WELL @ 2082.0usft (Original Well Elev)
MD Reference: WELL @ 2082.0usft (Original Well Elev)
North Reference: True
Survey Calculation Method: Minimum Curvature

Planned Survey									
Measured Depth (usft)	Inclination (°)	Azimuth (°)	Vertical Depth (usft)	+N/-S (usft)	+E/-W (usft)	Vertical Section (usft)	Dogleg Rate (°/100usft)	Build Rate (°/100usft)	Turn Rate (°/100usft)
14,400.0	89.90	90.00	10,770.9	-568.1	3,741.6	3,741.6	0.00	0.00	0.00
14,500.0	89.90	90.00	10,771.1	-568.1	3,841.6	3,841.6	0.00	0.00	0.00
14,600.0	89.90	90.00	10,771.2	-568.1	3,941.6	3,941.6	0.00	0.00	0.00
14,700.0	89.90	90.00	10,771.4	-568.1	4,041.6	4,041.6	0.00	0.00	0.00
14,800.0	89.90	90.00	10,771.6	-568.1	4,141.6	4,141.6	0.00	0.00	0.00
14,900.0	89.90	90.00	10,771.8	-568.1	4,241.6	4,241.6	0.00	0.00	0.00
15,000.0	89.90	90.00	10,771.9	-568.1	4,341.6	4,341.6	0.00	0.00	0.00
15,100.0	89.90	90.00	10,772.1	-568.1	4,441.6	4,441.6	0.00	0.00	0.00
15,200.0	89.90	90.00	10,772.3	-568.1	4,541.6	4,541.6	0.00	0.00	0.00
15,300.0	89.90	90.00	10,772.5	-568.1	4,641.6	4,641.6	0.00	0.00	0.00
15,400.0	89.90	90.00	10,772.6	-568.1	4,741.6	4,741.6	0.00	0.00	0.00
15,500.0	89.90	90.00	10,772.8	-568.1	4,841.6	4,841.6	0.00	0.00	0.00
15,600.0	89.90	90.00	10,773.0	-568.1	4,941.6	4,941.6	0.00	0.00	0.00
15,700.0	89.90	90.00	10,773.2	-568.1	5,041.6	5,041.6	0.00	0.00	0.00
15,800.0	89.90	90.00	10,773.3	-568.1	5,141.6	5,141.6	0.00	0.00	0.00
15,900.0	89.90	90.00	10,773.5	-568.1	5,241.6	5,241.6	0.00	0.00	0.00
16,000.0	89.90	90.00	10,773.7	-568.1	5,341.6	5,341.6	0.00	0.00	0.00
16,100.0	89.90	90.00	10,773.9	-568.1	5,441.6	5,441.6	0.00	0.00	0.00
16,200.0	89.90	90.00	10,774.0	-568.1	5,541.6	5,541.6	0.00	0.00	0.00
16,300.0	89.90	90.00	10,774.2	-568.1	5,641.6	5,641.6	0.00	0.00	0.00
16,400.0	89.90	90.00	10,774.4	-568.1	5,741.6	5,741.6	0.00	0.00	0.00
16,500.0	89.90	90.00	10,774.6	-568.1	5,841.6	5,841.6	0.00	0.00	0.00
16,600.0	89.90	90.00	10,774.7	-568.1	5,941.6	5,941.6	0.00	0.00	0.00
16,700.0	89.90	90.00	10,774.9	-568.1	6,041.6	6,041.6	0.00	0.00	0.00
16,800.0	89.90	90.00	10,775.1	-568.1	6,141.6	6,141.6	0.00	0.00	0.00
16,900.0	89.90	90.00	10,775.3	-568.1	6,241.6	6,241.6	0.00	0.00	0.00
17,000.0	89.90	90.00	10,775.4	-568.1	6,341.6	6,341.6	0.00	0.00	0.00
17,100.0	89.90	90.00	10,775.6	-568.1	6,441.6	6,441.6	0.00	0.00	0.00
17,200.0	89.90	90.00	10,775.8	-568.1	6,541.6	6,541.6	0.00	0.00	0.00
17,300.0	89.90	90.00	10,776.0	-568.1	6,641.6	6,641.6	0.00	0.00	0.00
17,400.0	89.90	90.00	10,776.1	-568.1	6,741.6	6,741.6	0.00	0.00	0.00
17,500.0	89.90	90.00	10,776.3	-568.1	6,841.6	6,841.6	0.00	0.00	0.00
17,600.0	89.90	90.00	10,776.5	-568.1	6,941.6	6,941.6	0.00	0.00	0.00
17,700.0	89.90	90.00	10,776.7	-568.1	7,041.6	7,041.6	0.00	0.00	0.00
17,800.0	89.90	90.00	10,776.8	-568.1	7,141.6	7,141.6	0.00	0.00	0.00
17,900.0	89.90	90.00	10,777.0	-568.1	7,241.6	7,241.6	0.00	0.00	0.00
18,000.0	89.90	90.00	10,777.2	-568.1	7,341.6	7,341.6	0.00	0.00	0.00
18,100.0	89.90	90.00	10,777.3	-568.1	7,441.6	7,441.6	0.00	0.00	0.00
18,200.0	89.90	90.00	10,777.5	-568.1	7,541.6	7,541.6	0.00	0.00	0.00
18,300.0	89.90	90.00	10,777.7	-568.1	7,641.6	7,641.6	0.00	0.00	0.00
18,400.0	89.90	90.00	10,777.9	-568.1	7,741.6	7,741.6	0.00	0.00	0.00
18,500.0	89.90	90.00	10,778.0	-568.1	7,841.6	7,841.6	0.00	0.00	0.00
18,600.0	89.90	90.00	10,778.2	-568.1	7,941.6	7,941.6	0.00	0.00	0.00
18,700.0	89.90	90.00	10,778.4	-568.1	8,041.6	8,041.6	0.00	0.00	0.00
18,800.0	89.90	90.00	10,778.6	-568.1	8,141.6	8,141.6	0.00	0.00	0.00
18,900.0	89.90	90.00	10,778.7	-568.1	8,241.6	8,241.6	0.00	0.00	0.00
19,000.0	89.90	90.00	10,778.9	-568.1	8,341.6	8,341.6	0.00	0.00	0.00
19,100.0	89.90	90.00	10,779.1	-568.1	8,441.6	8,441.6	0.00	0.00	0.00
19,200.0	89.90	90.00	10,779.3	-568.1	8,541.6	8,541.6	0.00	0.00	0.00
19,300.0	89.90	90.00	10,779.4	-568.1	8,641.6	8,641.6	0.00	0.00	0.00
19,400.0	89.90	90.00	10,779.6	-568.1	8,741.6	8,741.6	0.00	0.00	0.00
19,500.0	89.90	90.00	10,779.8	-568.1	8,841.6	8,841.6	0.00	0.00	0.00
19,600.0	89.90	90.00	10,780.0	-568.1	8,941.6	8,941.6	0.00	0.00	0.00
19,700.0	89.90	90.00	10,780.1	-568.1	9,041.6	9,041.6	0.00	0.00	0.00



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Measured Depth (usft)	Inclination (°)	Azimuth (°)	Vertical Depth (usft)	+N/-S (usft)	+E/-W (usft)	Vertical Section (usft)	Dogleg Rate (°/100usft)	Build Rate (°/100usft)	Turn Rate (°/100usft)
19,800.0	89.90	90.00	10,780.3	-568.1	9,141.6	9,141.6	0.00	0.00	0.00
19,900.0	89.90	90.00	10,780.5	-568.1	9,241.6	9,241.6	0.00	0.00	0.00
20,000.0	89.90	90.00	10,780.7	-568.1	9,341.6	9,341.6	0.00	0.00	0.00
20,100.0	89.90	90.00	10,780.8	-568.1	9,441.6	9,441.6	0.00	0.00	0.00
20,200.0	89.90	90.00	10,781.0	-568.1	9,541.6	9,541.6	0.00	0.00	0.00
20,300.0	89.90	90.00	10,781.2	-568.1	9,641.6	9,641.6	0.00	0.00	0.00
20,400.0	89.90	90.00	10,781.4	-568.1	9,741.6	9,741.6	0.00	0.00	0.00
20,500.0	89.90	90.00	10,781.5	-568.1	9,841.6	9,841.6	0.00	0.00	0.00
20,600.0	89.90	90.00	10,781.7	-568.1	9,941.6	9,941.6	0.00	0.00	0.00
20,708.4	89.90	90.00	10,781.9	-568.1	10,050.0	10,050.0	0.00	0.00	0.00
5300 41-18 14BX PBHL									

Design Targets									
Target Name - hit/miss target - Shape	Dip Angle (°)	Dip Dir. (°)	TVD (usft)	+N/-S (usft)	+E/-W (usft)	Northing (usft)	Easting (usft)	Latitude	Longitude
5300 41-18 14BX PBHL - plan misses target center by 0.1usft at 20708.4usft MD (10781.9 TVD, -568.1 N, 10050.0 E) - Point	0.00	0.00	10,781.9	-568.0	10,050.0	404,098.33	1,219,984.51	48° 4' 1.268 N	103° 33' 43.975 W

Casing Points				
Measured Depth (usft)	Vertical Depth (usft)	Name	Casing Diameter (")	Hole Diameter (")
11,037.0	10,765.0	7"	7	8-3/4



Ryan Directional Services
Planning Report



Database:	EDM 5000.1 Single User Db	Local Co-ordinate Reference:	Well Kline Federal 5300 41-18 14BX
Company:	Oasis Petroleum	TVD Reference:	WELL @ 2082.0usft (Original Well Elev)
Project:	Indian Hills	MD Reference:	WELL @ 2082.0usft (Original Well Elev)
Site:	153N-100W-17/18	North Reference:	True
Well:	Kline Federal 5300 41-18 14BX	Survey Calculation Method:	Minimum Curvature
Wellbore:	Kline Federal 5300 41-18 14BX		
Design:	Design #2		

Formations						
Measured Depth (usft)	Vertical Depth (usft)	Name	Lithology	Dip (°)	Dip Direction (°)	
1,972.0	1,972.0	Pierre				
4,646.1	4,646.0	Greenhorn				
5,057.1	5,057.0	Mowry				
5,471.1	5,471.0	Dakota				
6,486.2	6,486.0	Rierdon				
6,814.2	6,814.0	Dunham Salt				
6,929.2	6,929.0	Dunham Salt Base				
7,024.2	7,024.0	Spearfish				
7,283.2	7,283.0	Pine Salt				
7,318.2	7,318.0	Pine Salt Base				
7,374.2	7,374.0	Opeche Salt				
7,403.2	7,403.0	Opeche Salt Base				
7,605.2	7,605.0	Broom Creek (Top of Minnelusa Gp.)				
7,684.2	7,684.0	Amsden				
7,853.2	7,853.0	Tyler				
8,047.2	8,047.0	Otter (Base of Minnelusa Gp.)				
8,390.2	8,390.0	Kibbey Lime				
8,542.2	8,542.0	Charles Salt				
9,159.2	9,159.0	UB				
9,238.2	9,238.0	Base Last Salt				
9,301.2	9,301.0	Ratcliffe				
9,454.2	9,454.0	Mission Canyon				
10,022.2	10,022.0	Lodgepole				
10,209.2	10,209.0	Lodgepole Fracture Zone				
10,843.5	10,726.0	False Bakken				
10,867.6	10,735.0	Upper Bakken				
10,921.9	10,751.0	Middle Bakken				
10,968.7	10,760.0	Middle Bakken Sand Target				
13,888.8	10,770.0	Base Middle Bakken Sand Target				

Plan Annotations				
Measured Depth (usft)	Vertical Depth (usft)	Local Coordinates		Comment
		+N/-S (usft)	+E/-W (usft)	
10,287.8	10,287.6	-50.0	0.0	KOP Build 12°/100'
11,036.9	10,765.0	-263.8	426.0	EOC

**Oasis Petroleum
Well Summary
Kline Federal 5300 41-18 14BX
Section 18 T153N R100W
McKenzie County, ND**

SURFACE CASING AND CEMENT DESIGN

Size	Interval	Weight	Grade	Coupling	I.D.	Drift	Make-up Torque (ft-lbs)		
							Minimum	Optimum	Max
13-3/8"	0' to 2,072'	54.5	J-55	STC	12.615"	12.459"	4,100	5,470	6,840

Interval	Description	Collapse	Burst	Tension
		(psi) a	(psi) b	(1000 lbs) c
0' to 2,072'	13-3/8", 54.5#, J-55, STC, 8rd	1130 / 1.17	2730 / 2.82	514 / 2.61

API Rating & Safety Factor

- a) Based on full casing evacuation with 9 ppg fluid on backside (2,072' setting depth).
- b) Burst pressure based on 9 ppg fluid with no fluid on backside (2,072' setting depth).
- c) Based on string weight in 9 ppg fluid at 2,072' TVD plus 100k# overpull. (Buoyed weight equals 97k lbs)

Cement volumes are based on 13-3/8" casing set in 17-1/2" hole with 50% excess to circulate cement back to surface.
Mix and pump the following slurry.

Pre-flush (Spacer): **20 bbls** fresh water

Lead Slurry: **635 sks** (328 bbls) 2.9 yield conventional system with 94 lb/sk cement, .25 lb/sk D130 Lost Circulation Control Agent, 2% CaCl₂, 4% D079 Extender and 2% D053 Expanding Agent.

Tail Slurry: **349 sks** (72 bbls) 1.16 yield conventional system with 94 lb/sk cement, .25% CaCl₂ and 0.25 lb/sk Lost Circulation Control Agent

**Oasis Petroleum
Well Summary
Kline Federal 5300 41-18 14BX
Section 18 T153N R100W
McKenzie County, ND**

SURFACE CASING AND CEMENT DESIGN

Size	Interval	Weight	Grade	Coupling	I.D.	Drift	Make-up Torque (ft-lbs)		
							Minimum	Optimum	Max
13-3/8"	0' to 2,072'	54.5	J-55	STC	12.615"	12.459"	4,100	5,470	6,840

Interval	Description	Collapse	Burst	Tension
		(psi) a	(psi) b	(1000 lbs) c
0' to 2,072'	13-3/8", 54.5#, J-55, STC, 8rd	1130 / 1.17	2730 / 2.82	514 / 2.61

API Rating & Safety Factor

- a) Based on full casing evacuation with 9 ppg fluid on backside (2,072' setting depth).
- b) Burst pressure based on 9 ppg fluid with no fluid on backside (2,072' setting depth).
- c) Based on string weight in 9 ppg fluid at 2,072' TVD plus 100k# overpull. (Buoyed weight equals 97k lbs.)

Cement volumes are based on 13-3/8" casing set in 17-1/2" hole with 50% excess to circulate cement back to surface.
Mix and pump the following slurry.

Pre-flush (Spacer): **20 bbls** fresh water

Lead Slurry: **635 sks** (328 bbls) 2.9 yield conventional system with 94 lb/sk cement, .25 lb/sk D130 Lost Circulation Control Agent, 2% CaCl₂, 4% D079 Extender and 2% D053 Expanding Agent.

Tail Slurry: **349 sks** (72 bbls) 1.16 yield conventional system with 94 lb/sk cement, .25% CaCl₂ and 0.25 lb/sk Lost Circulation Control Agent

**Oasis Petroleum
Well Summary
Kline Federal 5300 31-18 14BX
Section 18 T153N R100W
McKenzie County, ND**

CONTINGENCY SURFACE CASING AND CEMENT DESIGN

Size	Interval	Weight	Grade	Coupling	I.D.	Drift	Make-up Torque (ft-lbs)		
							Minimum	Optimum	Max
9-5/8"	0' to 6,101'	40	HCL-80	LTC	8.835"	8.75"	5,450	7,270	9,090

Interval	Description	Collapse	Burst	Tension
		(psi) a	(psi) b	(1000 lbs) c
0' to 6,101'	9-5/8", 40#, HCL-80, LTC, 8rd	4230 / 2.13	5750 / 3.72	837 / 2.78

API Rating & Safety Factor

- a) Collapse pressure based on 11.5 ppg fluid on the backside and 9 ppg fluid inside of casing.
- b) Burst pressure calculated from a gas kick coming from the production zone (Bakken Pool) at 9,000 psi and a subsequent breakdown at the 9-5/8" shoe, based on a 13.5#/ft fracture gradient. Backup of 9 ppg fluid.
- c) Yield based on string weight in 10 ppg fluid, (207k lbs buoyed weight) plus 100k lbs overpull

Cement volumes are based on 9-5/8" casing set in 12-1/4" hole with 10% excess in OH and 0% excess inside surface casing. TOC at surface.

Pre-flush (Spacer): **20 bbls** Chem wash

Lead Slurry: **598 sks** (309 bbls) Conventional system with 75 lb/sk cement, 0.5 lb/sk lost circulation, 10% expanding agent, 2% extender, 2% CaCl₂, 0.2% anti-foam and 0.4% fluid loss agent

Tail Slurry: **349 sks** (72 bbls) Conventional system with 94 lb/sk cement, 0.3% anti-settling agent, 0.3% fluid loss agent, 0.3 lb/sk lost circulation control agent, 0.2% anti-foam and 0.1% retarder.

**Oasis Petroleum
Well Summary
Kline Federal 5300 41-18 14BX
Section 18 T153N R100W
McKenzie County, ND**

INTERMEDIATE CASING AND CEMENT DESIGN

Size	Interval	Weight	Grade	Coupling	I.D.	Drift	Make-up Torque (ft-lbs)		
							Minimum	Optimum	Max
7"	0' – 6,664'	29	P-110	LTC	6.184"	6.059"	5,980	7,970	8,770
7"	6,664' – 10,288'	32	HCP-110	LTC	6.094"	6.000***	6,730	8,970	9,870
7"	10,288' – 11,037'	29	P-110	LTC	6.184"	6.059"	5,980	7,970	8,770

***Special drift

Interval	Length	Description	Collapse	Burst	Tension
			(psi) a	(psi) b	(1000 lbs) c
0' – 6,664'	6,664'	7", 29#, P-110, LTC, 8rd	8530 / 2.46*	11220 / 1.19	797 / 2.09
6,664' – 10,288'	3,624'	7", 32#, HCP-110, LTC, 8rd	11820 / 2.20*	12460 / 1.29	
6,664' – 10,288'	3,624'	7", 32#, HCP-110, LTC, 8rd	11820 / 1.07**	12460 / 1.29	
10,288' – 11,037'	749'	7", 29 lb, P-110, LTC, 8rd	8530 / 1.52*	11220 / 1.16	

API Rating & Safety Factor

- a) *Assume full casing evacuation with 10 ppg fluid on backside. **Assume full casing evacuation with 1.2 psi/ft equivalent fluid gradient across salt intervals.
- b) Burst pressure based on 9000 psig max press for stimulation plus 10.2 ppg fluid in casing and 9 ppg fluid on backside-to 10,765' TVD
- c) Based on string weight in 10 ppg fluid, (280k lbs buoyed weight) plus 100k

Cement volumes are estimates based on 7" casing set in an 8-3/4" hole with 30% excess.

Pre-flush (Spacer): **50 bbls Saltwater**
 40 bbls Weighted MudPush Express

Lead Slurry: **157 sks** (72 bbls) 2.21 yield conventional system with 47 lb/sk cement, 37 lb/sk D035 extender, 3.0% KCl, 3.0% D154 extender, 0.3% D208 viscosifier, 0.07% retarder, 0.2% anti-foam, 0.5 lb/sk, D130 LCM.

Tail Slurry: **608 sks** (168 bbls) 1.54 yield conventional system with 94 lb/sk cement, 3.0% KCl, 35.0% Silica, 0.5% retarder, 0.2% fluid loss, 0.2% anti-foam and 0.5 lb/sk LCM.

**Oasis Petroleum
Well Summary
Kline Federal 5300 41-18 14BX
Section 18 T153N R100W
McKenzie County, ND**

PRODUCTION LINER

Size	Interval	Weight	Grade	Coupling	I.D.	Drift	Estimated Torque
4-1/2"	10,238' – 20,708'	13.5	P-110	BTC	3.92"	3.795"	4,500

Interval	Description	Collapse	Burst	Tension
		(psi) a	(psi) b	(1000 lbs) c
10,238' – 20,708'	4-1/2", 13.5 lb, P-110, BTC, 8rd	10680 / 2.00	12410 / 1.28	443 / 2.00

API Rating & Safety Factor

- a) Based on full casing evacuation with 9.5 ppg fluid on backside @ 10,782' TVD.
Burst pressure based on 9000 psi treating pressure with 10.2 ppg internal fluid gradient and 9 ppg external
- b) fluid gradient @ 10,782' TVD.
- c) Based on string weight in 9.5 ppg fluid (Buoyed weight: 121k lbs.) plus 100k lbs overpull.

Oasis Petroleum does not use Diesel Fuel, as defined by the US EPA in the list below, in our hydraulic fracture operations.

68334-30-5 (Primary Name: Fuels, diesel)
68476-34-6 (Primary Name: Fuels, diesel, No. 2)
68476-30-2 (Primary Name: Fuel oil No. 2)
68476-31-3 (Primary Name: Fuel oil, No. 4)
8008-20-6 (Primary Name: Kerosene)

OPERATOR			WELL NAME		COUNTY/STATE	
WELL TYPE			RIG		GROUND ELEV:	
LOCATION			Surface Location (survey plat):		Sub Height:	
EST. T.D.			TOTAL LATERAL		KB ELEV:	
Oasis Petroleum			Kline Federal 5300 41-18 14BX		Mckenzie Co. ND	
Horizontal Middle Bakken			237' pad		0	
SWSW 18-153N-100W			388' lat		2057' Finished Pad Elev	
20,708'			237' pad		2082'	
Based on 2,082' KB(est)			2082'		2082'	
PROGNOSIS:			LOGS:		Interval	
MARKER			DEPTH (Surf Loc)		DATUM (Surf Loc)	
Pierre			1,972		110	
Greenhorn			4,646		-2564	
Mowry			5,057		-2975	
Dakota			5,471		-3389	
Rierdon			6,486		-4404	
Dunham Salt			6,814		-4732	
Dunham Salt Base			6,929		-4847	
Spearfish			7,024		-4942	
Pine Salt			7,283		-5201	
Pine Salt Base			7,318		-5236	
Opeche Salt			7,374		-5292	
Opeche Salt Base			7,403		-5321	
Broom Creek (Top of Minnelusa Gp)			7,605		-5523	
Amsden			7,684		-5602	
Tyler			7,853		-5771	
Otter (Base of Minnelusa Gp)			8,047		-5965	
Kibbey Lime			8,390		-6308	
Charles Salt			8,542		-6460	
UB			9,159		-7077	
Base Last Salt			9,238		-7156	
Ratcliffe			9,301		-7219	
Mission Canyon			9,454		-7372	
Lodgepole			10,022		-7940	
Lodgepole Fracture Zone			10,209		-8127	
False Bakken			10,726		-8644	
Upper Bakken			10,735		-8653	
Middle Bakken			10,751		-8669	
Middle Bakken Sand Target			10,760		-8678	
Base Middle Bakken Sand Target			10,770		-8688	
Lower Bakken			10,785		-8703	
Three Forks			10,813		-8731	
Dip Rate:			0.1			
Max. Anticipated BHP:			4073			
MUD:			Interval		Type	
Surface:			0' -		2,072' FW/Gel - Lime Sweeps	
Intermediate:			2,072' -		11,037' Invert	
Lateral:			11,037' -		20,708' Salt Water	
Casing:			Size		WT ppf	
Surface:			13-3/8"		54 5#	
Dakota Contingency:			9-5/8"		40#	
Intermediate:			7"		29/32#	
Production Liner:			4.5"		13.5#	
PROBABLE PLUGS, IF REQ'D:			Hole		Depth	
Surface:			17-1/2"		2,072'	
Dakota Contingency:			12-1/4"		6,101'	
Intermediate:			8-3/4"		11,037'	
Production Liner:			8"		20,708'	
Cement			To Surface		3971	
WOC			12		100' into Pierre	
Remarks			Circ Mud Tanks		Circ Mud Tanks	
Circ Mud Tanks			Circ Mud Tanks		Circ Mud Tanks	
OTHER:			MD		TVD	
Surface:			2,072		2,072	
KOP:			10,288'		10,288'	
EOC:			11,037'		10,765'	
Casing Point:			11,037'		10,765'	
Middle Bakken Lateral TD:			20,708'		10,782'	
FNL/FSL			368' FSL		237' FSL	
FEL/FWL			237' FSL		237' FSL	
S-T-R			SEC 18-T153N-R100W		SEC 18-T153N-R100W	
AZI			Survey Company:		Build Rate:	
116 65			116 65		116 65	
90 00			90 00		90 00	
Comments:			Request a Sundry for an Open Hole Log Waiver			
Exception well:			Oasis Petroleum's Kline Federal 5300 11-18H (153N 100W 18 NW NW)			
35 packers, 35 sleeves, no frac string						
Oasis Petroleum does not use Diesel Fuel, as defined by the US EPA in the list below, in our hydraulic fracture operations.						
68334-30-5 (Primary Name: Fuels, diesel) 68476-34-6 (Primary Name: Fuels, diesel, No. 2) 68476-30-2 (Primary Name: Fuel oil No. 2)						
68476-31-3 (Primary Name: Fuel oil, No. 4) 8008-20-6 (Primary Name: Kerosene)						
Geology: M. Steed 4/3/2014						
Engineering: hbader rpm 7/20/14						



Oil and Gas Division

Lynn D. Helms - Director

Bruce E. Hicks - Assistant Director

Department of Mineral Resources

Lynn D. Helms - Director

North Dakota Industrial Commission

www.dmr.nd.gov/oilgas

28655

BRANDI TERRY
OASIS PETROLEUM NORTH AMERICA LLC
1001 FANNIN STE 1500
HOUSTON, TX 77002 USA

Date: 6/23/2014

RE: CORES AND SAMPLES

Well Name: **KLINE FEDERAL 5300 41-18 14BX** Well File No.: **28655**
Location: **LOT4 18-153-100** County: **MCKENZIE**
Permit Type: **Development - HORIZONTAL**
Field: **BAKER** Target Horizon: **BAKKEN**

Dear BRANDI TERRY:

North Dakota Century Code Section 38-08-04 provides for the preservation of cores and samples and their shipment to the State Geologist when requested. The following is required on the above referenced well:

- 1) All cores, core chips and samples must be submitted to the State Geologist as provided for under North Dakota Century Code: Section 38-08-04 and North Dakota Administrative Code: Section 43-02-03-38.1.
- 2) Samples: The Operator is to begin collecting sample drill cuttings no lower than the:
Base of the Last Charles Salt
 - Sample cuttings shall be collected at:
 - o 30' maximum intervals through all vertical and build sections.
 - o 100' maximum intervals through any horizontal sections.
 - Samples must be washed, dried, placed in standard sample envelopes (3" x 4.5"), packed in the correct order into standard sample boxes (3.5" x 5.25" x 15.25").
 - Samples boxes are to be carefully identified with a label that indicates the operator, well name, well file number, American Petroleum Institute (API) number, location and depth of samples; and forwarded in to the state core and sample library within 30 days of the completion of drilling operations.
- 3) Cores: Any cores cut shall be preserved in correct order, boxed in standard core boxes (4.5", 4.5", 35.75"), and the entire core forwarded to the state core and samples library within 180 days of completion of drilling operations. Any extension of time must have approval on a Form 4 Sundry Notice.

All cores, core chips, and samples must be shipped, prepaid, to the state core and samples library at the following address:

**ND Geological Survey Core Library
2835 Campus Road, Stop 8156
Grand Forks, ND 58202**

North Dakota Century Code Section 38-08-16 allows for a civil penalty for any violation of Chapter 38 08 not to exceed \$12,500 for each offense, and each day's violation is a separate offense.

Sincerely


Stephen Fried
Geologist

**SUNDRY NOTICES AND REPORTS ON WELLS - FORM 1**

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5749 (09-2006)



Well File No.

28655

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

☒ Notice of Intent	Approximate Start Date October 1, 2014
☐ Report of Work Done	Date Work Completed
☐ Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03.	Approximate Start Date

☐ Drilling Prognosis	☐ Spill Report
☐ Redrilling or Repair	☐ Shooting
☐ Casing or Liner	☐ Acidizing
☐ Plug Well	☐ Fracture Treatment
☐ Supplemental History	☐ Change Production Method
☐ Temporarily Abandon	☐ Reclamation
☒ Other	Waiver to rule Rule 43-02-03-31

Well Name and Number Kline Federal 5300 41-18 14BX					
Footages		Qtr-Qtr	Section	Township	Range
368 F S L 237 F W L		1034	18	153 N	100 W
Field		Pool		County	
		Bakken		McKenzie	

24-HOUR PRODUCTION RATE

Before		After	
Oil	Bbls	Oil	Bbls
Water	Bbls	Water	Bbls
Gas	MCF	Gas	MCF

Name of Contractor(s)			
Address	City	State	Zip Code

DETAILS OF WORK

Oasis Petroleum respectfully requests a waiver to Rule 43-02-03-31 in regards to running open hole logs for the above referenced well. Justification for this request is as follows:

The Oasis Petroleum's #20275 **Kline Federal 5300 11-18H** (153N 100W 18 NW NW) located within a mile of the subject well

If this exception is approved, Oasis Petroleum will run a CBL on the intermediate string, and we will also run GR to surface. Oasis Petroleum will also submit two digital copies of each cased hole log and a copy of the mud log containing MWD gamma ray.

Company Oasis Petroleum North America LLC		Telephone Number 281-404-9563	
Address 1001 Fannin, Suite 1500			
City Houston		State TX	Zip Code 77002
Signature 		Printed Name Heather McCowan	
Title Regulatory Assistant		Date May 29, 2014	
Email Address hmccowan@oasispetroleum.com			

FOR STATE USE ONLY

☐ Received	☒ Approved
Date 6-17-2014	
By 	
Title Stephen Fried Geologist	

**SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4**

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5749 (09-2006)

Well File No.
28655

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

☒ Notice of Intent	Approximate Start Date October 1, 2014
☐ Report of Work Done	Date Work Completed
☐ Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03.	Approximate Start Date

☐ Drilling Prognosis	☐ Spill Report
☐ Redrilling or Repair	☐ Shooting
☐ Casing or Liner	☐ Acidizing
☐ Plug Well	☐ Fracture Treatment
☐ Supplemental History	☐ Change Production Method
☐ Temporarily Abandon	☐ Reclamation
☒ Other	Suspension of Drilling

Well Name and Number Kline Federal 5300 41-18 14BX					
Footages	Qtr-Qtr	Section	Township	Range	
368 F S L 237 F WL	SWSW	18	153 N	100 W	
Field	Pool	County			
	Bakken	McKenzie			

24-HOUR PRODUCTION RATE			
Before		After	
Oil	Bbls	Oil	Bbls
Water	Bbls	Water	Bbls
Gas	MCF	Gas	MCF

Name of Contractor(s) Advanced Energy Services			
Address	City	State	Zip Code

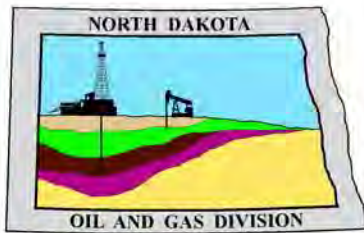
DETAILS OF WORK

Oasis Petroleum North America LLC requests permission for suspension of drilling for up to 90 days for the referenced well under NDAC 43-02-03-55. Oasis Petroleum North America LLC intends to drill the surface hole with freshwater based drilling mud and set surface casing with a small drilling rig and move off within 3 to 5 days. The casing will be set at a depth pre-approved by the NDIC per the Application for Permit to Drill NDAC 43-02-03-21. No saltwater will be used in the drilling and cementing operations of the surface casing. Once the surface casing is cemented, a plug or mechanical seal will be placed at the top of the casing to prevent any foreign matter from getting into the well. A rig capable of drilling to TD will move onto the location within the 90 days previously outlined to complete the drilling and casing plan as per the APD. The undersigned states that this request for suspension of drilling operations in accordance with the Subsection 4 of Section 43-02-03-55 of the NDAC, is being requested to take advantage of the cost savings and time savings of using an initial rig that is smaller than the rig necessary to drill a well to total depth but is not intended to alter or extend the terms and conditions of, or suspend any obligation under, any oil and gas lease with acreage in or under the spacing or drilling unit for the above-referenced well. Oasis Petroleum North America LLC understands NDAC 43-02-03-31 requirements regarding confidentiality pertaining to this permit. The drilling pit will be fenced immediately after construction if the well pad is located in a pasture (NDAC 43-02-03-19 & 19.1). Oasis Petroleum North America LL will plug and abandon the well and reclaim the well site if the well is not drilled by the larger rotary rig within 90 days after spudding the well with the smaller drilling rig.

Notify NDIC inspector **Richard Dunn** at 701-770-3554 with spud and TD info.

Company Oasis Petroleum North America LLC		Telephone Number (281) 404-9563	
Address 1001 Fannin, Suite 1500			
City Houston	State TX	Zip Code 77002	
Signature <i>Heather McCowan</i>	Printed Name Heather McCowan		
Title Regulatory Assistant	Date May 29, 2014		
Email Address hmccowan@oasispetroleum.com			

FOR STATE USE ONLY	
☐ Received	☒ Approved
Date	6/17/14
By	Nathaniel Erbele
Title	Petroleum Resource Specialist



Oil and Gas Division

Lynn D. Helms - Director

Bruce E. Hicks - Assistant Director

Department of Mineral Resources

Lynn D. Helms - Director

North Dakota Industrial Commission

www.oilgas.nd.gov

June 17, 2014

Heather McCowan
Regulatory Assistant
OASIS PETROLEUM NORTH AMERICA LLC
1001 Fannin Suite 1500
Houston, TX 77002

**RE: HORIZONTAL WELL
KLINE FEDERAL 5300 41-18 14BX
LOT4 Section 18-153N-100W
McKenzie County
Well File # 28655**

Dear Heather:

Pursuant to Commission Order No. 23752, approval to drill the above captioned well is hereby given. The approval is granted on the condition that all portions of the well bore not isolated by cement shall be located approximately down the east-west axis (**within 250' of the section line** per Commission policy) and no closer than **200 feet** to the east or west boundary within the 2560 acre spacing unit consisting of Sections 17, 18, 19, & 20 T153N R100W.

PERMIT STIPULATIONS: Effective June 1, 2014, a covered leak-proof container (with placard) for filter sock disposal must be maintained on the well site beginning when the well is spud, and must remain on-site during clean-out, completion, and flow-back whenever filtration operations are conducted. Due to the proximity of Lake Sakakawea to the well site, a dike is required surrounding the entire location. OASIS PETRO NO AMER must contact NDIC Field Inspector Richard Dunn at 701-770-3554 prior to location construction.

Drilling pit

NDAC 43-02-03-19.4 states that "a pit may be utilized to bury drill cuttings and solids generated during well drilling and completion operations, providing the pit can be constructed, used and reclaimed in a manner that will prevent pollution of the land surface and freshwaters. Reserve and circulation of mud system through earthen pits are prohibited. All pits shall be inspected by an authorized representative of the director prior to lining and use. Drill cuttings and solids must be stabilized in a manner approved by the director prior to placement in a cuttings pit."

Form 1 Changes & Hard Lines

Any changes, shortening of casing point or lengthening at Total Depth must have prior approval by the NDIC. The proposed directional plan is at a legal location. The minimum legal coordinate from the well head at casing point is: 118' south. Also, based on the azimuth of the proposed lateral the maximum legal coordinate from the well head is: 10052' east.

Location Construction Commencement (Three Day Waiting Period)

Operators shall not commence operations on a drill site until the 3rd business day following publication of the approved drilling permit on the NDIC - OGD Daily Activity Report. If circumstances require operations to commence before the 3rd business day following publication on the Daily Activity Report, the waiting period may be waived by the Director. Application for a waiver must be by sworn affidavit providing the information necessary to evaluate the extenuating circumstances, the factors of NDAC 43-02-03-16.2 (1), (a)-(f), and any other information that would allow the Director to conclude that in the event another owner seeks revocation of the drilling permit, the applicant should retain the permit.

Permit Fee & Notification

Payment was received in the amount of \$100 via credit card .The permit fee has been received. It is requested that notification be given immediately upon the spudding of the well. This information should be relayed to the Oil & Gas Division, Bismarck, via telephone. The following information must be included: Well name, legal location, permit number, drilling contractor, company representative, date and time of spudding. Office hours are 8:00 a.m. to 12:00 p.m. and 1:00 p.m. to 5:00 p.m. Central Time. Our telephone number is (701) 328-8020, leave a message if after hours or on the weekend.

Survey Requirements for Horizontal, Horizontal Re-entry, and Directional Wells

NDAC Section 43-02-03-25 (Deviation Tests and Directional Surveys) states in part (that) the survey contractor shall file a certified copy of all surveys with the director free of charge within thirty days of completion. Surveys must be submitted as one electronic copy, or in a form approved by the director. However, the director may require the directional survey to be filed immediately after completion if the survey is needed to conduct the operation of the director's office in a timely manner. Certified surveys must be submitted via email in one adobe document, with a certification cover page to certsurvey@nd.gov.

Survey points shall be of such frequency to accurately determine the entire location of the well bore.

Specifically, the Horizontal and Directional well survey frequency is 100 feet in the vertical, 30 feet in the curve (or when sliding) and 90 feet in the lateral.

Surface casing cement

Tail cement utilized on surface casing must have a minimum compressive strength of 500 psi within 12 hours, and tail cement utilized on production casing must have a minimum compressive strength of 500 psi before drilling the plug or initiating tests.

Logs

NDAC Section 43-02-03-31 requires the running of (1) a suite of open hole logs from which formation tops and porosity zones can be determined, (2) a Gamma Ray Log run from total depth to ground level elevation of the well bore, and (3) a log from which the presence and quality of cement can be determined (Standard CBL or Ultrasonic cement evaluation log) in every well in which production or intermediate casing has been set, this log must be run prior to completing the well. All logs run must be submitted free of charge, as one digital TIFF (tagged image file format) copy and one digital LAS (log ASCII) formatted copy. Digital logs may be submitted on a standard CD, DVD, or attached to an email sent to digitallogs@nd.gov

Thank you for your cooperation.

Sincerely,

Nathaniel Erbele

Petroleum Resource Specialist



APPLICATION FOR PERMIT TO DRILL HORIZONTAL WELL - FORM 1H

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 54269 (08-2005)

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

Type of Work New Location	Type of Well Oil & Gas	Approximate Date Work Will Start 10 / 1 / 2013	Confidential Status No
Operator OASIS PETROLEUM NORTH AMERICA LLC			Telephone Number 281-404-9563
Address 1001 Fannin Suite 1500		City Houston	State TX Zip Code 77002

☒ Notice has been provided to the owner of any permanently occupied dwelling within 1,320 feet.

☒ This well is not located within five hundred feet of an occupied dwelling.

WELL INFORMATION (If more than one lateral proposed, enter data for additional laterals on page 2)

Well Name KLINE FEDERAL				Well Number 5300 41-18 14BX			
Surface Footages 368 F S L 237 F W L		Qtr-Qtr LOT4	Section 18	Township 153 N	Range 100 W	County McKenzie	
Longstring Casing Point Footages 105 F S L 663 F W L		Qtr-Qtr LOT4	Section 18	Township 153 N	Range 100 W	County McKenzie	
Longstring Casing Point Coordinates From Well Head 263 S From WH 426 E From WH		Azimuth 116.6 °	Longstring Total Depth 11036 Feet MD 10765 Feet TVD				
Bottom Hole Footages From Nearest Section Line 200 F N L 202 F E L		Qtr-Qtr LOT2	Section 20	Township 153 N	Range 100 W	County McKenzie	
Bottom Hole Coordinates From Well Head 568 S From WH 10050 E From WH		KOP Lateral 1 10288 Feet MD	Azimuth Lateral 1 90 °		Estimated Total Depth Lateral 1 20708 Feet MD 10781 Feet TVD		
Latitude of Well Head 48 ° 04 ' 6.90 "		Longitude of Well Head -103 ° 36 ' 11.95 "		NAD Reference NAD83	Description of (Subject to NDIC Approval) Spacing Unit: Sections 18 & 17 T153N R100W		
Ground Elevation 2056 Feet Above S.L.	Acres in Spacing/Drilling Unit 2560	Spacing/Drilling Unit Setback Requirement 500 Feet N/S 200 Feet E/W			Industrial Commission Order 23752		
North Line of Spacing/Drilling Unit 10544 Feet		South Line of Spacing/Drilling Unit 10489 Feet		East Line of Spacing/Drilling Unit 10524 Feet		West Line of Spacing/Drilling Unit 10520 Feet	
Objective Horizons Bakken						Pierre Shale Top 1972	
Proposed Surface Casing	Size 9 - 5/8 "	Weight 36 Lb./Ft.	Depth 2072 Feet	Cement Volume 749 Sacks	NOTE: Surface hole must be drilled with fresh water and surface casing must be cemented back to surface.		
Proposed Longstring Casing	Size 7 - "	Weight(s) 29/32 Lb./Ft.	Longstring Total Depth 11036 Feet MD 10765 Feet TVD		Cement Volume 752 Sacks	Cement Top 3971 Feet	Top Dakota Sand 5471 Feet
Base Last Charles Salt (If Applicable) 9238 Feet		NOTE: Intermediate or longstring casing string must be cemented above the top Dakota Group Sand.					
Proposed Logs Triple Combo: KOP to Kibby GR/Res to BSC GR to surf CND through the Dakota							
Drilling Mud Type (Vertical Hole - Below Surface Casing) Invert				Drilling Mud Type (Lateral) Salt Water Gel			
Survey Type in Vertical Portion of Well MWD Every 100 Feet		Survey Frequency: Build Section 30 Feet		Survey Frequency: Lateral 90 Feet		Survey Contractor Ryan	

NOTE: A Gamma Ray log must be run to ground surface and a CBL must be run on intermediate or longstring casing string if set.

Surveys are required at least every 30 feet in the build section and every 90 feet in the lateral section of a horizontal well. Measurement inaccuracies are not considered when determining compliance with the spacing/drilling unit boundary setback requirement except in the following scenarios: 1) When the angle between the well bore and the respective boundary is 10 degrees or less; or 2) If Industry standard methods and equipment are not utilized. Consult the applicable field order for exceptions.

If measurement inaccuracies are required to be considered, a 2° MWD measurement inaccuracy will be applied to the horizontal portion of the well bore. This measurement inaccuracy is applied to the well bore from KOP to TD.

REQUIRED ATTACHMENTS: Certified surveyor's plat, horizontal section plat, estimated geological tops, proposed mud/cementing plan, directional plot/plan, \$100 fee.
See Page 2 for Comments section and signature block.

COMMENTS, ADDITIONAL INFORMATION, AND/OR LIST OF ATTACHMENTS

Documents forwarded by email: Drill plan with drilling fluids, Well Summary with casing/cement plans, Directional Plan & Plot, Plats

Lateral 2

KOP Lateral 2	Azimuth Lateral 2	Estimated Total Depth Lateral 2		KOP Coordinates From Well Head		
Feet MD	°	Feet MD	Feet TVD	From WH		From WH
Formation Entry Point Coordinates From Well Head		Bottom Hole Coordinates From Well Head				
From WH		From WH		From WH		From WH
KOP Footages From Nearest Section Line		Qtr-Qtr	Section	Township	Range	County
F	L			N	W	
Bottom Hole Footages From Nearest Section Line		Qtr-Qtr	Section	Township	Range	County
F	L			N	W	

Lateral 3

KOP Lateral 3	Azimuth Lateral 3	Estimated Total Depth Lateral 3		KOP Coordinates From Well Head		
Feet MD	°	Feet MD	Feet TVD	From WH	From WH	
Formation Entry Point Coordinates From Well Head		Bottom Hole Coordinates From Well Head				
From WH	From WH	From WH	From WH			
KOP Footages From Nearest Section Line		Qtr-Qtr	Section	Township	Range	County
F	L	F	L	N	W	
Bottom Hole Footages From Nearest Section Line		Qtr-Qtr	Section	Township	Range	County
F	L	F	L	N	W	

Lateral 4

KOP Lateral 4 Feet MD	Azimuth Lateral 4 °	Estimated Total Depth Lateral 4 Feet MD Feet TVD		KOP Coordinates From Well Head From WH From WH	
Formation Entry Point Coordinates From Well Head From WH From WH		Bottom Hole Coordinates From Well Head From WH From WH			
KOP Footages From Nearest Section Line F L F L		Qtr-Qtr	Section	Township N	Range W County
Bottom Hole Footages From Nearest Section Line F L F L		Qtr-Qtr	Section	Township N	Range W County

Lateral 5

KOP Lateral 5	Azimuth Lateral 5	Estimated Total Depth Lateral 5		KOP Coordinates From Well Head		
Feet MD	°	Feet MD	Feet TVD	From WH	From WH	
Formation Entry Point Coordinates From Well Head		Bottom Hole Coordinates From Well Head				
From WH	From WH	From WH	From WH			
KOP Footages From Nearest Section Line		Qtr-Qtr	Section	Township	Range	County
F	L	F	L	N	W	
Bottom Hole Footages From Nearest Section Line		Qtr-Qtr	Section	Township	Range	County
F	L	F	L	N	W	

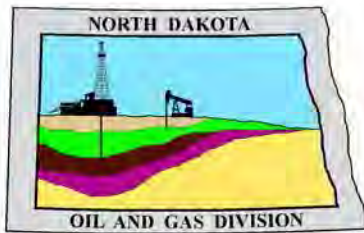
I hereby swear or affirm the information provided is true, complete and correct as determined from all available records.		Date 5 / 29 / 2014
ePermit	Printed Name Heather McCowan	Title Regulatory Assistant

FOR STATE USE ONLY

Permit and File Number 28655	API Number 33 - 053 - 06029
Field BAKER	
Pool BAKKEN	Permit Type DEVELOPMENT

FOR STATE USE ONLY

Date Approved 6 / 17 / 2014
By Nathaniel Erbele
Title Petroleum Resource Specialist



Oil and Gas Division

Lynn D. Helms - Director

Bruce E. Hicks - Assistant Director

Department of Mineral Resources

Lynn D. Helms - Director

North Dakota Industrial Commission

www.oilgas.nd.gov

April 9, 2014

**RE: Filter Socks and Other Filter Media
Leakproof Container Required
Oil and Gas Wells**

Dear Operator,

North Dakota Administrative Code Section 43-02-03-19.2 states in part that all waste material associated with exploration or production of oil and gas must be properly disposed of in an authorized facility in accord with all applicable local, state, and federal laws and regulations.

Filtration systems are commonly used during oil and gas operations in North Dakota. The Commission is very concerned about the proper disposal of used filters (including filter socks) used by the oil and gas industry.

Effective June 1, 2014, a container must be maintained on each well drilled in North Dakota beginning when the well is spud and must remain on-site during clean-out, completion, and flow-back whenever filtration operations are conducted. The on-site container must be used to store filters until they can be properly disposed of in an authorized facility. Such containers must be:

- leakproof to prevent any fluids from escaping the container
- covered to prevent precipitation from entering the container
- placard to indicate only filters are to be placed in the container

If the operator will not utilize a filtration system, a waiver to the container requirement will be considered, but only upon the operator submitting a Sundry Notice (Form 4) justifying their request.

As previously stated in our March 13, 2014 letter, North Dakota Administrative Code Section 33-20-02.1-01 states in part that every person who transports solid waste (which includes oil and gas exploration and production wastes) is required to have a valid permit issued by the North Dakota Department of Health, Division of Waste Management. Please contact the Division of Waste Management at (701) 328-5166 with any questions on the solid waste program. Note oil and gas exploration and production wastes include produced water, drilling mud, invert mud, tank bottom sediment, pipe scale, filters, and fly ash.

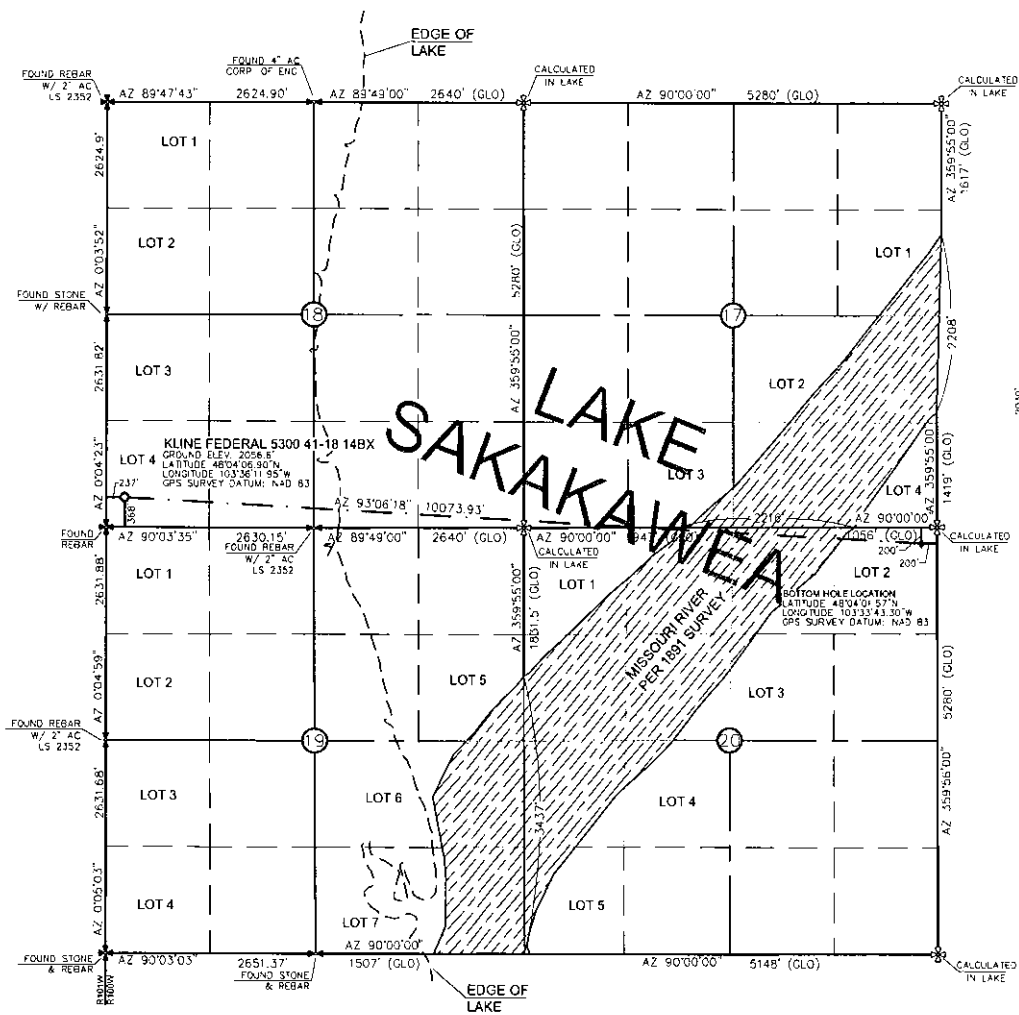
Thank you for your cooperation.

Sincerely,

Bruce E. Hicks

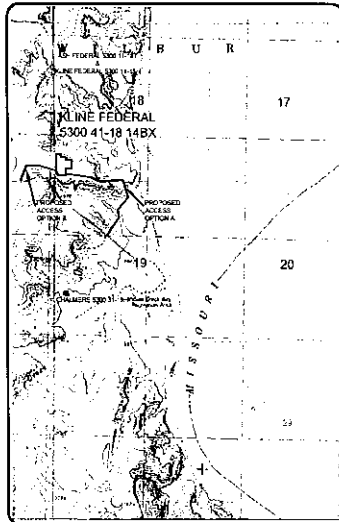
Assistant Director

WELL LOCATION PLAT
OASIS PETROLEUM NORTH AMERICA, LLC
 1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
 "KLINE FEDERAL 5300 41-18 148X"
 368 FEET FROM SOUTH LINE AND 237 FEET FROM WEST LINE
 SECTION 18, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA

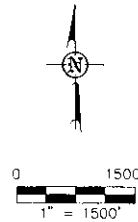


- ⊕ — MONUMENT — RECOVERED
- ⊗ — MONUMENT — NOT RECOVERED

VICINITY MAP



THIS DOCUMENT WAS ORIGINALLY ISSUED
 AND SEALED BY ROBERT L. PROOVE,
 PLS, REGISTRATION NUMBER 2884 ON
 4/15/14 AND THE ORIGINAL
 DOCUMENTS ARE STORED AT THE
 OFFICES OF INTERSTATE ENGINEERING,
 INC.



STAKED ON 4/15/14
 VERTICAL CONTROL DATUM WAS BASED UPON
 CONTROL POINT 4 WITH AN ELEVATION OF 2090.8'

THIS SURVEY AND PLAT IS BEING PROVIDED AT THE REQUEST
 OF ERIC BAYES OF OASIS PETROLEUM. I CERTIFY THAT THIS
 PLAT CORRECTLY REPRESENTS WORK PERFORMED BY ME OR
 UNDER MY SUPERVISION AND IS TRUE AND CORRECT TO THE
 BEST OF MY KNOWLEDGE AND BELIEF.

Robert L. Proove

ROBERT L. PROOVE 2884LS

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1/8

SHEET NO.



Professionals do not need, please do not.

Interstate Engineering, Inc.
 P.O. Box 648
 425 East Main Street
 Sidney, Montana 59270
 Ph (406) 433-5617
 Fax (406) 433-5618
 www.interstateeng.com

Other offices in Minnesota, North Dakota and South Dakota

OASIS PETROLEUM NORTH AMERICA, LLC
 WELL LOCATION PLAT
 SECTION 18, T153N, R100W
 MCKENZIE COUNTY, NORTH DAKOTA

Drawn By: B.H.H. Project No.: 514-09-001
 Checked By: B.H.H. Date: APRIL 2014

Revision	Date	By	Description

DRILLING PLAN										
OPERATOR	Oasis Petroleum					COUNTY/STATE	McKenzie Co., ND			
WELL NAME	Kline Federal 5300 41-18 14BX					RIG	0			
WELL TYPE	Horizontal Middle Bakken									
LOCATION	SWSW 18-153N-100W		Surface Location (survey plat): 368' fsl			237' fwl				
EST. T.D.	20,708'		GROUND ELEV:			2057		Finished Pad Elev.	Sub Height: 25	
TOTAL LATERA		9,671'		KB ELEV:			2082			
PROGNOSIS: Based on 2,082' KB(est)				LOGS: Type Interval						
MARKER DEPTH (Surf Loc) DATUM (Surf Loc)				OH Logs: Triple Combo KOP to Kibby (or min run of 1800' whichever is greater); GR/Res to BSC; GR to surf; CND through the Dakota						
				CBL/GR: Above top of cement/GR to base of casing						
				MWD GR: KOP to lateral TD						
Pierre NDIC MAP 1,972 110 Greenhorn 4,646 -2564 Mowry 5,057 -2975 Dakota 5,471 -3389 Rierdon 6,486 -4404 Dunham Salt 6,814 -4732 Dunham Salt Base 6,929 -4847 Spearfish 7,024 -4942 Pine Salt 7,283 -5201 Pine Salt Base 7,318 -5236 Opeche Salt 7,374 -5292 Opeche Salt Base 7,403 -5321 Broom Creek (Top of Minnelusa Gp.) 7,605 -5523 Amsden 7,684 -5602 Tyler 7,853 -5771 Otter (Base of Minnelusa Gp.) 8,047 -5965 Kibbey Lime 8,390 -6308 Charles Salt 8,542 -6460 UB 9,159 -7077 Base Last Salt 9,238 -7156 Ratcliffe 9,301 -7219 Mission Canyon 9,454 -7372 Lodgepole 10,022 -7940 Lodgepole Fracture Zone 10,209 -8127 False Bakken 10,726 -8644 Upper Bakken 10,735 -8653 Middle Bakken 10,751 -8669 Middle Bakken Sand Target 10,760 -8678 Base Middle Bakken Sand Target 10,770 -8688 Lower Bakken 10,785 -8703 Three Forks 10,813 -8731				DEVIATION: Surf: 3 deg. max., 1 deg / 100'; srvy every 500' Prod: 5 deg. max., 1 deg / 100'; srvy every 100'						
				DST'S: None planned						
Dip Rate: -0.1 Max. Anticipated BHP: 4673				CORES: None planned						
				MUDLOGGING: Two-Man: 8,342' ~200' above the Charles (Kibbey) to Casing point; Casing point to TD 30' samples at direction of wellsite geologist; 10' through target @ curve land						
				BOP: 11" 5000 psi blind, pipe & annular						
Surface Formation: Glacial till										
MUD:	Interval		Type		WT		Vis		WL	Remarks
Surface:	0' - 2,072'		FW/Gel - Lime Sweeps		8.4-9.0		28-32		NC	Circ Mud Tanks
Intermediate:	2,072' - 11,037'		Invert		9.5-10.4		40-50		30+HtHp	Circ Mud Tanks
Lateral:	11,037' - 20,708'		Salt Water		9.8-10.2		28-32		NC	Circ Mud Tanks
CASING:	Size	Wt ppf	Hole	Depth	Cement		WOC		Remarks	
Surface:	9-5/8"	36#	13-1/2"	2,072'	To Surface		12		100' into Pierre	
Intermediate:	7"	29/32#	8-3/4"	11,037'	3971		24		1500' above Dakota	
Production Liner:	4.5"	11.6#	6"	20,708'	TOL @ 10,238'				50' above KOP	
PROBABLE PLUGS, IF REQ'D:										
OTHER:		MD	TVD	FNL/FSL	FEL/FWL	S-T-R	AZI			
	Surface:	2,072	2,072	368' FSL	237' FWL	SEC 18-T153N-R100W	Survey Company:			
	KOP:	10,288'	10,288'	318' FSL	237' FWL	SEC 18-T153N-R100W	Build Rate: 12 deg /100'			
	EOC:	11,037'	10,765'	104' FSL	663' FWL	SEC 18-T153N-R100W				
	Casing Point:	11,037'	10,765'	104' FSL	663' FWL	SEC 18-T153N-R100W				
	Middle Bakken Lateral TD:	20,708'	10,782'	200' FNL	200' FEL	SEC 20-T153N-R100W				
Comments:										
Request a Sundry for an Open Hole Log Waiver										
Exception well: Oasis Petroleum's Kline Federal 5300 11-18H (153N 100W 18 NW NW)										
35 packers, 35 sleeves, no frac string										
Oasis Petroleum does not use Diesel Fuel, as defined by the US EPA in the list below, in our hydraulic fracture operations.										
68334-30-5 (Primary Name: Fuels, diesel) 68476-34-6 (Primary Name: Fuels, diesel, No. 2)68476-30-2 (Primary Name: Fuel oil No. 2)										
68476-31-3 (Primary Name: Fuel oil, No. 4)8008-20-6 (Primary Name: Kerosene)										
										
Geology: M. Steed 4/3/2014					Engineering: hbader rpm					

**Oasis Petroleum
Well Summary
Kline Federal 5300 41-18 14BX
Section 18 T153N R100W
McKenzie County, ND**

SURFACE CASING AND CEMENT DESIGN

Size	Interval	Weight	Grade	Coupling	I.D.	Drift	Make-up Torque (ft-lbs)		
							Minimum	Optimum	Max
9-5/8"	0' to 2,072'	36	J-55	LTC	8.921"	8.765"	3400	4530	5660

Interval	Description	Collapse	Burst	Tension	Cost per ft
		(psi) a	(psi) b	(1000 lbs) c	
0' to 2,072'	9-5/8", 36#, J-55, LTC, 8rd	2020 / 2.08	3520 / 3.63	453 / 2.76	

API Rating & Safety Factor

- a) Based on full casing evacuation with 9 ppg fluid on backside (2,072' setting depth).
- b) Burst pressure based on 9 ppg fluid with no fluid on backside (2,072' setting depth).
- c) Based on string weight in 9 ppg fluid at 2,072' TVD plus 100k# overpull. (Buoyed weight equals 64k lbs.)

Cement volumes are based on 9-5/8" casing set in 13-1/2" hole with 60% excess to circulate cement back to surface. Mix and pump the following slurry.

Pre-flush (Spacer): **20 bbls** fresh water

Lead Slurry: **449 sks** (232 bbls) Conventional system with 94 lb/sk cement, 4% extender, 2% expanding agent, 2% CaCl₂ and 0.25 lb/sk lost circulation control agent

Tail Slurry: **300 sks** (62 bbls) Conventional system with 94 lb/sk cement, 3% NaCl, and 0.25 lb/sk lost circulation control agent

**Oasis Petroleum
Well Summary
Kline Federal 5300 41-18 14BX
Section 18 T153N R100W
McKenzie County, ND**

INTERMEDIATE CASING AND CEMENT DESIGN

Size	Interval	Weight	Grade	Coupling	I.D.	Drift	Make-up Torque (ft-lbs)		
							Minimum	Optimum	Max
7"	0' – 6,614'	29	P-110	LTC	6.184"	6.059"	5,980	7,970	8,770
7"	6,614' – 10,288'	32	HCP-110	LTC	6.094"	6.000***	6,730	8,970	9,870
7"	10,288' – 11,037'	29	P-110	LTC	6.184"	6.059"	5,980	7,970	8,770

***Special drift

Interval	Length	Description	Collapse	Burst	Tension
			(psi) a	(psi) b	(1000 lbs) c
0' – 6,614'	6,614'	7", 29#, P-110, LTC, 8rd	8530 / 2.48*	11220 / 1.19	797 / 2.10
6,614' – 10,288'	3,674'	7", 32#, HCP-110, LTC, 8rd	11820 / 2.21*	12460 / 1.29	
6,614' – 10,288'	3,674'	7", 32#, HCP-110, LTC, 8rd	11820 / 1.07**	12460 / 1.29	
10,288' – 11,037'	749'	7", 29 lb, P-110, LTC, 8rd	8530 / 1.53*	11220 / 1.16	

API Rating & Safety Factor

- a) *Assume full casing evacuation with 10 ppg fluid on backside
 **Assume full casing evacuation with 1.2 psi/ft equivalent fluid gradient across salt intervals. (Bottom of last salt 9,238' TVD)
- b) Burst pressure based on 9000 psig max press for stimulation plus 10.2 ppg fluid in casing and 9 ppg fluid on backside-to 10,765' TVD.
- c) Based on string weight in 10 ppg fluid, (280k lbs buoyed weight) plus 100k

Cement volumes are estimates based on 7" casing set in an 8-3/4" hole with 30% excess.

Pre-flush (Spacer): **170 bbls** Saltwater
 20 bbls CW8 System
 10 bbls Fresh Water

Lead Slurry: **177 sks** (82 bbls) Conventional system with 47 lb/sk cement, 10% NaCl, 34 lb/sk extender, 10% D020 extender, 1% D079 extender, 1% anti-settling agent, 1% fluid loss agent, 0.2% anti-foam agent, 0.7% retarder, 0.125 lb/sk lost circulation control agent, and 0.3% dispersant

Tail Slurry: **575 sks** (168 bbls) Conventional system with 94 lb/sk cement, 10% NaCl, 35% Silica, 0.2% fluid loss agent, 0.8% dispersant, 0.125 lb/sk lost circulation control agent and 0.3% retarder

**Oasis Petroleum
Well Summary
Kline Federal 5300 41-18 14BX
Section 18 T153N R100W
McKenzie County, ND**

PRODUCTION LINER

							Make-up Torque (ft-lbs)		
Size	Interval	Weight	Grade	Coupling	I.D.	Drift	Minimum	Optimum	Max
4-1/2"	10,238' – 20,708'	11.6	P-110	BTC	4.000"	3.875"			

Interval	Description	Collapse	Burst	Tension	Cost per ft
		(psi) a	(psi) b	(1000 lbs) c	
10,238' – 20,708'	4-1/2", 11.6 lb, P-110, BTC, 8rd	7560 / 1.42	10690 / 1.11	385 / 1.89	

API Rating & Safety Factor

- a) Based on full casing evacuation with 9.5 ppg fluid on backside @ 10,765' TVD.
Burst pressure based on 9000 psi treating pressure with 10.2 ppg internal fluid gradient and 9 ppg external
- b) fluid gradient @ 10,765' TVD.
- c) Based on string weight in 9.5 ppg fluid (Buoyed weight: 104k lbs.) plus 100k lbs overpull.



Oasis Petroleum

Indian Hills

153N-100W-17/18

Kline Federal 5300 41-18 14BX

Kline Federal 5300 41-18 14BX

Plan: Design #2

Standard Planning Report

28 May, 2014

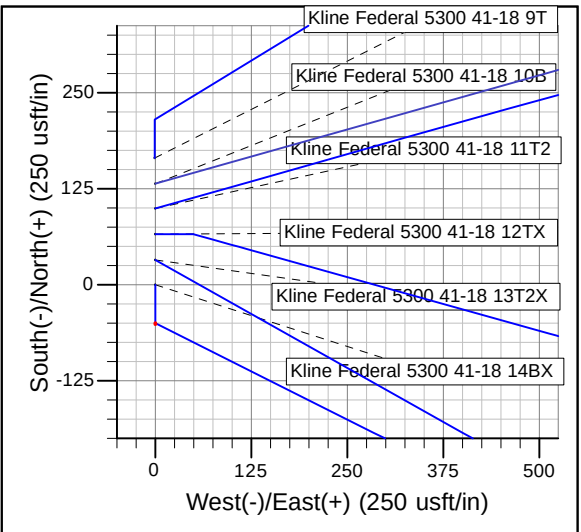
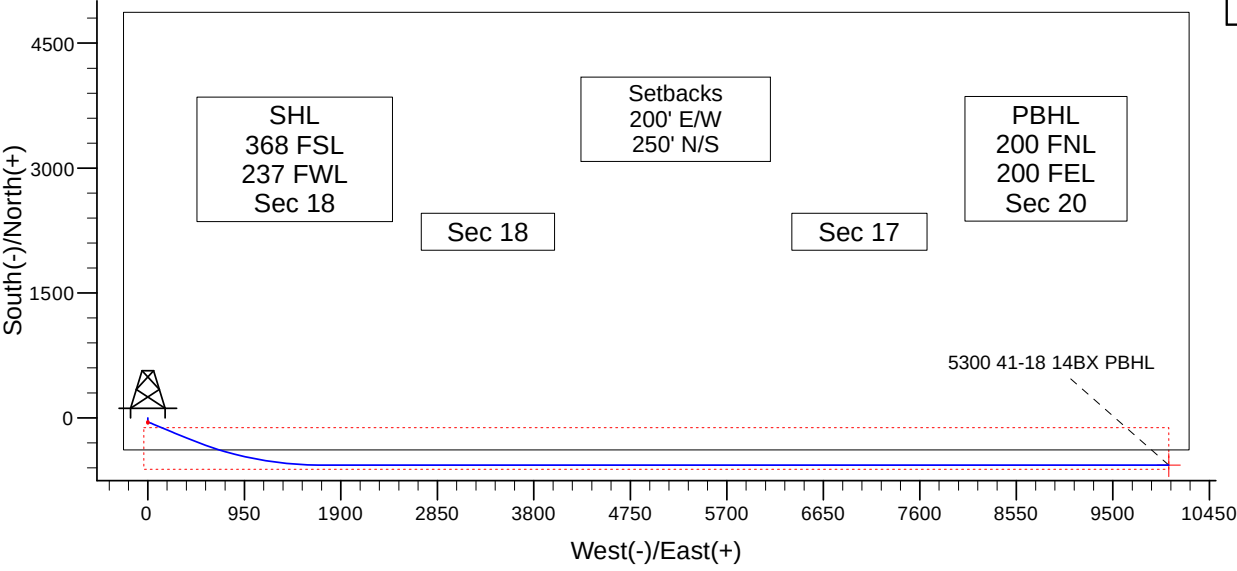


**RYAN DIRECTIONAL
SERVICES**
A NABORS COMPANY

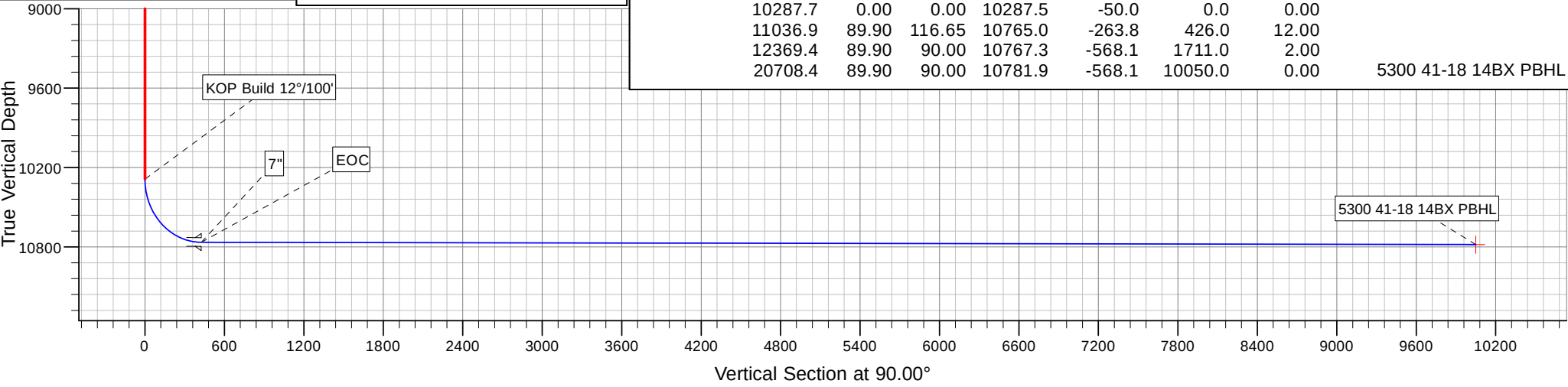
Project: Indian Hills
Site: 153N-100W-17/18
Well: Kline Federal 5300 41-18 14BX
Wellbore: Kline Federal 5300 41-18 14BX
Design: Design #2



WELL DETAILS: Kline Federal 5300 41-18 14BX			
Ground Level: 2057.0			
Northing	Easting	Latitude	Longitude
405070.82	1209965.56	48° 4' 6.900 N	103° 36' 11.950 W



CASING DETAILS				T M Azimuths to True North Magnetic North: 8.27° Magnetic Field Strength: 56437.0snT Dip Angle: 72.99° Date: 5/23/2014 Model: IGRF2010	SECTION DETAILS							
TVD	MD	Name	Size		MD	Inc	Azi	TVD	+N/-S	+E/-W	Dleg	Target
10765.0	11037.0	7"	7		0.0	0.00	0.00	0.0	0.0	0.0	0.00	
					2200.0	0.00	0.00	2200.0	0.0	0.0	0.00	
					2216.7	0.50	180.00	2216.7	-0.1	0.0	3.00	
					7926.7	0.50	180.00	7926.4	-49.9	0.0	0.00	
					7943.3	0.00	0.00	7943.1	-50.0	0.0	3.00	
					10287.7	0.00	0.00	10287.5	-50.0	0.0	0.00	
					11036.9	89.90	116.65	10765.0	-263.8	426.0	12.00	
					12369.4	89.90	90.00	10767.3	-568.1	1711.0	2.00	
				20708.4	89.90	90.00	10781.9	-568.1	10050.0	0.00	5300 41-18 14BX PBHL	



Database:	EDM 5000.1 Single User Db	Local Co-ordinate Reference:	Well Kline Federal 5300 41-18 14BX
Company:	Oasis Petroleum	TVD Reference:	WELL @ 2082.0usft (Original Well Elev)
Project:	Indian Hills	MD Reference:	WELL @ 2082.0usft (Original Well Elev)
Site:	153N-100W-17/18	North Reference:	True
Well:	Kline Federal 5300 41-18 14BX	Survey Calculation Method:	Minimum Curvature
Wellbore:	Kline Federal 5300 41-18 14BX		
Design:	Design #2		

Project	Indian Hills		
Map System:	US State Plane 1983	System Datum:	Mean Sea Level
Geo Datum:	North American Datum 1983		
Map Zone:	North Dakota Northern Zone		

Site		153N-100W-17/18			
Site Position:		Northing:	408,962.44 usft	Latitude:	48° 4' 45.380 N
From:	Lat/Long	Easting:	1,210,229.18 usft	Longitude:	103° 36' 10.380 W
Position Uncertainty:	0.0 usft	Slot Radius:	13-3/16 "	Grid Convergence:	-2.31 °

Well	Kline Federal 5300 41-18 14BX					
Well Position	+N/-S	-3,899.1 usft	Northing:	405,070.82 usft	Latitude:	48° 4' 6.900 N
	+E/-W	-106.6 usft	Easting:	1,209,965.56 usft	Longitude:	103° 36' 11.950 W
Position Uncertainty	2.0 usft		Wellhead Elevation:		Ground Level:	2,057.0 usft

Wellbore	Kline Federal 5300 41-18 14BX				
Magnetics	Model Name	Sample Date	Declination (°)	Dip Angle (°)	Field Strength (nT)
	IGRF2010	5/23/2014	8.27	72.99	56,437

Design	Design #2			
Audit Notes:				
Version:	Phase:	PROTOTYPE	Tie On Depth:	0.0
Vertical Section:	Depth From (TVD) (usft)	+N/-S (usft)	+E/-W (usft)	Direction (°)
	0.0	0.0	0.0	90.00

Plan Sections										
Measured Depth (usft)	Inclination (°)	Azimuth (°)	Vertical Depth (usft)	+N/-S (usft)	+E/-W (usft)	Dogleg Rate (°/100usft)	Build Rate (°/100usft)	Turn Rate (°/100usft)	TFO (°)	Target
0.0	0.00	0.00	0.0	0.0	0.0	0.00	0.00	0.00	0.00	
2,200.0	0.00	0.00	2,200.0	0.0	0.0	0.00	0.00	0.00	0.00	
2,216.7	0.50	180.00	2,216.7	-0.1	0.0	3.00	3.00	0.00	180.00	
7,926.7	0.50	180.00	7,926.4	-49.9	0.0	0.00	0.00	0.00	0.00	
7,943.3	0.00	0.00	7,943.1	-50.0	0.0	3.00	-3.00	0.00	180.00	
10,287.7	0.00	0.00	10,287.5	-50.0	0.0	0.00	0.00	0.00	0.00	
11,036.9	89.90	116.65	10,765.0	-263.8	426.0	12.00	12.00	0.00	116.65	
12,369.4	89.90	90.00	10,767.3	-568.1	1,711.0	2.00	0.00	-2.00	-90.02	
20,708.4	89.90	90.00	10,781.9	-568.1	10,050.0	0.00	0.00	0.00	0.00	5300 41-18 14BX PBI

Database:	EDM 5000.1 Single User Db	Local Co-ordinate Reference:	Well Kline Federal 5300 41-18 14BX
Company:	Oasis Petroleum	TVD Reference:	WELL @ 2082.0usft (Original Well Elev)
Project:	Indian Hills	MD Reference:	WELL @ 2082.0usft (Original Well Elev)
Site:	153N-100W-17/18	North Reference:	True
Well:	Kline Federal 5300 41-18 14BX	Survey Calculation Method:	Minimum Curvature
Wellbore:	Kline Federal 5300 41-18 14BX		
Design:	Design #2		

Planned Survey									
Measured Depth (usft)	Inclination (°)	Azimuth (°)	Vertical Depth (usft)	+N/-S (usft)	+E/-W (usft)	Vertical Section (usft)	Dogleg Rate (°/100usft)	Build Rate (°/100usft)	Turn Rate (°/100usft)
2,200.0	0.00	0.00	2,200.0	0.0	0.0	0.0	0.00	0.00	0.00
2,216.7	0.50	180.00	2,216.7	-0.1	0.0	0.0	3.00	3.00	0.00
2,300.0	0.50	180.00	2,300.0	-0.8	0.0	0.0	0.00	0.00	0.00
2,400.0	0.50	180.00	2,400.0	-1.7	0.0	0.0	0.00	0.00	0.00
2,500.0	0.50	180.00	2,500.0	-2.5	0.0	0.0	0.00	0.00	0.00
2,600.0	0.50	180.00	2,600.0	-3.4	0.0	0.0	0.00	0.00	0.00
2,700.0	0.50	180.00	2,700.0	-4.3	0.0	0.0	0.00	0.00	0.00
2,800.0	0.50	180.00	2,800.0	-5.2	0.0	0.0	0.00	0.00	0.00
2,900.0	0.50	180.00	2,900.0	-6.0	0.0	0.0	0.00	0.00	0.00
3,000.0	0.50	180.00	3,000.0	-6.9	0.0	0.0	0.00	0.00	0.00
3,100.0	0.50	180.00	3,100.0	-7.8	0.0	0.0	0.00	0.00	0.00
3,200.0	0.50	180.00	3,200.0	-8.7	0.0	0.0	0.00	0.00	0.00
3,300.0	0.50	180.00	3,300.0	-9.5	0.0	0.0	0.00	0.00	0.00
3,400.0	0.50	180.00	3,400.0	-10.4	0.0	0.0	0.00	0.00	0.00
3,500.0	0.50	180.00	3,500.0	-11.3	0.0	0.0	0.00	0.00	0.00
3,600.0	0.50	180.00	3,599.9	-12.1	0.0	0.0	0.00	0.00	0.00
3,700.0	0.50	180.00	3,699.9	-13.0	0.0	0.0	0.00	0.00	0.00
3,800.0	0.50	180.00	3,799.9	-13.9	0.0	0.0	0.00	0.00	0.00
3,900.0	0.50	180.00	3,899.9	-14.8	0.0	0.0	0.00	0.00	0.00
4,000.0	0.50	180.00	3,999.9	-15.6	0.0	0.0	0.00	0.00	0.00
4,100.0	0.50	180.00	4,099.9	-16.5	0.0	0.0	0.00	0.00	0.00
4,200.0	0.50	180.00	4,199.9	-17.4	0.0	0.0	0.00	0.00	0.00
4,300.0	0.50	180.00	4,299.9	-18.3	0.0	0.0	0.00	0.00	0.00
4,400.0	0.50	180.00	4,399.9	-19.1	0.0	0.0	0.00	0.00	0.00
4,500.0	0.50	180.00	4,499.9	-20.0	0.0	0.0	0.00	0.00	0.00
4,600.0	0.50	180.00	4,599.9	-20.9	0.0	0.0	0.00	0.00	0.00
4,646.1	0.50	180.00	4,646.0	-21.3	0.0	0.0	0.00	0.00	0.00
Greenhorn									
4,700.0	0.50	180.00	4,699.9	-21.7	0.0	0.0	0.00	0.00	0.00
4,800.0	0.50	180.00	4,799.9	-22.6	0.0	0.0	0.00	0.00	0.00
4,900.0	0.50	180.00	4,899.9	-23.5	0.0	0.0	0.00	0.00	0.00
5,000.0	0.50	180.00	4,999.9	-24.4	0.0	0.0	0.00	0.00	0.00
5,057.1	0.50	180.00	5,057.0	-24.9	0.0	0.0	0.00	0.00	0.00
Mowry									
5,100.0	0.50	180.00	5,099.9	-25.2	0.0	0.0	0.00	0.00	0.00
5,200.0	0.50	180.00	5,199.9	-26.1	0.0	0.0	0.00	0.00	0.00
5,300.0	0.50	180.00	5,299.9	-27.0	0.0	0.0	0.00	0.00	0.00
5,400.0	0.50	180.00	5,399.9	-27.9	0.0	0.0	0.00	0.00	0.00
5,471.1	0.50	180.00	5,471.0	-28.5	0.0	0.0	0.00	0.00	0.00
Dakota									
5,500.0	0.50	180.00	5,499.9	-28.7	0.0	0.0	0.00	0.00	0.00
5,600.0	0.50	180.00	5,599.9	-29.6	0.0	0.0	0.00	0.00	0.00
5,700.0	0.50	180.00	5,699.9	-30.5	0.0	0.0	0.00	0.00	0.00
5,800.0	0.50	180.00	5,799.9	-31.3	0.0	0.0	0.00	0.00	0.00
5,900.0	0.50	180.00	5,899.9	-32.2	0.0	0.0	0.00	0.00	0.00
6,000.0	0.50	180.00	5,999.9	-33.1	0.0	0.0	0.00	0.00	0.00
6,100.0	0.50	180.00	6,099.9	-34.0	0.0	0.0	0.00	0.00	0.00
6,200.0	0.50	180.00	6,199.8	-34.8	0.0	0.0	0.00	0.00	0.00
6,300.0	0.50	180.00	6,299.8	-35.7	0.0	0.0	0.00	0.00	0.00
6,400.0	0.50	180.00	6,399.8	-36.6	0.0	0.0	0.00	0.00	0.00
6,486.2	0.50	180.00	6,486.0	-37.3	0.0	0.0	0.00	0.00	0.00
Rierdon									
6,500.0	0.50	180.00	6,499.8	-37.5	0.0	0.0	0.00	0.00	0.00

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Company:	Oasis Petroleum	TVD Reference:	WELL @ 2082.0usft (Original Well Elev)
Project:	Indian Hills	MD Reference:	WELL @ 2082.0usft (Original Well Elev)
Site:	153N-100W-17/18	North Reference:	True
Well:	Kline Federal 5300 41-18 14BX	Survey Calculation Method:	Minimum Curvature
Wellbore:	Kline Federal 5300 41-18 14BX		
Design:	Design #2		

Planned Survey										
Measured Depth (usft)	Inclination (°)	Azimuth (°)	Vertical Depth (usft)	+N/-S (usft)	+E/-W (usft)	Vertical Section (usft)	Dogleg Rate (°/100usft)	Build Rate (°/100usft)	Turn Rate (°/100usft)	
6,600.0	0.50	180.00	6,599.8	-38.3	0.0	0.0	0.00	0.00	0.00	
6,700.0	0.50	180.00	6,699.8	-39.2	0.0	0.0	0.00	0.00	0.00	
6,800.0	0.50	180.00	6,799.8	-40.1	0.0	0.0	0.00	0.00	0.00	
6,814.2	0.50	180.00	6,814.0	-40.2	0.0	0.0	0.00	0.00	0.00	
Dunham Salt										
6,900.0	0.50	180.00	6,899.8	-40.9	0.0	0.0	0.00	0.00	0.00	
6,929.2	0.50	180.00	6,929.0	-41.2	0.0	0.0	0.00	0.00	0.00	
Dunham Salt Base										
7,000.0	0.50	180.00	6,999.8	-41.8	0.0	0.0	0.00	0.00	0.00	
7,024.2	0.50	180.00	7,024.0	-42.0	0.0	0.0	0.00	0.00	0.00	
Spearfish										
7,100.0	0.50	180.00	7,099.8	-42.7	0.0	0.0	0.00	0.00	0.00	
7,200.0	0.50	180.00	7,199.8	-43.6	0.0	0.0	0.00	0.00	0.00	
7,283.2	0.50	180.00	7,283.0	-44.3	0.0	0.0	0.00	0.00	0.00	
Pine Salt										
7,300.0	0.50	180.00	7,299.8	-44.4	0.0	0.0	0.00	0.00	0.00	
7,318.2	0.50	180.00	7,318.0	-44.6	0.0	0.0	0.00	0.00	0.00	
Pine Salt Base										
7,374.2	0.50	180.00	7,374.0	-45.1	0.0	0.0	0.00	0.00	0.00	
Opeche Salt										
7,400.0	0.50	180.00	7,399.8	-45.3	0.0	0.0	0.00	0.00	0.00	
7,403.2	0.50	180.00	7,403.0	-45.3	0.0	0.0	0.00	0.00	0.00	
Opeche Salt Base										
7,500.0	0.50	180.00	7,499.8	-46.2	0.0	0.0	0.00	0.00	0.00	
7,600.0	0.50	180.00	7,599.8	-47.1	0.0	0.0	0.00	0.00	0.00	
7,605.2	0.50	180.00	7,605.0	-47.1	0.0	0.0	0.00	0.00	0.00	
Broom Creek (Top of Minnelusa Gp.)										
7,684.2	0.50	180.00	7,684.0	-47.8	0.0	0.0	0.00	0.00	0.00	
Amsden										
7,700.0	0.50	180.00	7,699.8	-47.9	0.0	0.0	0.00	0.00	0.00	
7,800.0	0.50	180.00	7,799.8	-48.8	0.0	0.0	0.00	0.00	0.00	
7,853.2	0.50	180.00	7,853.0	-49.3	0.0	0.0	0.00	0.00	0.00	
Tyler										
7,900.0	0.50	180.00	7,899.8	-49.7	0.0	0.0	0.00	0.00	0.00	
7,926.7	0.50	180.00	7,926.4	-49.9	0.0	0.0	0.00	0.00	0.00	
7,943.3	0.00	0.00	7,943.1	-50.0	0.0	0.0	3.00	-3.00	0.00	
8,000.0	0.00	0.00	7,999.8	-50.0	0.0	0.0	0.00	0.00	0.00	
8,047.2	0.00	0.00	8,047.0	-50.0	0.0	0.0	0.00	0.00	0.00	
Otter (Base of Minnelusa Gp.)										
8,100.0	0.00	0.00	8,099.8	-50.0	0.0	0.0	0.00	0.00	0.00	
8,200.0	0.00	0.00	8,199.8	-50.0	0.0	0.0	0.00	0.00	0.00	
8,300.0	0.00	0.00	8,299.8	-50.0	0.0	0.0	0.00	0.00	0.00	
8,390.2	0.00	0.00	8,390.0	-50.0	0.0	0.0	0.00	0.00	0.00	
Kibbey Lime										
8,400.0	0.00	0.00	8,399.8	-50.0	0.0	0.0	0.00	0.00	0.00	
8,500.0	0.00	0.00	8,499.8	-50.0	0.0	0.0	0.00	0.00	0.00	
8,542.2	0.00	0.00	8,542.0	-50.0	0.0	0.0	0.00	0.00	0.00	
Charles Salt										
8,600.0	0.00	0.00	8,599.8	-50.0	0.0	0.0	0.00	0.00	0.00	
8,700.0	0.00	0.00	8,699.8	-50.0	0.0	0.0	0.00	0.00	0.00	
8,800.0	0.00	0.00	8,799.8	-50.0	0.0	0.0	0.00	0.00	0.00	
8,900.0	0.00	0.00	8,899.8	-50.0	0.0	0.0	0.00	0.00	0.00	

Database:	EDM 5000.1 Single User Db	Local Co-ordinate Reference:	Well Kline Federal 5300 41-18 14BX
Company:	Oasis Petroleum	TVD Reference:	WELL @ 2082.0usft (Original Well Elev)
Project:	Indian Hills	MD Reference:	WELL @ 2082.0usft (Original Well Elev)
Site:	153N-100W-17/18	North Reference:	True
Well:	Kline Federal 5300 41-18 14BX	Survey Calculation Method:	Minimum Curvature
Wellbore:	Kline Federal 5300 41-18 14BX		
Design:	Design #2		

Planned Survey									
Measured Depth (usft)	Inclination (°)	Azimuth (°)	Vertical Depth (usft)	+N/-S (usft)	+E/-W (usft)	Vertical Section (usft)	Dogleg Rate (°/100usft)	Build Rate (°/100usft)	Turn Rate (°/100usft)
9,000.0	0.00	0.00	8,999.8	-50.0	0.0	0.0	0.00	0.00	0.00
9,100.0	0.00	0.00	9,099.8	-50.0	0.0	0.0	0.00	0.00	0.00
9,159.2	0.00	0.00	9,159.0	-50.0	0.0	0.0	0.00	0.00	0.00
UB									
9,200.0	0.00	0.00	9,199.8	-50.0	0.0	0.0	0.00	0.00	0.00
9,238.2	0.00	0.00	9,238.0	-50.0	0.0	0.0	0.00	0.00	0.00
Base Last Salt									
9,300.0	0.00	0.00	9,299.8	-50.0	0.0	0.0	0.00	0.00	0.00
9,301.2	0.00	0.00	9,301.0	-50.0	0.0	0.0	0.00	0.00	0.00
Ratcliffe									
9,400.0	0.00	0.00	9,399.8	-50.0	0.0	0.0	0.00	0.00	0.00
9,454.2	0.00	0.00	9,454.0	-50.0	0.0	0.0	0.00	0.00	0.00
Mission Canyon									
9,500.0	0.00	0.00	9,499.8	-50.0	0.0	0.0	0.00	0.00	0.00
9,600.0	0.00	0.00	9,599.8	-50.0	0.0	0.0	0.00	0.00	0.00
9,700.0	0.00	0.00	9,699.8	-50.0	0.0	0.0	0.00	0.00	0.00
9,800.0	0.00	0.00	9,799.8	-50.0	0.0	0.0	0.00	0.00	0.00
9,900.0	0.00	0.00	9,899.8	-50.0	0.0	0.0	0.00	0.00	0.00
10,000.0	0.00	0.00	9,999.8	-50.0	0.0	0.0	0.00	0.00	0.00
10,022.2	0.00	0.00	10,022.0	-50.0	0.0	0.0	0.00	0.00	0.00
Lodgepole									
10,100.0	0.00	0.00	10,099.8	-50.0	0.0	0.0	0.00	0.00	0.00
10,200.0	0.00	0.00	10,199.8	-50.0	0.0	0.0	0.00	0.00	0.00
10,209.2	0.00	0.00	10,209.0	-50.0	0.0	0.0	0.00	0.00	0.00
Lodgepole Fracture Zone									
10,287.7	0.00	0.00	10,287.5	-50.0	0.0	0.0	0.00	0.00	0.00
10,287.8	0.00	116.65	10,287.6	-50.0	0.0	0.0	0.00	0.00	0.00
KOP Build 12°/100'									
10,300.0	1.47	116.65	10,299.8	-50.0	0.1	0.1	12.07	12.07	0.00
10,325.0	4.47	116.65	10,324.7	-50.6	1.3	1.3	12.00	12.00	0.00
10,350.0	7.47	116.65	10,349.6	-51.8	3.6	3.6	12.00	12.00	0.00
10,375.0	10.47	116.65	10,374.3	-53.5	7.1	7.1	12.00	12.00	0.00
10,400.0	13.47	116.65	10,398.8	-55.9	11.7	11.7	12.00	12.00	0.00
10,425.0	16.47	116.65	10,422.9	-58.8	17.5	17.5	12.00	12.00	0.00
10,450.0	19.47	116.65	10,446.7	-62.2	24.4	24.4	12.00	12.00	0.00
10,475.0	22.47	116.65	10,470.0	-66.2	32.4	32.4	12.00	12.00	0.00
10,500.0	25.47	116.65	10,492.9	-70.8	41.5	41.5	12.00	12.00	0.00
10,525.0	28.47	116.65	10,515.1	-75.9	51.6	51.6	12.00	12.00	0.00
10,550.0	31.47	116.65	10,536.8	-81.5	62.8	62.8	12.00	12.00	0.00
10,575.0	34.47	116.65	10,557.8	-87.6	74.9	74.9	12.00	12.00	0.00
10,600.0	37.47	116.65	10,578.0	-94.2	88.1	88.1	12.00	12.00	0.00
10,625.0	40.47	116.65	10,597.4	-101.2	102.1	102.1	12.00	12.00	0.00
10,650.0	43.47	116.65	10,616.0	-108.7	117.1	117.1	12.00	12.00	0.00
10,675.0	46.47	116.65	10,633.7	-116.6	132.8	132.8	12.00	12.00	0.00
10,700.0	49.47	116.65	10,650.4	-125.0	149.4	149.4	12.00	12.00	0.00
10,725.0	52.47	116.65	10,666.2	-133.7	166.8	166.8	12.00	12.00	0.00
10,750.0	55.47	116.65	10,680.9	-142.7	184.9	184.9	12.00	12.00	0.00
10,775.0	58.47	116.65	10,694.5	-152.1	203.6	203.6	12.00	12.00	0.00
10,800.0	61.47	116.65	10,707.0	-161.9	222.9	222.9	12.00	12.00	0.00
10,825.0	64.47	116.65	10,718.4	-171.8	242.8	242.8	12.00	12.00	0.00
10,843.5	66.69	116.65	10,726.0	-179.4	257.9	257.9	12.00	12.00	0.00
False Bakken									
10,850.0	67.47	116.65	10,728.5	-182.1	263.2	263.2	12.00	12.00	0.00

Database:	EDM 5000.1 Single User Db	Local Co-ordinate Reference:	Well Kline Federal 5300 41-18 14BX
Company:	Oasis Petroleum	TVD Reference:	WELL @ 2082.0usft (Original Well Elev)
Project:	Indian Hills	MD Reference:	WELL @ 2082.0usft (Original Well Elev)
Site:	153N-100W-17/18	North Reference:	True
Well:	Kline Federal 5300 41-18 14BX	Survey Calculation Method:	Minimum Curvature
Wellbore:	Kline Federal 5300 41-18 14BX		
Design:	Design #2		

Planned Survey									
Measured Depth (usft)	Inclination (°)	Azimuth (°)	Vertical Depth (usft)	+N/-S (usft)	+E/-W (usft)	Vertical Section (usft)	Dogleg Rate (°/100usft)	Build Rate (°/100usft)	Turn Rate (°/100usft)
10,867.6	69.59	116.65	10,735.0	-189.4	277.9	277.9	12.00	12.00	0.00
Upper Bakken									
10,875.0	70.47	116.65	10,737.5	-192.5	284.1	284.1	12.00	12.00	0.00
10,900.0	73.47	116.65	10,745.3	-203.2	305.3	305.3	12.00	12.00	0.00
10,921.9	76.10	116.65	10,751.0	-212.7	324.2	324.2	12.00	12.00	0.00
Middle Bakken									
10,925.0	76.47	116.65	10,751.7	-214.0	326.9	326.9	12.00	12.00	0.00
10,950.0	79.47	116.65	10,756.9	-225.0	348.8	348.8	12.00	12.00	0.00
10,968.7	81.72	116.65	10,760.0	-233.3	365.3	365.3	12.00	12.00	0.00
Middle Bakken Sand Target									
10,975.0	82.47	116.65	10,760.9	-236.1	370.8	370.8	12.00	12.00	0.00
11,000.0	85.47	116.65	10,763.5	-247.2	393.1	393.1	12.00	12.00	0.00
11,025.0	88.47	116.65	10,764.8	-258.4	415.4	415.4	12.00	12.00	0.00
11,036.9	89.90	116.65	10,765.0	-263.8	426.0	426.0	12.00	12.00	0.00
EOC									
11,037.0	89.90	116.65	10,765.0	-263.8	426.1	426.1	0.00	0.00	0.00
7"									
11,100.0	89.90	115.39	10,765.1	-291.4	482.7	482.7	2.00	0.00	-2.00
11,200.0	89.90	113.39	10,765.3	-332.7	573.8	573.8	2.00	0.00	-2.00
11,300.0	89.90	111.39	10,765.4	-370.8	666.2	666.2	2.00	0.00	-2.00
11,400.0	89.90	109.39	10,765.6	-405.7	760.0	760.0	2.00	0.00	-2.00
11,500.0	89.90	107.39	10,765.8	-437.2	854.8	854.8	2.00	0.00	-2.00
11,600.0	89.90	105.39	10,766.0	-465.4	950.8	950.8	2.00	0.00	-2.00
11,700.0	89.90	103.39	10,766.2	-490.3	1,047.6	1,047.6	2.00	0.00	-2.00
11,800.0	89.90	101.39	10,766.3	-511.7	1,145.3	1,145.3	2.00	0.00	-2.00
11,900.0	89.90	99.39	10,766.5	-529.7	1,243.7	1,243.7	2.00	0.00	-2.00
12,000.0	89.90	97.39	10,766.7	-544.3	1,342.6	1,342.6	2.00	0.00	-2.00
12,100.0	89.90	95.39	10,766.9	-555.4	1,442.0	1,442.0	2.00	0.00	-2.00
12,200.0	89.90	93.39	10,767.1	-563.1	1,541.7	1,541.7	2.00	0.00	-2.00
12,300.0	89.90	91.39	10,767.2	-567.3	1,641.6	1,641.6	2.00	0.00	-2.00
12,369.4	89.90	90.00	10,767.3	-568.1	1,711.0	1,711.0	2.00	0.00	-2.00
12,400.0	89.90	90.00	10,767.4	-568.1	1,741.6	1,741.6	0.00	0.00	0.00
12,500.0	89.90	90.00	10,767.6	-568.1	1,841.6	1,841.6	0.00	0.00	0.00
12,600.0	89.90	90.00	10,767.8	-568.1	1,941.6	1,941.6	0.00	0.00	0.00
12,700.0	89.90	90.00	10,767.9	-568.1	2,041.6	2,041.6	0.00	0.00	0.00
12,800.0	89.90	90.00	10,768.1	-568.1	2,141.6	2,141.6	0.00	0.00	0.00
12,900.0	89.90	90.00	10,768.3	-568.1	2,241.6	2,241.6	0.00	0.00	0.00
13,000.0	89.90	90.00	10,768.4	-568.1	2,341.6	2,341.6	0.00	0.00	0.00
13,100.0	89.90	90.00	10,768.6	-568.1	2,441.6	2,441.6	0.00	0.00	0.00
13,200.0	89.90	90.00	10,768.8	-568.1	2,541.6	2,541.6	0.00	0.00	0.00
13,300.0	89.90	90.00	10,769.0	-568.1	2,641.6	2,641.6	0.00	0.00	0.00
13,400.0	89.90	90.00	10,769.1	-568.1	2,741.6	2,741.6	0.00	0.00	0.00
13,500.0	89.90	90.00	10,769.3	-568.1	2,841.6	2,841.6	0.00	0.00	0.00
13,600.0	89.90	90.00	10,769.5	-568.1	2,941.6	2,941.6	0.00	0.00	0.00
13,700.0	89.90	90.00	10,769.7	-568.1	3,041.6	3,041.6	0.00	0.00	0.00
13,800.0	89.90	90.00	10,769.8	-568.1	3,141.6	3,141.6	0.00	0.00	0.00
13,888.8	89.90	90.00	10,770.0	-568.1	3,230.4	3,230.4	0.00	0.00	0.00
Base Middle Bakken Sand Target									
13,900.0	89.90	90.00	10,770.0	-568.1	3,241.6	3,241.6	0.00	0.00	0.00
14,000.0	89.90	90.00	10,770.2	-568.1	3,341.6	3,341.6	0.00	0.00	0.00
14,100.0	89.90	90.00	10,770.4	-568.1	3,441.6	3,441.6	0.00	0.00	0.00
14,200.0	89.90	90.00	10,770.5	-568.1	3,541.6	3,541.6	0.00	0.00	0.00
14,300.0	89.90	90.00	10,770.7	-568.1	3,641.6	3,641.6	0.00	0.00	0.00

Database:	EDM 5000.1 Single User Db	Local Co-ordinate Reference:	Well Kline Federal 5300 41-18 14BX
Company:	Oasis Petroleum	TVD Reference:	WELL @ 2082.0usft (Original Well Elev)
Project:	Indian Hills	MD Reference:	WELL @ 2082.0usft (Original Well Elev)
Site:	153N-100W-17/18	North Reference:	True
Well:	Kline Federal 5300 41-18 14BX	Survey Calculation Method:	Minimum Curvature
Wellbore:	Kline Federal 5300 41-18 14BX		
Design:	Design #2		

Planned Survey									
Measured Depth (usft)	Inclination (°)	Azimuth (°)	Vertical Depth (usft)	+N/-S (usft)	+E/-W (usft)	Vertical Section (usft)	Dogleg Rate (°/100usft)	Build Rate (°/100usft)	Turn Rate (°/100usft)
14,400.0	89.90	90.00	10,770.9	-568.1	3,741.6	3,741.6	0.00	0.00	0.00
14,500.0	89.90	90.00	10,771.1	-568.1	3,841.6	3,841.6	0.00	0.00	0.00
14,600.0	89.90	90.00	10,771.2	-568.1	3,941.6	3,941.6	0.00	0.00	0.00
14,700.0	89.90	90.00	10,771.4	-568.1	4,041.6	4,041.6	0.00	0.00	0.00
14,800.0	89.90	90.00	10,771.6	-568.1	4,141.6	4,141.6	0.00	0.00	0.00
14,900.0	89.90	90.00	10,771.8	-568.1	4,241.6	4,241.6	0.00	0.00	0.00
15,000.0	89.90	90.00	10,771.9	-568.1	4,341.6	4,341.6	0.00	0.00	0.00
15,100.0	89.90	90.00	10,772.1	-568.1	4,441.6	4,441.6	0.00	0.00	0.00
15,200.0	89.90	90.00	10,772.3	-568.1	4,541.6	4,541.6	0.00	0.00	0.00
15,300.0	89.90	90.00	10,772.5	-568.1	4,641.6	4,641.6	0.00	0.00	0.00
15,400.0	89.90	90.00	10,772.6	-568.1	4,741.6	4,741.6	0.00	0.00	0.00
15,500.0	89.90	90.00	10,772.8	-568.1	4,841.6	4,841.6	0.00	0.00	0.00
15,600.0	89.90	90.00	10,773.0	-568.1	4,941.6	4,941.6	0.00	0.00	0.00
15,700.0	89.90	90.00	10,773.2	-568.1	5,041.6	5,041.6	0.00	0.00	0.00
15,800.0	89.90	90.00	10,773.3	-568.1	5,141.6	5,141.6	0.00	0.00	0.00
15,900.0	89.90	90.00	10,773.5	-568.1	5,241.6	5,241.6	0.00	0.00	0.00
16,000.0	89.90	90.00	10,773.7	-568.1	5,341.6	5,341.6	0.00	0.00	0.00
16,100.0	89.90	90.00	10,773.9	-568.1	5,441.6	5,441.6	0.00	0.00	0.00
16,200.0	89.90	90.00	10,774.0	-568.1	5,541.6	5,541.6	0.00	0.00	0.00
16,300.0	89.90	90.00	10,774.2	-568.1	5,641.6	5,641.6	0.00	0.00	0.00
16,400.0	89.90	90.00	10,774.4	-568.1	5,741.6	5,741.6	0.00	0.00	0.00
16,500.0	89.90	90.00	10,774.6	-568.1	5,841.6	5,841.6	0.00	0.00	0.00
16,600.0	89.90	90.00	10,774.7	-568.1	5,941.6	5,941.6	0.00	0.00	0.00
16,700.0	89.90	90.00	10,774.9	-568.1	6,041.6	6,041.6	0.00	0.00	0.00
16,800.0	89.90	90.00	10,775.1	-568.1	6,141.6	6,141.6	0.00	0.00	0.00
16,900.0	89.90	90.00	10,775.3	-568.1	6,241.6	6,241.6	0.00	0.00	0.00
17,000.0	89.90	90.00	10,775.4	-568.1	6,341.6	6,341.6	0.00	0.00	0.00
17,100.0	89.90	90.00	10,775.6	-568.1	6,441.6	6,441.6	0.00	0.00	0.00
17,200.0	89.90	90.00	10,775.8	-568.1	6,541.6	6,541.6	0.00	0.00	0.00
17,300.0	89.90	90.00	10,776.0	-568.1	6,641.6	6,641.6	0.00	0.00	0.00
17,400.0	89.90	90.00	10,776.1	-568.1	6,741.6	6,741.6	0.00	0.00	0.00
17,500.0	89.90	90.00	10,776.3	-568.1	6,841.6	6,841.6	0.00	0.00	0.00
17,600.0	89.90	90.00	10,776.5	-568.1	6,941.6	6,941.6	0.00	0.00	0.00
17,700.0	89.90	90.00	10,776.7	-568.1	7,041.6	7,041.6	0.00	0.00	0.00
17,800.0	89.90	90.00	10,776.8	-568.1	7,141.6	7,141.6	0.00	0.00	0.00
17,900.0	89.90	90.00	10,777.0	-568.1	7,241.6	7,241.6	0.00	0.00	0.00
18,000.0	89.90	90.00	10,777.2	-568.1	7,341.6	7,341.6	0.00	0.00	0.00
18,100.0	89.90	90.00	10,777.3	-568.1	7,441.6	7,441.6	0.00	0.00	0.00
18,200.0	89.90	90.00	10,777.5	-568.1	7,541.6	7,541.6	0.00	0.00	0.00
18,300.0	89.90	90.00	10,777.7	-568.1	7,641.6	7,641.6	0.00	0.00	0.00
18,400.0	89.90	90.00	10,777.9	-568.1	7,741.6	7,741.6	0.00	0.00	0.00
18,500.0	89.90	90.00	10,778.0	-568.1	7,841.6	7,841.6	0.00	0.00	0.00
18,600.0	89.90	90.00	10,778.2	-568.1	7,941.6	7,941.6	0.00	0.00	0.00
18,700.0	89.90	90.00	10,778.4	-568.1	8,041.6	8,041.6	0.00	0.00	0.00
18,800.0	89.90	90.00	10,778.6	-568.1	8,141.6	8,141.6	0.00	0.00	0.00
18,900.0	89.90	90.00	10,778.7	-568.1	8,241.6	8,241.6	0.00	0.00	0.00
19,000.0	89.90	90.00	10,778.9	-568.1	8,341.6	8,341.6	0.00	0.00	0.00
19,100.0	89.90	90.00	10,779.1	-568.1	8,441.6	8,441.6	0.00	0.00	0.00
19,200.0	89.90	90.00	10,779.3	-568.1	8,541.6	8,541.6	0.00	0.00	0.00
19,300.0	89.90	90.00	10,779.4	-568.1	8,641.6	8,641.6	0.00	0.00	0.00
19,400.0	89.90	90.00	10,779.6	-568.1	8,741.6	8,741.6	0.00	0.00	0.00
19,500.0	89.90	90.00	10,779.8	-568.1	8,841.6	8,841.6	0.00	0.00	0.00
19,600.0	89.90	90.00	10,780.0	-568.1	8,941.6	8,941.6	0.00	0.00	0.00
19,700.0	89.90	90.00	10,780.1	-568.1	9,041.6	9,041.6	0.00	0.00	0.00

Database:	EDM 5000.1 Single User Db	Local Co-ordinate Reference:	Well Kline Federal 5300 41-18 14BX
Company:	Oasis Petroleum	TVD Reference:	WELL @ 2082.0usft (Original Well Elev)
Project:	Indian Hills	MD Reference:	WELL @ 2082.0usft (Original Well Elev)
Site:	153N-100W-17/18	North Reference:	True
Well:	Kline Federal 5300 41-18 14BX	Survey Calculation Method:	Minimum Curvature
Wellbore:	Kline Federal 5300 41-18 14BX		
Design:	Design #2		

Planned Survey										
Measured Depth (usft)	Inclination (°)	Azimuth (°)	Vertical Depth (usft)	+N/-S (usft)	+E/-W (usft)	Vertical Section (usft)	Dogleg Rate (°/100usft)	Build Rate (°/100usft)	Turn Rate (°/100usft)	
19,800.0	89.90	90.00	10,780.3	-568.1	9,141.6	9,141.6	0.00	0.00	0.00	
19,900.0	89.90	90.00	10,780.5	-568.1	9,241.6	9,241.6	0.00	0.00	0.00	
20,000.0	89.90	90.00	10,780.7	-568.1	9,341.6	9,341.6	0.00	0.00	0.00	
20,100.0	89.90	90.00	10,780.8	-568.1	9,441.6	9,441.6	0.00	0.00	0.00	
20,200.0	89.90	90.00	10,781.0	-568.1	9,541.6	9,541.6	0.00	0.00	0.00	
20,300.0	89.90	90.00	10,781.2	-568.1	9,641.6	9,641.6	0.00	0.00	0.00	
20,400.0	89.90	90.00	10,781.4	-568.1	9,741.6	9,741.6	0.00	0.00	0.00	
20,500.0	89.90	90.00	10,781.5	-568.1	9,841.6	9,841.6	0.00	0.00	0.00	
20,600.0	89.90	90.00	10,781.7	-568.1	9,941.6	9,941.6	0.00	0.00	0.00	
20,708.4	89.90	90.00	10,781.9	-568.1	10,050.0	10,050.0	0.00	0.00	0.00	
5300 41-18 14BX PBHL										

Design Targets									
Target Name	Dip Angle (°)	Dip Dir. (°)	TVD (usft)	+N/-S (usft)	+E/-W (usft)	Northing (usft)	Easting (usft)	Latitude	Longitude
- hit/miss target									
- Shape									
5300 41-18 14BX PBHL	0.00	0.00	10,781.9	-568.0	10,050.0	404,098.33	1,219,984.51	48° 4' 1.268 N	103° 33' 43.975 W
- plan misses target center by 0.1usft at 20708.4usft MD (10781.9 TVD, -568.1 N, 10050.0 E)									
- Point									

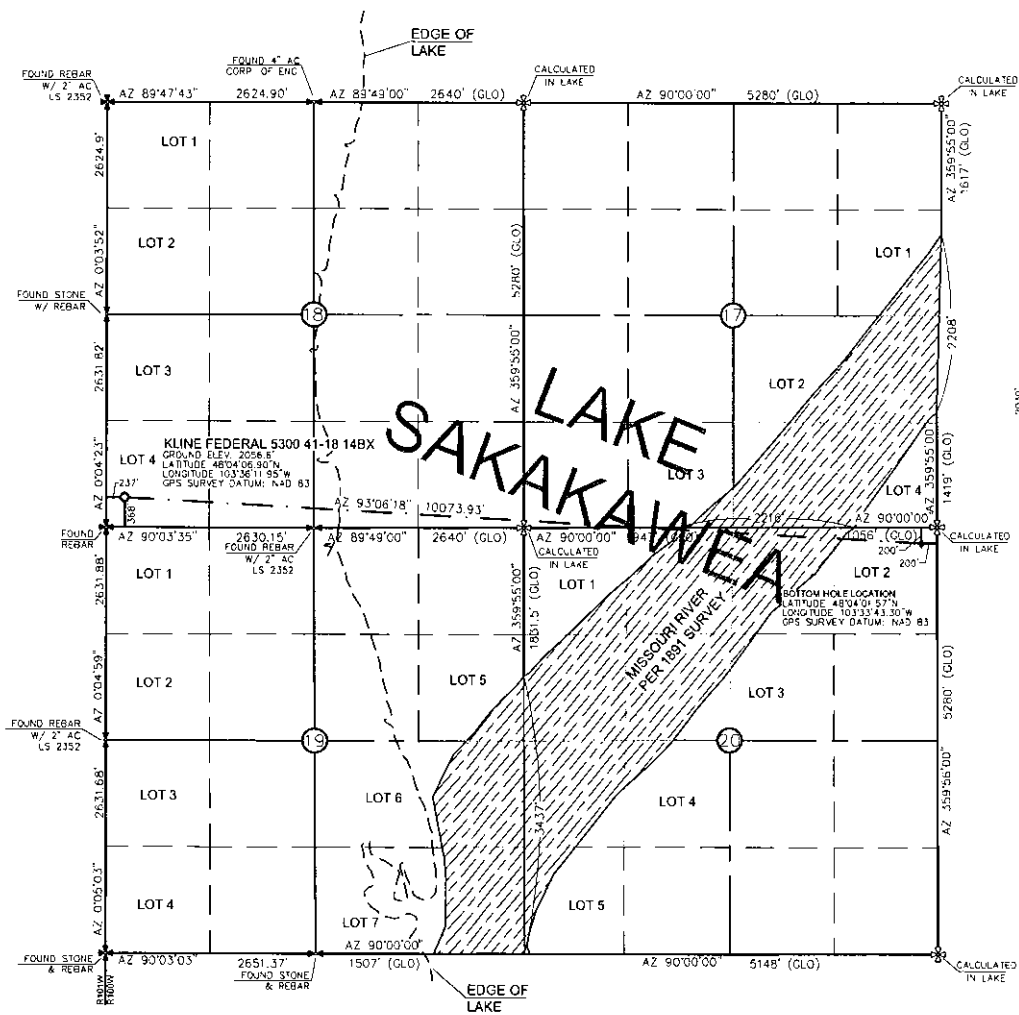
Casing Points					
Measured Depth (usft)	Vertical Depth (usft)	Name	Casing Diameter (")	Hole Diameter (")	
11,037.0	10,765.0	7"	7	8-3/4	

Database:	EDM 5000.1 Single User Db	Local Co-ordinate Reference:	Well Kline Federal 5300 41-18 14BX
Company:	Oasis Petroleum	TVD Reference:	WELL @ 2082.0usft (Original Well Elev)
Project:	Indian Hills	MD Reference:	WELL @ 2082.0usft (Original Well Elev)
Site:	153N-100W-17/18	North Reference:	True
Well:	Kline Federal 5300 41-18 14BX	Survey Calculation Method:	Minimum Curvature
Wellbore:	Kline Federal 5300 41-18 14BX		
Design:	Design #2		

Formations						
Measured Depth (usft)	Vertical Depth (usft)	Name	Lithology	Dip (°)	Dip Direction (°)	
1,972.0	1,972.0	Pierre				
4,646.1	4,646.0	Greenhorn				
5,057.1	5,057.0	Mowry				
5,471.1	5,471.0	Dakota				
6,486.2	6,486.0	Rierdon				
6,814.2	6,814.0	Dunham Salt				
6,929.2	6,929.0	Dunham Salt Base				
7,024.2	7,024.0	Spearfish				
7,283.2	7,283.0	Pine Salt				
7,318.2	7,318.0	Pine Salt Base				
7,374.2	7,374.0	Opeche Salt				
7,403.2	7,403.0	Opeche Salt Base				
7,605.2	7,605.0	Broom Creek (Top of Minnelusa Gp.)				
7,684.2	7,684.0	Amsden				
7,853.2	7,853.0	Tyler				
8,047.2	8,047.0	Otter (Base of Minnelusa Gp.)				
8,390.2	8,390.0	Kibbey Lime				
8,542.2	8,542.0	Charles Salt				
9,159.2	9,159.0	UB				
9,238.2	9,238.0	Base Last Salt				
9,301.2	9,301.0	Ratcliffe				
9,454.2	9,454.0	Mission Canyon				
10,022.2	10,022.0	Lodgepole				
10,209.2	10,209.0	Lodgepole Fracture Zone				
10,843.5	10,726.0	False Bakken				
10,867.6	10,735.0	Upper Bakken				
10,921.9	10,751.0	Middle Bakken				
10,968.7	10,760.0	Middle Bakken Sand Target				
13,888.8	10,770.0	Base Middle Bakken Sand Target				

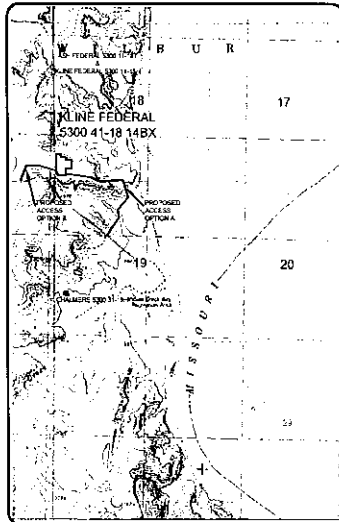
Plan Annotations				
Measured Depth (usft)	Vertical Depth (usft)	Local Coordinates		Comment
		+N/-S (usft)	+E/-W (usft)	
10,287.8	10,287.6	-50.0	0.0	KOP Build 12"/100'
11,036.9	10,765.0	-263.8	426.0	EOC

WELL LOCATION PLAT
 OASIS PETROLEUM NORTH AMERICA, LLC
 1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
 "KLINE FEDERAL 5300 41-18 148X"
 368 FEET FROM SOUTH LINE AND 237 FEET FROM WEST LINE
 SECTION 18, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA

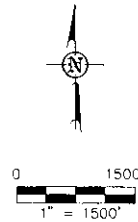


- ⊕ — MONUMENT — RECOVERED
- ⊗ — MONUMENT — NOT RECOVERED

VICINITY MAP



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STAKED ON 4/15/14
 VERTICAL CONTROL DATUM WAS BASED UPON
 CONTROL POINT 4 WITH AN ELEVATION OF 2090.8'

THIS SURVEY AND PLAT IS BEING PROVIDED AT THE REQUEST
 OF ERIC BAYES OF OASIS PETROLEUM. I CERTIFY THAT THIS
 PLAT CORRECTLY REPRESENTS WORK PERFORMED BY ME OR
 UNDER MY SUPERVISION AND IS TRUE AND CORRECT TO THE
 BEST OF MY KNOWLEDGE AND BELIEF.

Robert L. Prociwe

ROBERT L. PROCIWE 2884LS

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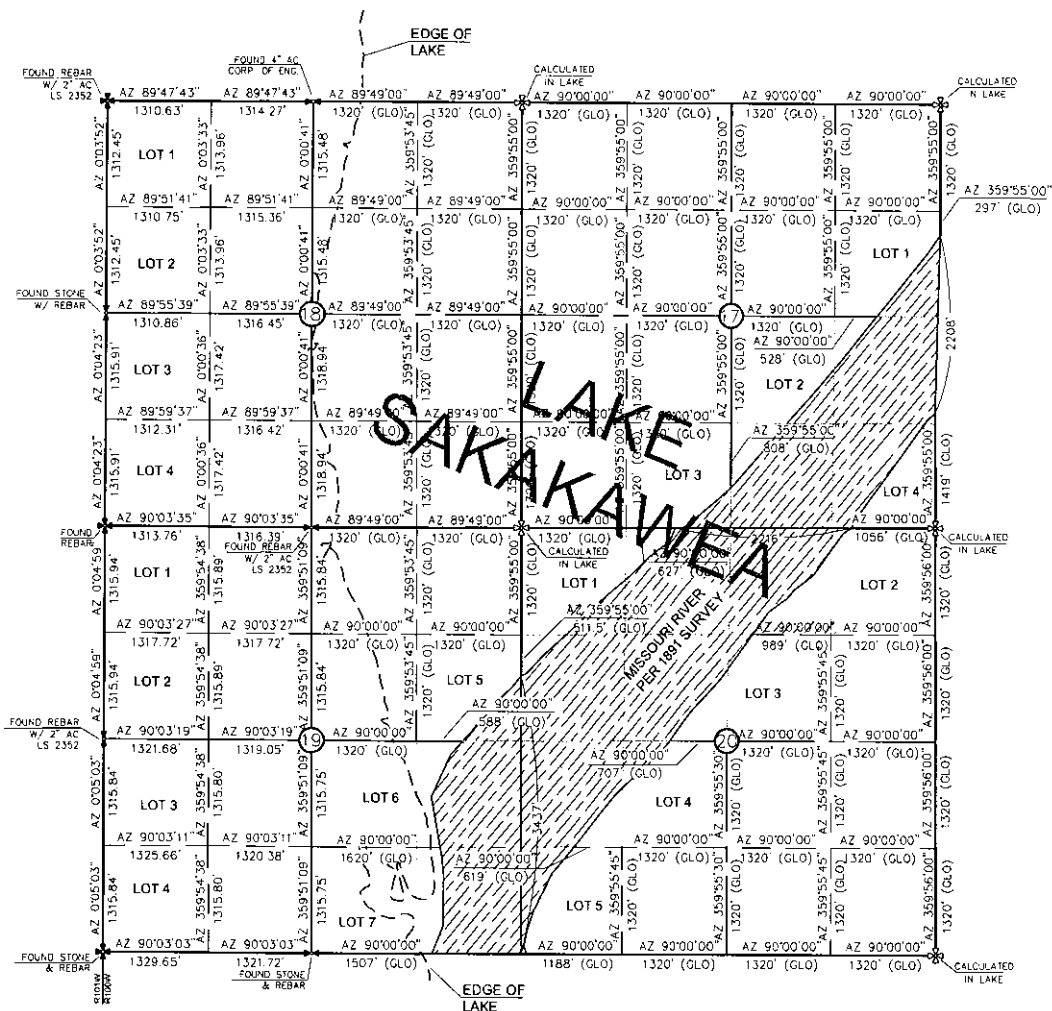
Other offices in Minnesota, North Dakota and South Dakota

OASIS PETROLEUM NORTH AMERICA, LLC
 WELL LOCATION PLAT
 SECTION 18, T153N, R100W
 MCKENZIE COUNTY, NORTH DAKOTA

Drawn By: B.H.H. Project No.: 514-09-001
 Checked By: B.H.H. Date: APRIL 2014

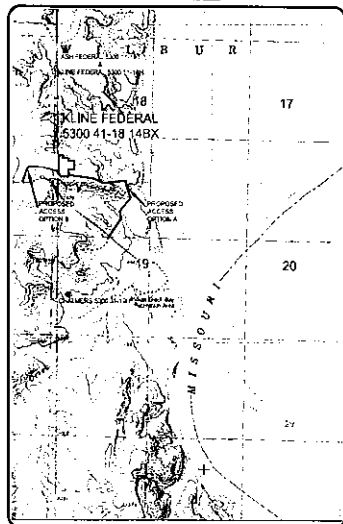
Revision	Date	By	Description

SECTION BREAKDOWN
OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
"KLINE FEDERAL 5300 41-18 148X"
368 FEET FROM SOUTH LINE AND 237 FEET FROM WEST LINE
SECTIONS 17, 18, 19, & 20, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



- MONUMENT - RECOVERED
 - MONUMENT - NOT RECOVERED

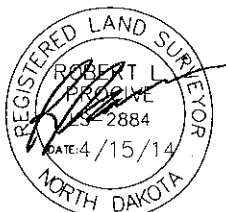
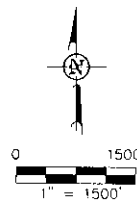
VICINITY MAP



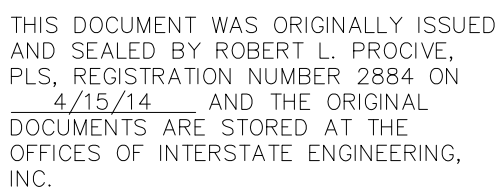
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ALL AZIMUTHS ARE BASED ON G.P.S.
 OBSERVATIONS. THE ORIGINAL SURVEY OF THIS
 AREA FOR THE GENERAL LAND OFFICE (G.L.O.)
 WAS 1891. THE CORNERS FOUND ARE AS
 INDICATED AND ALL OTHERS ARE COMPUTED FROM
 THOSE CORNERS FOUND AND BASED ON G.L.O.
 DATA. THE MAPPING ANGLE FOR THIS AREA IS
 APPROXIMATELY 0°03'.

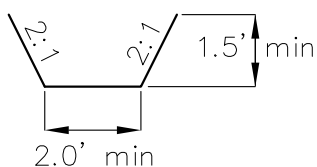
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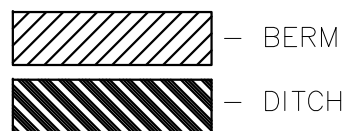
OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
"KLINE FEDERAL 5300 41-18 9T"
533 FEET FROM SOUTH LINE AND 237 FEET FROM WEST LINE
SECTION 18, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



NOTE: Pad dimensions shown are to usable area, the v-ditch and berm areas shall be built to the outside of the pad dimensions.

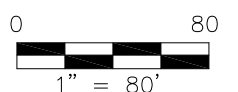
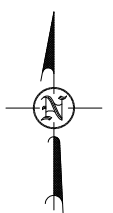


———— Proposed Contours
- - - - - Original Contours



NOTE: All utilities shown are preliminary only, a complete utilities location is recommended before construction.

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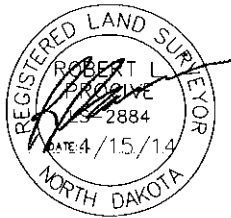
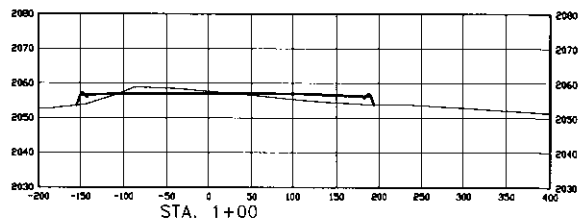
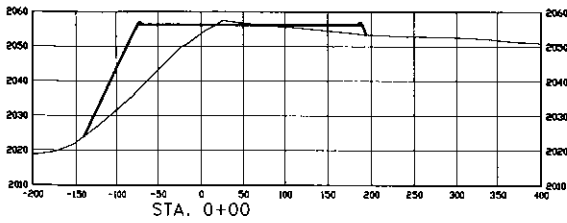
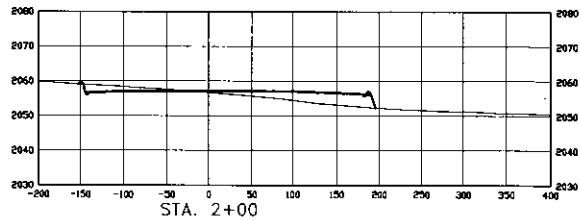
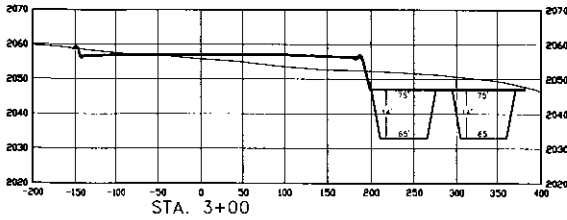
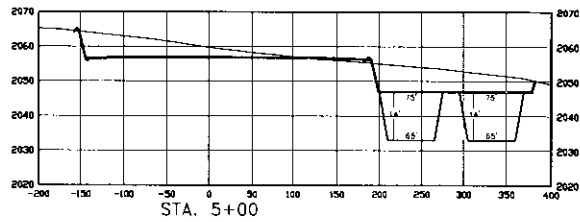
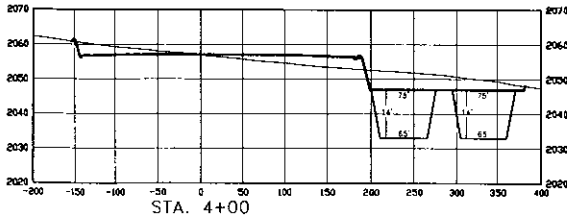
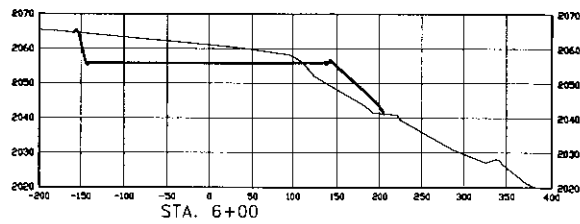
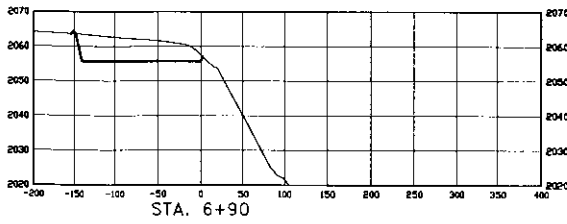
OASIS PETROLEUM NORTH AMERICA, LLC
PAD LAYOUT
SECTION 18, T153N, R100W
MCKENZIE COUNTY, NORTH DAKOTA

Drawn By: <u>B.H.H.</u>	Project No.: <u>S14-09-081.05</u>
Checked By: <u>D.D.K.</u>	Date: <u>APRIL 2014</u>

[illegible]

Q:\2014\14-09-081.05 Oasis Petroleum - Kline Federal 5300 41-18 91

CROSS SECTIONS
 OASIS PETROLEUM NORTH AMERICA, LLC
 1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
 "KLINE FEDERAL 5300 41-18 14BX"
 368 FEET FROM SOUTH LINE AND 237 FEET FROM WEST LINE
 SECTION 18, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



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SCALE
 HORIZ 1"=140'
 VERT 1"=35'

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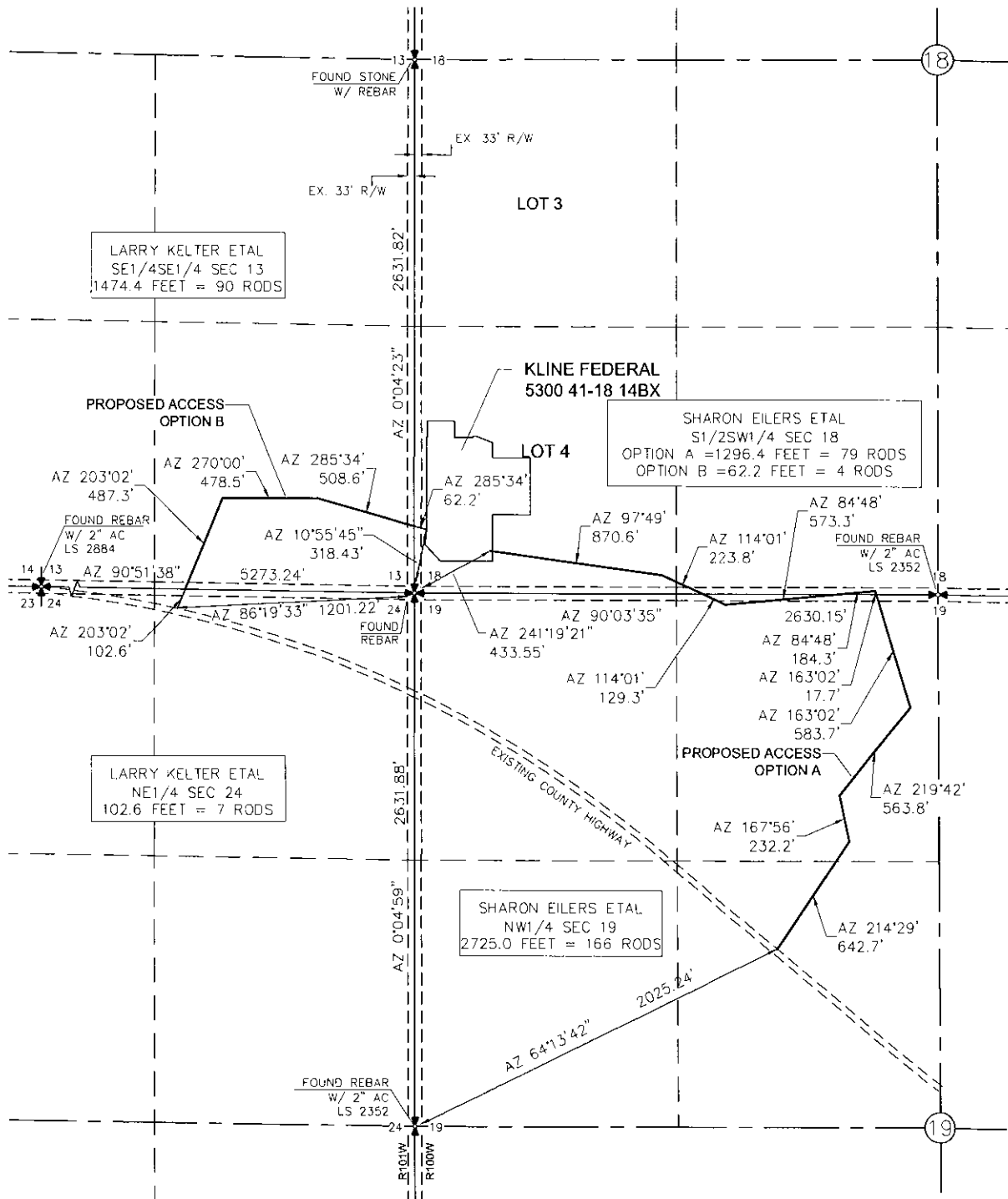


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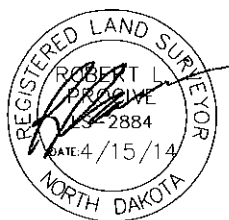
OASIS PETROLEUM NORTH AMERICA, LLC
 CROSS SECTIONS
 SECTION 18, T153N, R100W
 MCKENZIE COUNTY, NORTH DAKOTA
 Drawn By: B.J.M.
 Checked By: B.J.P.
 Project No.: 514-09081
 Date: APRIL 2014

Revised No.	Date	By	Description

ACCESS APPROACH
 OASIS PETROLEUM NORTH AMERICA, LLC
 1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
 "KLINE FEDERAL 5300 41-18 14BX"
 368 FEET FROM SOUTH LINE AND 237 FEET FROM WEST LINE
 SECTION 18, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA

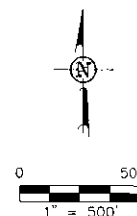


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OASIS PETROLEUM NORTH AMERICA, LLC
 ACCESS APPROACH
 SECTION 18, T153N, R100W
 MCKENZIE COUNTY, NORTH DAKOTA
 Drawn By: B.L.P.
 Checked By: B.L.P.
 Project No.: 514-09281
 Date: APRIL 2014

Revision No.	Date	By	Description

OASIS PETROLEUM NORTH AMERICA, LLC
 KLINE FEDERAL 5300 41-18 14BX
 368' FSL/237' FWL
 QUAD LOCATION MAP
 SECTION 18, T153N, R100W
 MCKENZIE COUNTY, NORTH DAKOTA

W I L L B U R

ASH FEDERAL 5300 11-181
 &
 KLINE FEDERAL 5300 11-18H

18
 KLINE FEDERAL
 5300 41-18 14BX

PROPOSED
 ACCESS
 OPTION B

PROPOSED
 ACCESS
 OPTION A

19
 CHALMERS 5300 31-19 Indian Creek Bay
 Recreation Area

MISSOURI

5/8



OASIS PETROLEUM NORTH AMERICA, LLC
QUAD LOCATION MAP
SECTION 18, T153N, R100W
MCKENZIE COUNTY, NORTH DAKOTA

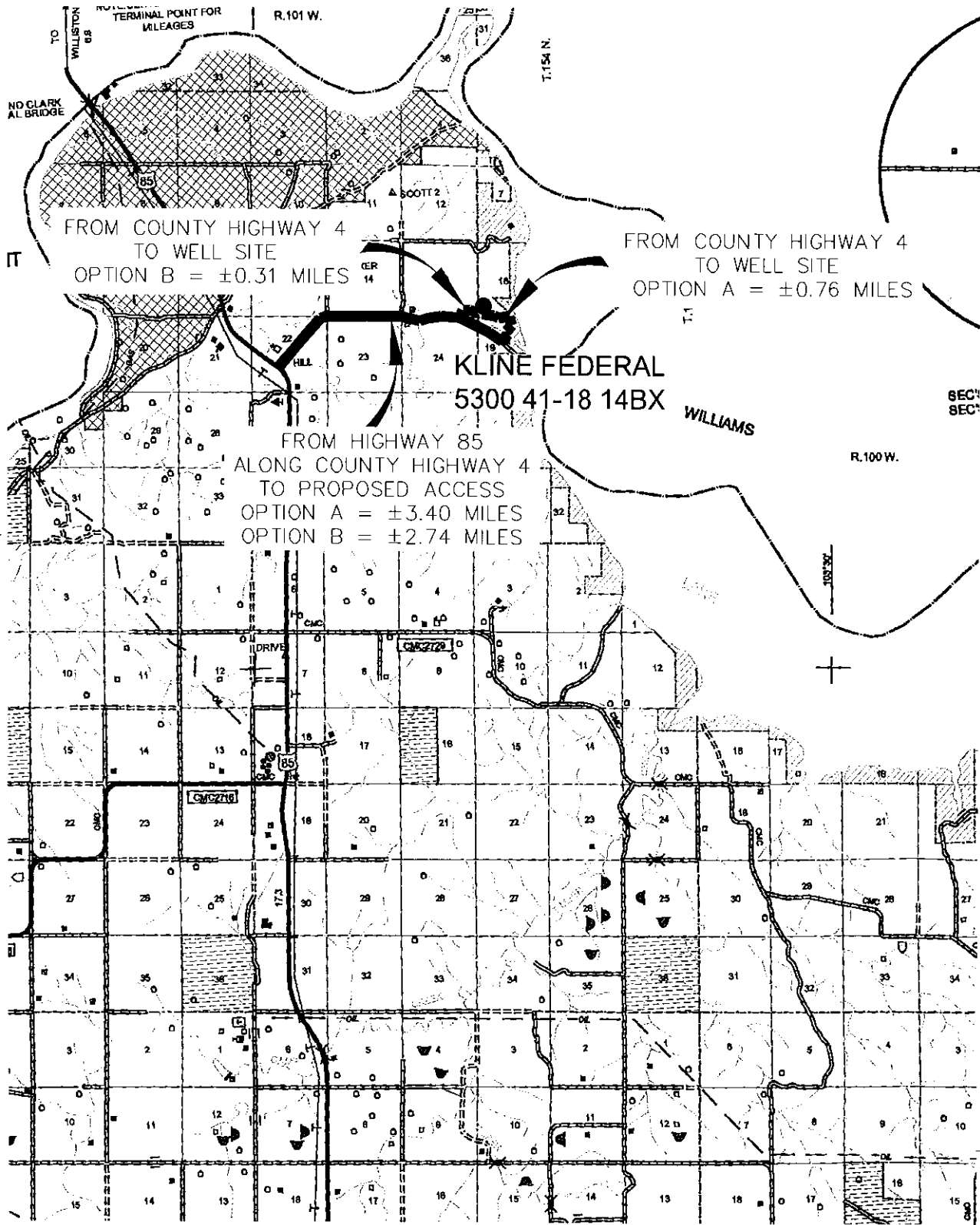
Drawn By: <u>B.H.H.</u>	Project No.: <u>S14-09-081</u>
Checked By: <u>R.L.P.</u>	Date: <u>APRIL 2014</u>

[illegible]

COUNTY ROAD MAP

OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
"KLINE FEDERAL 5300 41-18 14BX"

368 FEET FROM SOUTH LINE AND 237 FEET FROM WEST LINE
SECTION 18, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



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SCALE: 1" = 2 MILE

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OASIS PETROLEUM NORTH AMERICA, LLC
COUNTY ROAD MAP
SECTION 18, T153N, R100W
MCKENZIE COUNTY, NORTH DAKOTA

Drawn By: B.H.H. Project No.: S14-09-081
Checked By: R.L.P. Date: APRIL 2014

Revision No.	Date	By	Description

WELL LOCATION SITE QUANTITIES
 OASIS PETROLEUM NORTH AMERICA, LLC
 1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
 "KLINE FEDERAL 5300 41-18 14BX"
 368 FEET FROM SOUTH LINE AND 237 FEET FROM WEST LINE
 SECTION 18, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA

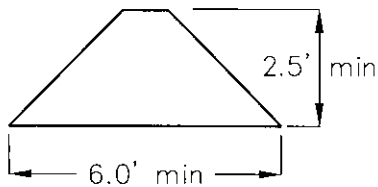
WELL SITE ELEVATION 2056.6
 WELL PAD ELEVATION 2057.0

EXCAVATION	20,682
PLUS PIT	18,900
	<u>39,582</u>
EMBANKMENT	11,538
PLUS SHRINKAGE (30%)	3,461
	<u>14,999</u>
STOCKPILE PIT	18,900
STOCKPILE TOP SOIL (6")	5,261
BERMS	1,889 LF = 612 CY
DITCHES	1,649 LF = 252 CY
DETENTION AREA	2,341 CY
STOCKPILE MATERIAL	2,403
DISTURBED AREA FROM PAD	6.52 ACRES

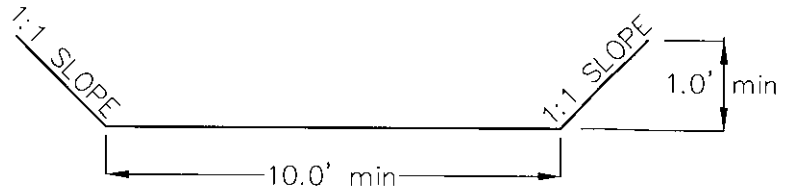
NOTE: ALL QUANTITIES ARE IN CUBIC YARDS (UNLESS NOTED)
 CUT END SLOPES AT 1:1
 FILL END SLOPES AT 1.5:1

WELL SITE LOCATION
 368' FSL
 237' FWL

BERM DETAIL



DITCH DETAIL



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OASIS PETROLEUM NORTH AMERICA, LLC
 QUANTITIES
 SECTION 18, T153N, R100W
 MCKENZIE COUNTY, NORTH DAKOTA
 Drawn By: B.H.H. Project No: S14-09-081
 Checked By: R.L.P. Date: APRIL 2014

Revision No.	Date	By	Description



5/29/2014

Mineral Resources Permit Manager
North Dakota Industrial Commission
600 East Boulevard Avenue Dept. 405
Bismarck, ND 58505-0840

RE: **Kline Federal 5300 41-18 9T**
 Kline Federal 5300 41-18 10B
 Kline Federal 5300 41-18 11T2
 Kline Federal 5300 41-18 12TX
 Kline Federal 5300 41-18 13T2X
 Kline Federal 5300 41-18 14BX
 Request for a legal street address

Dear NDIC:

Oasis Petroleum has requested a physical street address for the Kline Federal 5300 41-18 9T, Kline Federal 5300 41-18 10B, Kline Federal 5300 41-18 11T2, Kline Federal 5300 41-18 12TX, Kline Federal 5300 41-18 13T2X, and Kline Federal 5300 41-18 14BX. The request was made to Aaron Chisolm, in McKenzie County. Upon receiving a legal street address, Oasis will submit the address to the NDIC on a Sundry Notice (form 4) pursuant to 43-02-03-28.

Thank you for your consideration.

Respectfully,

A handwritten signature in black ink, appearing to read "Heather McCowan". The signature is fluid and cursive, with the first name and last name clearly distinguishable.

Heather McCowan
Regulatory Assistant
Oasis Petroleum North America, LLC

Surface Damage Affidavit

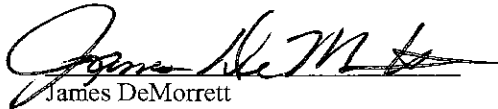
STATE OF NORTH DAKOTA)
) ss.
COUNTY OF WILLIAMS)

James DeMorrett, being duly sworn, states as follows:

The below described lands are not located in a wellhead protection/source water protection area per the website provided by the NDIC

and the adjacent landowner is fully aware that Oasis Petroleum will be drilling and operating multiple wells on spacing unit from a surface location in Section 18: Lot 4 a/k/a SW¼SW¼, Township 153 North, Range 100 West, 5th P.M., McKenzie County, North Dakota. (totaling 6.52 acres). The adjacent surface landowner is fully aware that the above referenced location will consist of the well bores and pumping units only and will not have cuttings pits, storage tanks or flare pits.

Dated this 16th day of June, 2014

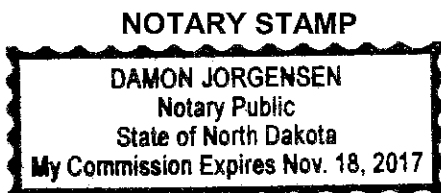

James DeMorrett

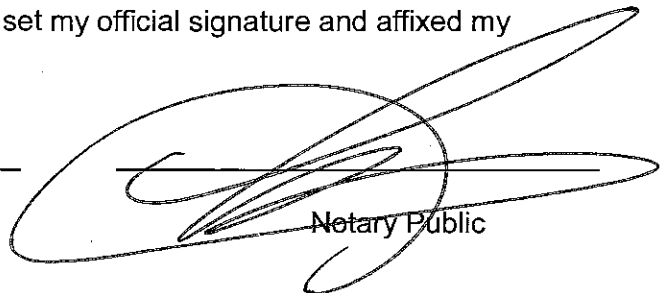
STATE OF NORTH DAKOTA }
 } ss.
COUNTY OF WILLIAMS }

BE IT REMEMBERED, That on this 16 day of June, 2014 before me, a Notary Public, in and for said County and State, personally appeared James DeMorrett, to me known to be the identical persons described in and who executed the within and foregoing instrument and acknowledged to me that he executed the same as his free and voluntary act and deed for the uses and purposes therein set forth.

IN WITNESS WHEREOF, I have hereunto set my official signature and affixed my notarial seal, the day and year first above written.

My commission expires: Nov 18, 2017




Notary Public